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Advancing interdisciplinarity in empirical translation research

Edited by

Tomáš Svoboda

Olga Nádvorníková

Věra Kloudová

Translation and Multilingual Natural Language Processing

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Translation and Multilingual Natural Language Processing

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Chapter 1

Introduction

Authorname

Abstract

Translation studies has become a thriving and autonomous research discipline. This development was fueled by the emergence of Descriptive Translation Studies in the 1980s and 1990s (Toury 1995), as well as the so-called turns that solidified the field's merit (Snell-Hornby 2006). The significant evolution of parallel and comparable corpora in the 1990s (Baker 1993, Johansson 1998) allowed for enhancing empirical research in the field and the rise of Corpus-Based Translation Studies (CBTS). The establishment of translation studies as a highly developed field is evident not only in the wealth of publications, but also in a vivid research community that regularly convenes at well-established conferences.[1] Among them, it is important to cite UCCTS (*Using Corpora in Contrastive and Translation Studies)*, reflecting the close connection between CBTS and contrastive linguistics (see Bernardini & Ferraresi 2023, Czulo & Hansen-Schirra 2017), and especially the *Translation in Transition* series, initiated in 2014 in Copenhagen, focusing on empirical approaches to translation studies. The idea for the present volume emerged from the sixth edition of this conference, held in Prague in September 2022.

As such, the remarkable evolution of translation studies is largely fueled by its inherently interdisciplinary nature (as noted by Granger & Lefer 2022), combining applied fields such as translator training, audiovisual translation, and machine translation with solid empirical methodologies, including corpora and psycholinguistic experiments. More importantly, this interdisciplinary approach is

[1]For an overview, see the Translation Studies Bibliography – https://benjamins.com/online/etsb; the BITRA database https://aplicacionesua.cpd.ua.es/tra_int/usu/buscar.asp?idioma=en; or journals such as *Meta, Perspectives,* and *Target*).

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grounded in broader linguistic frameworks, reflecting the socio-cultural and cognitive nature of translation, both its product and its process. Notable among these frameworks are cognitive linguistics and psycholinguistics (e.g., Halverson 2013, Carl 2021, and this volume), construction grammar (e.g., Czulo et al. 2023, etc.), systemic functional linguistics (e.g., Neumann et al. 2017), and variationist sociolinguistics (e.g., Prieels & de Sutter 2018). The two conference series exemplify this interdisciplinary potential, and the present volume seeks to highlight this aspect as well.

The publication consists of ten chapters: this introduction by the editors and nine separate contributions. In total, seventeen authors contributed to the volume. The core content originates from the 2022 *Translation in Transition* conference mentioned above, showcasing the methodological and interdisciplinary diversity in contemporary translation studies. This is complemented by the opening and closing contributions that were not presented at the conference but address two current trends of translation studies. The opening chapter explores the research potential of advanced statistical methods and data visualization in the analysis of translation data in multilingual corpora, which are inherently multidimensional (see also De Sutter & Lefer 2019). The closing chapter addresses the growing field of interpreting studies, particularly in the context of the technological turn (Jimenez-Crespo 2020), highlighting the paradigm shift brought about by ubiquitous machine translation and enabled by evolving translation technologies.

1 General overview

Across all chapters of the volume, interdisciplinarity is clearly a common thread. Readers are taken on a journey from a methodology focus (Chapter 2) through corpus-based translation studies (Chapters 3–5), corpus-based contrastive linguistics (Chapters 6 and 7), audiovisual translation (Chapter 8) and specialized translation (nanomedicine in Chapter 5; legal translation in Chapter 7) to machine translation (Chapters 9 and 10). In Chapter 4, a cognitive approach to translation appears, adding a process-oriented viewpoint on translation to the pool.

Another unifying factor of all chapters is the coverage of novel topics, approaches, and methods. Each author brings a fresh perspective to their respective fields of study. At the same time, the novelty in each case is expressed differently. Some chapters, such as Chapter 1, introduce innovative methods in translation research; others offer new approaches in terms of thematic focus. Chapters 8 and 9 cover a novel theme in the study of audiovisual translation and machine translation, respectively. The overlapping interest in translation studies among the

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chapters goes hand in hand with a shared concern about how translation theory intersects with practical application (see particularly Chapters 4, 5, and 7, which focus on translator training).

From the point of view of methodology, chapters dealing specifically with contrastive analysis (Chapters 2, 6, 7, and 8) present qualitative assessments of translation and interpreting. Chapters 5, 6, 8, and 10 predominantly employ a quantitative approach to analyze linguistic data patterns and link those to translation research.

Beyond that, the volume covers a wide range of languages, including Arabic, Czech, English, French, German, Italian, Simplified Chinese, and Spanish – with a variety of pairings and directions of the languages involved.

2 Outlines of individual chapters

The first chapter, written by Volker Gast and Vahram Atayan, deals with data visualization in translation studies as well as in corpus-based contrastive linguistics. Thus, it primarily represents a methodology-focused contribution. The authors begin by noting that the increasing availability of large multilingual corpora has created new opportunities for empirical research in these fields. They point out, however, that these "possibilities come with challenges" and that researchers need to filter relevant information and interpret complex data extracted from parallel (translation) corpora, in which the linguistic choices may be influenced by the input from the source language. This chapter aims to address these challenges by presenting and evaluating various visualization techniques.

After an overview of the importance of data visualization in understanding comparative linguistic data, the authors categorize comparative corpus data into three types: language-specific data, undirected comparative data, and directed comparative data. The chapter discusses three main visualization techniques: proportional visualizations, graph-based visualizations, and ordination techniques. Each technique is illustrated with concrete examples based on the authors' previous research into so-called adverbials of immediate posteriority (IP-adverbials) in French and German. The authors also discuss the strengths and limitations of each technique and provide scripts (in R and Python) for readers to use in their own research.

One of the innovative aspects of the chapter is the introduction of "context-conditional translation graphs," a new method for visualizing translation choices under specific contextual conditions. Arguably, this technique, along with Bayesian networks for modeling bilingual language production, represents a significant advancement in the field.

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The second chapter, "Navigating translation effort through translation process research database to unravel dimensions of script and text direction" by Ali Saeedi, Longhui Zou, and Michael Carl, investigates the impact of script and text direction differences on the translation process, using eye-tracking and keylogging data to measure the technical, temporal, and cognitive effort exerted by translators. By analysing translations from English into Spanish, Simplified Chinese, and Arabic, the study reveals significant challenges posed by different scripts and text directions. The findings show that while script differences increase orthographic, lexical, and segment-level complexities, text-direction differences increase spatial orientation demands. Interestingly, no interaction effect between script and text-direction differences was observed, suggesting distinct cognitive processes for each challenge.

The study uses data from the CRITT Translation Process Research Database, focusing on from-scratch translation sessions. Indicators such as insertion, deletion, pause-to-word ratio, total number of fixations, and reading times are examined to gauge translation effort. The results indicate that translating between languages with similar scripts (English into Spanish) requires less cognitive effort than translating between languages with different scripts (English into Chinese and Arabic). Similarly, the temporal effort is higher for different-script translations, reflecting the additional cognitive load.

The findings of this study have significant implications for translation pedagogy, advocating for tailored training approaches that address script and text direction challenges separately. The study emphasizes the need for specific strategies to navigate these difficulties, such as exercises focusing on visual-spatial processing and spatial attention adaptation. By highlighting the nuanced cognitive demands of translation across different scripts and text directions, this chapter contributes valuable empirical insights and a deeper understanding of the translation process.

The third chapter, "A corpus-driven investigation of translation strategies for prenominal complex adjectives," by Rossella Resi explores the translation of prenominal complex adjectives (PCAs) from German to Italian within a tourism-focused parallel corpus. Initially, German PCAs are categorized into seven groups: predicative structures with past and present participles, compound adjectives, multiple non-coordinated prenominal adjectives, adjectives modified by adverbs, adjectives modified by complex structures, and infinitive structures. The study uses regression analysis to assess the frequency of translation strategies, finding a strong correlation between PCA categories and translation strategies, with a notable tendency to preserve the original syntactic structures.

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The chapter's qualitative analysis focuses on the more transformational translation strategy of transposition, which involves changing grammatical categories without altering the meaning. Results indicate that the cognitive complexity inherent in German PCAs can encourage or trigger the use of transposition. Transposition, though less commonly applied, seems particularly effective for translating complex syntactic structures due to its ability to handle bidirectional syntactic reference and elevate semantic information within sentences.

The findings suggest that while preserving original syntactic structures is a common strategy, transposition offers a viable alternative when dealing with intricate adjectival constructions. This highlights the importance of understanding the cognitive and syntactic challenges involved in translating PCAs and suggests that translation strategies should be adaptable to the specific complexities of the source text. Overall, the chapter provides valuable insights into the translation of complex adjectival structures and the application of different strategies to maintain meaning and coherence in the target language text.

Chapter four, "*Schodiště z mramoru* and 'the Marble Staircase': English divergent counterparts of Czech postmodifying prepositional phrases," is authored by Markéta Malá. This chapter focuses on noun-phrase modification, similar to the preceding chapter, but adopts a markedly different approach. In fact, large corpus-based studies have previously indicated that the majority of Czech postmodifying prepositional phrases, such as *sochy z bílého mramoru* [statues of white marble], have a translation counterpart in English that employs a prepositional phrase as well (Klégr & Malá 2009). Using Johansson's terminology (Johansson 2007), widely adopted in contrastive linguistics, Malá thus asserts that most counterparts are *congruent* rather than *divergent*. This aligns with observations made about English, where postmodifying prepositional phrases represent the most common type of noun modification (Biber et al. 2021). Malá's methodological innovation thus lies in her focus on the less frequent divergent counterparts.

The linguistic material used in this study is derived from a large multilingual corpus named InterCorp, which is limited to fiction (novels and short stories) and non-fiction in the Czech-English language pair. Adhering to the methodological requirements of translation studies, Malá meticulously considers the direction of translation in her analyses: she separately examines Czech postmodifying prepositional phrases in original Czech texts compared to their English translations (CZ-EN) and in Czech translations from English source texts (EN-CZ). A two-step analysis enables Malá to derive significant insights into the impact of register (text type) and direction of translation. However, the primary contribution of her study is the fine-grained formal and semantic typology of divergent English

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counterparts to the Czech postmodifying prepositional phrases. Her wealth of examples pave the way for future research in both contrastive linguistics and translation studies.

Chapter five, authored by Cécile Cuchet and Pascaline Dury, explores short diachronic corpora as a tool while translating emerging and new specialized fields, particularly in medical translation. It argues that diachronic corpora – texts collected over a few decades – can provide translators with valuable conceptual and terminological information to understand the evolution of terms in rapidly developing scientific fields, such as nanomedicine for cancer treatment. The authors performed a study using a corpus of scientific texts published in this domain from 1993 to 2017. The study aimed to identify useful diachronic information, such as term variations and the emergence of new terms, to aid translators in understanding complex, specialized terminology that is still evolving. The research highlights the challenges translators face when dealing with unstable terminology in emerging fields, where traditional lexicographic resources may prove insufficient.

In terms of methodology, the study emphasizes simplicity in corpus analysis to ensure that the approach is accessible to practicing translators, who may lack the time or technical expertise for complex analyses. The findings suggest that diachronic information can provide translators with a "minimal cognitive knowledge" necessary for translating new scientific concepts.

This approach holds potential for applications in translator training and other specialized fields where understanding the historic evolution of terminology is crucial. It also opens new avenues for research in translation studies, particularly in how translators can use diachronic data to improve translation accuracy in rapidly changing scientific domains.

Chapter six, "Legal language in employment contracts and offer letters" by Patrizia Giampieri, investigates the differences in legal language and style between employment contracts and employment offers. Through constructing and analysing two specific corpora — one for contracts and the other for offer letters — the study aims to determine the extent to which these documents employ legalese or plain English. Legalese, characterized by complex prepositions, passive constructions, and polysemy, is compared to plain English, which strives for simplicity and accessibility. Findings suggest that employment offer letters use plain English more frequently, whereas employment contracts predominantly feature legalese, indicating different levels of formality and drafting styles.

The chapter covers the intricacy of legal language and the efforts by the Plain English movement to simplify the former. It highlights the use of corpora in legal

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language research, noting the challenges of accessing private legal documents and the necessity of DIY corpora for specific studies.

From the point of view of its methodology, the chapter describes the manual creation of corpora and the use of AntConc software for detailed linguistic analysis. Results show significant differences in legalese patterns between the two document types, such as the use of 'shall,' passive forms, and deictic elements.

There are considerable implications for translation studies, translator practice as well as training in that the research demonstrates how corpus-driven approaches can inform translation practices by highlighting language nuances and patterns. The chapter advocates for the application of rigorous corpus-based methods across disciplines to enhance understanding and translation of legal texts. Future research directions include exploring legal language in non-native contexts and analyzing other types of legal documents.

The seventh chapter is entitled "Translating Rick & Morty's audiovisual novum parody: strategies and effects" and its author is Alice Ray. The chapter studies the popular genre of audio-visual translation, specifically the postmodern science fiction TV show originally appearing in English. The research analyses the series' neologisms together with their counterparts in the French dubbed version. It considers the multi-faceted constraints of audiovisual translation, one of which is the relationship between text and image, focusing on parody and inherent humor in the combination of textual elements and visual content. The research applies a hybrid model involving contrastive linguistic methods and translation studies approaches.

The results of the analysis allow the reader to gain a better understanding of the challenges involved in translating what is referred to as 'novum parody' – the fictive terms that parody the sci-fi genre, resulting in humorous coinages. As a result of the complexity of the task, translators often have to prioritize some aspects of the audiovisual message contained in the invented terms over other aspects. Research findings involve the observation that translation of novum parody must be studied with a view to both narrative and visual contexts.

The analysis model put forward by the author is subsequently validated by way of analysing *Futurama*, yet another audiovisual work. The chapter is a valuable addition to the volume in terms of both methodology and field as it focuses on a niche topic, namely that of humor in sci-fi TV series. Apart from that, it presents a customized analytical model that can become useful in similar future research.

The final two chapters present research drawn from a similar triad of subjects, as they both compare the respective performance of professionals, trainees, and NMT systems. Whereas Kamenická's chapter (eight) focuses on translation,

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Kloudová et al. (Chapter nine) focus on both interpreting and real-time MT of speech transcripts.

Kamenická's chapter, entitled "'Default translation' – how much free of fault? An exploratory study of NMT and human translation comparison," compares human and machine translation output. It looks into cognitive modeling of translation and, at the same time, assesses neural machine translation (NMT) performance in terms of how successful the latter is at imitating human translation. Crucially, the chapter employs and adapts Halverson's concept of default translation as a relatively routinized and uninterrupted mode of translating – a concept touching upon the widely used notion of translation universals.

As for methodology, four NMT engines' performance is compared to that of six translation trainees and six professional translators on texts in the English-Czech language pair. Translations are compared in both quantitative terms – using a custom metric – and qualitative terms as well.

From among the MT engines, DeepL was found to approximate human translation most successfully at the level of collocations. The chapter also puts forward a hypothesis that concerns the division of attention and effort of human translators. It stipulates that in their default mode – in order to avoid literalness – translators often favor translation solutions that are more readily available in terms of cognitive load, yet less literal than the ones considered to be optimal.

The main goal of the last chapter is well summed up in its title: "A comparison of human simultaneous interpreting and automatic simultaneous speech translation as a basis for further research." The four authors, Věra Kloudová, Petra Mračková-Vavroušová, Dávid Javorský, and Ondřej Bojar, summarize the complexities of simultaneous interpreting (SI) before covering automatic simultaneous speech translation (ASST), before comparing these two services.

The study analyzes translation departures in SI on the one hand and ASST systems on the other. For SI, three interpreters were recruited; for the translation system, ELITR – a custom speech translation system – was used. Two speeches in English were analyzed as source texts, the target language being Czech in both cases. For the purpose of analysing content quality, a hybrid error-based QA model was implemented.

The results show that the two trainee interpreters surprisingly outperformed a professional one. Compared to the interpreters the ASST model translated in a more literal fashion and its output involved fewer omissions and other translation departures than human interpreted renderings.

The challenges of the research involved the fact that two different services are compared here (interpreting activity results in a spoken output, whereas the output of ASST is a translated transcription of the original), and the focus on

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content in the comparison inevitably leads to partial results. The evaluation of omissions proved to be a challenge, too, knowing that in high-quality SI they are part of a deliberate strategy to produce a user-friendly rendering. The reflection of these research limitations represents a valuable methodological aspect of the chapter in its own right. With its well-resourced summary of evaluating interpreting quality, its methodological focus, and its novel thematic coverage, the chapter complements the other approaches well in that it provides a foundation for further research.

3 Research limitations and potentials

The research limitations of this volume logically stem from its greatest asset – interdisciplinarity. Tackling a great many areas often sacrifices depth for breadth. We believe that we have struck the right balance between both. Another challenge of the volume's agenda has been the dynamic nature of some areas covered, such as emerging methodologies and the development of machine translation engines. We acknowledge this and feel the volume may pave the way for further research in the respective sub-fields.

Current developments in translation studies are brimming with possibilities. The vitality of the field is evident in the diversity of topics covered in the forthcoming *International Encyclopedia of Language and Linguistics* (Nesi & Milin, forthcoming), which spans 78 different areas, across domains such as localization, translation policy, collaborative translation, cognitive aspects in translation, and neurolinguistics in translation studies. New perspectives are also emerging through innovative types of corpora, such as the MUST corpus (Granger & Lefer 2020), which offers a wealth of metadata for translation studies research, and multilingual translation corpora enriched with specific linguistic information – for example, the InterCorp multilingual corpus includes Universal Dependencies annotation and syntactic complexity measures (Rosen et al. 2024). However, as highlighted during the round table discussion at the Translation in Transition conference in Prague in 2022, the future of translation studies will largely depend on researchers' ability to comprehend the theoretical grounding of the domain, allowing them a better understanding of the nature of translation and the sociocultural, technological, and cognitive factors that shape it. We hope to have added yet another example of the interdisciplinary potential of translation studies and provided stimuli in search for its common ground in the future. It may well be that this common ground will involve a rediscovery of some grand and ambitious attempts at charting the territory the founding figures of the discipline,

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such as Jiří Levý. In 1965, in respect to literary theory and literary history he had the following to say:

> Currently, particularly favourable conditions have emerged for the refinement of literary scholarship, since most of the adjacent social sciences have come within the reach of exact methods. … The sciences that achieve far more exact and indisputable results [compared to statements of some literary historians], formulate their straightforward conclusions from a certain set of observations, based on which they formulate nothing but further hypotheses, in a series of increasingly adequate ones. (1965: 11, 18, our translation)

Interdisciplinarity and respect for exact methods seem to be two of the best ways forward in our "interdiscipline".

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Chapter 2

Data visualization in corpus-based contrastive linguistics and translation and interpreting studies

Volker Gast[a], Vahram Atayan[b] & Martin Schwingenheuer[c]
[a]Friedrich Schiller University Jena [b]Heidelberg University [c]FOSBOS Rosenheim

While inferential statistics are indispensable in comparative corpus studies such as those conducted in contrastive linguistics (CL) and translation and interpreting studies (TIS), it is equally important for researchers to develop an intuitive understanding of the data. Visualizations serve as a key tool for this purpose. In this contribution we distinguish three types of comparative corpus data commonly encountered by contrastive linguists and translation scholars: language-specific data, undirected comparative data, and directed comparative data. We discuss three main visualization techniques: proportional visualizations, graph-based visualizations and ordination. We explore the strengths and limitations of these approaches, point out some caveats and provide scripts (in R and Python) to allow readers to replicate the methods.

1 Introduction

The availability of large amounts of multilingual corpus data has opened up new possibilities for empirical approaches to contrastive linguistics and translation and interpreting studies. For example, we can empirically determine the contexts in which comparable linguistic expressions from different languages are used (e.g. Atayan 2021b, Gast 2015, 2017a,b, 2019, 2022), and we can investigate the conditions under which specific expressions are translated in a specific way, (e.g. Marco 2018, Crible et al. 2019, Defrancq & Magnifico 2020, Defrancq 2021,

Volker Gast, Vahram Atayan & Martin Schwingenheuer. 2026. Data visualization in corpus-based contrastive linguistics and translation and interpreting studies. In Tomáš Svoboda, Olga Nádvorníková & Věra Kloudová (eds.), *Advancing interdisciplinarity in empirical translation research*, 11–54. Berlin: Language Science Press. DOI: ??

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Atayan 2021a, Yung et al. 2023). These possibilities come with new challenges, however. From the perspective of the researcher, it is sometimes hard to filter out more relevant from less relevant information, to understand the interplay of the variables used in a study, and to interpret the results in a theoretically meaningful way, beyond statements of the type "variable *x* has a significant effect on variable *y*".

While inferential statistics are indispensable in the comparative analysis of corpora, it is important for researchers (and readers) to also develop an intuitive understanding of the data (cf. also Gries et al. 2020). Visualizations are a useful tool for this purpose. In this chapter we discuss the use of specific tools for visualizing bilingual data commonly encountered in contrastive linguistics and translation studies (cf. also Sönning & Schützler 2023 on the visualization of corpus data in general).

An obvious particularity of contrastive studies based on translation corpora is the relational nature of the data, which often consists of pairs of an original sentences and its translation or interpretation. While the original sentence is an instance of monolingual production, the translated sentence is determined both by the co(n)text[1] and the input from the source language. We thus distinguish between three ways of analyzing frequency distributions. First, before carrying out a comparative study we need to understand language-specific distributions, using common tools of corpus linguistics; second, we can compare distributions from different languages in a symmetric way ('undirected comparison'); and third, we can study the choices made in one language under the influence of another language, e.g. translations of a given source language expression in a target language ('directed comparison').

For illustration we use data analysed in previous studies by us (Atayan et al. 2019, Gast & Atayan 2019, Gast et al. 2019, Atayan 2021b,a) on what we call adverbials of 'immediate' or 'proximal posteriority'. These adverbials are introduced in Section 2, where we also provide a brief description of the data. In Section 3, we introduce techniques commonly used for categorical (count) data in monolingual studies, presenting some extensions for higher-dimensional data. These techniques can also be used for undirected comparative data, as is illustrated in Section 4. In Section 5, we demonstrate ways of visualizing translation data. Sankey diagrams are useful for this purpose, but have certain limitations. We therefore propose a new method, 'context-conditional translation graphs', which

[1]'Cotext' is often used for the textual surrounding of a sentence and 'context' for the extralinguistic circumstances of production. We will use 'context' as a cover term for 'cotext' and 'context' narrowly defined in the following.

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visualize translation choices made under specific contextual conditions. Section 5 moreover contains some comments on Bayesian networks, which can be used to factorize joint probability distributions, and to visualize these factorizations, enabling researchers to test models of bilingual language production. Section 6 contains the conclusions.

2 Background on the domain of inquiry and the data

2.1 Adverbials of immediate posteriority in French and German

In earlier work we studied ways of expressing 'immediate posteriority' in several languages, including adverbials such as Engl. *immediately*, Germ. *sofort* and *gleich*, Fr. *immédiatement* and *tout de suite* (Atayan et al. 2019, Gast & Atayan 2019, Gast et al. 2019, Atayan 2021a,b). We will use the abbreviation 'ImPost-adverbial' in the following. ImPost-adverbials locate an event e relative to some reference point R in such a way that e follows R without a relevant temporal gap. For example, in (1) Jane's arriving at home (R) and her going to bed (e) are temporally adjacent, which is here expressed by *immediately*.

(1) When $[_R$ Jane arrived at home$]$ $[_e$ she *immediately* went to bed$]$.

Adverbials like *soon* (cf. (2)), *bientôt* (French) and *bald* (German) are related to ImPost-adverbials insofar as their use implies a relatively short interval between events or points in time, but the interval is larger than in the case of ImPost-adverbials.

(2) *Soon* after $[_R$ she arrived$]$, $[_e$ Jane went to bed$]$.

We will refer to the class of adverbials covering both those of immediate posteriority and *soon*/*bald*/*bientôt* as 'adverbials of proximal posteriority', or ProxPost-adverbials for short. ImPost-adverbials are regarded as a special case of ProxPost-adverbials.[2]

Expressions of proximal posteriority tend to be context-sensitive, insofar as their use is (partially) determined by specific properties of the context. In a (simplifying) model of language production we can assume that there is a given function f that a speaker wants to express – say, immediate posteriority – in a context

[2]Note that arguably, there are also differences in the lengths of the intervals between, for instance, Germ. *sofort* (no or very short interval) and *gleich* (somewhat longer interval). We disregard these differences in the following, as they are not of primary interest to the questions addressed in our projects.

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C with context variables $\{C_1 \ldots C_m\}$, each of which comprises a certain number of levels (e.g. 'directive', 'commissive' and 'representative' for the variable ILLOCUTIONARY FORCE). Moreover, there is a set of expression candidates $W = \{w_1 \ldots w_n\}$ which are compatible with function f. One aspect of use that we are interested in is the probablities with which the candidates $W = \{w_1 \ldots w_n\}$ are used in context C to express function f. These probabilities can be regarded as conditional probabilities of the form $p(w|f,c)$: the probability of using expression w, given that we want to express the function f in context c.

To illustrate some relevant context parameters determining the distribution of ImPost-adverbials, let us consider some examples. ImPost-adverbials may be used in utterances with representative illocutionary force, as in (1) above, or with directive illocutionary force, as in (3) below; they can be used anaphorically, with another event e as a reference point (cf. (1)), or deictically, with the moment of speech as a reference point (cf. (3)); and they can be used with intentional (controlled) actions, as in (1) and (3), or with unintentional/spontaneous events, as in (4).

(3) [$_e$ Get out] *immediately*! (directive, deictic, intentional; implicit R)

(4) [$_e$ The house of cards collapsed] *immediately* when [$_R$ I touched it]. (representative, anaphoric, spontaneous)

ImPost-adverbials often exhibit particularities in their distribution amounting to interactions between their distributional variables. For example, Engl. *immediately*, while being compatible with both deictic reference points (as in (3)) and unintentional actions (as in (4)), is not normally used when these two context parameters co-occur, as in (5a), in a reading that is both spontaneous and deictic. This sentence is not normally understood as a warning about an immanent, spontaneous collapse, in which case we might utter something like (5b); rather, (5a) suggests that the collapse will follow some other event triggering it, as in (4) above and (5c) below.

(5) a. Be careful, it will collapse *immediately*!
    b. Be careful, it will collapse *any moment*! (deictic)
    c. Be careful, it will collapse *immediately* if you touch it! (anaphoric)

In translation and interpreting, there are, obviously, additional factors determining the probability of a specific form being used to express some function f in a

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given context c. First, it is well-known that translated (or, more generally speaking, mediated) language production differs systematically from monolingual language production (cf. for instance the well-known translation universals investigated in the tradition of Baker 1993, 1995). Second, there are interference effects, i.e., the choice of a target language form is partially determined by the form found in the input (often leading to "shining through" effects, cf. Teich 2003). There is, thus, an additional variable to be considered. The probability of using a given expressions is conditioned by both the context type $c_i \in C$ and the source language expression $s_j \in S$. We are thus dealing with a conditional probability of the form $p(w|f, c, s)$: the probability of using expression w, given that we want to express function f in context c, with expression s in the source text. For example, if we translate (5c) into German, we can determine probabilities for using *gleich* or *sofort* to express immediate posteriority in the specific context, given that the English original contains *immediately*. This probability may differ if the source sentence contained a different adverbial with a similar meaning, e.g. *right away*.

2.2 The data

For illustration we use two data sets, both of them from the Europarl corpus (Koehn 2005, Cartoni & Meyer 2012, Cartoni et al. 2013). The first data set was created in the context of a joint project on ImPost-adverbials carried out at the Universities of Heidelberg and Jena. The main results of this project have been published in Atayan et al. (2019), Gast et al. (2019), Gast & Atayan (2019), Atayan (2021a) and Atayan (2021b), to which the reader is refered for more information (e.g. concerning the annotation procedure). The data used for this chapter is based on two originally distinct data sets: (i) 508 annotated observations from German and French in which the ImPost-adverbials *gleich*, *sofort*, *immédiatement* and *tout de suite* occur either in originals or in translations, annotated by a team of annotators as mentioned in the aforementioned publications; and (ii) 100 observations each of the two adverbials of proximal posteriority *bientôt* (French) and *bald* (German), annotated by V. Atayan. The data as well as all R- and Python-scripts used for the analyses and visualizations are available in the OSF-repository https://osf.io/nufpv/.

The data were annotated for the following variables (the specific levels of the variables will be briefly explained when examples are discussed):

- Adverbial in original and translation
- French and German tense and mood
- Illocutionary force/modality

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- Presence of a syntactic modifier
- Intentionality of the predicate
- Agent of the main predicate
- Reference point

3 Visualizing language-specific distributions

Even in a contrastive study, the first step consists in the inspection of the language-specific distributions. We mostly deal with four such distributions if we study two languages L_1 and L_2: L_1/original, L_1/translated (from L_2), L_2/original, and L_2/translated (from L_1). By inspecting these distributions we can gain a first impression of the differences between L_1 and L_2, as well as between the original and translated versions of both languages.

In this section we consider three types of visualization techniques, summarized under 'proportional geometric visualizations' (Sect. 3.1), 'graph-based visualizations' (Sect. 3.2), and 'ordination' (Sect. 3.3).

3.1 Proportional geometric visualizations

Among the techniques most commonly used for the visualization of monolingual data are what we can loosely call 'proportional plots', where frequencies observed in data are graphically represented in such a way that the size of some geometric shape is in some way proportional to the frequencies. The most widely used plot types are probably mosaic plots and Cohen-Friendly association plots (Cohen 1980, Friendly 1982). We will focus on mosaic plots, using the `vcd` package for R.[3] Consider Figure 1. The contingency table at the top shows how frequently the French ProxPost-adverbials *immédiatement*, *tout de suite* and *bientôt* are used with chronological or deictic reference points (i.e., with reference points in the surrounding text, as in (1), or with the moment of speech as the reference point, as in (3)). Each cell in the table corresponds to a combination of levels from the two variables ProxPost-adverbial and Reference Point. The rectangles in the mosaic plot at the botttom are proportional in size to the frequencies shown in the table at the top along both dimensions. Along the x-axis their length corresponds to the proportion of the relevant level within the category French ProxPost-adverbial, and along the y-axis the rectangles reflect the proportion of deictic and chronological observations. The mosaic plot shows, among other things,

[3] Mosaic plots can also be generated with the Python module `statsmodels`.

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	immédiatement	tout de suite	bientôt
chronological	32	15	3
deictic	29	41	97

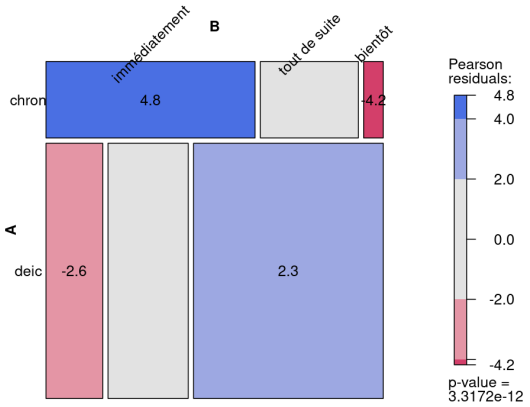


Figure 1: Contingency table (top) and mosaic plot (bottom) representing the frequencies as well as degrees of over- or underrepresentation for three French ProxPost-adverbial-adverbials relative to the variable REFERENCE POINT (*chron*: chronological, *deic*: deictic). The plot was created with the package `vcd` for `R` (Zeileis et al. 2007); cf. the script `01_proportional_visualizations.R` in the Supplementary Materials.

that the cells corresponding to chronological/*immédiatement* (top left) and deictic/*immédiatement* (bottom left), while exhibiting rather similar counts (32 and 29), exhibit very different relative frequencies in the rows for chronological and deictic cases.

Figure 1 also provides information about levels of deviation from statistical independence (a homogeneous distribution). The colour of a rectangle corresponding to a cell $C_{m,n}$ of a contingency table C reflects the magnitude of the Pearson residuals $E_{m,n}$ for $C_{m,n}$ (cf. Agresti 2002: Ch. 3 for details). Pearson residuals can be regarded as a standardized measure of deviation from statistical independence. They are calculated as the difference between the observed and expected frequency, divided by the square root of the expected frequency (see Sect. A.1 in the Appendix for some explanations). Pearson residuals approximately follow a

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standard normal distribution (but see below for a caveat), which means that we can estimate the probability of a cell count, given the assumption of statistical independence (homogeneity): ≈ 68% of the data are located within one standard deviation of the distribution, and ≈ 95% are located within two standard distributions. As we are typically interested in results beyond the 95% threshold ($\alpha = 0.05$, for an undirected hypothesis), the thresholds 2 (for overrepresentation) and −2 for underrepresentation are particularly relevant.

As the legend to the right of the plot shows, the colours indicate the following magnitudes: light blue: $E_{m,n} > 2$, dark blue: $E_{m,n} > 4$, light red: $E_{m,n} < -2$, dark red: $E_{m,n} < -4$. In cells with a significant over- or underrepresentation the Pearson residuals are moreover specified in the relevant rectangle. The plot suggests that *immédiatement* tends to co-occur with chronological reference points, while *bientôt* is attracted by deictic reference points. *Tout de suite* does not seem to show a particular association with either type of reference point.

An important point to note is that Pearson residuals do not exactly follow a standard normal distribution and "[provide] conservative indications of cells having lack of fit" (Agresti 2002: 81). Alternatively, an "adjusted" residual, which "results from dividing it by its standard error", can be used (Agresti 2002: 81). In the context of mosaic plots, Friendly (1994: 7) notes that "the adjusted residuals are always larger in magnitude than the unadjusted values", and "adjusted residuals have the property that their standard errors are equal, so their values are more comparable over cells in the contingency table". He also holds that "experience shows that they provide *better visual display of the patterns of association* than do ordinary Pearson …residuals".

As we do not focus on inferential statistics in this contribution, we will not go into any further details at this point (but see Sect. A.1 in the Appendix for some comments). Readers and users should be aware of the "meaning" of the colours in a mosaic plot, however, and take it into account in their interpretation of the data. In the following, we will continue to use Pearson residuals for the sake of simplicity, because they also play an important role in ordination (where they are mostly just called 'standardized residuals', see Sect. 3.3).

So far we have only crossed two variables. Figure 2 shows a mosaic plot where three variables have been crossed: (French) ProxPost-adverbial, Reference Point and Intentionality. In this case the two-dimensional representation provided by a mosaic plot has its limitations, and the graphic effectively shows two plots, one at the top, where Intentionality is crossed with ProxPost-adverbial for chronological reference points, and one at the bottom, where this combination is shown for deictic reference points. We can see that three combinations have Pearson residuals higher than 2: {chronological, intentional,

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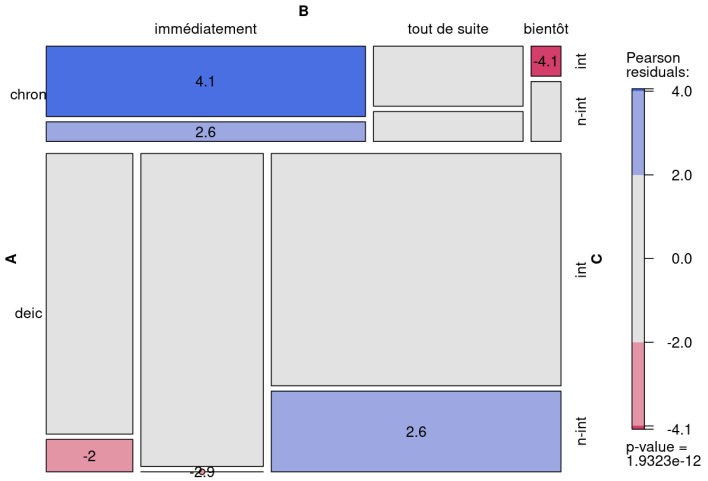


Figure 2: Three-dimensional mosaic plot representing the frequencies as well as degrees of over- or underrepresentation for three French ProxPost-adverbial-adverbials relative to the variables REFERENCE POINT (*chron*: chronological, *deic*: deictic) and INTENTIONALITY (*int*: intentional, *n-int*: non-intentional). The plot was created with the package `vcd` for `R` (Zeileis et al. 2007); cf. the script `01_proportional_visualizations.R` in the Supplementary Materials.

immédiatement}, {chronological, non-intentional, *immédiatement*}, and {deictic, non-intentional, *bientôt*}. {Deictic, non-intentional, *immédiatement*} and {deictic, non-intentional, *tout de suite*} and {chronological, intentional, *bientôt*} have residuals lower than −2.

An alternative way of visualizing three-dimensional distributions is shown in Figure 3, in what we call a 'feature combination frequency grid' (FCFG) (see Gast 2017b). Each node in the grid corresponds to one feature combination, and the size of a sphere represents the observed or frequency of the relevant feature combination (cell). The colours of the spheres reflect Pearson residuals, as in Figure 3.

3.2 Context-conditional correlation graphs

With feature-combination frequency grids (FCFGs) we can maximally visualize data along three dimensions. More dimensions are possible with 'stacked' mosaic plots as shown in Figure 3, but the information becomes increasingly difficult to process as the number of dimensions increases. In order to represent pairwise

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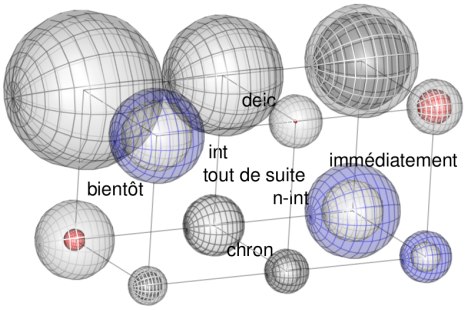


Figure 3: Feature-Combination Frequency Grid (FCF-grid, bottom) representing the frequencies as well as degrees of over- or underrepresentation for three French ProxPost-adverbials relative to the variables Illocutionary force and Intentionality (see Fig. 2 for the levels). The visualization was created using the package `rgl` for R; cf. the script `01_proportional_visualizations.R`.

associations of variables in a single plot we can make use of graphs. We call graphs of the type shown in Figure 4 'context-conditional correlation graphs' (CCCG). The graph illustrates (significant) imbalances in the distribution of German ProxPost-adverbials relative to four variables (Intentionality, Agent, Reference point and Illocutionary force/Modality).[4] The arcs show, for each pair of variables, the levels for which there is a significant overrepresentation in a pairwise cross-tabulation, along with the Pearson residual (following the underscore). For example, the arc at the very top shows that in a cross-table T of the variables ProxPost-adverbial and Agent, the Pearson residual for the cell $T_{3rd,bald} \approx 4$. The location of 'Ag3ps_4' (closer to *bald* than to *gleich*) indicates that this feature is overrepresented for *bald*.

The graph in Figure 4 only shows pairwise associations and, in a way, summarizes the information distributed over a large number of 2×2 cross-tables. In order to understand the interplay and mutual dependencies between the variables in a multivariate data set like ours, it often makes sense to use ordination techniques, to which we turn now.

[4]Variables and levels: Agent: *AgxPs* indicates that Agent is *x*. person; Illocutionary force/modality: *dir.boul*: directive/bouletic, *n.illoc*: not indicated, *poss*: possibility; Intentionality: *int*: intentional predicate, *n.int*: non-intentional predicate; Reference point: *chron*: chronological, *deic*: deictic; Modification: *modf*: adverbial is syntactically modified, *n-modf*: adverbial is not syntactically modified.

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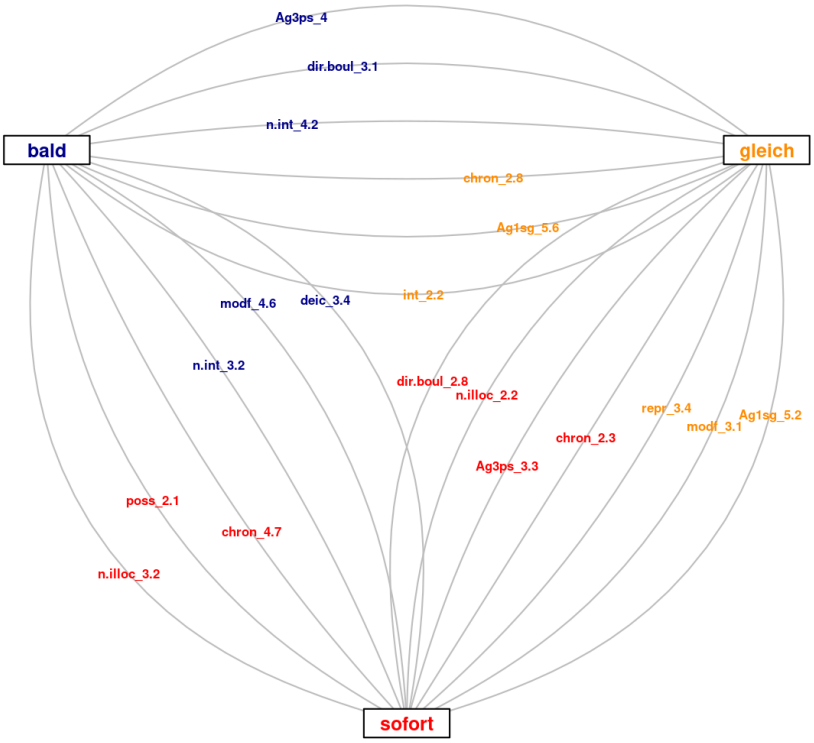


Figure 4: Context-conditional correlation graph; the graph visualizes the distribution of German ProxPost-adverbials relative to four variables (Intentionality, Agent, Reference point and Illocutionary force; see Note 4 for the levels). It was created with the package `qgraph` for R (Epskamp et al. 2012); cf. the script `02_CCCG.R` in the Supplementary Materials.

3.3 Ordination

The term 'ordination' subsumes a number of methods that make use of 'latent space', i.e., space constituted by latent variables – variables that are not directly observable but that can be infered from observable ('manifest') variables. In psychology and the social sciences researchers often work with variables that are not measurable but reflected in attitudes or behaviour. For example, 'sense of belonging' cannot be measured directly, but it can be infered from responses to stimuli such as "Spain is my nation" and "I feel Spanish" (Bollen & Medrano 1998, Bollen 2002).

In this section we discuss two types of ordination techniques that are useful for the analysis of categorical data, i.e., (simple symetric) Correspondence Analysis

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Table 1: Table with row profiles (left) and standardized residuals (right) for German ImPost-adverbials in translations from French, relative to the variable ILLOCUTIONARY FORCE

	row profiles			standardized residuals		
	comm	dir/boul	repr	comm	dir/boul	repr
gleich	0.16	0.12	0.72	0.11	-0.23	0.13
sofort	0.06	0.41	0.53	-0.1	0.09	-0.03
unverzüglich	0.11	0.67	0.22	0.01	0.21	-0.17

(Sects. 3.3.1) and Multiple Correspondence Analysis (Sect. 3.3.2).

3.3.1 Simple symmetric Correspondence Analysis

Correspondence Analysis (CA) is probably the most basic type of ordination (see Greenacre 2016 for a comprehensive overview of the ordination techniques discussed in this section). It is based on the idea that contingency tables can be interpreted as coordinate systems. Consider the left-hand side of Table 1. The table shows rowwise relative frequencies of the German adverbials *gleich*, *sofort* and *unverzüglich* as translations of the French ImPost-adverbials *immédiatement* or *tout de suite* (so essentially it shows the distribution of German ImPost-adverbials in translations from French). Each row corresponds to a 'row profile', with the percentages adding up to 1. The columns represent three levels of the variable ILLOCUTIONARY FORCE/MODALITY, i.e., 'comm[issive]' (promises), 'dir[ective]/boul[etic]' (utterances with directive illocutionary force or bouletic modality, where the speaker expresses a desire, often amounting to an indirect directive), and 'repr[esentative]' (descriptions of the world).

Without providing any information about significant deviations from the independence model the row profiles illustrate associations between levels. For example, the table suggests that *gleich* is associated with representative illocutionary force, as 72% of its cases occur with that level, while for the other adverbials this column only shows 53% (*sofort*) and 22% (*unverzüglich*).

We can regard the columns of the left-hand side of Table 1 as constituting a coordinate system within which we can locate the three adverbials of interest. This is illustrated in Figure 5. The three axes originating at (0 0 0) (the origin) represent the three columns of the left-hand side of Table 1 and constitute a three-dimensional space. The adverbial *unverzüglich*, for instance, is located at the coordinate (0.11 0.67 0.22), corresponding to the third row profile in Table 1 (cf. the orthogonal lines emanating from the relevant point). The plot also shows

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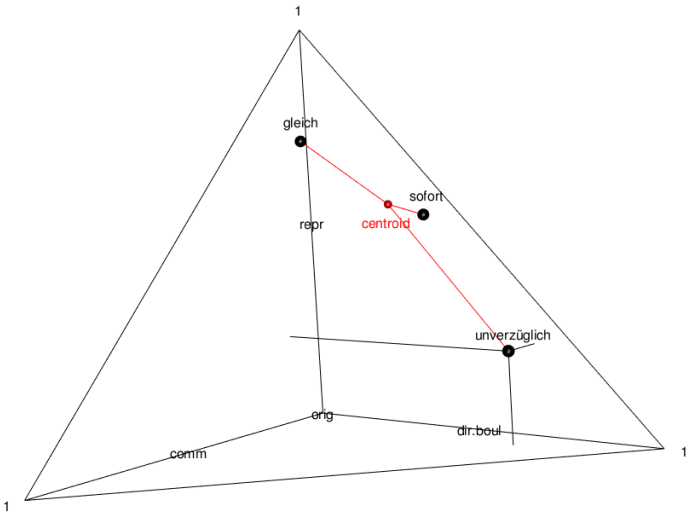


Figure 5: Visualizing Table 1, with the columns constituting a coordinate system for the rows. The plot was generated with the package `rgl` for `R` (Murdoch & Adler 2022).

the centroid of the three row points (see Sect. A.2 in the Appendix on how the centroid is calculated).

The type of visualization shown in Figure 5 constitutes the basis for a lower-dimensional visualization using CA, which fits a new coordinate system to the data, in such a way that the largest amount of 'inertia' is captured on a low number of (so-called 'principal') axes – ideally two, as two axes can be easily visualized. Inertia is a measure of spread for count data around the independence model, corresponding to variance, which is a measure of spread around the mean value for continuous data. Inertia thus measures the degree to which a contingency table deviates from statistical independence. The term is inspired by the 'moment of inertia' in mechanics, which quantifies the distribution of mass around an axis of rotation.

CA proceeds as follows: first, the original contingency table (see Table 9 on p. 53) is converted into a percentage table, and then into a table of standardized residuals through division by the square root of the expected percentage (see the right-hand side of Table 1; cf. also Sect. 6.0.2). As mentioned in Sect. 3.1, Pearson residuals indicate deviation from independence. The resulting matrix thus indicates to what extent a cell deviates from the independence model (assuming a homogeneous distribution). The centroid of the rows constituting this matrix is

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the origin of a three-dimensional coordinate system, which is rescaled along the three axes as a result of the standardization (division by the square root of the expected frequency). The standardized residuals from Table 1 are visualized in Figure 6.

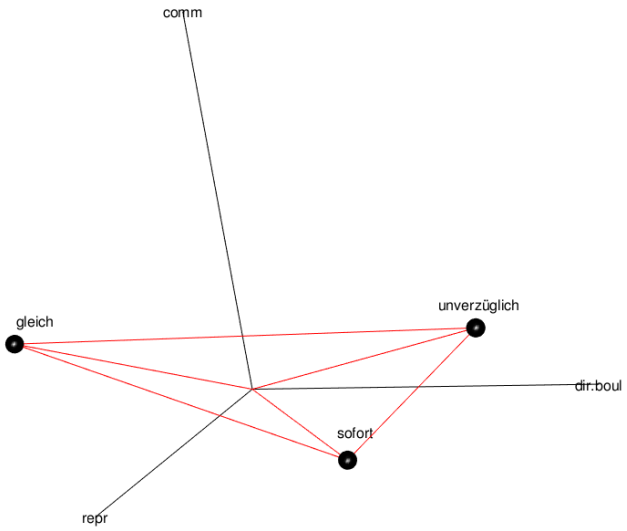


Figure 6: Visualizing the standardized residuals as shown on the right-hand side of Table 1. The plot was generated with the package `rgl` for `R` (Murdoch & Adler 2022).

In a second step, the matrix of standardized residuals is subject to an operation called 'singular value decomposition' (SVD). The mathematical procedure of SVD is outlined in Sect. A.2 in the Appendix (based on Greenacre 2016). In the main text we will focus on the results.

SVD fits a new coordinate system to the data, in such a way that the first ('principal') axis captures most of the inertia, the second axis, which is orthogonal to the first, captures most of the remaining inertia, and so on until all data points can be precisely located in the newly created space. In the case of the data visualized in Figure 6, we can find a coordinate system with only two dimensions that preserves all the information contained in the table of standardized residuals, because three points are always located on a (two-dimensional) plane in 3-dimensional space. If a table has been centered, as in the case of Pearson residuals, this plane moreover runs through the origin (cf. the triangle in Figure 6).

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The principal axes determined with SVD constitute the axes of a new coordinate system in which we can locate both the data points corresponding to the rows of the table of standardized residuals – in our case, the ImPost-adverbials *gleich*, *sofort* and *unverzüglich* – and the original axes corresponding to the levels of the variable Illocutionary Force ('comm', 'dir.boul', 'repr'). A CA created with the package `ca` for `R` (Nenadic & Greenacre 2007) is visualized in Figure 7.[5]

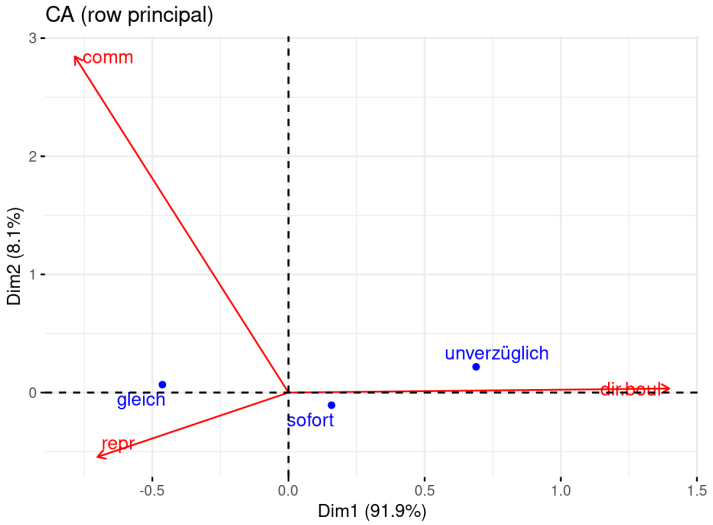


Figure 7: Correspondence Analysis for the data in Table 1. The CA was created with the package `ca` for `R` (Nenadic & Greenacre 2007). For the visualization we used the package `factoextra` (Kassambara & Mundt 2020). The plot shows a 'row analysis', with row points in principal coordinates and column vectors in standard coordinates.

Along with the two-dimensional representation of the standardized residuals Figure 7 shows percentages on both axes. These percentages indicate how much inertia the relevant axis covers. Inertia has a direct connection to the Chi-squared test: it is identical to the Chi-squared statistic, divided by the number of observations; it is also the square root of the ϕ-coefficient, a correlation measure for contingency tables; and it relates directly to the Chi-squared distances of the row coordinates to the origin. The Chi-squared distance is defined as the square root of the sum of the squared differences between the observed values and the expected values, divided by the expected values. It is a measure of deviation from independence. The inertia is the sum of the Chi-squared distances squared, and weighted by the row masses.

[5]In Python, the module `prince` can be used for CA.

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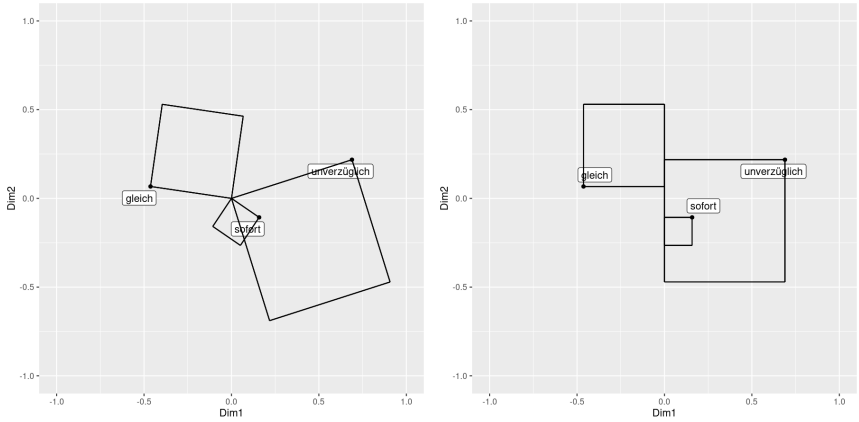


Figure 8: Squared Chi-squared distances (left) and squared Dimension 1 coordinates (right). The plot was generated with the packages `ggplot2` and `ggrepel` for `R` (Wickham 2016, Slowikowski 2024).

Consider Figure 8 for illustration. The left-hand side shows the squared distances of the three coordinates representing the ImPost-adverbials to the origin, corresponding to the inertia (without scaling by the row masses). The right plot shows squares based on the lines that run parallel to the x-axis (the first principal axis) from the three ImPost-adverbials (also unweighted by the row masses). The area covered by the squares in the right plot amounts to 91.9% of the area covered by the total inertia as shown in the left plot. This is what the percentage on the x-axis in Figure 7 indicates.

The inertia not only gives us information about the degree of truthfulness of the dimensionality-reduced representation relative to the original, higher-dimensional one, it also tells us how 'important' specific axes are. When looking at Figure 7 one might be led to believe that the two axes explain a similar amount of inertia; but Axis 1 alone explains ≈ 92% of it. Note that all adverbials as well as the two levels of ILLOCUTIONARY FORCE ('repr', 'dir/boul') are very close to the *x*-axis – with low values on the *y*-axis –, implying that the *y*-axis explains a low amount of their row inertia.

So far we have focused on the position of the points representing the rows (ImPost-adverbials) in what is called a "row analysis" of CA. Remember that our starting point was the location of ImPost-adverbials relative to distributional variables, which constitute a space within which we can locate the latter. In principle, CA is symmetrical and we could just as well locate columns relative to rows. In fact, there are many ways of visualizing CAs, implemented, for instance, in the `fviz_ca_biplot` function of the `factoextra` package (Kassambara & Mundt

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2020), or the `plot.ca` function of the `ca` package for `R` (Nenadic & Greenacre 2007). In a row analysis the rows (ImPost-adverbials) are represented in so-called 'principal coordinates', which preserve Chi-squared distances between the observations. The columns, represented as vectors in Figure 7, are displayed in 'standard coordinates', which are so-called because they have a mean value of 0 and a variance of 1 (Greenacre 2016: 62).

In this display the relationship between an ImPost-adverbial (row) and a variable (column) is not represented by the distances between row and column vectors; it (ideally) corresponds to the inner products of these vectors ("ideally" because information may be lost as a result of the dimensionality reduction). This means that the degree of association between a row and a column is a function of the degree to which the two vectors point in the same direction. If all of the information has been preserved, we can actually reconstruct the values of the original table from the map shown in Figure 7. For example, the standardized residual for the combination *gleich*/'repr' is the inner product of the two vectors for *gleich* (scaled by the square root of its row mass) and the vector pointing at 'repr' (scaled by the square root of the average profile; see Greenacre 2016 for more information and Sect. A.2 for the calculation of this example).

While not everybody may be interesed in the mathematical details, it is important to be aware of the specific type of map that is used for visualization. The "row principal' representation chosen in Figure 7 is rather transparent, but often has the disadvantage of visual imbalance between the row points, which are closer to the centre, and the column points, which are spread out more. Users should be aware, however, that a representation that condenses the data – while visually being more pleasing – potentially does not have the inner product interpretation of an asymmetric "row principal" plot, where the rows are represented in principal coordinates and the columns in standard coordinates (see Greenacre 2016: Ch. 13 for discussion).

Representations like the one in Figure 7 give us an idea of the relationship between the levels of two variables. However, with a multivariate data set like ours, we need to be able to represent relationships between more than one variable. A very simple way of visualizing more than one variable is provided by a 'stacked' CA (Greenacre 2016: Ch. 17). We can simply concatenate contingency tables (horizontally or vertically) and carry out a CA on that table. Horizontal concatenation is illustrated for our data in Table 2, which shows the percentages for the variable Illocutionary force on the left, and for Agent on the right.

Given that the number of observations per variable is invariant, the Chi-squared distances between the levels of each variable are preserved in this type of stacked CA. The positions of the various levels relative to each other are

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Table 2: Stacked contingency table for German ImPost-adverbials (in translations from French) and the explanatory variables ILLOCUTIONARY FORCE and AGENT

	comm	dir.boul	repr	1pl	1sg	2ps	3ps
gleich	0.08	0.06	0.36	0.06	0.22	0.06	0.16
sofort	0.03	0.21	0.26	0.07	0.01	0.06	0.35
unverzüglich	0.06	0.33	0.11	0.06	0.00	0.00	0.44

thus not affected by the stacking (cf. Figure 9). However, it is clear that some information is lost in a stacked CA, as we do not get any information about the relationship between the levels of the explanatory variables (here, ILLOCUTIONARY FORCE and AGENT). Table 2 suggests that a certain number of ocurrences of *gleich* have commissive illocutionary force, and another set of observations has a first person agent. The table does not tell us how many of the commissive observations have first person agents. In other words, the distributional variables are treated as being independent. While this is a clear drawback of the stacked CA, the method can be useful in order to determine the degree of independence of predictor variables. In particular, it can be instructive to compare a stacked CA to a Multiple Correspondence Analysis, to which we turn next.

3.3.2 Multiple Correspondence Analysis

From a mathematical point of view, Multiple Correspondence Analysis (MCA) functions very much like (simple) CA. However, it is based on a representation of the data that takes the relationships between all (explanatory as well as response) variables into account. There are two (near equivalent) ways of accomplishing this: (i) carrying out a CA on the indicator matrix of a data set, or (ii) using the Burt table instead.

An indicator matrix is a binary matrix that indicates whether or not an observation exhibits a given property. Consider Table 3, which shows a random fragment of our data frame (German translations of French ImPost-adverbials). Each row corresponds to one example, and the columns specify the illocutionary force, the Agent, and the German adverbial. The indicator matrix corresponding to Table 3 is shown in Table 4. Here, each column corresponds to a level of a variable, not a variable. While Table 3 specifies the levels themselves in the cells, Table 4 shows whether (1) or not (0) the observation in question is an instance of the relevant level.

A Burt matrix is a crosstabulation of the frequencies of all levels of all variables with all other levels of all other variables. Table 5 shows a fragment of our data in

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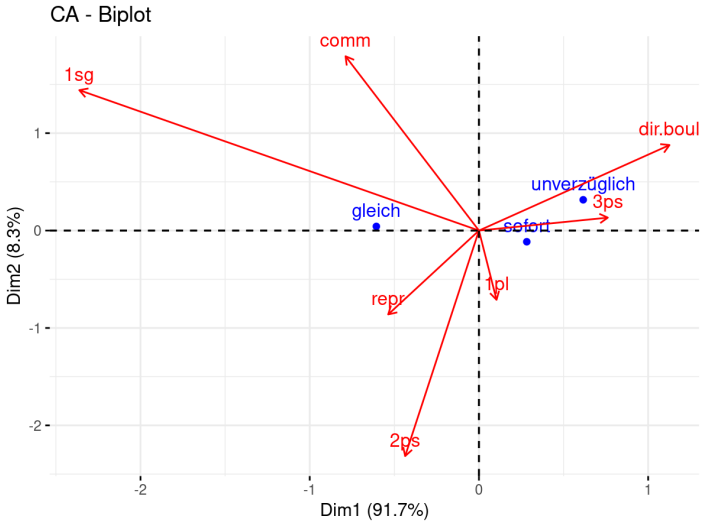


Figure 9: Stacked Correspondence analysis showing the location of three German ProxPost-adverbials in latent space constituted by the variables ILLOCUTIONARY FORCE and AGENT. The CA was fitted with the package `ca` for `R` (Nenadic & Greenacre 2007) and the plot was created with the package `factoextra` for `R` (Kassambara & Mundt 2020).

Table 3: Random fragment of the data frame

ID	ILLOC	AGENT	German IMPOST-ADVERBIAL
214	repr	1sg	*sofort*
216	dir.boul	1pl	*sofort*
217	repr	3ps	*gleich*
218	repr	3ps	*sofort*
221	dir.boul	3ps	*unverzüglich*

the form of a Burt matrix. Burt matrices and indicator matrices are systematically related insofar as the Burt matrix is the inner product of the transpose of the indicator matrix with itself ($B = I^{\mathsf{T}}I$, for an indicator matrix I).

While MCA is "mathematically" quite similar to CA, from the point of view of interpretation the two techniques are quite different. In CA, the rows correspond to categories of interest an the columns to the levels of a variable. In MCA, the rows correspond to individual observations (in the indicator matrix) and the columns correspond to variables that the individuals do (not) exhibit. Unlike CA, MCA thus provides information about individual observations. Moreover, in MCA *all* (levels of) variables are related to each other via Chi-squared distances.

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A downside of MCA is that the underlying data tables – the indicator matrix (Table 4) or the Burt matrix (Table 5) – are heavily redundant. The indicator matrix contains a lot of zeros, which turn into negative values when the residuals are calculated. The Burt matrix contains a lot of uninformative material on the diagonal, where levels are crossed with themselves. Because of the redundancy, both tables inflate the overall inertia. As a result, MCAs typically appear to explain a very low percentage of inertia on their first principal axes. Greenacre (2016) has proposed a way of 'adjusting' the inertia in MCAs, which is implemented in the package `ca` for `R` (Nenadic & Greenacre 2007).

Figure 10 shows an MCA for German translations of French ImPost-adverbials, with three variables: ImPost-adverbial, Illocutionary force and Agent.[6] Without adjustment, the percentage of inertia captured by the first two principal axes is 26.3% (Axis 1) and 19.2% (Axis 2). It is much higher when computed with Greenacre's (2016) adjustment (72.7% [Axis 1] and 11.0% [Axis 2]).

An MCA creates a new (latent) space that is structured by all variables relative to all other variables. It creates a mapping from frequency distributions to positions in this space. We can therefore also locate variables (and observations) in the MCA-space without them being used to create the latent space. In MCA speech, this means that we use 'supplementary' (or 'passive') variables. In a data set like ours it often makes sense to treat the target variable – in our case, expressions of immediate or proximal posteriority – as a supplementary variable. The idea is that the latent space is constituted by explanatory variables only, and that the adverbials in question can be located relative to that space. This is shown in Figure 11 for the same data that was used to generate the stacked CA in Figure 9, and the MCA without supplementary variables in Figure 10.

[6]We used the package `FactoMineR` for `R` (Lê et al. 2008). In Python, the module `prince` can be used for MCAs.

Table 4: Indicator matrix corresponding to Table 3

	Illoc			Agent				ImPost-adverbial		
	comm	dir.boul	repr	1pl	1sg	2ps	3ps	*gleich*	*sofort*	*unvzgl.*
214	0	0	1	0	1	0	0	0	1	0
215	0	1	0	1	0	0	0	0	1	0
216	0	0	1	0	0	0	1	1	0	0
217	0	0	1	0	0	0	1	0	1	0
218	0	1	0	0	0	0	1	0	0	1

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Table 5: Burt matrix corresponding to Table 3

	gleich	*sofort*	*unverzüglich*	comm	dir.boul	repr
gleich	25	0	0	4	3	18
sofort	0	34	0	2	14	18
unverzüglich	0	0	9	1	6	2
comm	4	2	1	7	0	0
dir.boul	3	14	6	0	23	0
repr	18	18	2	0	0	38

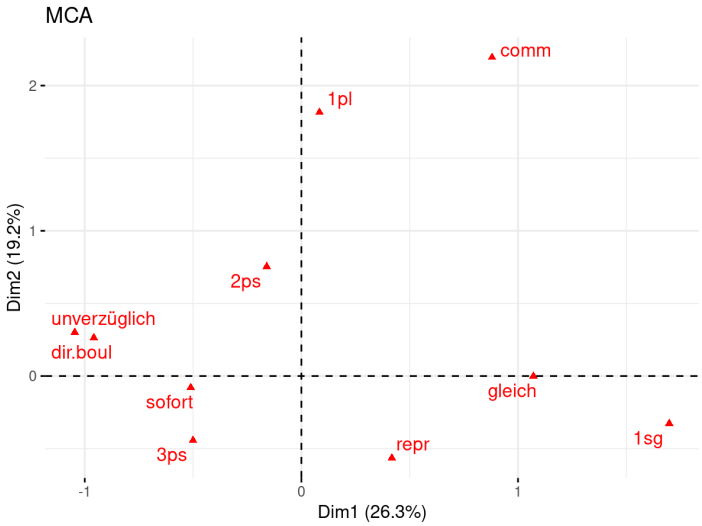


Figure 10: Multiple Correspondence Analysis for the variables Illocutionary force, Agent and ImPost-adverbial (German translations of French ImPost-adverbials). The MCA was fitted with the package `FactoMineR` for R (Lê et al. 2008), the plot was generated using the package `factoextra` (Kassambara & Mundt 2020).

3.4 Comparing CA and MCA

The three ordination methods mentioned above provide ways of displaying different types of interdependencies between variables. A stacked CA creates a latent space constituted by the columns (distributional variables) and locates the rows (adverbials) relative to the columns, or vice versa (it is symmetric, but our focus has been on the 'row' analysis). CA visualizes relations between context variables and adverbials, but no relations between pairs of context variables. An MCA with supplementary variables, in a way, does the opposite: it visualizes relationships between context variables, and then locates the adverbials relative to

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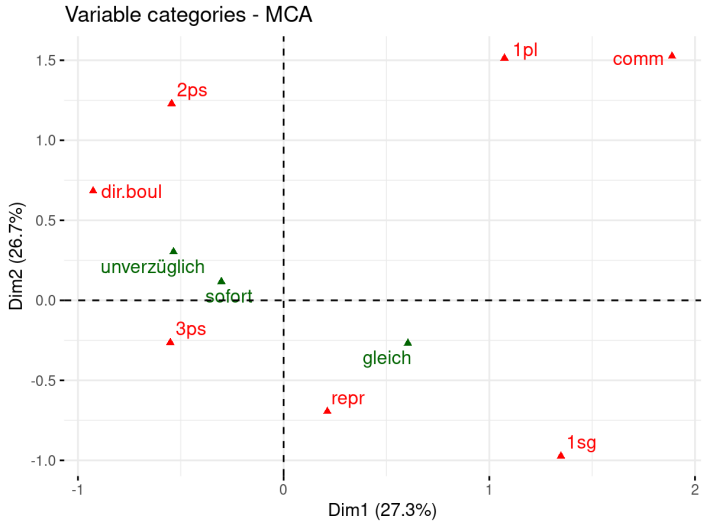


Figure 11: MCA for the variables Illocutionary force and Agent, with ImPost-adverbial as a supplementary variable (German translations of French ImPost-adverbials).) The MCA was fitted with the package `FactoMineR` for `R` (Lê et al. 2008), the plot was generated using the package `factoextra` (Kassambara & Mundt 2020).

the latent space constituted by the context variables. An MCA without supplementary variables visualizes relationships between all variables, i.e., it does not distinguish between context variables and adverbials.

The differences between the three types of visualizations can be illustrated with the relationship between one German adverbial, *unverzüglich*, and two context variables, Illocutionary force and Agent. Consider first the (stacked) CA shown in Figure 9. The ImPost-adverbial *unverzüglich* and the category '2nd person' are located on opposite ends of the origin, indicating a negative association. As a matter of fact, the variables are complementary, in the sense that *unverzüglich* is never used with the second person in our sample. In Figures 10 (MCA without supplementary variables) and 11 (MCA with supplementary variables), the second person is located in the same quadrant as *unverzüglich*. The reason is that both categories are distributionally related to directive illocutionary force or bouletic modality ('dir.boul'), though not to each other. In a way, the MCAs suggests a deceiving distributional association between *unverzüglich* and the second person. What this example illustrates is that none of the three methods considered above – stacked CA, MCA with or without supplementary variables – can, in itself, provide a complete picture of the data. It is therefore

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instructive to experiment with different methods in order to develop a better understanding of the data. This does not mean, of course, that readers should feel encouraged to cherry pick those plots that they like best. Remember that the primary focus of this contribution is to support researchers in understanding their data, not in presenting the results to an audience or readership.

4 Visualizing undirected comparative distributions

In order to compare two languages in terms of the way a given functional domain is structured, a common procedure is to use a semantic or functional criterion as a third of comparison, and then determine the way the semantic space in question is mapped to formal linguistic expressions (Gast 2012). In our case, this means that we use the functional annotations as a *tertium comparationis*, and we locate the expressions from different languages relative to these annotations.

In Sect. 3 we used four methods to visualize form-function mappings in specific languages: CA – with and without stacked tables – and MCA – with and without supplementary. In this section we will focus on only one of these methods, MCA with supplementary variables. The advantage of an MCA over a CA (with stacked tables) is that it allows us a per-item analysis, as each observation, and not just the variables, are located in latent space. The advantage of an MCA with supplementary variables is that it creates a latent space that is independent of the specific expressions used in a language, so we can (comparatively) locate expressions from different languages in that space. Put differently, the latent space constituted by the distributional variables functions as the *tertium comparationis* and the expressions as the *comparanda*.

An MCA with 'IP-adverbial' as a supplementary variables is shown for original occurrences of *gleich*, *sofort*, *immédiatement* and *tout de suite* in Figure 12, relative to four distributional variables (INTENTIONALITY, AGENT, ILLOCUTIONARY FORCE and REFERENCE POINT).

Though no doubt quite informative, Figure 12 is hard to read. A general observation that stands out is that the four adverbials are located in different quadrants of the plot. Another relevant observation is that *tout de suite* is considerably closer to the centre than the other adverbials, which suggests a less specialized distribution for this adverbial.

One way of simplifying a multivariate distribution as shown in Figure 12 is to apply clustering. As an MCA locates every single observation in the (latent) space that it creates, we can identify clusters of similar observations. We used PAM (partitioning around medods) clustering for this purpose, which identifies

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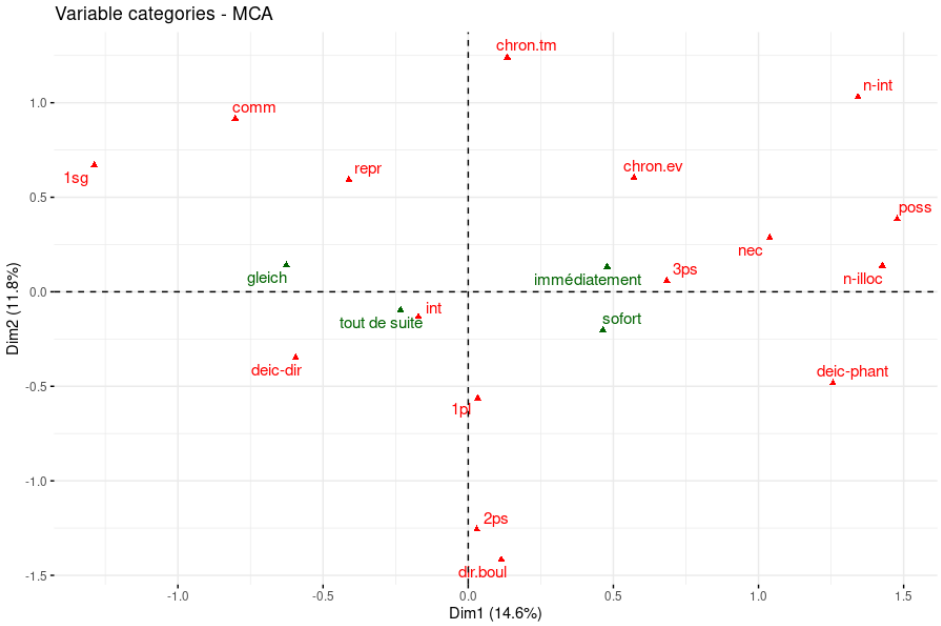


Figure 12: MCA with IP-adverbial as supplementary variable (German and originals), relative to the variables 'intentionality', 'agent', 'illocutionary force' and 'reference point'. The CA was created with the package `FactoMineR` for `R` (Lê et al. 2008), for the visualization we used the package `factoextra` (Kassambara & Mundt 2020). The adjusted percentages of explained inertia as returned by the package `ca` for `R` (Nenadic & Greenacre 2007) are 48.5% (Axis 1) and 15.0% (Axis 2).

clusters on the basis of distances between observations – as the name suggests, around a medoid, i.e., a (representative) observation in the centre of the cluster. [7] Clusters with 3 or 6 medoids were chosen because the average silhouette scores showed visible leaps at these points (in comparison to solutions with 2 and 5 clusters; see Figure 25, Sect. A.3, in the Appendix). The location of clusters of individual sentences in the latent space constituted by the MCA is visualized in Figure 13.

The annotations for the medoid observations of the clusters are shown in Table 6. The corresponding examples are provided in (6)–(8).

(6) a. Je lui donne bien volontiers tout de suite la parole. (original)
b. Ich erteile ihm daher gern sogleich das Wort.

[7] We used the package `cluster` for `R`, cf. Maechler et al. (2023). We validated the clustering solution using the silhouette method (see the package documentation for more information).

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Table 6: Medoid observations for cluster solutions with 3 (left) and 6 (right) clusters

ID	agent	int	mod	ref	cl
448	1sg	int	repr	deic-dir	1
16	3ps	int	repr	chron.ev	2
443	3ps	int	dir.boul	deic-dir	3

ID	agent	int	mod	ref	cl
448	1sg	int	repr	deic-dir	1
41	3ps	int	repr	deic-dir	2
443	3ps	int	dir.boul	deic-dir	3
441	3ps	int	repr	chron.ev	4
455	3ps	int	dir.boul	chron.ev	5
280	3ps	n-int	repr	deic-phant	6

c. I am very happy to give you the floor now.

(7) a. C'est une expérience amère, lorsque l'on fait des propositions de réforme, de constater que d'autres les bloquent constamment avec des référence à autre chose.

b. Ich finde es bitter, wenn man Reformvorschläge macht, dass die gleich immer so abgeblockt und mit etwas anderem in Bezug gebracht werden. (original)

c. It is a bitter experience, when one makes reform proposals, to find that others are constantly blocking them with reference to something else.

(8) a. Il faut mettre ce régime sous tutelle internationale et il faut le faire tout de suite!

b. Dieses Regime muß unter internationale Aufsicht gestellt werden, und zwar sofort! (original)

c. This regime needs to be placed under international supervision, and this should be done straight away!

In the French and German versions of the medoid of Cluster 1 (cf. (6)) we find *tout de suite* and *sogleich*, arguably a stylistic variant of *gleich*. The medoid of Cluster 2 (cf. (7)) also has *gleich* in German but no expression in French – a not uncommon situation. The medoid of Cluster 3 (cf. (8)) has *sofort* in German and *tout de suite* in French. This illustrates that the clusters are not actually constituted by German or French expressions. They represent characteristic utterance types defined by the distributional variables: (i) a (typically deictic) declaration of future action by the speaker (Cluster 1); (ii) a (typically chronological) statement about a third person (Cluster 2); and (iii) a call for action (Cluster 3).

In order to understand associations between clusters (major use types) and ImPost-adverbials we can cross-tabulate the ImPost-adverbials with the clusters of the two cluster solutions. This is shown in Figure 14. Both plots suggest a positive association of Cluster 1 with *gleich*. The plot with six clusters moreover

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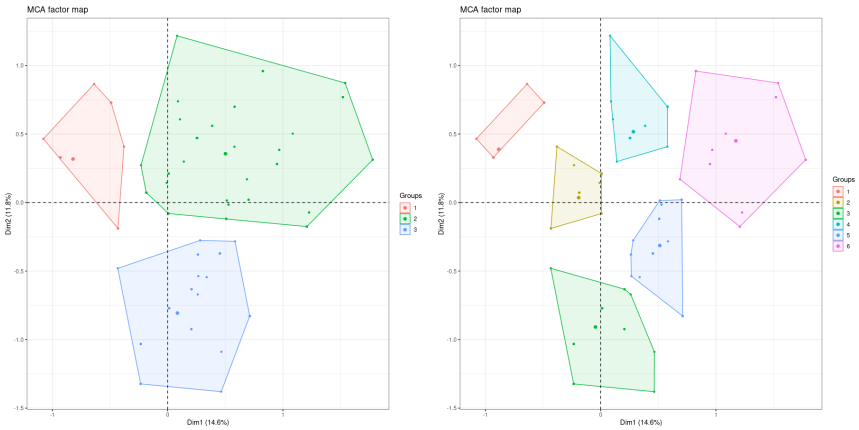


Figure 13: Clusters based on MCA-coordinates, using Partitioning around medoids with three (left) and six (right) clusters, using the package `cluster` for `R` (Maechler et al. 2023). The plots were generated with the package `factoextra` for R (Kassambara & Mundt 2020).

shows positive associations between *immédiatement* and Cluster 6, and between *sofort* and Cluster 5. Another important observation is that *tout de suite* does not show an association with any of the clusters. This confirms our impression that it is the most generic/least specialized expression of the ImPost-adverbials under investigation.

The visualization methods that can be used for undirected comparative data are thus essentially the same as those that were discussed for language-specific data, and all of the other methods that were discussed under 'proportional visualizations' are of course also in principle available (a mosaic plot was used above).

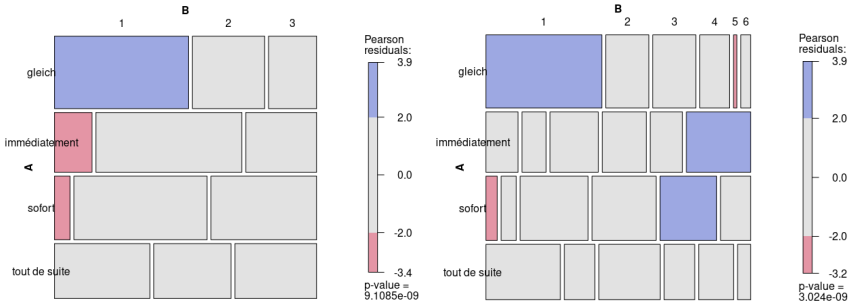


Figure 14: Mosaic plots for cross-tabulations of German and French ImPost-adverbials, for cluster solutions with three (left) and six (right) clusters. the plots were generated with the package `vcd` for `R` (Zeileis et al. 2007).

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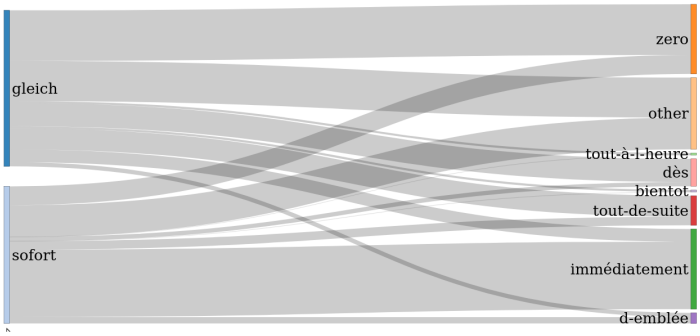


Figure 15: Sankey plot showing French translations of the German ImPost-adverbials *gleich* and *sofort*. The plot was generated with the package `networkD3` for `R` (Allaire et al. 2017).

What distinguishes comparative data from language-specific data is primarily the use of an appropriate *tertium comparationis*. This is a conceptual rather than technical matter. Note also that – as was pointed out above – in undirected comparative (e.g., contrastive) studies, it is always helpful to visualize language-specific distributions as well. For example, it is of course conceivable for context variables to behave differently (e.g., in their clustering behaviour) across languages. Such differences can best be identified by inspecting and comparing language-specific distributions, potentially identifying hypotheses that can be tested with inferential statistics.

5 Visualizing directed comparative distributions

5.1 Sankey diagrams and translation graphs

One of the most simple ways of representing translation choices visually is by using Sankey diagrams, named after H.R. Sankey (Kennedy & Sankey 1898). Figure 15 (generated with the package `networkD3` for `R`, cf. Allaire et al. 2017) shows a Sankey plot visualizing the frequencies of (French) target language choices for the two German source language expressions *gleich* and *sofort*.[8]

We can add more information to 'plain' Sankey diagrams by including an intermediate layer of nodes showing the frequencies of the levels of some variable relative the source and target expressions, as in Figure 16. It shows the frequencies of source and target language correspondences mediated by the

[8]In Pyhon, the `plotly` module can be used to create Sankey diagrams.

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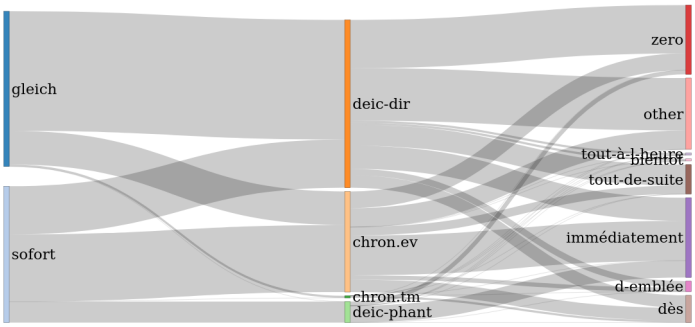


Figure 16: Comparative Sankey plot showing associations of IP-adverbials with French IP-adverbials, given the value for the variable Reference point. The plot was generated with the package `networkD3` for `R` (Allaire et al. 2017).

variable Reference point. We can see that most cases of 'deic[tic]-dir[ect]' come from *gleich* and go to 'zero' or 'other' (in French), while almost all cases of 'deic[tic]-phant[asma]' come from *sofort* and go to *immédiatement*. Cases of 'chron[ological]-ev[ent]' seem to be typically associated with *sofort* in German originals, and with *immédiatement* in French translations.

To obtain a more global view, consider Figure 17, which shows the correspondences between source and target language expressions relative to the clusters mentioned in Sect. 4 (solution with three clusters). We already know that Cluster 1 is positively associated with *gleich* in German. The plot shows that most cases of this category go to 'zero' or 'other' in French. Clusters 2 and 3 are more evenly distributed among *gleich* and *sofort* in German, but they also seem to exhibit clear asymmetries.

While Sankey plots contain useful (relational) information about frequency distributions in the source and target language, they do not contain by-item information and are potentially deceptive. The edges conecting the nodes reflect frequencies, but the left edges are not connected by-item to the right edges. For example, as mentioned above, if we look at the middle node 'chron.ev' (for 'chronological relative to an event'), the plot suggests that those cases that are rendered with *gleich* in German go to 'zero' and 'other' in French. While this is probably true, the underlying data does not actually contain that information. Like a CA, Sankey diagrams are built on cross-tables and merely represent frequencies of combinations of levels for pairs of crossed variables. For example, it would be conceivable that most or even all occurrences of *sofort* going to 'deic-dir' actually end up at 'zero'.

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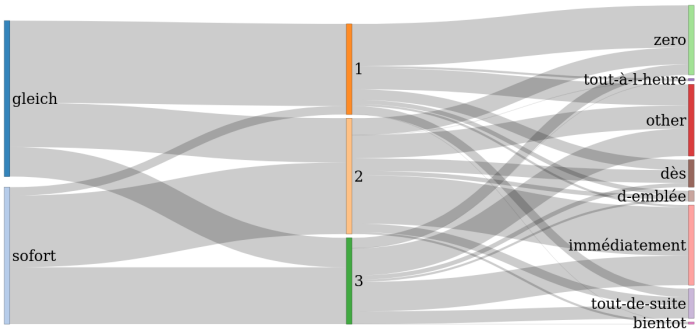


Figure 17: Comparative Sankey plot showing associations of IP-adverbials with French IP-adverbials, given a specific context cluster (solution with three clusters). The plot was generated with the package `networkD3` for `R` (Allaire et al. 2017).

Translation choices can be represented more precisely with visualizations that we call 'conditional translation graphs' (see also Gast & Atayan 2019). Consider the graph in Figure 18. It shows frequencies of choices made in German-to-French translation, under specific contextual conditions. The colours of the edges correspond to the three clusters of context variables (see also the caption of the graph).

Figure 18 shows that *gleich* is indeed mostly not translated at all in Context 1. *Sofort* typically goes to *immédiatement* in Contexts 2 and 3. Unlike suggested by Figure 17, there does not seem to be any particular association between *gleich* and 'other', as the three arrows between these nodes have approximately the same width.

Once again, we wish to emphasize that impressions based on visualizations like the one in Figure 18 should of course be supported by statistical evidence. We will now sketch out ways of handling translation data using methods from Bayesian statistics.

5.2 Bayesian networks

In Section 2 it was mentioned that choices in translation can be regarded as coming with a specific conditional probability. For example, we can compare the conditional probability of finding *immédiatement* in Context 2 if there is *gleich* in the German original with its conditional probability if there is *sofort* in the original. That would tell us something about the relationships between context types in the original, expressions in the original and expressions in the target language (note that translators may of course also choose to change the context type, in which case we have to deal with yet another variable determining the

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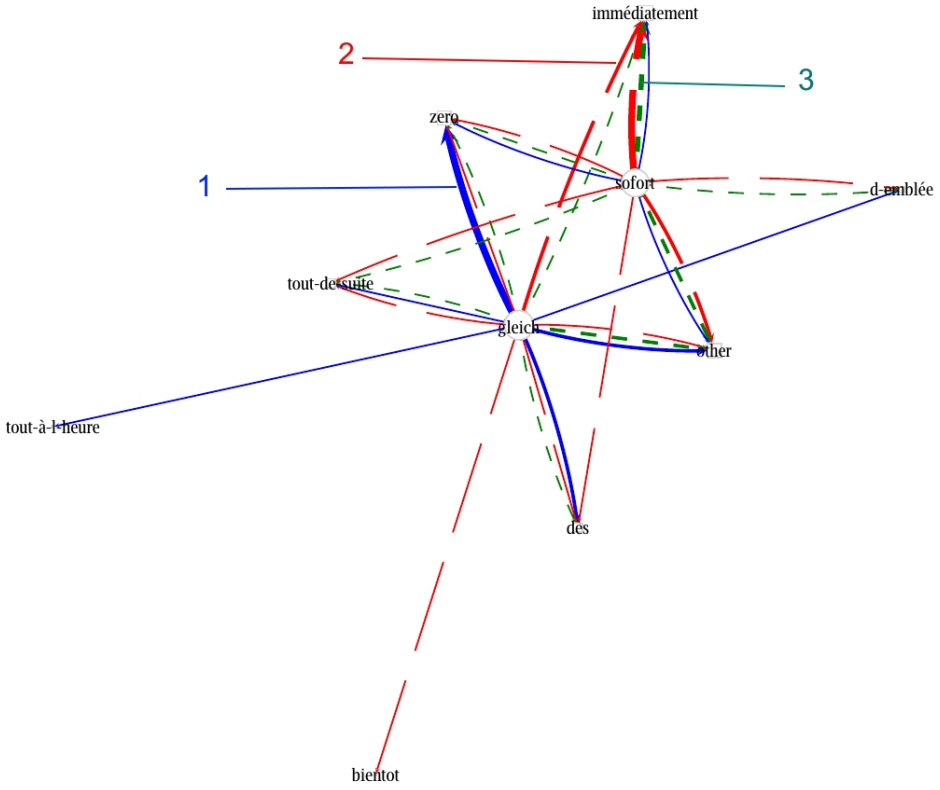


Figure 18: Conditional translation graph showing French translation choices for German *gleich* and *sofort*, given specific context clusters. The (solid) blue edge indicates translation in Context 1, the (long-dashed) red line represents translations in Context 2, and the (short-dashed) green line corresponds to Context 3. The graph was generated with the package `graph_tool` for `Python` (https://graph-tool.skewed.de/static/doc/).

probabilities of using a specific expression in the target language). While we cannot offer a complete and statistically sound analysis of this type, we will present a general framework which is flexible enough to accommodate such cases, and that makes use of visualizations, more specifically, graphical models.

Graphical models capture the type of complex data structure as encountered in translation studies in the form of networks or graphs representing dependencies between random variables (cf. Koller & Friedman 2009). We will use Bayesian networks, a type of graphical model that represents the (assumed, hypothesized) causal relations between variables. We will not be concerned with the causal re-

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lationships between specific distributional variables. For example, it seems reasonable to assume that ILLOCUTIONARY FORCE has a causal effect on the variables AGENT and INTENTIONALITY; if a speaker wants an addressee to do something, that implies a certain amount of control on the part of the addressee. This way of causal modeling is not specific to the type of distribution under discussion in this section, i.e., choices made in a target language under the influence of a source language expression.

It should be noted that corpus linguistic methods are not adequate for investigating causal relationships in the process of bilingual language production. We would need experimental evidence for that. What we can do with corpus data, however, is model the cognitive processes during translation and test the predictions derived from this modeling. A simple example of a model of bilingual language production is shown in Figure 19 in the form of a directed acyclic graph (DAG).[9] It represents the choice of a French ImPost-adverbial in a translation from German. The nodes represent variables, and they are associated with local probability distributions:

The distribution of **Context types** is represented at the top of the graph. We assume the three types distinguished by the 'smaller' PAM-cluster solution (cf. Sect. 4). Each of the context types has a certain (relative) frequency. We estimated the frequencies of the three context types from our sample.

The specific context type conditions a frequency distribution in the originals. We can, again determine this distribution on the basis of our sample. The node **OrigIPDe** is associated with conditional probabilities of the form $p(w_{de}|c)$ – the probability of using a specific German ImPost-adverbialw_{de}, given a context c.

The node **HypIpFr** stands for a hypothetically used French ImPost-adverbial – the ImPost-adverbial that *may* have been used in monolingual (French) language production, in the specific context. Like the node 'OrigIPDe', this node is associated with a conditional frequency distribution, $p(w_{fr}|c)$. This distribution can be estimated from the original French data.

Finally, the node **TrlIPFr** stands for the ImPost-adverbial used in the French translation from German. According to the DAG in Figure 19, this choice is conditioned by the probability of using a specific French expression (as in monolingual language production), and the influence from the German ImPost-adverbial, reflecting form-to-form associations in translation. The node 'TrlIPFr' thus represents a distribution over French ImPost-adverbials, conditioned by the hypothetical French ImPost-adverbial and the German ImPost-adverbial in the original

[9]The graph was generated with the package `bnlearn` for `R`. In Python, the module `pgmpy` can be used for Bayesian networks.

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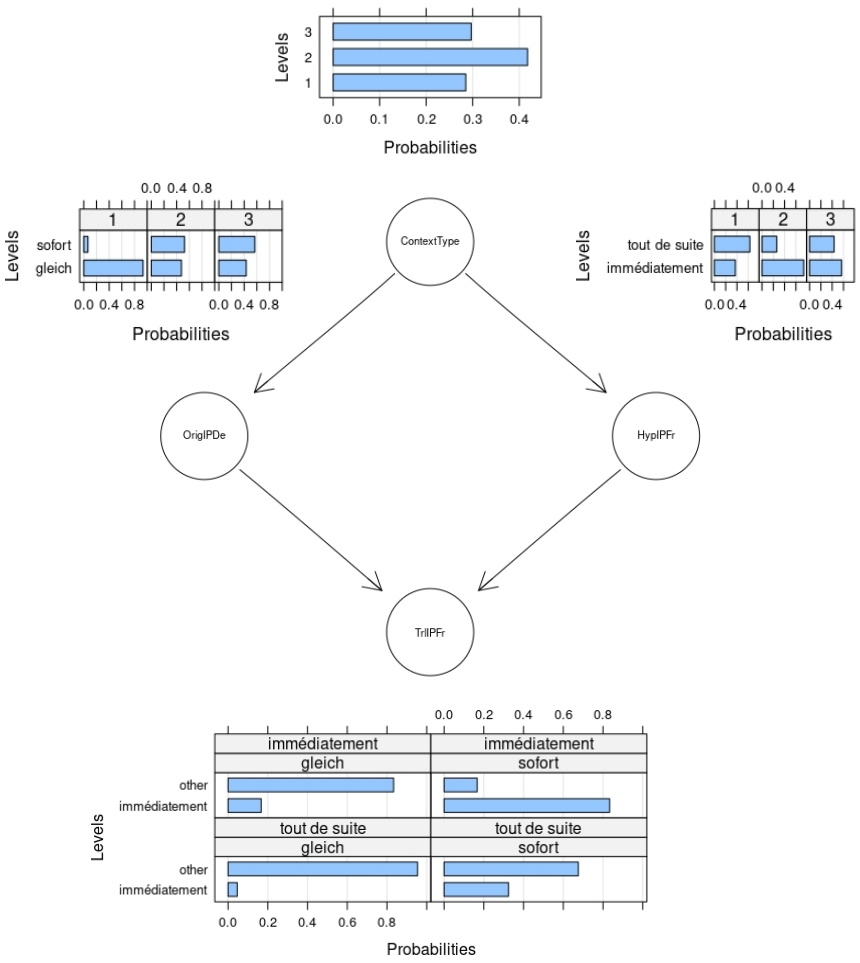


Figure 19: Directed acyclic graph for IP-adverbials (comparative/directional, with clustered observations). The graph was created with the package `bnlearn` for `R` (Scutari 2010).

sentence. This distribution cannot be observed, as no (hypothetical) monolingual ImPost-adverbial is ever produced in French translation from German. In order to infer the hypothetical French expressions, we trained a classifier (a support vector machine/SVM) on all explanatory variables, predicting a given ImPost-

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Table 7: Empirical probabilities (left) and theoretical probabilities according to the Bayesian network (right) for the occurrence of *immédiatement* in French translations, in specific contexts, given a specific German ImPost-adverbial in the original.

	empirical		theoretical	
ContextType	*gleich*	*sofort*	*gleich*	*sofort*
1	0	0.25	0.09	0.51
2	0.24	0.53	0.13	0.7
3	0.07	0.48	0.11	0.61

adverbial for each translated French sentence.[10]

The network shown in Figure 19 represents a hypothesis about mediated language production. By breaking down the joint probability distribution into local, partially independent distributions, we can model components of the highly complex process of mediated language production. As our data set is quite small and the estimates for some parameters are not very precise, we will illustrate this procedure with a binary choice, by modeling the use of *immédiatement* in French translations. All French ImPost-adverbials other than *immédiatement* were subsumed under a category of 'other'.

The Bayesian network in Figure 19 allows us to determine theoretical distributions of French ImPost-adverbials under different conditions, and to compare them to the empirical distributions. This can give us an estimate of the plausibility of the causal model. Tables 7 shows the empirical probabilities on the left and the theoretical probabilities (according to the Bayesian network) on the right. While the ranks of the expressions – by context type and ImPost-adverbial – are identical, the two distributions are considerably different. Even though the model in the form displayed in Figure 19 may not be entirely realistic, it might provide a starting point for a more elaborate model. This is one of the main assets of Bayesian networks: they can be extended and built up step by step, and their graphical nature helps researchers to develop an understanding of the data.

6 Conclusions and outlook

The present chapter has provided an overview of various types of methods used in comparative corpus linguistics and translation and interpreting studies. Some

[10]The model has an accuracy of 88.2% for the entire training set, and of 79.5% with a 0.9/0.1 training/test split, averaging over ten-fold cross-validation. We used a radial kernel, with $C = 3$ and $\gamma = 0.1$.

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of the methods are well known, – e.g., mosaic plots (as in Figure 1) – while others may be less familiar to parts of the community of contrastive linguists and translation scholars.

We discussed two major types of ordination: CA (with and without stacking), and MCA (with and without supplementary variables). We made an effort to convey an intuitive understanding of the mathematics underlying these techniques. Users have to make specific choices and decision, e.g., CA vs. MCA, the type of map (symmetric, asymmetric, row principal, etc.), and such decisions need to be informed. We also highlighted the connection between (M)CA and the familiar Chi-squared statistic (e.g., the connections between this statistic and Chi-squared distance, inertia, and the ϕ-coefficient).

It seems to us that graph-based methods have some potential that has not fully been exploited. Context-conditional correlation graphs (CCCGs, cf. Sect. Fig. 4 in Sect. 3) allow us to represent a large amount of information about pairwise associations between variables in a single plot. Sankey diagrams (Figures 15, 16 and 17), which are commonly used in other fields, do not seem to be widespread in translation studies, though they offer a very good way of summarizing directional data. Translation graphs visualize choices made in translation, under specific conditions. While the type of graph that we provided (Figure 18 in Sect. 5) does not contain any type of statistical information, such information could be added, potentially using Bayesian statistical reasoning as mentioned in Sect. 5.2, where we presented a type of graphical model, i.e., Bayesian networks.

Once again, we would like to remind the readers that the purpose of the present chapter was not to present research results about ProxPost-adverbials. The data was mainly used for the purpose of illustration. As previously mentioned, more thorough studies on ProxPost-adverbials were presented in earlier publications of ours (Atayan et al. 2019, Gast & Atayan 2019, Gast et al. 2019, Atayan 2021a,b), where some of the visualizations presented in this chapter were also used.

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Appendix A Appendix

A.1 Notes on Sect. 3

When dealing with categorical data as discussed in this chapter, we typically want to know if two or more variables correlate. In other words, we want to

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know if a property of some observation – a level of a variable – allows us to make inferences about some other property of that observation. According to the null hypothesis (of statistical independence), which says that the two variables are unrelated, the conditional probability of observing a specific property A in some observation x, given some other property B of x, is identical to the probability of observing A in x independently of B (cf. (9a)). Put differently, the probability of any x exhibiting both A and B is the product of the probabilities of A and B (cf. (9b)).

(9) Assumption of independence:
   a. $p(A|B) = p(A)$
   b. $p(A \wedge B) = p(A) \times p(B)$

If two variables do not correlate, the frequencies in a contigency table will be proportional along both dimensions. Consider Table 8.

Table 8: Schematic contingency table

	A: level 1	A: level 2	Σ
B: level 1	a	b	a+b
B: level 2	c	d	c+d
Σ	a+c	b+d	a+b+c+d

If the variables A and B are independent of each other, we expect the ratio of any two pairs of cells $\frac{c_1}{c_2}$ and $\frac{c_3}{c_4}$ to be the same if c_1 and c_2, and c_3 and c_4, are in the same column, and c_1 and c_3, and c_2 and c_4, are in the same row (including the marginal row and column sums). For instance:

(10)
   a. $\frac{a}{c} = \frac{b}{d}$
   b. $\frac{a}{a+c} = \frac{b}{b+d}$
   c. $\frac{a}{c} = \frac{a+b}{a+b+c+d}$
   d. $\frac{a}{b} = \frac{c}{d}$
   e. $\frac{a}{a+b} = \frac{c}{c+d}$
   f. $\frac{a}{a+c} = \frac{a+b}{a+b+c+d}$

Based on such analogies, we expect the following count for cell a, under the assumption of statistical independence of the variables. The equation in (11) follows from (10f) by multiplication with $(a+c)$ (the symbol $\hat{}$ on the $\hat{a}$ indicates that this variable represents an expected value).

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(11) $\hat{a} = \frac{(a+c)(a+b)}{a+b+c+d}$

The difference between the observed frequency a and the expected frequency $\hat{a}$ provides a way of determining deviations from statistical independence.

In what follows we use the matrix formalism, which provides a compact way of expressing and generalising the above example. C represents the original contingency table, and $\hat{C}$ corresponds to the matrix of the expected frequencies as calculated from the independence condition as stated in 9. The matrix of **Pearson residuals** is then calculated as:

(12) Pearson residuals
$R = (C - \hat{C})/\sqrt{\hat{C}}$

The matrix $\sqrt{\hat{C}}$ contains the square roots of the cells in $\hat{C}$. Division by the square root of the expected value is a standardization step. We assume Poisson sampling (independent selection of subjects), so each cell is assumed to follow a Poisson distribution. In a Poisson distribution, the variance is equal to the expected value (both values are represented by the parameter λ). We divide the difference between the expected and the observed value by the square root of the variance – the standard deviation – to standardize the values.

In matrix notation, the matrix of expected values $\hat{C}$ can be represented as follows:

(13) a. $\hat{C} = (m_m \otimes m_n)/N$, (the outer product of row and column margins, divided by the number of observations)
b. $m_m = 1C, m_n = 1C^\mathsf{T}$, (row and column sums)
c. $N = 1^\mathsf{T}C1$ (the sum of all cells)

The **adjusted Pearson residuals** are obtained as shown in (14). We divide the difference between the observed and the expected value by an estimate of the standard error based on the assumption that the marginal sums are fixed, therefore assuming a hypergeomteric distribution. Adjusted Pearson residuals are always larger than unadjusted Pearson residuals, as the additional product in the demoninator – $\sqrt{(1 - m_m/N)(1 - m_n/N)}$ – is always smaller than 1.

(14) $\tilde{r} = \frac{C - \hat{C}}{\sqrt{\hat{C}(1-m_m/N)(1-m_n/N)}}$

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A.2 Notes on ordination

The information in this section is mainly intended to facilitate an intuitive understanding of the methods. For more complete and more accurate information, please consult Greenacre (2016).

6.0.1 Some terms and concepts

A **contingency table** is a table with counts of observations with specific properties, cf. Table 9.

Table 9: Contingency table

	comm	dir.boul	repr	Σ
gleich	4	3	18	25
sofort	2	14	18	34
unverzüglich	1	6	2	9
Σ	7	23	38	68

A **percentage table** shows the relative frequency of each cell, i.e., the cell value divided by the number of observations (cf. Table 10). The **row** and **column masses** (the margins of Table 10) are the relative frequencies of all observations in specific rows or columns.

Table 10: Percentage table

	comm	dir.boul	repr	Σ
gleich	0.06	0.04	0.26	**0.37**
sofort	0.03	0.21	0.26	**0.50**
unverzüglich	0.01	0.09	0.03	**0.13**
Σ	**0.11**	**0.34**	**0.56**	1

The **profile** of a given category constituting the row of a contingency or percentage table is the rowwise proportion of that category (cf. Table 11).

Table 11: Row profiles

	comm	dir.boul	repr	Σ
gleich	0.16	0.12	0.72	1
sofort	0.06	0.41	0.53	1
unverzüglich	0.11	0.67	0.22	1

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The **average profile** – geometrically speaking the **centroid** – is the average of the row profiles, weighted by the row masses. For the first column:

(15) $\hat{c_1} = 0.37 * 0.16 + 0.5 * 0.06 + 0.13 * 0.11 = 0.1$

The average profile for Table 9 is thus:

(16) $\hat{C}_{j+} = (0.1\ 0.34\ 0.56)$

We can obtain a **standardized profile table** by scaling the columns of the profile table (Table 11) by the inverse of the square root of the average profile values (i.e., we divide each value by the square root of the corresponding average profile value). For example, the average profile value of the first column is 0.1, and its square root is ≈ 0.32. If we divide 0.16 (the first value of the profile of *gleich*) by 0.32, we obtain a value of 0.5:

Table 12: Standardized profile table

	1	2	3
gleich	0.5	0.21	0.96
sofort	0.18	0.71	0.71
unverzüglich	0.35	1.15	0.30

In a space scaled by the inverse of the square root of the expected value Euclidean distances are equivalent with Chi-squared distances.

The **Chi-squared distance** χ^2_Δ is the square root of the sum of the squared differences between the observed values and the expected values, divided by the expected values. For each row m of a matrix $C_{m\times n}$:

(17) $\chi^2_{\Delta,m} = \sqrt{\sum_{i=1}^{n} \frac{(C_{m,i}-\hat{C}_{m,i})^2}{\hat{C}_{m,i}}}$

($\hat{C}$ being the matrix of expected values, see (13))

The sum of the Chi-squared distances of a matrix provides the basis for the concept of **inertia** in Correspondence Analysis. The inertia of a contingency table is "the weighted average of the squared χ^2 -distances between the row profiles and their average profile" (Greenacre 2016: 29). It is weighted by the row masses M. For a matrix $C_{m\times n}$,

(18) $I_C = \sum_{i=1}^{m} \sum_{j=1}^{n} M_i \frac{(C_{i,j}-\hat{C}_{i,j})^2}{\hat{C}_{i,j}}$.

Note that the Chi-squared distance is identical to the Chi-squared value of a table, divided by the number of observations, and that it is equivalent to the ϕ-coefficient squared, so $\phi = \sqrt{\chi^2/n}$.

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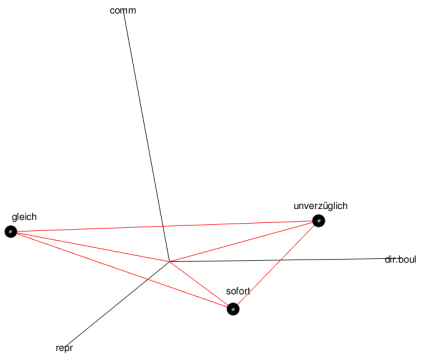


Figure 20: Visualizing the standardized residuals of Table 1

6.0.2 Singular value decomposition

As a starting point of singular value decomposition (Sect. 6.0.2), we use a table of standardized residuals. It is obtained just was in described in Sect. A.1 for the Pearson residuals. However, in Correspondence Analysis we calculate standardized residuals from a percentage table as shown in Table 10, not from a contingency table as shown in Table 9. The two tables differ by a factor of $\sqrt{N}$, i.e., in our case, 8.246 (=$\sqrt{68}$). Table 13 contains standardized residuals.

Table 13: Table of standardized residuals

	comm	dir.boul	repr
gleich	0.11	-0.23	0.13
sofort	-0.10	0.09	-0.03
unverzüglich	0.01	0.21	-0.16

We can represent Table 13 in a three-dimensional coordinate system as shown in Figure 6 in the main text, which is repeated for convenience in Figure 20.

Alternatively, we can think of Table 13 as a mapping from one three-dimensional space to another three-dimensional space. Let us assume a source space that consists of three unit vectors only, i.e., a standard basis (cf. Figure 20). When regarded as a transformation map, Table 13 (via matrix multiplication) maps the standard basis to the residuals shown in red. What matters here is that by regarding the matrix as a transformation map we cannot only map the three standard vectors to new coordinates, we can map *any* point in the source space to the new space. For example, the point located at (0.5 0 0), which is halfway between the origin and the unit vector (1 0 0) in the source space, will

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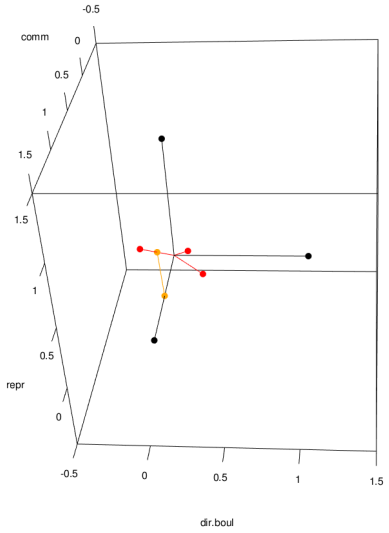


Figure 21: Table 13 as a mapping from a standard basis to a new space

be mapped to a point that is halfway between the origin and the target vector of (1 0 0) in the target space (see the two orange points in Figure 21).

Under the transformation view of the matrix in Table 13 we can perform a 'global' transformation of the source space to the target space. Consider Figure 22, which shows a random sample of vectors (represented as points) of unit length. The cloud of points shown in Figure 22 can be mapped to the space of standardized residuals in the same way in which we mapped the three unit vectors to that space in Figure 20 with matrix multiplication. All points will be mapped to a 'disc' which runs through the origin, see Figure 23.

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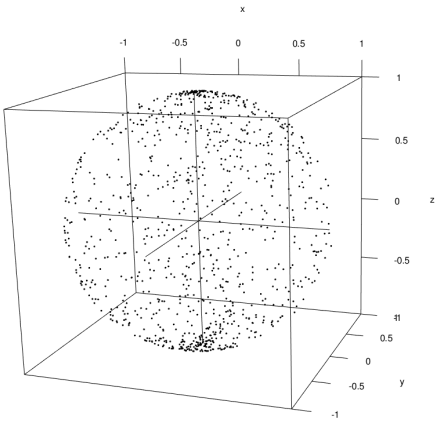


Figure 22: A random sample of 1,000 vectors of length 1

We can now look at the 'disc' of Figure 23 from a different point of view, see Figure 24. This plot shows two straight lines. The horizontal line points in the direction in which the points of the new cloud are spread out most. It is the first **singular vector**. The vertical line is orthogonal to the horizontal line. It shows the direction in which the data is spread out most, on the condition of being orthogonal to the first line. (Note that there appears to be only one line that is orthogonal to the first line, but remember that the disc is actually located in three-dimensional space, see Figure 23, so there is in fact an infinite number of

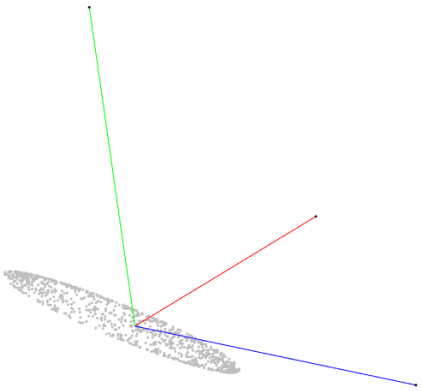


Figure 23: The random sample of 1,000 vectors from Figure 22 mapped to residual space

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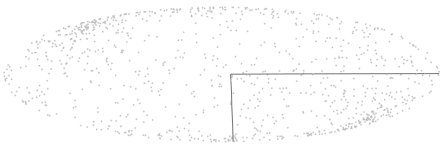


Figure 24: The random sample of 1,000 vectors from Figure 22 mapped to residual space, from a different perspective and with singular vectors

vectors that are orthogonal to the first singular vector, pointing away from the disc).

Singular value decomposition is a method to find singular vectors and their lengths, the singular values. It decomposes a matrix M into two rectangular matrices U and V and a (diagonal) scaling matrix Σ such that:

(19) $M = U\Sigma V^*$

where * denotes the conjugate transpose of a matrix. For matrices of the type dealt with in the present context, the conjugate transpose is identical to the transpose, i.e., $V^* = V^\mathsf{T}$.

The matrices U and V form orthonormal bases, which means that they contain vectors that are orthogonal to each other, and each vector is of unit length. In the case of U, the row vectors are orthogonal, in the case of V it is the column vectors, hence:

(20) $UU^T = I$

(21) $V^TV = I$

In our case, the SVD delivers the following factorization (obtained with the function `svd` in R):

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(22)
$$\text{U} \quad \begin{bmatrix} -0.72 & 0.35 & 0.61 \\ 0.28 & -0.65 & 0.71 \\ 0.64 & 0.68 & 0.36 \end{bmatrix} \qquad \text{V} \quad \begin{bmatrix} -0.25 & 0.91 & -0.32 \\ 0.81 & 0.02 & -0.58 \\ -0.52 & -0.41 & -0.75 \end{bmatrix} \qquad \Sigma \quad \begin{bmatrix} 0.39 & 0.00 & 0.00 \\ 0.00 & 0.12 & 0.00 \\ 0.00 & 0.00 & 0.00 \end{bmatrix}$$

To see that the first two row vectors of U are orthogonal (they stand in a 90° angle to each other), consider their dot product:

(23)
$$\begin{aligned} & u_1 \cdot u_2 \\ & = (-0.715 \cdot 0.285) + (0.348 \cdot -0.647) + (0.606 \cdot 0.707) \\ & = -0.204 - 0.225 + 0.429 = 0 \end{aligned}$$

To see that the length of the first vector is 1:

(24)
$$\begin{aligned} & \sqrt{-0.715^2 + 0.348^2 + 0.606^2} \\ & = \sqrt{0.511 + 0.121 + 0.367} \\ & = \sqrt{1} = 1 \end{aligned}$$

The matrices U and V constitute the bases for the coordinates of the rows and columns in CA. Both matrices are scaled by the inverse of the square root of the average profile values. The matrix of interest is moreover scaled by the singular values in matrix Σ. If we want to locate rows in space constituted by the columns, we will only scale the left singular values, i.e., matrix U. If we want to locate columns in space constituted by rows, by contrast, we scale the right singular values, i.e., matrix V. Matrices of coordinates, here represented as C, can thus be obtained as follows (in the text we were only interested in the row analysis):

(25) row analysis
   a. $C_{row} = W_m U \Sigma$
   b. $C_{col} = W_n V$

(26) column analysis
   a. $C_{col} = W_n V \Sigma$
   b. $C_{row} = W_m U$

In an asymmetric map, where rows are represented in principal coordinates and columns in standard coordinates, we can reconstruct the original matrix from the map via the inner product between row and column vectors. For example, the ImPost-adverbial *gleich* and the column 'repr' have the following coordinates:

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(27) *gleich*: $(-0.46\ 0.07)$

(28) 'repr': $(-0.70 - 0.54)$

These coordinates need to be scaled by the square root of the row masses/average profile (to 'undo' the scaling that was applied previously):

(29) *gleich*: $0.606\ (-0.463\ 0.067) = (-0.281\ 0.041)$

(30) 'repr': $0.748\ (-0.702\ -0.545) = (-0.525\ -0.407)$

Their inner product of these vectors is:

(31) $-0.281 \times -0.525 + 0.041 \times -0.407 = 0.131$

0.131 is also the standardized residual of the cell for *gleich* and 'repr'.

A.3 Notes on Sect. 4

Figure 25 shows the silhouette plot for the clustering reported on in Sect. 4.

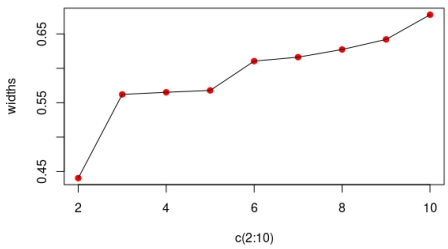


Figure 25: Silhouette widths for cluster solutions

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Chapter 3

Navigating translation effort through translation process research database to unravel dimensions of script and text direction

XYZ Saedi

This chapter delves into the multifaceted interplay between different script and text direction in translation, exploring the nuanced technical, temporal, and cognitive effort exerted by translators. We utilize a methodological approach that combines eye-tracking and keylogging data with 2-way ANOVA, using a series of empirical features (such as insertion, deletion, pause-to-word ratio, total number of fixations, etc.) to measure indicators of effort in translation. The study reveals elevated effort when translating between languages with different scripts and text directions. The findings underscore that script differences may introduce challenges at orthographic, lexical, and segment levels, while text direction differences augment spatial orientation demands. Notably, while both factors individually increased translation effort, an interaction effect between script difference and text direction difference was not observed, suggesting distinct cognitive processes for each challenge. These revelations hold profound implications for translation pedagogy, advocating for tailored training approaches and the integration of specific translation strategies to navigate these difficulties. The chapter contributes empirical insights to translation research, emphasizing the pivotal role of scriptural and spatial attributes in the translation process.

XYZ Saedi. 2026. Navigating translation effort through translation process research database to unravel dimensions of script and text direction. In Tomáš Svoboda, Olga Nádvorníková & Věra Kloudová (eds.), *Advancing interdisciplinarity in empirical translation research*, 55–82. Berlin: Language Science Press. DOI: ??

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1 Introduction

1.1 Research background

The process of translation is influenced by a myriad of factors (Marmaridou 1996; Campbell 1999; Sun 2015; Screen 2016; Rojo López et al. 2021; Zou et al. 2022), among which script and text direction of the language pairs involved play a role (Saeedi & Zou 2022). Nevertheless, most research that relates script and text direction to the cognitive process mainly revolves around topics of bilingualism, second language learning, reading comprehension, and information processing, among others, but not translation studies (Aro 2004; Barrett et al. 2002; Boroditsky 2001; Chatterjee et al. 1999; Spalek & Hammad 2005). For instance, in bilingualism research, Duñabeitia et al. (2009) reported increased priming effects when dealing with words of similar orthographical representations in two languages of the similar script compared to dealing with words of dissimilar orthographical representations in these two languages (e.g., Arabic (en)-> Arabica (es); Dutch (en)-> Holandesa (es)). Other findings from many studies in the fields of reading and information processing have shown effects of native language orientation and text direction on spatial bias in reading comprehension. For instance, readers accustomed to left-to-right-oriented scripts, such as English, and those familiar with right-to-left-oriented scripts, like Arabic, exhibit biases that correspond to their respective text directions (Mashat 2017).

As for translation process research, much research has focused on the relationship between different aspects of language specificity (i.e., unique characteristics, nuances, and idiosyncrasies of a particular language pair that can influence the translation process) (see Campbell 1999; Gile 2005, 2008; Wang & Zou 2018; Zou et al. 2022) and translation difficulty but not on the language script and text direction. Some of these studies linked specific linguistic features like lexical and syntactic complexity to translation difficulty (e.g., Vanroy et al. 2021; Zou et al. 2021). However, little work has focused on how differences in writing systems and text orientations influence the translation process. Whenever mentioned, studies state that language pairs with different scripts or text directions are expected to entail greater efforts, compared to languages with similar script and same text direction. Lacruz et al. (2016) for instance, posited that language pairs with differing scripts involve increased cognitive effort during the translation process compared to language pairs with similar scripts. The underlying rationale might be that the visual-spatial differences in scripts can introduce additional layers of complexity, requiring translators to navigate the apparent differences in the orthographic features of each script. Hence, there remains a

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conspicuous gap, specifically, on the interplay between script, text direction, and translation difficulty. This gap is even more pronounced when juxtaposing different language pairs, such as translations from English into Spanish, English into Chinese, and English into Arabic, in the context of their script and directional attributes. Building on this gap, the nuanced interrelations between script, text direction, and exerted effort in translation warrant deeper exploration.

1.2 Research purpose

This study aims to examine how different scripts and text direction between language pairs impact the effort involved in translation. This would reveal the inherent complexities faced by translators when bridging languages that employ diverse symbolic representations and reading and writing directions. Specifically, we compare the translation process from English into three different target languages: Spanish, Simplified Chinese, and Arabic. Spanish shares the same Modern Latin alphabet with English and has left-to-right text direction. Simplified Chinese utilizes logographic characters and left-to-right text direction. Arabic has a cursive Abjad script written right-to-left. By comparing the translation process of different language pairs, we can determine if variations in script and text direction increase effort.

Script disparities require utilizing distinct visual-spatial pathways and cognitive resources (Kandhadai & Federmeier 2010; Perfetti et al. 2007). This likely increases the cognitive load for translators, but there is a lack of research directly testing this hypothesis. Beyond scripts, flipping text direction may involve extra spatial processing and attention shifting, such as left-to-right (e.g., English) versus right-to-left (e.g., Arabic). Contrasting our three language pairs allows investigation of these two variables. Spanish acts as a control, while Chinese exemplifies different script but same text direction as English and Arabic both different script and different text direction.

Demonstrating increased difficulty for the Chinese and Arabic pairs would have important theoretical and practical implications. It would empirically validate the influence of script and text direction on the cognitive translation process. Theoretically, this helps advance models of how language systems interact during translation. Practically, it suggests value in developing specialized pedagogical approaches tailored to teaching translation between dissimilar scripts and text directions. For Chinese, this may involve an added focus on strategies for converting between logographic and alphabetic systems during the translation process, like engaging in holistic visual-spatial processing of extended chunks of Chinese logograms during translation and mapping meaning across writing

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systems. For Arabic, emphasis could be placed on mastering the intricacies of its right-to-left script progression, coupled with exercises that delve into the nuances of its phonetic, morphological and syntactic structures, and activities designed to enhance spatial attention adaptation to its unique cursive and diacritical features.

Our study utilizes eye-tracking and keylogging data of from-scratch translation experiments. We hypothesize English-into-Spanish translation would show relatively less effort due to shared script and text direction. English into Chinese and English into Arabic are expected to show an increased effort due to different scripts and, in case of Arabic, additionally, reversed text direction. By pinpointing the challenges and proposing tailored strategies, our study represents an essential step toward better understanding and facilitating the quality and efficiency of translations across genetically distant languages (Xiao 2010). Consequently, this chapter attempts to address the following research questions:

(1) Does translating from a source text (ST) to a target text (TT) of different scripts (e.g., English into Chinese/Arabic) require more translation effort than translating between languages with similar scripts (e.g., English into Spanish)?

(2) Does the difference in text direction between the languages (e.g., English into Arabic) contribute to an increased effort in translation?

(3) Is there an interaction effect between the script difference and text direction difference on the cognitive effort during the translation process?

1.3 Methodology

1.4 Datasets description

This study utilizes eye-tracking and keylogging data from three existing datasets available at the Center for Research and Innovation in Translation and Translation Technology's Translation Process Research Database (CRITT TPR-DB) (Carl et al. 2016).[1] The datasets are BML12 (English into Spanish), RUC17 (English into Simplified Chinese), and AR19 (English into Arabic). The three datasets contain

[1]The CRITT TPR-DB consists of studies that were conducted with Translog-II, Trados or with the CASMACAT workbench. Behavioral and textual data from translation experiments is publicly available, and a list of publications utilizing this database is accessible on the CRITT@kent website: https://sites.google.com/site/centretranslationinnovation/tpr-db/public-studies?authuser=0.

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sessions of different tasks (from-scratch translation, post-editing, sight translation, etc.). We only use the from-scratch translation sessions for the purpose of this study. These three datasets involve an identical set of English STs, thus the linguistic complexity of the STs is identical among all datasets. The STs include six English texts from Encyclopedia (four news articles and two sociology articles). The STs comprise a total of 847 words over 41 segments on the sentence level. A detailed breakdown of segments and words for each ST is outlined in Table 1.

Table 1: Breakdown of segments and words for the six source texts.

Source text	1	2	3	4	5	6	Total
Source segments	11	7	5	5	6	7	41
Source words	160	153	146	110	139	139	847

English-into-Spanish dataset (BML12): This dataset encompasses 32 translators, all of whom are native Spanish speakers. It serves as the baseline condition, since Spanish shares the same Latin alphabet and left-to-right direction as English. The gaze data was collected using a Tobii T60 remote eye-tracker.

English-into-Simplified Chinese dataset (RUC17): This dataset contains data from 21 translators, all native Chinese speakers. It represents a language pair with different scripts but the same text direction since simplified Chinese uses logographic script and generally maintains left-to-right text direction. The gaze data was recorded using a Tobii TX300 eye-tracker.

English-into-Arabic dataset (AR19): This dataset includes 22 translators, all of whom are native Arabic speakers. The gaze data was collected with an SMI iView eye-tracker.

The keystroke logging data was collected using Translog-II for all datasets. All translators in these three datasets have English as their second language and are native speakers of their respective target language. In total, there are 162 translation sessions consisting of 1,116 individual segments in this research. As can be seen in Table 2, the number of sessions and segments of English-into-Spanish (BML12) and English-into-Chinese (RUC12) datasets are in closer range compared to a smaller number for the English-into-Arabic dataset (AR19). Therefore, we normalize the data by the number of ST words on the segment level for comparable analysis. An interesting observation here is that, proportionally speaking, the number of target words and characters for the English-into-Spanish dataset (BML12) is remarkably higher than both the English-into-Chinese dataset

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(RUC17) and the English-into-Arabic dataset (AR19). We delve into a discussion about the correlation of such divergence with technical and temporal effort in section 1.7 (Results and Discussion) of this chapter (see subsections 1.8., 1.8.2., 1.10., and 2.).

Table 2: Detailed description of the three datasets.

Datasets	Number of sessions	Number of segments	Number of target words	Number of target characters
BML12 (en-es)	60	411	10,050	56,936
RUC17 (en-zh)	62	428	8,507	14,470
AR19 (en-ar)	40	277	6,131	33,276
Total	162	1,116	24,688	104,682

1.5 Grouping language pairs for analysis

One of the primary aims of this chapter is to dissect the influence of script and text direction on translation effort. To this end, we initially compare the translation process of English-into-Spanish against the other two language pairs (English-into-Chinese and English-into-Arabic) to isolate and examine the impact of script differences. This comparison is pivotal as English and Spanish have similar scripts and the same text direction, offering a baseline for assessing the additional effort introduced by different scripts. Subsequently, we combine English-into-Spanish and English-into-Chinese into a group to focus on the impact of text direction differences in comparison to English-into-Arabic. This methodological decision stems from the need to separately evaluate the influence of the two independent variables, script, and text direction, where English-into-Arabic is distinct in both aspects. By comparing this combined group against English-into-Arabic, we aim to isolate and scrutinize the translation effort attributed solely to the opposite text direction. This strategic grouping allows for a nuanced analysis of how each factor—script and text direction—contributes to the overall translation effort, while also considering their interaction effect.

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1.6 Indicators of translation effort

Several indicators have been meticulously examined to gauge the intricacies of translation effort at the segment level in this chapter. These indicators are selected according to Krings' categorization of technical, temporal, and cognitive efforts (2001). While this framework was developed to better understand and measure the challenges faced by translators, especially in the context of post-editing machine translation outputs, they collectively provide a holistic view of the multifaceted effort involved in the translation process (Carl et al. 2022). Therefore, this framework is utilized herein for from-scratch translation data. These effort indicators are measured relying on several features available in the CRITT TPR-DB (Carl et al. 2016). Technical effort primarily focuses on the tangible modifications made during the translation process. This includes the insertion keystrokes ("Ins"), the deletion of existing characters ("Del"), and the overall number of revisions made to the segment ("Nedits"). These metrics provide a straightforward measure of the manual adjustments a translator makes at the segment level. Temporal effort is encapsulated by the "DurLog" metric, which is calculated by the logarithm of the duration spent on translating the segment.

The cognitive effort involved in translation is perhaps the most intricate to measure, given its intangible nature. One of the primary metrics we use in this chapter is pause-to-word ratio ("PWR"), which stands for the number of pauses per word in a segment. PWR is a measure of cognitive effort mainly developed in machine translation post-editing, although it has been applied to assess cognitive effort in from-scratch translation (Lacruz et al. 2016; Lacruz et al. 2018). Another set of metrics that offer insights into cognitive effort are "FixS" and "FixT", which represent the total number of fixations on the STs and TTs. These fixation counts offer a quantitative measure of the attention and focus a translator directs towards the STs and TTs during the translation process. Complementing the fixation metrics are "TrtS" and "TrtT", which stand for the total reading time on the STs and TTs, respectively. These metrics capture the total of all fixation durations on the ST side or the TT side of certain segment, irrespective of when these occurred during the session. In essence, they provide a measure of the time a translator spends comprehending and processing both the STs and TTs.

Lastly, the concept of translation entropy is introduced through the "HTot" metric. HTot, at the segment level, denotes the total translation entropy of a source segment (Carl 2021). HTot measures segment-level entropy by integrating HTra (word translation entropy) and HCross (entropy of the Cross value). It assesses the variation concerning both word choice and word ordering (Carl & Schaeffer 2017). For the indicators of cognitive effort in translation, two-way

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ANOVAs are conducted with script (similar vs. different) and text direction (same vs. opposite) serving as independent variables. This allows analysis of the main effects of script and text direction and their interaction effect on translation effort. To this end, the dependent variable HTot is chosen as the primary measure for our analysis due to its comprehensive nature, reflecting both word translation entropy and the entropy of Cross value. This measure's capability to integrate various aspects of cognitive effort renders it particularly suitable for examining translation difficulty. In addition to HTot, we consider PWR, FixS, FixT, TrtS, and TrtT. The selection of HTot does not preclude the relevance of these indicators; rather, it underscores our methodological decision to anchor our analysis in a variable that encapsulates the multifaceted nature of translation effort.

1.7 Results and discussion

In our analysis, we used the Mann-Whitney U test to assess the statistical significance of the differences observed between groups across our measures of effort. In this chapter, we focus on discussing results from measures that demonstrated significant differences, as they are of relevance to our research questions and contribute meaningful insights into the impact of script and text direction on translation effort. Non-significant results, while part of the comprehensive analysis, are not elaborated upon here, as they do not substantively alter the interpretation of our primary findings.

1.8 Script

1.8.1 Script and technical effort

In examining the technical effort associated with script, we analyzed metrics of insertion (Ins), deletion (Del), and the number of revisions (Nedit) per ST token as Figure 1 shows. For the English-into-Spanish dataset with similar script, these metrics are represented in the blue plot boxes (SScript), while the English-into-Chinese and English-into-Arabic datasets with different scripts are depicted in the orange plot boxes (NScript). Notably, the data indicates that the language pair with similar scripts exhibits a higher frequency of insertions, deletions, and number of revisions compared to the insertions, deletions, and number of revisions of language pairs with different scripts. This observation aligns with the disparities in target word and character counts among the datasets, as shown in Table 2. Such findings underscore an increased technical effort inherent to the similar-script dataset relative to its different-script counterparts.

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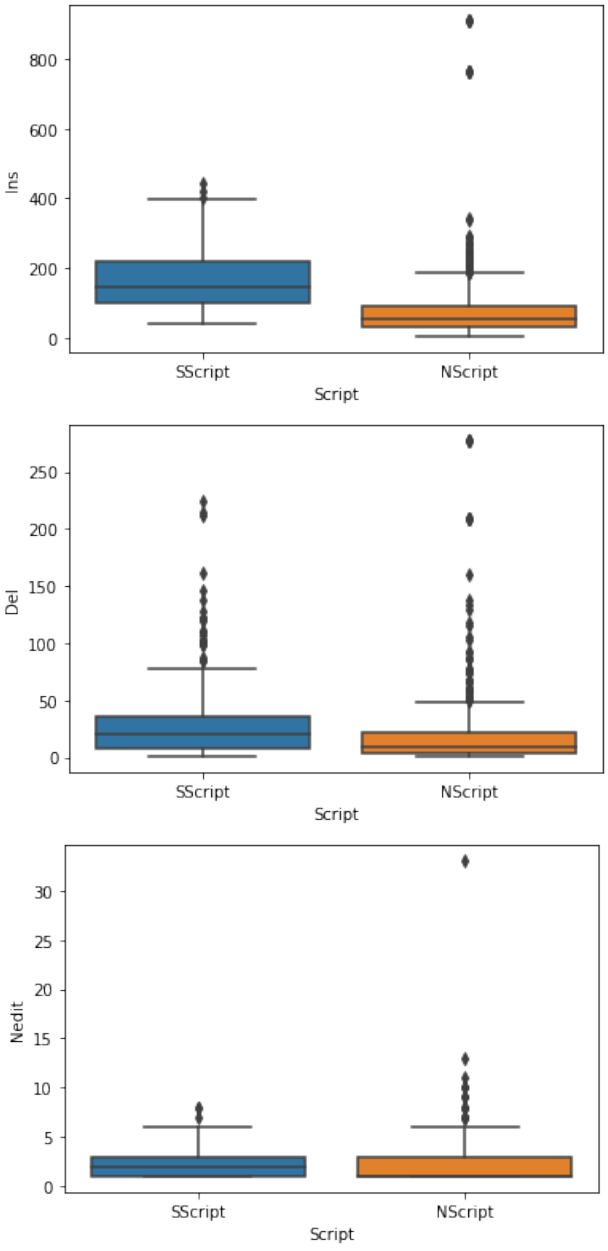


Figure 1: Insertion, deletion, and number of revisions for similar-script and different-script language pairs.

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At first glance, one might assume that translations between languages with similar script would be more straightforward, given the shared orthographic features. Yet, our results challenge this assumption, showing higher technical effort in terms of insertions, deletions, and revisions for the similar-script English-into-Spanish dataset compared to the different-script datasets. In addition to the variations in word length among languages, a possible reason is that the higher degree of lexical similarity found between languages that are closely related (such as English and Spanish) introduces more opportunities for ambiguity that needs to be resolved in the translation process. Many words in both languages have similar orthography and phonetics. This can lead to negative priming effect manifested in a myriad of phenomena such as "false friends" – words that look or sound similar in both languages but have different meanings. Such similarities can lead to confusion and require careful handling during translation (Granger & Swallow 1988). In our English-into-Spanish dataset (BML12), the English source word 'increases' for instance was translated by half of the translators into Spanish using an iteration of the target word '*incrementos*' and later edited into an iteration of the target word '*aumen*'. This introduced an average of 12.25 insertions, 3.13 deletions, and 15.38 edits per English-into-Spanish translator for the translation of this source word in our dataset.

In contrast, when translating between languages with different scripts (e.g., English-into-Chinese and English-into-Arabic), there is less likelihood of confusion due to "false friends" or orthographically or phonetically similar words. The challenges here lie more in understanding the structure, context, and nuances of the source and target languages. The English-into-Arabic dataset (AR19) shows that the source word 'increases' was translated by all the translators into an iteration of the target word '□□□□□' ('increase'). The average number of insertions was 6.9, 1.42 for deletions, and 8.32 for edits per English-into-Arabic translator for this word, which are significantly lower averages than English-into-Spanish. By the same token, English-into-Chinese translators translated the source word 'increases' into an iteration of the target word '□□' ('rise') with minimal edits. The average number of insertions per translator was 2.75, 0 for deletions, and 2.75 for edits for this word. Additionally, the divergence in target word and character counts, as mentioned, might also reflect the inherent challenges of linguistic transfer between similar-script languages. This echoes findings from previous studies like (Rojo López et al. 2021), which suggest that higher familiarity can sometimes lead to increased challenges in translation.

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1.8.2 Script and temporal effort

As for the analysis of temporal effort in regard to script, we took the DurLog metric as the indicator of effort. In Figure 2, the red line in the histogram refers to the logarithm of production duration for the similar-script English-into-Spanish dataset (Sscript), while the blue line refers to the different-script English-into-Chinese and English-into-Arabic datasets (Nscript). We can see that translation with the similar-script language pair generally takes less time and eventually less temporal effort than the different-script language pairs. This is an interesting finding given the fact that English-into-Spanish sessions tend to have more insertions, deletions, and number of revisions, still they are faster.

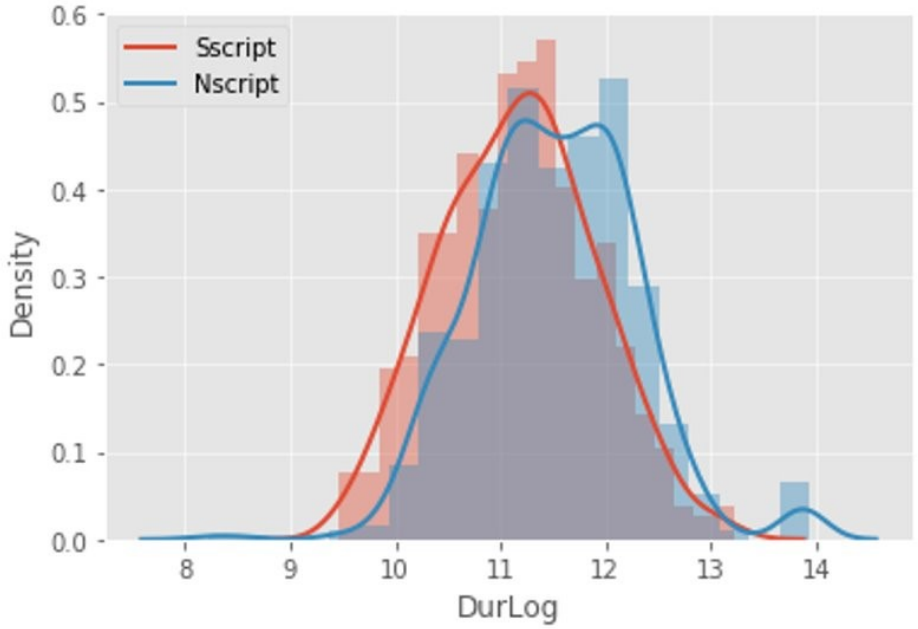


Figure 2: Production duration for similar-scripts and different-scripts language pairs.

The temporal effort analysis adds another layer to our understanding of the nuances involved in translation across different scripts. While the similar-script English-into-Spanish dataset exhibits higher technical effort in terms of insertions, deletions, and overall revisions, it intriguingly requires less temporal effort. This finding might be reminiscent of the "familiarity and fluency" observed in cognitive psychology, where familiarity with a task can lead to quicker completion, even if the task involves more steps or corrections (Alter & Oppenheimer

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2008). In the context of translation, translators working with similar scripts might experience an increased fluency in reading and processing the ST, allowing them to translate faster despite the increased technical effort. Moreover, these findings might suggest different cognitive load on translators (Sweller 2011). The different scripts of Chinese and Arabic could introduce an additional layer of cognitive load on the translators, slowing down their translation speed. On the other hand, while Spanish and English have similar scripts, the translators might be leveraging their familiarity to process information quickly, even if it entails more edits. This dynamic interplay between technical and temporal efforts underscores the multifaceted nature of translation processes and highlights the importance of considering multiple dimensions of effort when analyzing translation tasks.

1.9 Script and cognitive effort

Focusing on the cognitive effort, we take several indicators for our analysis. The first of which is pause-to-word ratio (PWR). As seen in Figure 3, the blue plot box of the similar-script English-into-Spanish (SScript) pause-to-word ratio (PWR) is lower than that of the orange plot box of the different-script English-into-Chinese and English-into-Arabic (NScript). To determine the statistical significance of the differences in PWR, we conducted the Mann-Whitney U test. The results revealed a significant difference ($p < 0.05$) between the SScript and NScript groups, suggesting that script differences indeed contribute to the cognitive effort in translation tasks. This would be an indicator of more cognitive effort exerted by the different-script language-pair translators compared to the similar-script language-pair translators.

Pause durations in translation have been frequently associated with cognitive effort, with longer pauses typically reflecting increased cognitive effort or decision-making processes (Dragsted 2005). The elevated PWR observed for the different-script language pairs (English-into-Chinese and English-into-Arabic) can be seen as a manifestation of the added cognitive challenges that these translators encounter, potentially due to the script differences. The lower PWR for the similar-script English-into-Spanish translations, despite its increased technical effort as discussed earlier, might indicate a smoother cognitive flow in terms of processing and decision-making. This smoother flow could be attributed to the orthographic familiarity between English and Spanish, which reduces the cognitive load of script decoding and allows translators to focus more on meaning transfer. Such findings underscore the nuanced ways in which script differences can modulate the cognitive load of translation, reaffirming the importance of

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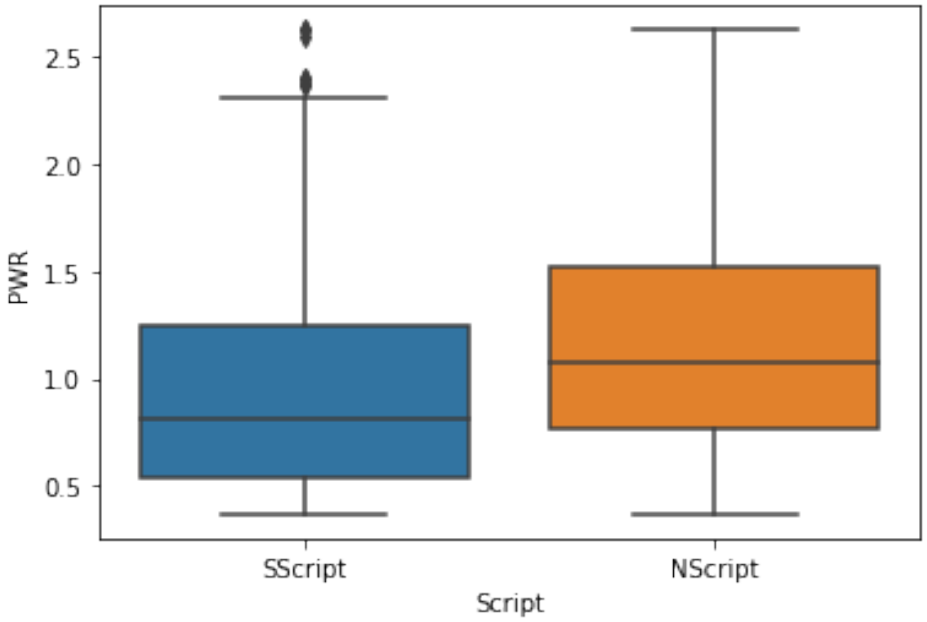


Figure 3: Pause-to-word ratio for similar-script and different-script language pairs

orthographic considerations in translation research and training (Saeedi & Zou 2022).

The next cognitive effort indicators are the total number of fixations on the ST (FixS) and the total number of fixations on the TT (FixT). On the left side of Figure 4, FixS results align with PWR results and more visibly suggest less cognitive effort in general for the similar-script language pair in blue plot box (SScript) compared to the different-script language pairs in green plot box (NScript). As for FixT, as shown on the right side of Figure 4, the similar-script language pair in blue in the histogram (Sscript) show more variant effort than the different-script language pairs in orange (Nscript).

This suggests that the different-script language pairs require more cognitive effort, particularly evident in the increased fixations on the ST (FixS). This may indicate the nuanced challenges of processing a source language that diverges in script from the target language, especially given the fact that translators' second language of command (L2) is English (the ST). On the other hand, the variability in fixations on the TT (FixT) for the similar-script language pair suggests that translators might be engaging in meticulous self-monitoring and revisions.

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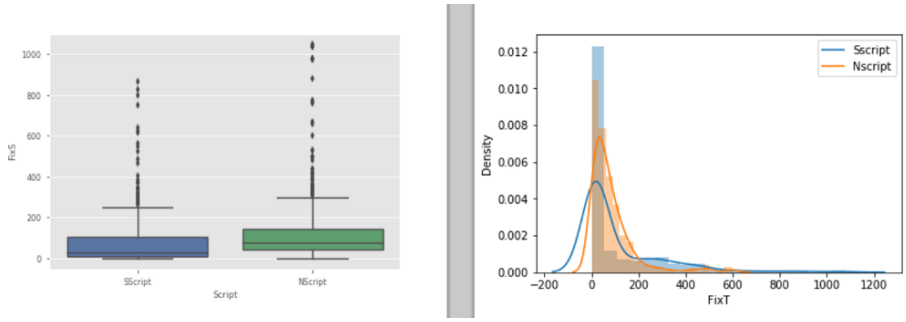


Figure 4: The total number of fixations on the source text and target text for similar-script and different-script language pairs

Conversely, the less variation of fixations on the TT for different-script language pairs might suggest the translators' overall familiarity with TT scripts given the TTs are their first language of command (L1). It may also underscore a consistent pattern of cognitive effort exerted in navigating between the source and target scripts during the translation process.

Another set of cognitive effort indicators is the total reading time on the ST (TrtS) showing on the left side of Figure 5 and the total reading time on the TT (TrtT) showing on the right side of Figure 5. For both, we can still see lower values for the similar-script English-into-Spanish language pair as shown on the left blue plot boxes in each graph (SScript) compared to the different-script language pairs on the right side of each graph (NScript). This is yet another marker of differences in cognitive effort that might be attributed to orthographical effects.

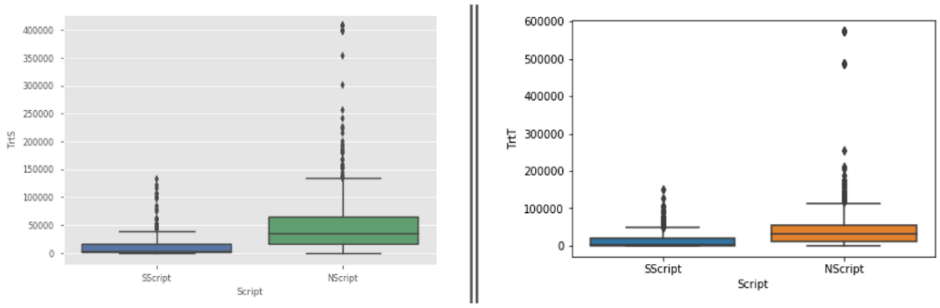


Figure 5: Total reading time on the source and target for similar-scripts and different-scripts language pairs

Extended reading time has been associated with increased cognitive effort in translation (Gile & Lei 2020). The shorter reading time for the similar-script

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English-into-Spanish translations, as compared to the different-script pairs, reinforces our earlier observations of reduced cognitive effort due to script familiarity. The consistency across multiple indicators of cognitive effort—PWR, fixations, and now reading times—suggests that the degree of orthographic similarity significantly impacts the cognitive effort of translation, streamlining processes when scripts are aligned and adding layers of translation difficulty when they differ.

The last indicator of cognitive effort for script incorporated here is the normalized HTot. Figure 6 shows a lower variation in the translation (HTotN) for the similar-script English-into-Spanish language pair in blue (SScript) compared to the different-script language pairs in orange (NScript).

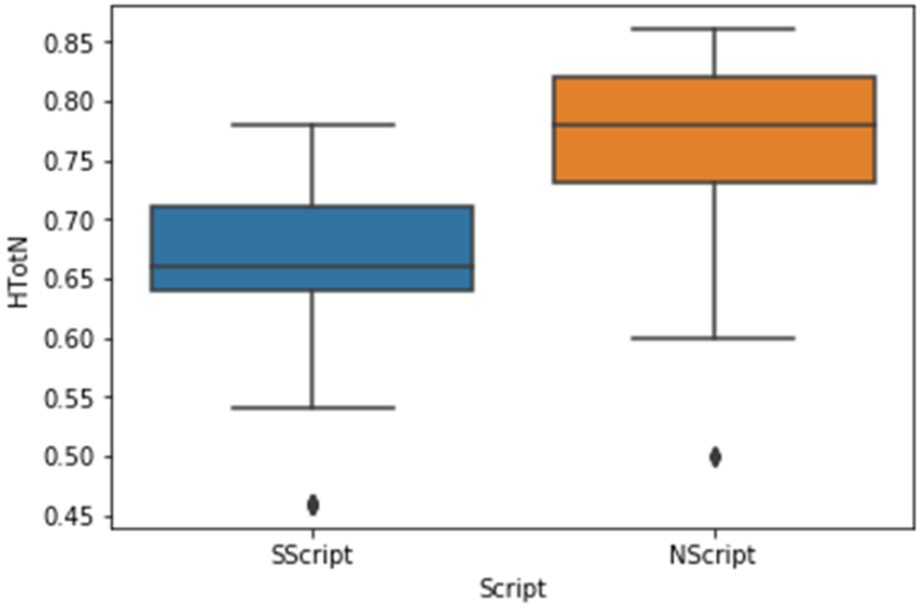


Figure 6: Normalized HTot for similar-scripts and different-scripts language pairs

The normalized HTot represents the average joint probability over alternative translations of a segment. As seen in Figure 6, the similar-script English-into-Spanish translations demonstrate less variation in HTotN, suggesting a more consistent and potentially less effortful translation process when compared to different-script pairs. Results from all these measures of cognitive effort consistently point toward the pivotal role of orthographic similarities and differences

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in translation difficulty. Increased variation in such probabilistic measures indicates a broader range of potential translations or more cognitive effort in selecting among alternatives (Schaeffer et al. 2016). The reduced variation for the similar-script pair might indicate that overall, translators face fewer ambiguities or alternatives, potentially due to the script and linguistic similarities.

In analyzing the translations of our source English sentence "Sociology is a relatively new academic discipline" into Spanish, Chinese, and Arabic, we observe notable differences impacted by word choice rather than word order. The Spanish translation, "*La sociología es una disciplina académica relativamente nueva,*" closely mirrors the English sentence in vocabulary but differs in word order, Conversely, the Chinese translation "□□□□□□□□□□□□□□" (Shèhuìxué shì yī mén xiāngduì jiào xīn de xuékē) diverges in word choice when compared to English although order is quite similar. The Arabic translation (□□□ □□□□□□□□ □□ □□ □□□□ □□□□□□□ □□□□ □□□□□□ ʻIlm al-ijtimāʻ huwa min al-ʻulūm al-akādīmīyah al-ḥadīthah nasbiyan) also introduces significant variations in lexical choice but not in word order. Juxtaposing these observations with the findings of HTotN may underscore how lexical variations influence translation effort, irrespective of the syntactic alignment between the target languages and the source language. The similarities and differences in word choice can be attributed to those in the scripts of these languages.

From our various cognitive indicators, the consistent findings of higher cognitive effort associated with differences in scripts between the language pair underscore the distinct challenges faced when translating between different scripts. These insights have direct implications for translator training. Given the pronounced cognitive effort associated with different-script language pairs, it is imperative for translation curricula to incorporate tailored modules that equip student translators with strategies to address these challenges. Educators must highlight the cognitive intricacies linked to script variations, ensuring student translators are aware of and become proficient in tackling this layer of translation difficulty both technically and cognitively. Moreover, integrating practical sessions, for instance intensive training on the proficiency of the L2 and transfer speed between the L1 and L2 on both lexical and phrasal level, can better prepare student translators for real-world translation tasks. Embracing advanced technological tools, such as specific terminology management tools, translation memories or AI-assisted translation tools, can further ease the challenges posed by script differences.

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1.10 Text direction

1.10.1 Text direction and technical effort

We repeat the same analysis procedures with text direction by comparing the datasets with language pairs sharing the same left-to-right text direction (English-into-Spanish and English-into-Chinese) to the dataset with language pair of different text directions (English-into-Arabic) since Arabic has a right-to-left writing system. For the technical effort indicators, all the indicators such as insertions (Ins), deletions (Del), and number of revisions (Nedit) generally show less variant technical effort for the same-direction datasets shown on the left side of each graph (SDirec) in Figure 7 when compared to the different-direction dataset shown on the right side of each graph (NDirec).

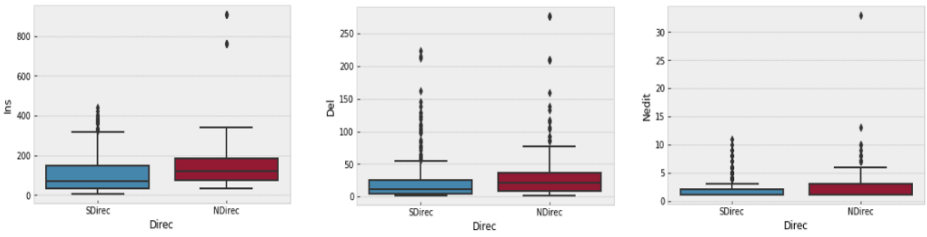


Figure 7: Insertion, deletion, and number of revisions for same-text-direction and different-text-direction language pairs

Variations in text direction, particularly between left-to-right and right-to-left writing systems, have been posited to introduce unique cognitive challenges (Fernandes et al. 2021). Our findings suggest that translating between languages with the same text direction (English-into-Spanish and English-into-Chinese) leads to a less varied technical effort, in terms of insertions, deletions, and overall revisions, compared to translations involving directionally divergent scripts (English-into-Arabic). Text direction has been observed to influence various cognitive tasks, with right-to-left readers often employing different spatial and sequential processing strategies compared to left-to-right readers (Chokron & Imbert 1993). The less variance in technical effort for the same-direction datasets might be attributed to the relative ease of transferring information within the same spatial framework. In contrast, translating into a language with a divergent text direction, such as Arabic, might introduce additional cognitive load, necessitating more varied technical interventions to translation production.

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In addition to analyzing metrics such as insertion (Ins), deletion (Del), and number of revisions (Nedit) per ST token, we further examined the technical effort using a measure of the number of insertions divided by the number of deletions per segment. We found that language pairs with the same text direction exhibited a higher ratio of insertions to deletions per segment (10.31) compared to language pairs with different text directions (9.95). This result suggests that maintaining consistent spatial frameworks in translation, such as left-to-right or right-to-left orientation, can impact the distribution of technical effort. The lower ratio observed for language pairs with different text directions may reflect the additional cognitive load associated with transferring information across divergent spatial frameworks, highlighting the complexity inherent in translations involving directional divergence.

2 Text direction and temporal effort

For the temporal effort, we see the same-direction language pairs in blue (SDirec) in Figure 8 generally take less time (DurLog) than the different-direction language pair in orange (NDirec). This might suggest that Spanish and Chinese translators have less temporal effort than the Arabic translators.

The more condensed durations for the same-direction language pairs (English-into-Spanish and English-into-Chinese) in comparison to the different-direction English-into-Arabic pair substantiates the argument that text direction can significantly modulate the time dimensions of translation processes. Prior research has highlighted that text direction can impact reading and processing speeds (Eviatar 1997). Mismatches in text direction would therefore introduce additional cognitive load, especially during the drafting and revision stages of translation. The observed temporal disparities might be attributed to the inherent challenges of adjusting from a left-to-right writing system to a reversed spatial framework in right-to-left system, such as from English into Arabic. The additional translation durations for English-into-Arabic translations may reflect the extra effort required for spatial orientation adjustments while drafting right-to-left TT. Moreover, translators may have iteratively relocated between the ST and TT during translation refinement to ensure accuracy and fluency when translating into the opposite text direction.

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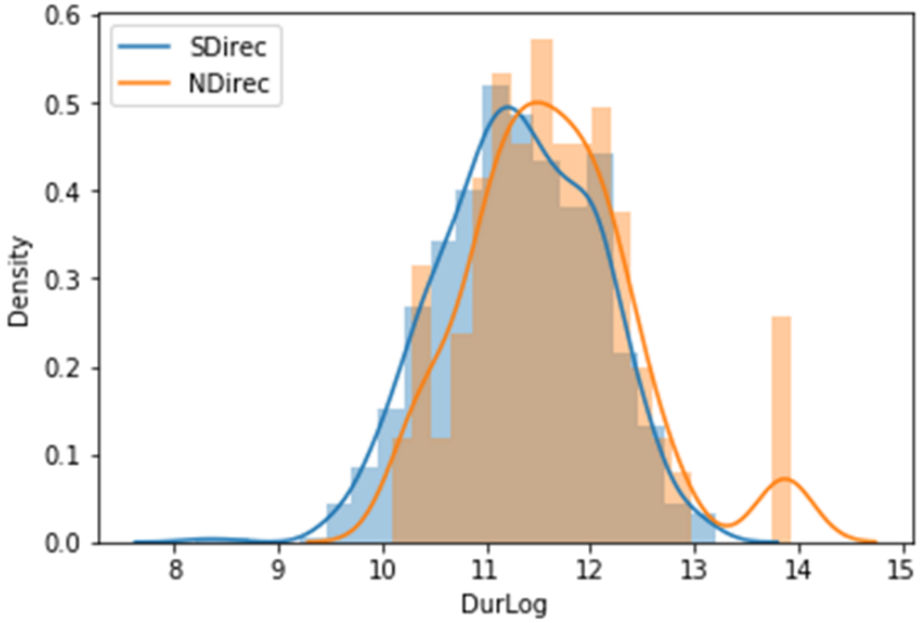


Figure 8: Production duration for same-direction and different-direction language pairs

3 Text direction and cognitive effort

Like our analysis of script effects, the first cognitive effort measure we examine in relation to text direction is PWR. As can be seen in Figure 9, the PWR shows less variation for the same-direction language pairs in blue (SDirec) and more variation for the different-direction language pair in red (NDirec).

The attenuated variation in PWR for the same-direction language pairs (English-into-Spanish and English-into-Chinese) suggests a more streamlined cognitive trajectory during the translation process. In contrast, the heightened variation for the different-direction English-into-Arabic language pair may be an indicative of the additional cognitive challenges posed by directional disparities. Our findings suggest that translating between different text directions (left-to-right into right-to-left) poses asymmetric cognitive challenges for translators. They must grapple with lexical and syntactic transfers while also realigning spatially to the opposite right-to-left text direction. This appears to create fluctuations in pause duration as translators navigate these linguistic and spatial difficulties. The additional cognitive effort required to translate from a different

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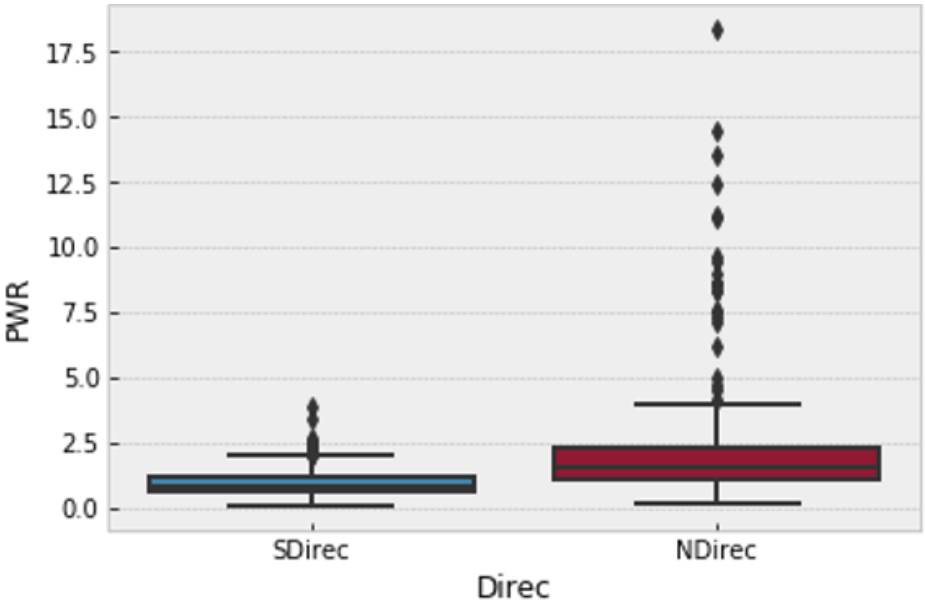


Figure 9: Pause-to-word ratio for same-direction and different-direction language pairs

text direction emphasizes that spatial and directional attributes, beyond just linguistic complexity, can alter the cognitive demands of translation.

As for the total number of fixations on the source (FixS) as shown on the left-side graph in Figure 10, the English-into-Spanish and English-into-Chinese datasets have less variation than the English-into-Arabic dataset. On the other hand, for the total number of fixations on the target (FixT) shown on the right-side graph, the pattern is like that of script where the same-direction language pairs (SDirec) have more variations than the different-direction language pair (NDirec).

The diminished variation in fixations on the ST for the same-direction language pairs contrasts with the heightened variation for the English-into-Arabic pair, potentially hinting at the complexities introduced by the shift in text orientation during the ST comprehension phase. On the TT front, the patterns intriguingly parallel those observed in the context of script differences. Such elevated variation for the same-direction language pairs might be emblematic of the intricate balance translators strive to achieve between familiarizing themselves with the ST and crafting an accurate rendition in the target language. These findings

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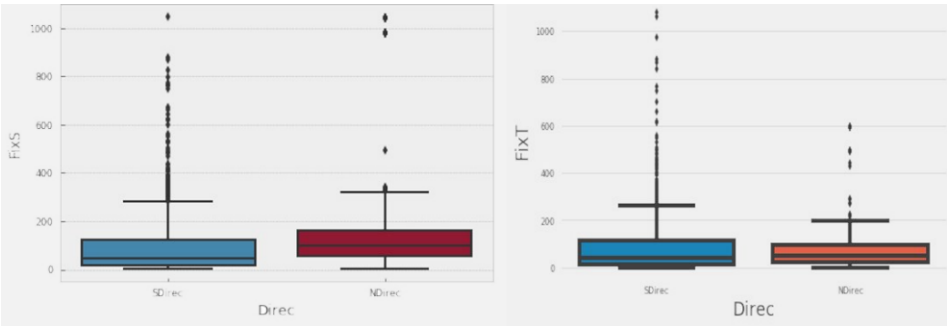


Figure 10: The total number of fixations on the source text and target text for same-direction and different-direction language pairs

underscore the profound impact of both script and text direction on the multifarious cognitive processes at play during translation.

The total reading time on the source (TrtS) is shown on the left-side graph in Figure 11 and the total reading time on the target (TrtT) is shown on the right-side graph of the same figure. For both, the same-direction language pairs (SDirec) show less variation compared to the different-text-direction language pair (SDirec).

The total reading times, as depicted in Figure 11, elucidate another dimension of the cognitive processes influenced by text direction during translation. The constricted variation in reading times for the same-direction language pairs contrasts starkly with the broader spectrum exhibited by the English-into-Arabic language pair. Such a distinction intimates a more uniform and possibly streamlined engagement with the ST and TT for translations that maintain consistent text direction. In contrast, the English-into-Arabic pair, with its shift to a right-to-left writing system, presents a more varied engagement, potentially reflective of challenges or pauses for comprehension and production as translators negotiate the shift in text orientation. This variance in reading times, particularly on the ST, might also suggest different translation practices, wherein the same-direction language pairs allow for a steadier translation flow, while the different-direction language pair necessitates longer orientation phase and periodic online revision and final reassessments of the TT based on the ST to ensure translation quality.

The last indicator of cognitive effort for text direction is the normalized HTot. Figure 12 shows more translation variation (HTotN) for the same-text-direction English-into-Spanish and English-into-Chinese language pairs in blue than the different-text-direction English-into-Arabic language pair in red.

The heightened variation observed in the same-direction language pairs (espe-

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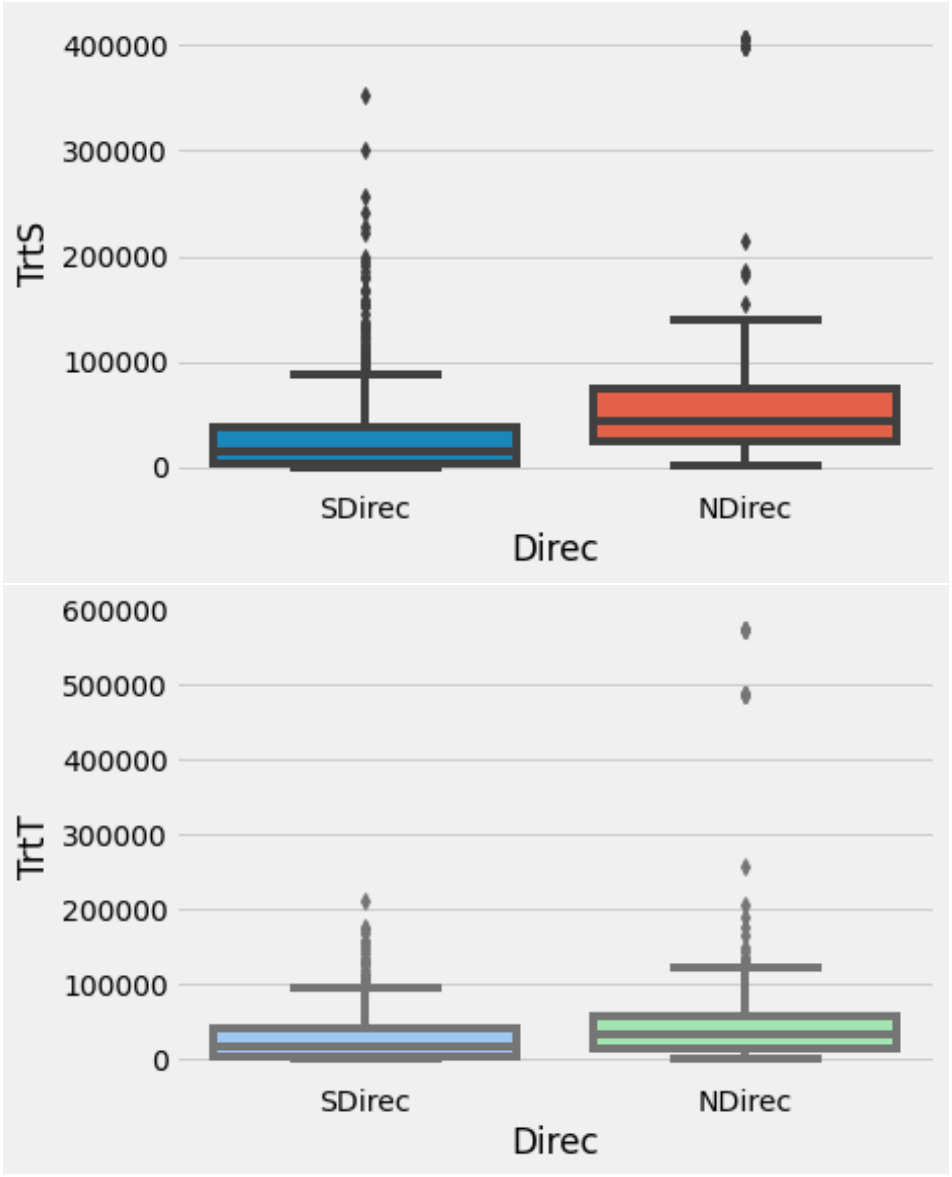


Figure 11: Total reading time on the source and target for same-direction and different-direction language pairs

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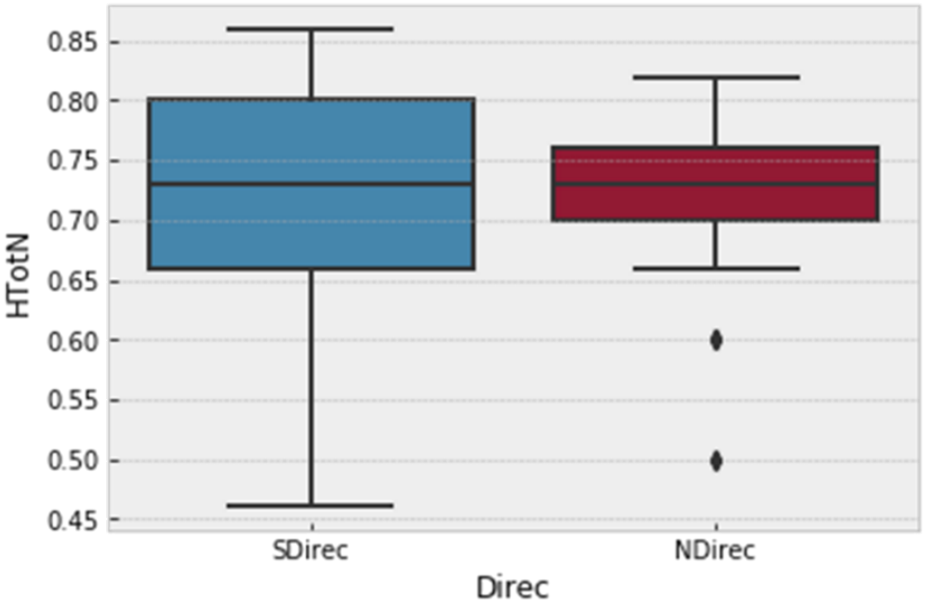


Figure 12: Normalized HTot for same-direction and different-direction language pairs

cially English-into-Chinese) could be indicative of a broader spectrum of translational alternatives or strategies employed by translators. This might suggest that when the spatial framework remains consistent across source and target languages, translators might be more inclined to explore a range of interpretative possibilities, leading to a more varied translational output. Conversely, the constricted variation for the English-into-Arabic pair, characterized by its right-to-left text direction, could intimate a more uniform translation approach. Such uniformity might arise from the inherent challenges of reconciling the spatial disparities between ST and TT, compelling translators to adopt a more constrained set of translational strategies to ensure coherence and quality. This dichotomy between the two groups underscores the profound influence of text direction on the cognitive and strategic landscape of translation, suggesting that spatial frameworks significantly modulate translational choices and cognitive engagement.

The discernible variations in cognitive effort across different text directions, as elucidated through multiple indicators, underscore the imperative to adapt translation pedagogy accordingly. Recognizing the heightened cognitive demands when translating into languages with different text directions, curricula should

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incorporate tailored exercises that specifically target the challenges of different text direction, helping students acclimatize to the spatial adjustments necessitated by such translations. Moreover, given the more constrained translational approach observed for the English-into-Arabic pair, there is a pedagogical opportunity to introduce tools or techniques that aid in navigating the unique challenges of such directionally divergent translations. For instance, intensive training on segmenting the ST into larger chunks during translation might reduce the cognitive load associated with navigating between different text directions. Finally, integrating these insights into classroom discussions can foster an environment of informed awareness, ensuring that future translators are cognitively attuned to the nuanced demands posed by text direction in translation.

3.1 Interaction effect

After verifying that both script and text direction have influence on translation effort, we test to see if there is an interaction effect between script and direction affecting effort exerted by translators. This is carried out by conducting a 2-way ANOVA with script and text direction as independent variables and the normalized HTot as the dependent variable. As can be seen in Table 3, the P values (PR(>F) for both script (Script) and text direction (Direc) are below 0.05. However, the P value for script and text direction interaction (Script:Direc) is above 0.05, which suggests no interaction influence.

Table 3: The interaction effect of differences in script and text direction in language pairs on cognitive effort

	df	sum_sq	mean_sq	F	PR(>F)
Script	1.0	3.079041	3.079041	455.786561	5.736319e-85
Direc	1.0	0.658369	0.658369	97.457595	4.355329e-22
Script:Direc	1.0	3.5e-05	3.5e-05	0.005203	9.425092e-01
Residual	1107.0	7.478278	0.006755		

In addition, the analysis yielded significant main effects for both script and text direction on all other indicators of cognitive effort utilized for purposes of this study, suggesting that these factors independently contribute to increased translation effort. However, there was no significant interaction effect between script and text direction, implying that the cognitive challenges posed by script differences and text direction do not amplify each other when they co-occur (see Table 4 below).

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Table 4: Two-way ANOVA results for cognitive effort indicators

Indicator	Script (p-value)	Direc (p-value)	Script:Direc (p-value)
PWR	< 0.00001	< 0.00001	0.950
FixS	0.00001	0.00004	0.389
FixT	0.00199	0.40293	0.75881
TrtS	< 0.00001	< 0.00001	0.77947
TrtT	< 0.00001	0.0000003	0.88705

The results from the 2-way ANOVAs indicate significant main effects for both script and text direction. This suggests that translating between different-script language pairs (e.g., English into Chinese) and different-direction language pairs (which are also different-script, e.g., English into Arabic) both independently contribute to increased translation effort when compared to translations between similar-script pairs (e.g., English into Spanish). However, the interaction effect between script and text direction is found to be insignificant. This finding is particularly intriguing given that different-text-direction language pair, English into Arabic, inherently involves different scripts as well. One might expect the challenges of script differences to be compounded when paired with different text directions. However, the lack of a significant interaction suggests that the challenges introduced by differing scripts and differing text directions might not amplify each other in a combined scenario. The cognitive processes associated with navigating script differences may be distinct from those required to manage changes in text direction. For instance, while different scripts introduce challenges related to lexical and orthographic shifts, changes in text direction might pose distinct challenges in spatial orientation and processing. The translators might be adept at compartmentalizing these challenges, tackling each in isolation even when both are present, thereby preventing an interaction effect in translation effort.

3.2 Final remarks

In the realm of translation studies, the intricate interplay between script and text direction, and the ensuing effort, has remained a relatively uncharted territory. Our exploration into this nexus sought to shed light on effort complexities that translators grapple with when confronted with language pairs of different scripts and text direction. Our empirical findings suggest that translating between lan-

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guages of different scripts, such as English into Chinese or Arabic, indeed imposes heightened cognitive and technical demands, as compared to translations between orthographically similar languages like English into Spanish. This observation aligns with previous assertions, which emphasize the cognitive intricacies tied to orthographic transfer and semantic processing. A different script introduces challenges at orthographic, lexical, and syntactic levels, demanding more intricate cognitive engagement for accurate and fluent translation. Furthermore, the distinction in text direction, as observed in the English-into-Arabic translation scenario, augments demand. Such findings underscore the profound role of spatial orientation in translation. Different text directions compel translators to reconcile not just linguistic but also spatial disparities, thereby magnifying the cognitive load.

However, one of the most intriguing revelations from our study was the absence of a significant interaction between script difference and text direction difference. While both factors individually heightened translation effort, they did not amplify each other's impact. This suggests that the cognitive processes associated with navigating script differences might be largely distinct from those managing changes in text direction. Such a distinction has profound implications for translation pedagogy. It suggests that training programs should address these challenges separately, offering strategies tailored to each challenge rather than a one-size-fits-all approach. From a pedagogical standpoint, our findings hold considerable ramifications. Given the delineated challenges, translation curricula could benefit from modules focusing on strategies specific to navigating script and text direction differences. This might include exercises tailored to familiarizing students with the nuances of different scripts and spatial orientations and leveraging translation techniques to help manage these challenges.

Our investigation contributes to translation research by providing empirical insights into areas previously underexplored. It elucidates the multifaceted nature of cognitive engagement in translation, emphasizing that beyond linguistic intricacies, spatial and scriptural attributes of languages play a pivotal role. For practice, our findings advocate for a more targeted approach to translator training and professional development. Recognizing the distinct challenges introduced by script and text direction differences, translation agencies might consider offering specialized training or resources to translators handling such language pairs. Another consideration is to account for these challenges in discussions about translator compensation, reflecting the varying complexities introduced by such differences in translation when dealing with such language pairs. Moreover, integrating certain computer-assisted translation (CAT) tools or translation technology could be invaluable in assisting translators in navigating these

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challenges and improving productivity. For instance, optimal segmentation at a granular level could potentially mitigate the cognitive effort associated with differences in text direction when navigating between ST and TT. Another example is the provision of bilingual term bases for specialized terminologies, which may be particularly beneficial in reducing effort when engaging with language pairs of different scripts and text directions.

Nonetheless, our investigation possesses certain limitations. The specificity of our selected language pairs could circumscribe the broader applicability to a more diverse linguistic spectrum. These delineated constraints underscore pertinent directions for subsequent research endeavors. While the challenges of script and text direction differences are formidable, understanding them paves the way for more informed, effective translation practices and pedagogical approaches. Future research endeavors might delve deeper into the underlying cognitive mechanisms, further enriching our understanding with more factors that might influence translation difficulty. Given the discussion points deliberated in the present study, detailed, granular qualitatively driven analyses could offer enhanced insights, further corroborating our findings and elucidating the subtleties of effort associated with script and text direction in contrast to other linguistic characteristics inherent to language specificity. Future investigations could also explore whether technical effort is a pivotal factor in measuring translation difficulty.

Abbreviations

ST	Source text	FixS	The total number of fixations on the source text
TT	Target text		
L1	First language	FixT	The total number of fixations on the target text
L2	Second language		
Ins	Insertions	TrtS	Tthe total reading time on the source text
Del	Deletions		
Nedits	The overall number of edits	TrtT	The total reading time on the target text
DurLog	The logarithm of the production duration	HTot	The total translation entropy
PWR	Pause-to-word ratio		

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Chapter 4

A corpus-driven investigation of translation strategies for prenominal complex adjectives

Resi

This chapter investigates a parallel corpus comprising original German texts and Italian translations from the field of tourism, with the objective of examining different translation strategies applied to Prenominal Complex Adjectives (PCAs). The study first categorizes German PCAs into seven groups. A regression analysis is then conducted to assess translation strategy frequencies. The results display a strong statistical correlation between categories and translation strategies, with a marked tendency to preserve the original syntactic structures. A qualitative analysis is carried out on the more unusual and transformational translation strategy of transposition in order to test its effectiveness for a complex syntactic structure characterized by two-directional syntactic reference. Results show that certain aspects of cognitive complexity in the original PCAs may encourage or trigger the application of transposition as a translation strategy for PCAs.

1 Introduction

This chapter investigates a parallel corpus of original German texts and their corresponding Italian translations to explore different translation strategies for German Prenominal Complex Adjectives (PCAs) into a language characterized by post-nominal attribution.

Section 2 introduces the phenomenon of prenominal complex adjectives as discussed in the literature and contrasts it with the definition adopted in this work, which is based on comparative reasoning tailored to the analytical goals

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of the study. Section **??** outlines the methodology, presenting parallel corpora as suitable sources for the quantitative and descriptive investigation of syntactic phenomena both within a single language and in contrastive terms. This follows the framework of corpus-based contrastive linguistics, as described in Laviosa (1998, 2003), Ebeling & Hasselgård (2018), and Johansson (1997, 2003, 2011), and includes a description of the corpus used. Section **??** presents the results of the extraction of PCA occurrences from the source corpus, leading to a linguistic characterology in the sense of Mathesius (1975). Section **??** provides a corpus-driven translational analysis aimed at identifying statistically significant translation strategies and evaluating the consistency of exceptional translation patterns, along with their potential generalization. This analysis follows the framework of Descriptive Translation Studies (Baker 1993; Toury 1991; Tognini Bonelli 2000, Tognini-Bonelli 2001; Laviosa 2002). Finally, some conclusions are drawn in sectrefsec:resi:6.

The complexity of the German structure between determiner and substantive requires syntactic referencing in both directions. It is assumed that the semantic content embedded in complex adjectival constructions is, in some cases, elevated to a higher semantic level within the sentence. This necessitates specific translation strategies when rendering such structures into Italian. Among these strategies, transposition is not frequently employed for PCAs, yet it remains a valid, and at times the most suitable, option.

2 The phenomenon of prenominal complex adjectives

A Prenominal Complex Adjective (PCA) can be described as a compound or phrase that appears before a noun and modifies it, similar to a regular attributive adjective. These constructions are considered ”complex” because they consist of more than one word, yet function syntactically as a single attributive adjective—that is, they are placed prenominally and follow the rules of adjective inflection (i.e., endings vary according to case, gender, and number). For a construction to qualify as a PCA, it must meet the following conditions: it must be prenominal (preceding the noun), complex (composed of more than one word), and adjectival (serving a noun-modifying function).

SOV languages, including German, tend to exhibit a wide range of constructions that fall under this definition, although the notion of complexity is not always applied consistently. In particular, morphological complexity in adjectival compounds (1), or the structural peculiarity of lining more than one adjective before the noun (2), is often not regarded as a PCA to the same extent as a participial structure like (3).

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(1) die subnationale Ebene nimmt wichtige ausgabenrelevante
the subnational level takes important relevant_for_expenses
Aufgaben wahr
tasks.

(2) ist eine Anerkennung ihres außergewöhnlichen universellen Wertes
is a acknowledgment its$_{GEN}$ extraordinary universal value

(3) dass die auf der Grundlage der Marketingverträge erworbenen
that the on the basis of marketing_contracts acquired
Dienstleistungen
services

Research on the complexity of prenominal adjectives across German and other languages tends to be analyzed in terms of syntactic operations, often excluding the morphological complexity of compounds. Moreover, lists of prenominal adjectives that are not syntactically coordinated by *und* 'and' are generally not considered complex, although the contrastive perspective provided in sectref-sec:resi:3.3 and recalled in Section 4.4 suggests otherwise. An idea of that is provided by Kotowski & Härtl (2019) who claim that multiple adjective sequences are shaped not only by syntactic rules but also by pragmatic and cognitive factors governing information structure and linearization preferences.

German provides valuable insight into the structure of prenominal adjectives. Some of these adjectives show properties similar to full sentences, such as case assignment and scope, which suggests that they may have originally come from or represent more complex propositional structures (Fanselow, 1986). Their placement within noun phrases is also shaped by interactions between agreement rules, inflectional endings, and word order, often going beyond the patterns seen in simpler attributive constructions (van Riemsdijk 1983).

Under a contrastive point of view, the work of Cinque (2010) provides a comparative account of prenominal and postnominal adjectives across Germanic and Romance language families, emphasizing structural differences rooted in syntactic hierarchies. The way adjective syntax has developed in these two language families shows both clear differences and shared patterns (Van de Van De Velde et al. 2014). Within Germanic languages, the variation in adjective endings, like the use of strong vs. weak forms, helps organize how adjectives appear before nouns, revealing a systematic pattern across these languages (Roehrs 2015).

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German grammars aimed at both native speakers and learners of German as a foreign language (DaF) (Götze & Hess-Lüttich 2002; Engel 1988; Dreyer & Schmitt 2001; Scollo 2020, among others) describe what we refer to as prenominal complex adjectives (PCAs), although they do so under various chapter headings and terminologies. A particularly detailed and systematic account of the structure between determiner and noun in German is provided by Eichinger (2006), who distinguishes three noun phrase positions—left, centre, and right—while designating the position to the left of the noun as the flexion point. Eichinger's framework accounts for several syntactic configurations, including present or past participles modified by materials, and adjectives modified by adverbs. However, it does not encompass the full range of PCA structures identified in our corpus, notably excluding compound adjectives and non-coordinated multiple adjectives.

Research on prenominal complex adjectives reveals a rich interplay between syntactic structure, morphosyntactic inflection, and typological variation across languages. The aspect concerning compounding between the determiner and the noun is typically addressed separately and not treated as a case of complexity within prenominal adjectival constructions. Also the situation of lining up more than one adjectives before noun is not always interpreted as complex.

To establish a consistent framework, we therefore examined the corpus of German texts using the approach of "linguistic characterology" (Mathesius 1975). This method aims to identify language-specific structural properties to enable a more precise understanding of their internal differentiation and typological relevance. The rationale for introducing a general label that encompasses all PCA types is grounded in a contrastive perspective. Specifically, we found no evidence in Italian of prenominal structures exhibiting a comparable degree of complexity between the determiner and the noun. Therefore, in addition to the requirements of prenominality, complexity, and adjectival modification outlined above, the linguistic characterology of German for the purpose of our study includes another distinctive feature that sets these structures apart: the absence of a structurally equivalent pattern in the Italian language.

3 Methodology

3.1 3.1. The corpus

The intuitive view about translation strategies for such a complex German language structure was derived largely from recursive, albeit independent, observations during translation from German into Italian in different text typologies. A

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more systematic test was deemed necessary for how PCAs are translated into a language that is generally unreceptive to this kind of structure. In order to achieve detachment from any personal introspections, it was decided to address a corpus of texts completely separate from the translation work of the present authors.

The corpus was compiled on Sketch Engine (Kilgarriff et al. 2004) as a parallel corpus in the sense of Hartmann (1980: 37) and consists of original German texts and their translations into Italian. Following the definitions provided by Granger (2003: 19), such a corpus can also be defined as a translation corpus. The original corpus comprises 24 informative texts from the sphere of tourism with translations intended for an equivalent target. Tourism was chosen as a common but entirely arbitrary field to constrain the selection of texts within certain limits, but without restricting analysis to a specific text typology. Original texts with their translation were collected from the Internet mostly from the European Union website in order to maintain a certain translation standard and include both informative and promotional texts. The corpus includes legal acts (e.g. EU regulations), tourism brochures issued by the European Commission, communications promoting, amongst others, social tourism, and special reports on EU financed projects in the field of tourism. This miscellaneous field made it possible to compile a heterogeneous corpus as regards text typology, without confining research to trends in a specific kind of language, and so attempting to lay the foundations for more universal observations. The corpus includes a total of amount of 700,000 original words.

Combining the study corpus with comparable corpora (in the sense of Johansson & Hasselgard 1999) was avoided for two explicit reasons. Firstly, because providing evidence of syntactic differences between the two languages was not a primary goal but instead a presupposition. It was assumed that the Italian language does not accept this category of complex adjectival forms before nouns. The main aim of the preliminary analysis is a systematic categorization of this German syntactical phenomenon, which descriptive grammars do not treat as a unified whole. However, considered from a contrastive point of view, it can be treated as a single category: the inclusion of complex adjectival information between determiner and main noun. Secondly, there is no need to compare original and translated language, with instead the aim of identifying and investigating possible translation strategies into Italian. All that is required are parallel texts that convey exactly the same meanings by exploiting the syntactic possibilities of the target language. Moreover, the translated texts were also considered within the framework of Translation Studies, classing them as texts in their own right in order to "understand what translation is and how it works" (Baker 1993: 243).

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3.2 3.2. The software

The *Sketch Engine* software was used to compile the corpus in order to be able to navigate the two languages separately. The corpus was intended for the two different purposes of clearly categorizing this syntactical phenomenon in the German language, and then for analysing the translation of the same phenomenon into Italian in relation to the defined classification.

Despite having an electronic corpus and a search engine to navigate it, a certain amount of manual investigation was still required to establish what was needed. There is a clear difference between using corpora to search for lexical items and using corpora to find syntactic phenomena. Syntactic features like the one under investigation here cannot be reduced to adequate algorithms because they embrace too many syntactical variables. While *Sketch Engine* was able to successfully identify all adjectives in the corpus and list them coherently according to frequency or alphabetical order, manual analysis was necessary to extract the smaller set of PCAs object of the study. Automatic searches retrieved a lot of irrelevant material but fortunately this was easy to eliminate because *Sketch Engine* provides a very clear overview of occurrences.

The analysis of Italian translations also required some time for the manual extraction of correspondences. *Sketch Engine* was not always able to spot the exact textual segment that included the translation of PCAs. The reason for such failures is presumably because the software in unable not spot a syntactically equivalent structure in the target language. Unlike lexical corpus analysis, you cannot look for a syntactic feature in a language that does not provide for it. Syntactic relationships do not covey specific semantic ideas, but rather semantic relationships within the original language, making automatic comparisons much more difficult. The use of *Sketch Engine* did however mean that for every occurrence it was possible to generate a list of sentences containing the item by simply by clicking on it. It still required some times to extract the complex adjectives from the lists, but less than would have been necessary working entirely manually. Each case was then schematized in an Excel table according to characteristic features, which will be explained in the next section.

4 Linguistic Characterology of PCAs in German

The way *Sketch Engine* works with its limits in matching syntactic features made it necessary to export data to another file format and reconstruct a subcategorization of the phenomenon of Prenominal Complex Adjectives (PCAs).

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As defined in sectrefsec:resi:2 a PCA includes more than a single lexical adjective normally positioned between the determiner and the noun, but in all cases before the target noun. Among all the German PCAs identified, seven different types were observed:

- predicative structures with past participles (4);

(4) dass die auf der Grundlage der Marketingverträge erworbenen
that the on the basis of marketing_contracts acquired
Dienstleistungen
services

- predicative structures with present participles (5);

(5) die unter dem EU-Durchschnitt liegenden Beschäftigungsquoten
the under the EU_average lying employment_rates

- compound adjectives (6);

(6) die subnationale Ebene nimmt wichtige ausgabenrelevante
the subnational level takes important relevant_for_expenses
Aufgaben wahr
tasks

- multiple non-coordinated prenominal adjectives (7)

(7) ist eine Anerkennung ihres außergewöhnlichen universellen Wertes
is a acknowledgment its$_{GEN}$ extraordinary universal value

- adjectives modified by adverbs (8);

(8) Für neu eingestellte Piloten
For new employed pilots

- adjectives modified by more complex structures (9);

(9) die für künftiges Wachstum verfügbaren Qualifikationen
the for future growth available qualifications

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- infinitive structures (10).

(10) in Form von vollständig zu überprüfenden und
in form of completely to_test and to_prepare
aufzubereitenden Daten
data

Moreover, it is considered necessary to set some rules for data extrapolation in order to establish a coherent data set and avoid an excessive number of exceptions that would add further variables to the investigation of an already extended phenomenon. The rules are as follows:

- Sentences including two categories of PCAs are counted twice.
- Only qualifying adjectives with semantic meanings are considered for categorization of adjectival constructs. So when considering at the category "multiple non-coordinated adjectives", negations like *kein* and possessive adjectives are excluded from the count.
- Texts or headers in tables and graphics are excluded. Only syntactic structures embedded in complete clauses, sentences, and paragraphs are considered because contextualization plays a major role in translation strategies and it is important to consider at least what comes before and after any analyzed elements before making a translation decision.

The list is the result of corpus extraction, but the broadness of the corpus, and knowledge of German grammar, makes it plausible to assume that it represents a comprehensive characterology of the syntactic phenomenon in question. In the following subchapters categories are discussed individually.

4.1 Partizip I and Partizip II before the noun

A contrastive German-Italian analysis of so-called "enlarged" German constructions is provided by Kather (2011: 67–73) and Coletsos (2007: 65–86), in which a lot of examples of both Partizip I and Partizip II in attributive function are described together with their possible translations into Italian. The "costruzoni attributive allargate" are presented as a single attributional phenomenon and not as one possible expression of the syntactical complexity of attribution. A first look at the list of examples of PCAs constructed with past or present participles modified by syntactical elements demonstrates that this is a much more prolific structure than previously imagined, not only in more normative texts as expected but also in informative and popular texts including promotional materials for tourist destinations.

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4.2 Past participles before nouns

In descriptive grammars of the German language, past participles before nouns are treated in the same section as normal adjectives, which traditionally assume a prenominal position in German. Observed contrastively, it is seen that in the Italian language PCAs are limited to specific situations related to subjectivity, and past participles never meet the semantic requirements to stand before nouns (because resistant to subjective interpretation). Past participles, even as adjectives on their own, do not occur as prenominal adjectives in standard use, although the influence of the English language seems to have changed that recently (see sectrefsec:resi:4).

When extracting occurrences of this category, simple prenominal past participles were excluded because of the lack of syntactic complexity. Prenominal simple past participles can be treated in translation like simple adjectives and do not need further investigation other than an awareness of a contrastive difference in the positioning of adjective forms.

Much more relevant is the analysis of prenominal past participles preceded by arguments like (4), (11), (12), and (13) which are sometimes considered reductions of relative clauses (Heidolph et al. 1984).

(11) Die Art der von der Kommission und anderen Nutzern der
The type of by the commission and other users of_the
europäischen Tourismus-Statistik benötigten Daten
European tourism-statistics needed data

(12) die entsprechenden Informationen wurden direkt bei 206 nach
The relevant information were directly from 206 by
dem Zufallsprinzip ausgewählten Projektträgern
random_principle selected Project_sponsors taken
eingeholt

(13) 94% der durch die Projekte geschaffenen oder erhaltenen
94% of_the by the projects created or received
Arbeitsplätze waren weiterhin vorhanden
workplaces were still available

The corpus provides a total of 310 cases of prenominal complex past participles.

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4.3 Present participles before nouns

A similar situation is seen in syntactic situations like (5), (14), and (15) in which a present participle adjective assigns one or more arguments to the left.

(14) die mit ihm zusammenhängenden *Branchen*
the with it linked sectors

(15) wegen nach den letzten Entwicklungen unzureichender
due_to following the recent developments transparency
Transparenz der Methodik
of the methodology

Present participles are sometimes difficult to identify because many of them have become lexicalized and acquired the status of adjectives. It was nevertheless decided to consider all adjectives derived from present participles, so as not to have to differentiate as regards the process of adjectivization, which is already rather controversial. Adjectives like *folgend* are counted as present participles and only considered in the analysis when preceded by other syntactical elements that increase the overall complexity.

The corpus provides a total of 205 cases of prenominal complex past participles.

4.4 Compound adjectives

German is a highly productive language as far as compounds are concerned. No syntactic categories are excluded from the compounding mechanism although some are more productive than others. As a category, adjectives are very productive, not only by combining an adjectival head with different syntactic categories, but also because the syntactic relationship between individual elements are not limited to specification. Non-endocentric compounds involve complex relationship as in (6) or (16) and so even if compound adjectives are single word adjectives they can be contrastively considered to be complex structures.

(16) die tourismusstatistischen Ausgangsdaten
the tourism_statistics starting_data

(17) um den haushaltspolitischen Rahmen Österreichs zu
in_order_to the fiscal_policy framework of_Austria to
verbessern
improve

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(18) eine gesamtstaatliche Strategie zur digitalen Sicherheit
a national_level strategy for digital security

The extensive analysis of Pümpel-Mader et al. (1992) on different kinds of syntactical relationship between the individual elements of adjectival compounds, including those constructed with past and present participles, demonstrates how prolific and complex the German language is in this respect. Two specific types of compounds stand out in our translation analysis. The first category encompasses compound adjectives, such as (17), where the head lacks inherent semantic content and primarily assumes a functional role. Although individually the head possesses lexical meaning, it essentially functions as a suffix to derive the adjective from the noun that precedes it. In terms of translation strategy, the focus should be on encapsulating the lexical meaning within the adjective, primarily derived from the first element of the compound, like *quadro fiscale*. Moreover, there are adjectival compounds that carry the most important lexical information in the head of the compound, as in (18). These compounds usually require a more elaborate translation strategy to account for the entire meaning of the adjective.

The corpus contains 96 examples of compound adjectives.

4.5 Multiple non-coordinated prenominal adjectives

The Italian language does accept two or more prenominal adjectives, however this is not a common construction and only appears under rare circumstances determined by exceptional adjectival meanings. It was therefore decided to consider this as a syntactic difference. Prenominal adjectives are used when they provide a "natural" classification in combination with the noun: hence the oddness of **indaffarato vecchio uomo* compared to *un simpatico vecchio uomo*. It appears that prenominal adjectives have to denote some plausibly characteristic property of the noun they modify. In German by contrast, two or more adjectives are standard and admissible for any type of adjective, with postnominal adjectives completely excluded (7, 19, 20 and 21).

(19) dass die erlassenen delegierten Rechtsakte für die Mitgliedsstaaten
that the issued delegated legal_acts for the Member

States

(20) angesichts des reichsten architektonischen Erbes der Welt
given the richest architectural heritage in_the world

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und einer bahnbrechenden städtebaulichen Entwicklung
and a groundbreaking urban development

(21) so dass insgesamt 145 Projekte zu einer erhöhten touristischen Aktivität
so that a_total_of 145 projects to a increased tourist activity
beitragen
contribute

It is therefore necessary to exclude adjectives that are linked by coordination (22), because they do not satisfy the requirement of complexity.

(22) um wirtschaftlichen, sozialen und technologischen
in_order_to economic, social and technological
Entwicklungen Rechnung zu tragen
developments into_account to take

Pümpel-Mader et al. (1992) includes adjectives linked by hyphens in the analysis of adjectival compounds, stating that the hyphen indicates a status of semantic parity/equivalence with the noun, hence coordination. Few cases of this hyphen relation are found in the corpus, confirming the hypothesis of coordination. Since coordination does not indicate for complexity, such cases are excluded from the categorization of PCAs.

(23) grün-goldene Landschaft: eine Landschaft, die grün und golden ist
green-gold landscape a landscape which green and gold is

When it comes to non-coordinated adjectives like (19), (20), and (21) there are rules governing the sequencing of adjectives, with those expressing opinions coming before those expressing facts, with the latter also subject to additional ordering. However, there are also syntactic relationships between non-coordinated prenominal adjectives that need to be more closely investigated. A good way of explaining these relationships is a branching tree structure, especially from a perspective of contrast, since long sequences of prenominal or postnominal adjectives are not an option in Italian, and translation strategies like nominalization were considered even if marginally. In the case of (7) for example, the adjective *außergewöhlich* does not refer directly to the noun but rather to the subsequent adjective *universell*, with the extraordinary nature referred to the universal character of the meaning, functioning almost like an adverb. The second adjective, which characterizes the noun, was in fact nominalized in an attempt to make this relation clear (see sectrefsec:resi:4.3 for more on this translation strategy)

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(7′) ihres außergewöhnlichen universellen Wertes la straordinaria
the extraordinary
universalità del suo valore
universality of its value

In most cases (19, 20, and 21) the first adjective refers to the entire following sequence of adjective+noun and only rarely does the first adjective refer directly to the noun.

(20′) bahnbrechenden städtebaulichen Entwicklung
breakingthrough urban development

(21′) erhöhten touristischen Aktivität
increased tourist activity

The corpus contains 390 cases of multiple non-coordinated adjectives before nouns.

4.6 Adjectives modified by adverbs

In a similar way that English adverbs can modify adjectives, in German prenominal adjectives can be modified or intensified with adverbs. When adverbs are used to modify adjectives, they provide more information about the quality or degree of the adjective and are positioned before the adjectives (8, 24, 25, and 26).

(24) Im Gegensatz dazu könnten die stetig steigenden
On the contrary could the steadily increasing
Wohnimmobilienpreise…
residential_real_estate_prices…

(25) kostenlos zur Verfügung gestellte Unterkünfte
free_of_charge provided accomodation

(26) auch zu teilweise oder vollständig subventionierten Preisen
also at partially or fully subsidized prices

The adverbs *stetig*, *kostenlos*, *teilweise* and *vollständig* modify the following adjective. These adverbs help convey the degree or intensity of the quality described by the adjective. It is important to note that while the positioning of adverbs can vary in German sentences, they generally come before adjectives

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when they function as intensifiers of the same. As in the category described before, the syntactical relationship is not directly with the noun, but rather with the adjective which acquires a pivotal importance in the NP structure.

The corpus contains 208 cases of prenominal adjectives modified by adverbs.

4.7 Adjectives modified by more complex structures

Most other languages like English that accept premodification systematically exclude adjectives with complements from the prenominal modifier position (Sadler & Arnold 1994). German does allow for such constructions, sometimes even with long complex structures between determiner and noun (9, 27, 28).

(27) bei einer an dem besuchten Ort ansässigen Einheit
by a in the visited place living subject
alle dipendenze di un soggetto residente nel luogo visitato
by a subject living in_the place visited

(28) eine nicht mit dem Binnenmarkt vereinbarte staatliche Beihilfe
a not with the internal_market agreed_upon state aid
darstellt
represent

In the literature on these prenominal complex structures generated in situ, there are three main hypotheses: prenominal adjectives are heads, adjuncts, or specifiers (Roehrs 2023). On the other hand, since German Aps, unlike other languages, are able to take as many arguments as desired in the prenominal position, a clausal analysis of adjectives is also generally represented (since Kayne 1994). PCAs are considered the result of relative clauses undergoing movement and deletion (Cinque 2010). According to Resi (2011), the Head Raising Analysis appears more promising for German restrictive relative clauses, when regardless of position with respect to the head noun, they are considered the complement of D° within a DP and can move to the left into a higher position. Alternatively a Matching Analysis is preferred for appositive relative clauses which are adjuncts to the right to full DPs and are therefore islands for movement to the left. In the case of appositive relative clauses, embedment between the determiner and the noun is in fact excluded.

The corpus contains 51 cases of prenominal adjectives modified by adverbs.

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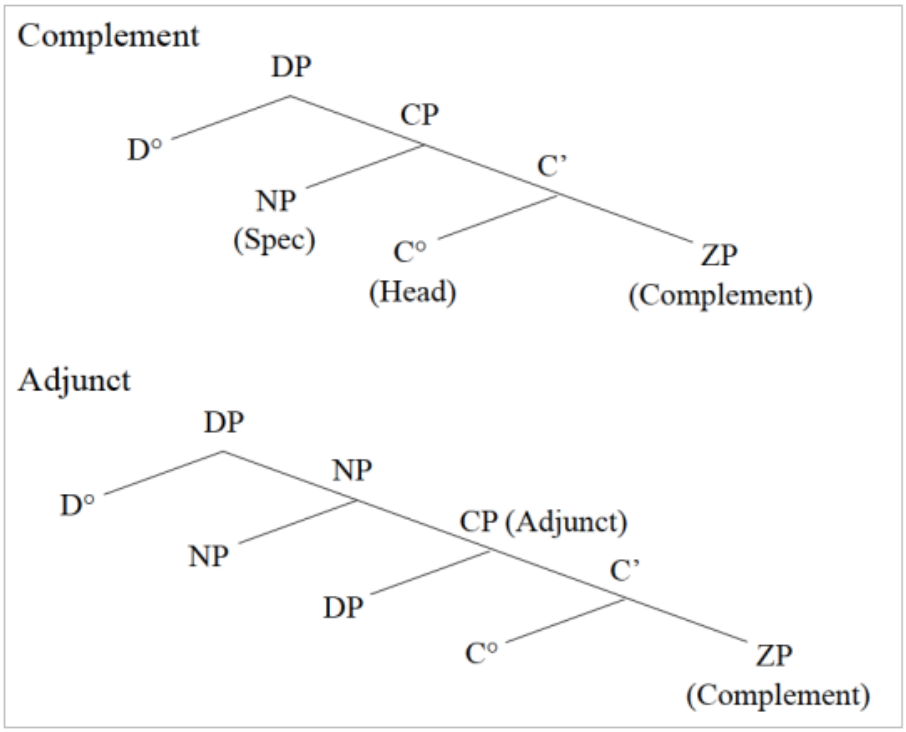


Figure 1: : derivation of German restrictive relative clauses and appositive relative clauses (Resi 2011)

4.8 Infinitive structures

German can provide extensive information about a noun simply by positioning an entire phrase as a modifier before a noun. In the above example there is no finite (conjugated) verb and the participles have endings, indicating that they are functioning as adjectives. The phrase therefore modifies a noun and is not an independent sentence. The present corpus provided few examples of infinitive structures embedded as modifiers of nouns, in which adjectivization of the verb in its infinitive form is required, as in (10), and (29).

(29) die von den Mitgliedstaaten gemäß Artikel 9 zu übermittelnden
the by the Member_States according_to Article 9 to be_transmitted
Daten
data

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This formulation occurs 72 times in the corpus and it is contrastively very complex because it often combines an infinite clause with multiple modifying phrases, all working together to provide detailed information about the noun.

Figure 2 provides a graphical overview of the distribution of PCAs.

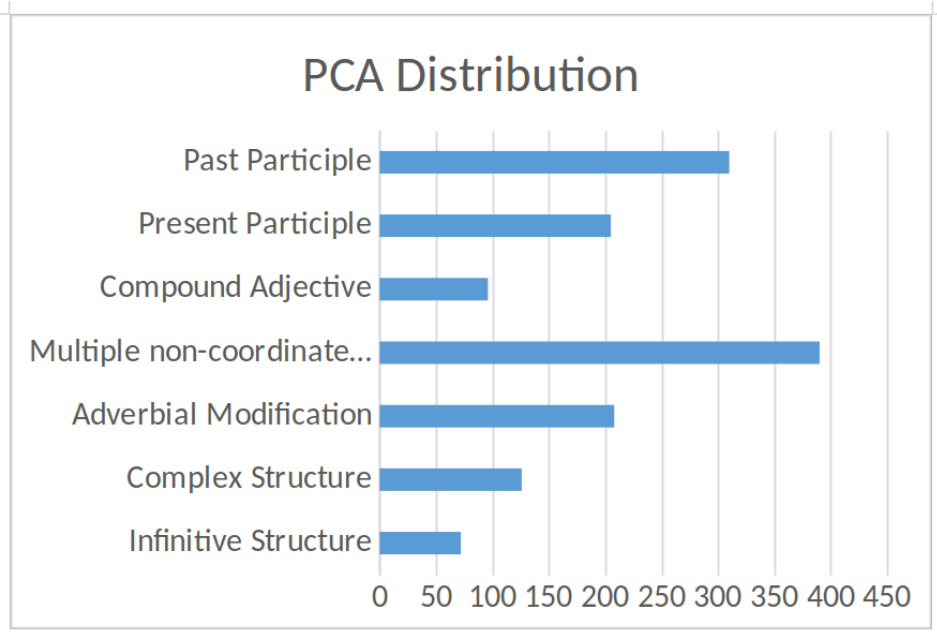


Figure 2: : Graphic distribution of PCAs

5 PCAs in Italian

5.1 Restrictions on PCAs in Italian

According to Cerruti (2015), prenominal positioning of complex adjectival phrases in a more restricted sense – hence ruling out adjectival compounds and infinite structures – is not entirely excluded in Italian, but it is more common in some contexts than in others. It occurs more frequently in the presence of internal factors such as prepositional phrases, relative clauses, or the coexistence of multiple modifiers of different types, to achieve a more balanced distribution of adjectival information within the nominal phrase.

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(30) menomato e perciò costretto a una per lui insopportabile
impaired and therefore forced into an for him unbearable
inattività
inactivity

(31) il mero e odierno schieramento della sinistra ufficiale
the mere and current alignment of_the left official

(Cerruti 2015)

The structures in (30) and (31) have increased significantly in recent years in journalistic language under the influence of massive exposure to English. However, despite the possibility of PCAs in Italian identified by Cerruti in a corpus of press articles and revealing clearly increasing trend, Italian is still strongly characterized by postnominal attribution (Berruto 1998; **Garzone2005**). In the translation corpus of the present study, no examples of PCAs were found in Italian. This absence can be attributed to two main causes. First and foremost, the corpus does not include press articles, effectively excluding the journalistic style from consideration. Secondly, the intricacy of these structures in the original language did not significantly influence the translation decision-making process.

Considering the exceptionality of PCAs in Italian, it can effectively be sustained that there is no direct equivalent for PCAs in Italian. References to "preserving the syntactical structure in translation" should be understood as the most similar structure, regardless of its position relative to the head noun. So an embedded structure could be preserved by translating it with a relative clause or reduced relative clause after the noun, or a list of prenominal adjectives could become a list of postnominal adjective, and so on.

5.2 Analysis of translation strategies for PCAs in Italian

Considering the abovementioned lack of direct equivalence for PCAs in Italian, translations of German texts were examined in an effort to identify different translation strategies dealing with the various kinds of PCAs. In this phase the use of Sketch Engine was fundamental but not enough in itself. It was easy to identify paragraphs in which these structures were translated, but the ordering of elements in Italian was completely different, making further manual analysis necessary.

All the adopted strategies resulted in postnominal attribution of the semantic meanings found in the original German PCAs, which was relatively predictable. The translation strategies were categorized according to frequency. Overall, a

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statistically significant equivalence for these structures in the Italian texts was the positioning of the attribution with its modifiers and arguments after the head noun in 45% of cases (32), postnominal full relative clauses in about 30% of cases (33), and post-nominal adjective lists in around 20% of cases.

(32) Die Art der von der Kommission und den Nutzern benötigten
The type of the from_the commission and the users required
Daten
data
Il tipo di dati richiesti dalla commissione e dagli utenti
The type of data required from_the commissione and from_the users

(33) Die entscheidende dem Tourismus bei der Förderung von Wachstum
The crucial to_the tourism in the promoting the growth
und Beschäftigung zukommende Rolle
and employment given role
Il ruolo fondamentale che il turismo acquisisce nella promozione di
The role crucial that the tourism acquires in promoting of
crescita e occupazione.
growth and employment

(34) eine Bandbreite möglicher marktkonformer jährlicher Prämien
a range of_possible market-compliant annual premiums
Una paletta di premi annuali plausibili conformi al mercato
A range of premiums annual possible compliant with_the market

The most frequent strategy in the study corpus is a postnominal adjectival construction, encompassing all adjectives modified to the right by syntactic elements. In this strategy it is important to note that the boundaries of the constructions to the right of the nouns are not always clearly defined in Italian. An analysis of translations from Italian to German should be able to reveal examples of this ambiguity. Indeed, the primary challenge associated with this strategy lies in the ambiguity of reference for adjectival modification. In fact, a sentence like (35) can be transferred into German either with the arguable and ambiguous (35a) or with the fully acceptable (35b), which comprises all arguments in the PCA.

(35) I contratti stipulati nel quadro del programma di
The contracts stipulated in the framework of the
promozione
program of promotion

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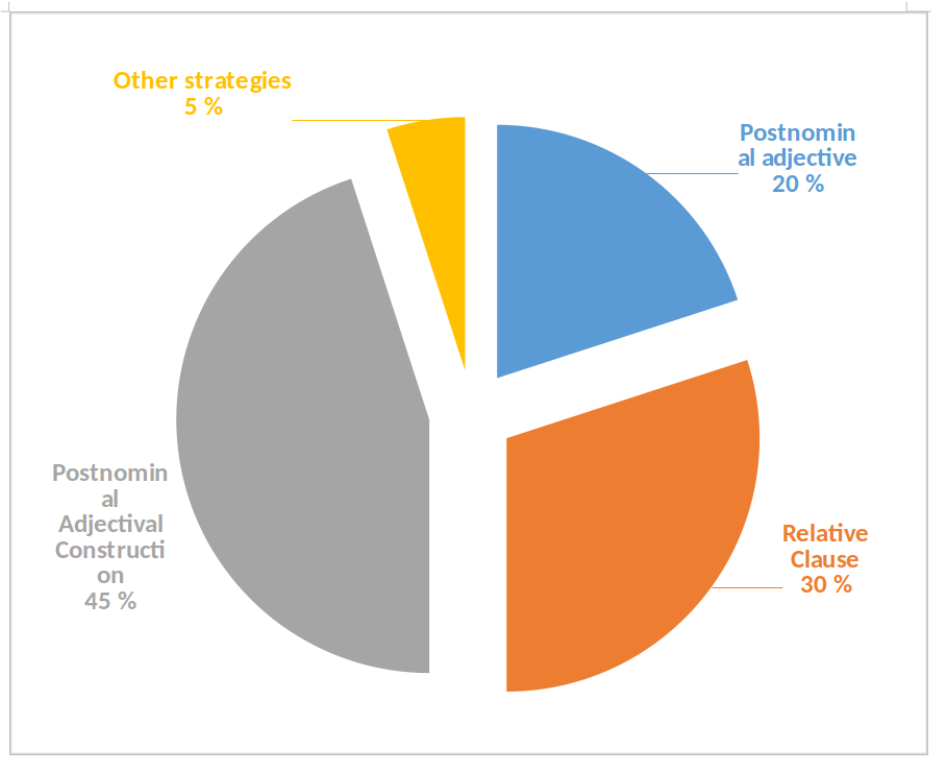


Figure 3: Percentage of different translation strategies

a. ?Die abgeschlossenen Verträge im Rahmen des
The stipulated contracts in_the framework of_the
Förderungsprogramms
promotion_program

b. ?Die im Rahmen des Förderungsprogramms
The in_the framework of_the promotion_program
abgeschlossenen Verträge
stipulated contracts

The fact that the grammaticality of (35a) is not completely excluded is not surprising since the original corpus contains cases of discontinuity between the elements making up the adjectival phrase (36) resulting in decontextualized ambiguous interpretation of the temporal argument *für 2012 und 2013*. Our proposal in (36a) would have solved the ambiguous reference of the argument, but (36) was preferred in the original text.

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(36) ?Das Burgenland und Vorarlberg wurden aufgrund fehlender
The Burgenland and Voralberg were due_to missing
vorliegender Date für 2012 und 2013 ausgeschlossen
actual data for 2012 and 2013 excluded

a. ?Das Burgenland und Vorarlberg wurden aufgrund für 2012 und 2013
The Burgenland and Voralberg were due_to for 2012 and 2013
fehlender vorliegender Date ausgeschlossen
missing actual data excluded

The second translation strategy is full postnominal relative clauses, in most cases (but not exclusively) used to translate embedded constructions of the sixth category like (37), which recalls example (9).

(37) könnten die für künftiges Wachstum verfügbaren Qualifikationen
could the for future growth available qualifications
begrenzen
limit
potrebbe finire per limitare le qualifiche che andrebbero a
could end_up for limiting the qualifications that would_go to
supportare una crescita futura
support a growth future

Full postnominal relative clauses are also the second most common strategy for translating both past (30%) and present participles (28%) as in (38) and (39).

(38) da die von Ryanair an den Flughafen Frankfurt-Hahn gezahlten
since the from Ryanair to the airport Frankfurt-Hahn paid
Flughafenentgelte
airport_taxes
le tasse aeroportuali che Ryanair pagò all' aeroporto di
the taxes airport which Ryanair paid to_the airport of
Frankfurt-Hahn
Francoforte-Hahn

(39) womit der seit 2012 bestehende Trend anhält
with_whom the since 2012 persisting tendency continues
che preserverebbe la tendenza che perdura dal 2012
which would_preserve the tendency, that has_been_persisting since 2012

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Surprisingly, the relative clauses used in translation do not always appear to be restrictive, even though appositive relative clauses are excluded from the embedded position between determiner and head noun in German. Using a postnominal full relative clause as in (40) does not rule out the appositive meaning generating ambiguity, which was unintended in the original text. Again, it is the position of the sentence within the text that disambiguates the meaning.

(40) Mehrere vor kurzem umgesetzte Maßnahmen zielen darauf ab
Several till recently implemented measures aim at
Molte misure che sono state implementate di recente hanno
Several measures which were implemented of recently have
come obiettivo
as aim

The translation strategy of postnominal full relative clauses is not however exclusive to embedded participle constructions, but was also applied for other categories of complex adjectives. In some cases of compound adjectives (41) or lists of non-coordinate adjectives (42), the appositive meaning of the relative clause is required.

(41) dass stärker auf wachstumsfreundlichere Einnahmequellen
that stronger on growth_friendlier revenue_sources
zurückgegriffen wird
accessed is
che si possa fare sempre più affidamento su fonti di
that one can increasingly rely on sources of revenue which
guadagno, le quali favorirebbero la crescita
would favor the growth

(42) Dank Österreichs allgemein gutem wirtschaftlichen Abschneiden
Thanks_to Austria's general good economic performance,
ergibt sich die Gelegenheit
it arises the opportunity
Grazie allo sviluppo economico dell' Austria, che in generale
Thanks to_the development economic of Austria, which in general
risulta positivo, si avrebbe la possibilità
is positive, one would_have the possibility

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The third most frequent translation strategy involves positioning adjectives postnominally, when adjectives are simply listed without any subsequent complement or adjunct (43). Completely postnominal attribution (43a) was often preferred rather than balancing adjectivization between the prenominal and postnominal positions as in Cerruti's analysis (2015) and proposed in (43b).

(43) hinter diesen Entwicklungen verbergen sich große regionale
behind these developments hide big regional
Unterschiede
differences

gloss sich als self

a. dietro a questi andamenti si celano divergenze regionali grandi
behind these developments hide differences regional big

b. ?dietro a questi andamenti si celano grandi divergenze regionali.
behind these developments hide big differences regional

Each PCA example was matched to one of the translation strategies utilized within the translation corpus. This aimed to pinpoint the most commonly employed strategies within each category and present them quantitatively in a graphic format.

A regression analysis (Gries & Wulff 2012) was conducted on the frequency of translation strategies for each category, considering approaches ranging from closest possible preservation of syntactic structure, to complete reformulation. The results reveal a strong overall statistical significance of association between categories and translation strategies (P-value: 0.00001859) with an odds ratio far below the threshold of 1. A general tendency is also observed for preserving the original syntactic structure, resulting in positioning of attributions after the head noun in the form of relative clauses, post-nominal adjective lists, or other post-nominal constructions (e.g. adverbs), mainly adopting the most similar possible prenominal structure to the source text category.

Postnominal adjectival constructions emerged as the most frequently used approach across six out of seven categories of PCAs (from 30 to 50%). This indicates that the translators consistently employed the technique of constructing more complex noun phrases by positioning adjectives, potentially modified by other elements, after the noun (44).

(44) die unter dem EU-Durchschnitt liegenden Beschäftigungsquoten
the under the EU-average lying employment rates
I tassi di occupazione al di sotto della media EU…
the rates of employment at of under the average EU…

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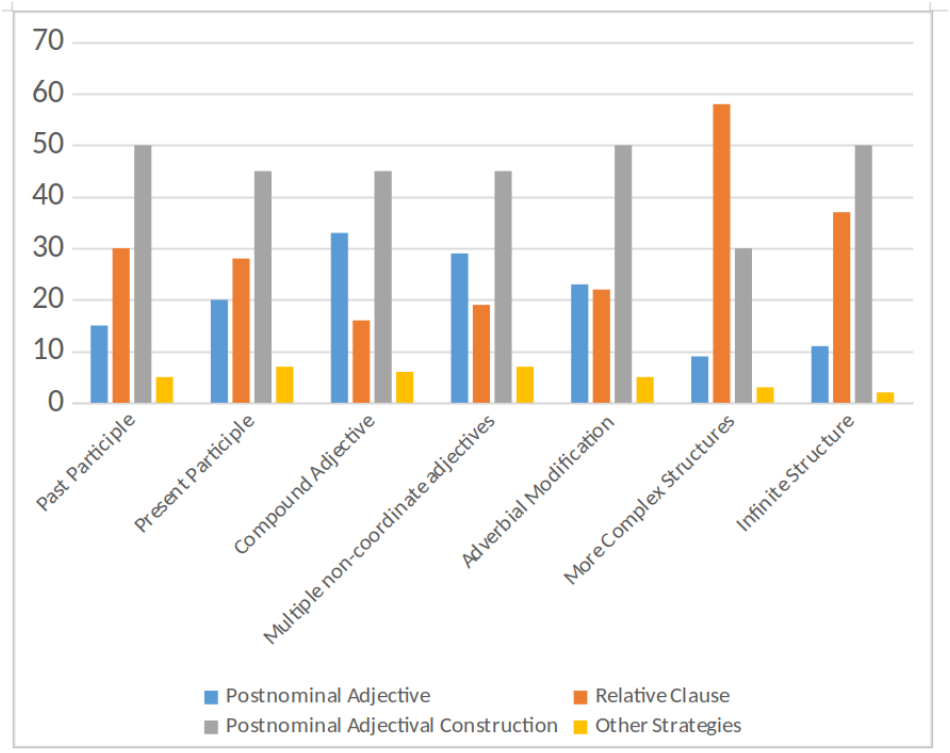


Figure 4: Distribution of translation strategies according to PCAs

This strategy, which also includes what could be defined as reduced relative clauses is explained later with (50) and (51), is the preferred choice for conveying the intended meaning in all contexts apart from adjectives modified by complex structures.

With compound adjectives, the translators consistently chose to position the adjectives or lists of adjectives after the noun, often simplifying the compound's nuances of meaning (33%). This straightforward approach was evidently effective for conveying the intended meaning of compound adjectives by taking into consideration the overall context (45). When the context is less clear and more precision is required in the translation of the compound, the same structure is translated without preserving the original grammatical categories (45b).

(45) unter normalen marktwirtschaftlichen Bedingungen
under normal market_economical conditions

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a. in condizioni economiche normali
   under conditions economical normal
b. in condizioni di normalità economica in un mercato..
   under conditions of normality economical in a market

When dealing with multiple non-coordinated adjectives, the translators most often opted for simplicity by placing the adjectives after the noun (29%) as in (46), or by using relative clauses as in (47), a less common approach (19%)

(46) nicht zuletzt auch die unzureichenden digitalen Kompetenzen
not least also the insufficient digital competences
non per ultimo anche le competenze digitali insufficienti
not for least also the competences digital insufficient

(47) Die subnationale Ebene nimmt wichtige ausgabenrelevante
The subnational level takes important relevant_for_expenses
Aufgaben wahr
tasks
A livello subnazionale vengono acquisiti incarichi che richiedono
At level subnational are acquired tasks that require
grosse spese
big expenses

In cases of adverbial modification, the translators employed both relative clauses (22%) and postnominal adjectives (23%) with almost equal frequency after the primary choice of postnominal adjectival constructions. Interestingly, when translating complex structures preceding adjectives, relative clauses accounted for a larger portion of equivalence in meaning (58%), possibly due to the explicitness of their structures, this being a common strategy to effectively manage complexity. This preference for explicitation using relative clauses may derive from the specific cognitive complexity of the translation task. In the case of (48), it is surprising to see that the relative clause was not reduced to a postnominal adjectival structure by eliding the relative pronoun together with the functional verb like in (48a) compared to (48b).

(48) dass die auf der Grundlage der Marketingverträge erworbenen
that the on the basis of marketing_contracts acquired
Dienstleistungen
services

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a. I servizi che vengono acquistati con contratti di marketing
The services that are acquired with contracts of marketing

b. I servizi acquistati con contratti di marketing
The services acquired with contracts of marketing

This is an indication of the general tendency to preserve the original structure of the sentence when possible (Cardinaletti & Garzone 2005) even after shifting to a position syntactically coherent with the grammar of the target language. The reduction of relative clauses is however not always excluded, as already seen. In the following example the relative pronoun is also in nominative form but the adjectivized verb in infinitive form takes a direct object, probably reducing the complexity of the relationship within the adjectival construction (49).

(49) die auf die steigenden Treibhausgase zurückzuführenden
the to the increasing greenhouse_gases to_attribute
Umweltrisiken
environmental_risks
I rischi ambientali riconducibili ai crescenti gas
The risks environmental attributable to_the increasing gases
serra
greenhouse

Both the complexity of the relationship between the NP and the relative pronoun, and between the adjective and its preceding modifiers, tends generally to be decisive for the use or deletion of the relative pronoun and consequent reduction of relative clause in Italian. The few exceptions include (50) and (51).

(50) Auf ihren Märkten gut positionierte Reiseveranstalter
In their markets well positioned travel_companies
Tour Operator ben posizionati sui loro mercati
Travel companies well positioned in their markets

(51) Die 2013, 2015 und 2016 zwischen der FFHG und Ryanair geschlossenen
The 2013, 2015 and 2016 between the FFHG and Ryanair stipulated
Verträge
contracts
I contratti stipulati tra FFHG e Ryanair nel 2013, 2015 e
The contracts stipulated between FFHG and Ryanair in 2013, 2015 and
2016
2016

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In summary, the reason for choosing one or another strategy probably lies in the semantics of each individual sentence, but with a clear preference for retaining the grammatical category of the original text, while shifting elements according to the rules of the target language. When preservation becomes marked, a certain degree of reformulation is implemented, accounting for about 5% of the translation strategies. Considering (52a), proposed here for the sake of comparison, it becomes clear that postposition of a list of adjectives would not be a particularly elegant solution. Since *öffentliche Finanzen* is considered to be a single technical term with a standard translation, reformulation can only affect the adjective *solider*. *Solider* is in fact the only part of the NP that can be modulated for a more elegant translation approach. Nominalization of the adjective as in (52b) helps to resolve the oddness of the list of adjectives. In this specific case it offers a matching syntactic structure with "*stabilità*", which is also a noun. The translation in (52b) from the corpus appears more straightforward, although deviating from the grammatical categories of the original words.

(52) Gewährleistung makrofinanzieller Stabilität und solider öffentlicher
Guarantee macrofinancial$_{GEN}$ stability and solid$_{GEN}$ public$_{GEN}$
Finanzen
finances

a. ?Garanzia di stabilità macrofinanziaria e finanze pubbliche più
Guarantee of stability macrofinancial and finance public more
solide
solid

b. Garanzia di stabilità macrofinanziaria e solidità nelle finanze
Guarantee of stability macrofinancial and solidity in finances
pubbliche
public

Exceptional translation strategies hinted at in examples (45a) and (52b) will be further discussed in the next section.

5.3 Exceptional translation strategies: 5%

The present analysis of translation strategies reveals a particular subset comprising approximately 5% of total cases. Within this subset instances are identified in which adjectival structures undergo significant transformations during translation without any effort to preserve the original patterns. A qualitative analysis of this exceptional 5% brings some noteworthy scenarios to light.

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Firstly, there are cases of total reformulation, with the original adjectival structures being entirely reconstrued in the target language, leaving no discernible trace of the original wording. For example the source text in example (53) is transformed without using adjectives, preserving the meaning by verbalizing the embedded information. The adjectival structure is completely reformulated, resulting in a distinct expression.

(53) auf eine durch die Mitgliedstaaten fristgerechte Umsetzung
toward a by the Member_States timely implementation
der Agenda für einen nachhaltigen und wettbewerbsfähigen
of_the agenda for a sustainable and competitive
Tourismus
tourism
affinché l' agenda per un turismo sostenibile e competitivo possa
so_that the agenda for a tourism sustainable and competitive can
essere tempestivamente attuata dagli Stati membri
be promptly implemented by_the Member States

Another case reveals a different approach, with adjectives being entirely omitted in the target language, although they can still be inferred from the surrounding context. For instance, the phrase in (54) is translated into Italian by eliminating the present participle *handelnder* completely and relying on the context to convey the intended meaning.

(54) ob ein unter normalen marktwirtschaftlichen Bedingungen
whether a under normal market_economic conditions
handelnder privater Anleger von vergleichbarer Größe
operating private investor of comparable size
se un investitore dello stesso calibro in condizioni
whether a private_investor of comparable size under conditions
normali…
normal

What raised most interest in the study are instances of transposition, when the adjectival structures are not omitted or completely reformulated but instead shifted in placement and form. These cases added an intriguing layer of complexity to the translation process, as they required a nuanced understanding of both source and target languages to ensure accuracy while accommodating the structural differences.

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Traditionally, transposition involves moving from one grammatical category to another without altering the meaning of the text. As a translation technique it introduces a change in grammatical structure (Alcaraz Varó & Hughes 2002; Kather 2011). In the following examples from the corpus in this section, the translators change the grammatical category of the adjective, restructuring the dependencies between various syntactical elements. A German adjective usually becomes a noun phrase and a German noun phrase becomes its prepositional phrase. Other information is added later in the form of additional syntactical structures. Cases of transpositions are found both as PCAs with past participles (55) and present participles (56), when adjectives like *festgelegt* or *zunehmend* are nominalized while nouns like *Kategorien* or *Zahl* become prepositional phrases of those nominalizations.

(55) die im Anhang II festgelegten Kategorien
the in attachment II defined categories
la definizione delle categorie di cui all' allegato II
the definition of_the categories of which in attachment II

(56) Eine stetig zunehmende Zahl
A constant increasing number
Un aumento costante del numero
an increase constant of the number

The complexity of the structure between determiner and noun requires a syntactical reference in both directions, and so it can be assumed that the semantic information within the complex structure in some cases is raised to a higher semantic level within the sentence, which must be taken into account with a specific strategy when translating into Italian.

Other cases of transposition are found within the set of prenominal adjectival compounds (57). Here the pattern is less well defined, as in (55) and (56), with the inflected component of the compound head becoming a noun, and other information attached afterwards.

(57) Tourismusstatistische Ausgangsdaten
tourism_statistic base_data
Le statistiche di base sul turismo
The statistics of base on tourism

Other transpositions are found in the category of multiple non-coordinated adjectives. As already seen with example (**??**), the syntactical relationship between the adjectives is guaranteed if we consider "*valore universale*" a single

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unit referred to by *straordinario*. The syntactical link is preserved without ambiguity again by choosing non-preservation of the grammatical category of the original text, hence nominalization. The same occurred in example (52) where nominalization of *solider* was favoured by the symmetry with the preceding NP *macrofinanzieller Stabilität*.

This way of treating PCAs is very interesting and may be based on an underlying logical syntactical mechanism. It was decided to investigate the possibility of applying the transposition strategy to other cases besides the few already identified in the corpus. This involved identifying the most important semantic information in the adjectival structure and nominalizing it in order to balance the structural shift. The transposition strategy was therefore tested on numerous examples that were translated using the more statistically significant strategies, a sample of which is provided here as (58) to (61), where examples (a) display more frequent translation startegies compared to examples (b) which display transposition.

(58) dass die auf der Grundlage der Marketingverträge erworbenen
that the on the basis of marketing_contracts acquired
Dienstleistungen
services

a. che i servizi che vengono acquistati con contratti di marketing
that the services that are acquired with contracts of marketing

b. che l' acquisizione di servizi tramite contratti di marketing.
that the acquisition of services through contracts of marketing

(59) zur Anpassung dieser Bedingungen an geänderte internationale
for adjusting $\text{these}_{\text{GEN}}$ conditions to revised international
Definitionen
definitions

a. per adattare queste condizioni alle definizioni internazionali
to adjust these conditions to_the definitions international
modificate
changed

b. per adattare queste condizioni alle modifiche delle definizioni
to adjust these conditions to_the changes of_the definitions
internazionali
international

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(60) die stetig steigenden Wohnimmobilienpreise
the continually increasing real_estate_prices

a. I prezzi del mercato immobiliare costantemente in crescita
The prices of market real_estate continually in increase

b. la costante crescita dei prezzi del mercato immobiliare
the constant growth of prices of the market real estate

(61) qualitativhochwertige Arbeitsplätze gewährleisten
high_qualitative work_places guarantee
garantire posti di lavoro di qualità garantire la qualità dei posti di lavoro
guarantee place of work of quality guarantee the quality of_the place of work

The possibility of applying a transposition strategy derives from the intricate nature of the structural relationship between determiners and nouns in complex adjectival phrases. This intricacy requires simultaneous syntactical reference in both directions: from the adjective to the determiner, and from the noun to the adjective. The idea that sequences of adjectives can also be rendered with transposition supports the argument of Kotowski & Härtl (2019) that multiple adjective sequences are shaped by syntactic rules. This justify our categorisation of such structures as complex. The semantic information encapsulated within the complex adjectival structure sometimes can assums heightened significance within the overall sentence context, which must be considered when rendering such constructs into Italian.

It is also supposed that the complexity of the adjectival structure may influence the adoption of such translation strategies. It appears that the more complex the structure, the more frequently transposition can be considered an option. A set of sentences was defined, drawn from various categories of PCAs, and further scrutinized to assess the presence of distinct factors contributing to cognitive complexity (Rohdenburg 1996; Gries & Wulff 2011). These factors encompassed several elements, including:

- The occurrence of genitive case prepositions within the intricate adjectival structure.
- The presence of lengthy adjectival structures that create greater separation between the determiner and the noun phrase. This extension could result from the inclusion of more than two non-coordinated adjectives or the adverbial modification of adjectives (LenghtDB>3).

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- The identification of relationships other than mere specification between the elements within an adjectival compound.
- Adjectives taking arguments and not merely complements within the complex structure. Arguments consist primarily of temporal (62) and local phrases (63) with a smaller portion of agent phrases (64).

(62) das 1998 gestartete Programm
the 1998 started program

(63) das in Österreich geltende Abkommen
the in Austria valid agreement

(64) den vom Verein vergebenen Preis
the from_the association given prize

The distribution of different factors within the selected subset of sentences is shown in the following graph:

Testing transposition on this subset of sentences suggests that genitive case prepositions within PCAs have a significant effect, not only enabling transposition but even triggering it. The translation provided in the corpus does not apply transposition (65) and is questionable (65a) while the proposed nominalization of the adjective *fehlend* following the preposition *wegen* is without doubt acceptable (65b).

(65) wegen fehlender gemeinsamer statistischer Merkmale und
due_to missing common statistical features and
Qualitätsanforderungen
quality_requirements

a. ?a causa di caratteristiche statistiche e requisiti di qualità
due to features statistic and requirement of quality
comuni mancanti
common missing

b. a causa della mancanza di caratteristiche e requisiti di qualità
due_to the lack of features and requirement of quality
comuni
common

The presence of adverbial modification with a long distance between the determiner and the noun phrase induces adjuvating effects (66a vs 66b). A combination of the two factors definitely requires transposition (67).

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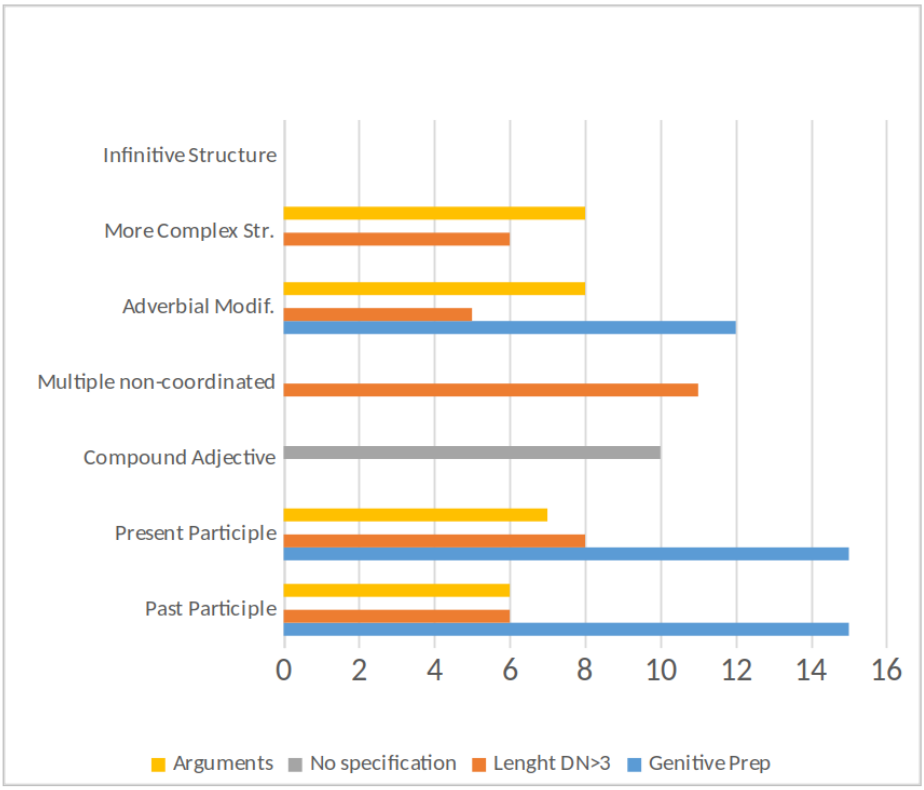


Figure 5: Complexity factors in different categories of PCAs

(66) so dass insgesamt 145 Projekte zu einer stark erhöhten touristischen
so that total 145 projects to a strong increased touristic
Aktivität beitragen
activity contribute

a. ?cosicché 145 progetti hanno contribuito ad un' attività turistica
so_that 145 projects have contributed to a touristic activity
fortemente in crescita
strongly in increase

b. cosicché 145 progetti hanno contribuito ad un forte aumento
so_that 145 projects have contributed o a strong growth
dell' attività turistica
of_the activity tourist

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(67) trotz eines leicht rückläufigen Beitrags der Investitionen
despite a slight decreasing contribute of investments
*nonostante un contributo leggermente in calo degli investimenti
despite a contribute light in decrease of the

investments
nonostante un leggero calo dei contributi per gli investimenti
despite a light decrease of_the contributions for the investments

This strategy simplifies the sentence structure and makes it more natural in the target language, while preserving the original meaning. However, there is no straightforward correlation between the possibility of transposition and the other two complexity factors tested. Compound adjectives can thus be transposed with a nominalization strategy, but this possibility has more to do both with the complexity of relationships between individual elements and the semantics of the elements involved outside the compound itself. The same compound like *geschlechtspezifisch* comprises a specifying relationship which might require transposition in one sentence (68) but rule it out if combined with a different NP (69).

(68) die geschlechtsspezifische Segmentierung auf dem Arbeitsmarkt
the gender-specific segmentation on the job market
la specificità di genere nella segmentazione del mercato
the specificity of gender in_the segmentation of the market

(69) geschlechtsspezifischen Lohngefälles für Österreich
gender-specific wage_gap for Austria
?specificità di genere nel divario degli stipendi in Austria
specificity of gender in_the gap of_the wages in Austria

The potential of transposition as a translation strategy was also explored by examining sentences without complicating factors, with the aim of determining whether transposition could be a viable alternative in these scenarios. Transposition is observed to be generally suitable as a translation approach in a wide range of cases, although it is worth noting that, particularly in cases of low complexity, preserving the original structure is just as good.

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6 Conclusion

The essence of this study is based on an undeniable distinction between German and Italian, with the former predominantly employing prenominal attribution, and the latter strongly inclined toward postnominal attribution. The contrast emerges most prominently in the way that German allows substantial syntactic complexity between determiners and associated nouns. Italian has recently been adopting increasingly complex prenominal adjectival structures, especially lists of adjectives, under the influence of English. Nevertheless, PCAs are still confined to exceptional cases. Starting from this disparity, some pivotal questions for translators were investigated. How should they adapt to this linguistic divergence? What are the most successful translation strategies for navigating the intricacies of adjectival constructions?

This analysis ventures into some uncharted territory, particularly the comprehensive exploration of adjectival complexity, extending beyond participial adjectival phrases or reduced relative clauses. The applied conception of complexity was shaped by a cross-linguistic perspective, encompassing scenarios in which an equivalent level of complexity cannot be seamlessly transposed into the Italian language. Previous studies have focused primarily on dissecting specific aspects of adjectival phrase structure from both the monolingual and contrastive perspectives. The authors propose that the present study can extend the current understanding of German prenominal adjectivization and its Italian equivalent from a translational perspective.

The assembled corpus proved invaluable for identifying the phenomenon under study. Categorization was not an initial goal, but as study of the corpus proceeded, it became clear that a systematic classification was essential to thoroughly examine the different translation strategies. The classification also facilitated quantitative analyses, of both original texts and associated translation strategies. The quantitative analyses were aided by statistical calculations, revealing a significant correlation between German forms and translation strategies. It was also possible to corroborate the hypothesis that preserving the original structure is sometimes favoured over reformulation, even when the latter may seem more fluid.

After quantitative analysis, a qualitative exploration of exceptional translation strategies was conducted. Transposition emerged as particularly noteworthy due to its unexpected versatility. Transposition was assessed first within corpus examples, and then in controlled contexts, subsequently extending the examination to cases in which it was not used even though it could have been. Guided by the intuition that greater complexity in prenominal adjectives might lead to less

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conservative translation strategies, several factors were postulated as contributing to the cognitive complexity of the structure. Among these, the presence of prepositions with indirect cases was seen to encourage nominalization of adjectives in translation, especially when a modifying adverb was enchained within the adjectival construction and the distance between the determiner and noun was substantial.

The intricate nature of German prenominal adjectival structures between determiners and nouns would suggest that a bidirectional syntactical reference is necessary. Consequently, the semantic information within these complex structures is sometimes raised to a higher semantic level within the sentence. In standard constructions the pivotal meaning is typically carried by the adjective. However, in adjectival compounds and prenominal adjective lists, discerning the highest level information requires a thorough comprehension of the semantics within the overall syntactic context. When this kind of complexity with bidirectionality is established, then transposition is an option. This phenomenon must be duly considered, prompting the development and application of targeted translation strategies when rendering these structures into Italian. On the basis of qualitative analysis of examples from the corpus and elsewhere, there is reason to believe that transposition represents a viable strategy for addressing this shift in attribution of meaning.

However, it is important to adopt a nuanced approach to translation strategies without being overly prescriptive. It is not possible to establish rigid rules for transposition as a translation strategy for PCAs. This depends also from the fact that syntactic complexity in prenominal environments is not only structural but also interfaces with semantic and discourse-related constraints.

Nevertheless, it is very helpful to be aware of the existence of transposition as a potential TS, particularly in some specific syntactical contexts, allowing translators to exploit its potential when appropriate. Transposition should be consciously included within every translator's toolkit, as a versatile mechanism for use when initial more spontaneous or straightforward translations fail to fully capture the essence of a source text.

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Chapter 5

Schodiště z mramoru and *the marble staircase*: English divergent counterparts of Czech postmodifying prepositional phrases

Mala

The chapter explores the English translation correspondences of Czech postmodifying prepositional phrases, focussing on the rare cases of formal divergence. The contrastive analysis relies on material drawn from the parallel corpus InterCorp. The results reveal the impact of the register (fiction vs. non-fiction) and direction of translation on the degree of congruence between the counterparts. Drawing on both Czech texts translated into English (CS(en)) and English-to-Czech translations (cs(EN)), the study catalogues the recurrent patterns of correspondence and describes their formal and semantic properties. The choice of a particular divergent counterpart was found to be related mostly to the typological differences between Czech and English, as manifested, for instance, in the correspondence between Czech postmodifying prepositional phrases and English premodifying nouns or postmodification by participial clauses with semantically weak predicates.

1 Postmodifying prepositional phrases

1.1 Introduction

The study[1] examines the extent and types of correspondence between Czech postmodifying prepositional phrases (henceforth PPs) and their English translation

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Mala. 2026. *Schodiště z mramoru* and *the marble staircase*: english divergent counterparts of czech postmodifying prepositional phrases. In Tomáš Svoboda, Olga Nádvorníková & Věra Kloudová (eds.), *Advancing interdisciplinarity in empirical translation research*, 121–143. Berlin: Language Science Press. DOI: ??

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counterparts. Dušková's (2015) research into syntactic constancy of clause elements has shown that not only is the syntactic function of a modifier preserved in about 90% of noun phrases translated from Czech to English, but also the form of the modification remains largely constant.[2] Given the fact that in English PPs are "by far the most common type of postmodification" (Biber et al. 2021: 601), the choice of a non-prepositional English translation counterpart of a Czech post-modifying PP, such as *postava v černé kombinéze* […] *s dlouhým úzkým mečem v ruce* ('*figure* in black suit […] with long narrow sword in hand') corresponding to *a black-suited figure* […] *holding a long narrow sword* in (1), may appear quite surprising. The present study sets out to explore the factors leading to such unusual correspondences.

(1) CS. Z vody vystoupila postava *v černé kombinéze* [figure in black overalls], s kyslíkovou bombou na zádech a *s dlouhým úzkým mečem v ruce* [with a long narrow sword in hand]. (Michal Ajvaz, Druhé město, fiction)[3]
en. Out of the water stepped *a black-suited* figure with an oxygen cylinder on its back and *holding a long narrow sword.* (transl. by Gerald Turner)

1.2 Prepositions and prepositional phrases

Prepositions and PPs are often relegated to the periphery of the grammatical description of a language, and denied "any real intrinsic syntactic interest other than as an annoying little surface peculiarity". (Jackendoff 1973: 345) Nor do they appear to be a topic worth exploring from the lexical point of view – together with other synsemantic words they just "waste space in a dictionary". (Sinclair 1991: 81) They have, however, received increasing attention since the 1990s with "the emergence of the field of lexico-grammar […], which recognizes the existence of a continuum between lexicon and grammar" (Cosme & Gilquin 2008: 260), and corpus-driven phraseological research. Due to their high frequency, little formal variability, and wide distribution across various registers, prepositions can serve as a useful starting point to "reveal phraseologies that are linked to recurrent meanings and functions rather than to subject-matter". (Hunston

[2] In the opposite, English > Czech direction, the ratio of noun modifiers translated by other clause elements was higher in Dušková's (2015) sample, mainly due to factors connected with the head of the noun phrase and different forms of predication. The study did not focus on the degree of correspondence at the level of formal realization of the clause element.

[3] The most relevant parts of the examples are highlighted in italics. All the examples were excerpted from the parallel translation corpus InterCorp (see Sources and Section 2).

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2008: 292) Prepositions have been found to constitute parts of multi-word lexicogrammatical sequences, or patterns, which can, for instance, distinguish among registers (e.g., Scott & Tribble 2006; Hunston 2008; Groom 2010), or reveal disciplinary variation in academic writing (Groom 2010).

Prepositions are a synsemantic word class. In both Czech and English, they are most typically complemented by nouns or noun phrases (*in a street*, *na ulici*). Other types of complements include adverbs (*from abroad*, *na potom* 'for afterwards'), and in English also nominal clauses, finite and non-finite (*instructions on where they are used*, *three days after leaving Darwin*), or other PPs (*from behind the wire fencing*). In Czech, the preposition determines the case of its complement. In English, this applies only to pronouns; "those pronouns that have different (non-genitive) case forms almost invariably appear in the accusative after prepositions". (Huddleston & Pullum 2002: 598) While in Czech the preposition always precedes its complement, in English, the preposition can be 'stranded', i.e. not followed by its complement, in such constructions as interrogative or relative clauses (e.g., *What more could a child ask for?*) (Biber et al. 2021: 108–110, Cvrček et al. 2010: 333), and "occasionally, a preposition is postposed, as in the case of English *ago* (e.g., *three weeks ago*)". (Hasselgård 2021: 147)

From the syntactic point of view, prepositions perform three basic functions, as illustrated in (2). They express the relation

- between the verb or adjective and a nominal expression, where the nominal element typically constitutes a valency complement of the verb (the object) or the adjective, e.g., the relation between *wondered* and the noun phrase *the acuteness of the symptoms*;
- between two nominal expressions, e.g., between the noun phrases *the acuteness* and *the symptoms*; or
- between the adverbial and the proposition conveyed by the clause, e.g., between the adverbial *at the time* and the rest of the clause (cf. Čermák 1996: 32, Biber et al. 2021: 109).

(2) EN. *At the time*, I […] merely wondered *at the acuteness of the symptoms* (Robert Louis Stevenson, The Strange Case of Dr Jekyll and Mr Hyde, fiction)

Biber et al. (2021: 76) use the term 'bound prepositions'[4] to refer to the prepositions which "have little independent meaning, and the choice of the preposi-

[4]More precisely, the terms 'bound' and 'free' prepositions refer to uses of prepositions, since some prepositions can be used either as 'bound' valency complements or as 'free' prepositions (cf. one had to rely *on* intuition – we were sitting *on* the terrace).

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tion depends upon some other word (often the preceding verb)"[5] (i.e. type a, and partly b above). 'Free prepositions' on the other hand, "have an independent meaning; the choice of the preposition is not dependent upon any specific words in the context" (type c above). Bound prepositions, therefore, enter a bidirectional relationship – their choice is dependent on the valency of the verb, adjective or noun, and at the same time, they determine the choice and form of their own complement (Čermák 1996: 33).

The present chapter focuses on the adnominal uses of prepositions, namely PPs which postmodify a noun. In Czech grammar, the modifier in the noun phrase is termed 'přívlastek' or 'atribut' ('attribute'), an optional element modifying a syntactic noun (Dušková 1999: 209, cf. Štícha et al. 2013: 725). PPs are a form of the 'non-congruent attribute', i.e. a postmodifier which does not display grammatical agreement with the head noun (Grepl et al. 1995: 503). The choice of the preposition can be dependent on the valency of the head noun, e.g., *přednáška o syntaxi* 'a lecture on syntax'. Free postmodifying PPs express various circumstances, characterizing the head noun, such as the place (*slavnost na zahradě* 'a party in the garden'), time (*noci v létě* 'nights in summer'), or origin (*svetr z vlny* 'sweater from wool'). Grepl et al. (1995: 509) point out that the postmodifying phrase often corresponds to adjectival premodification (e.g., *zahradní slavnost* 'garden party', *letní noci* 'summer nights', *vlněný svetr* 'woolen sweater'). However, the relationship between the noun and the modifier is expressed more explicitly by the preposition, cf. *kuchyňský nůž* ('a kitchen knife') and *nůž v kuchyni / z kuchyně / pro kuchyň* ('a knife in/from/for the kitchen') (Čermák 1996: 33, cf. Štícha et al. 2013: 730–734, Karlík 2017).

In English, "[p]repositional phrases are by far the most common type of postmodification", making up "65–80% of all postmodifiers in all registers". (Biber et al. 2021: 601) As mentioned above, the choice of the preposition can be constrained by the preceding noun valency, e.g., *her reliance on public support.*

1.3 A cross-linguistic view of postmodifying prepositional phrases

As shown by Hasselgård (2021: 147), "[c]ontrastive studies of the syntactic functions of prepositional phrases indicate that languages may have different restrictions and preferences regarding their use even when the linguistic resources are

[5]The close relationship between the verb and the preposition is also reflected in the fact that the 'verb + preposition' combinations "such as *confide in*, *rely on*, and *part with* are usually regarded as forming a multi-word unit and are called prepositional verbs". (Biber et al. 2021: 76)

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similar". Considering the postmodifying PPs, the differences may involve, for instance, their overall frequency (Dušková 2015, Hasselgård 2021), their degree of flexibility (Moreira-Rodríguez 2006), the restrictions on the use of a particular semantic group of postmodifiers (Mott 2013), or the polysemy and phraseology of individual prepositions (Cosme & Gilquin 2008, Schmied 1998).

The degree of translational equivalence of PPs is likely to be influenced by the syntactic function of the PP (Cosme & Gilquin 2008, Schmied 1998). For example, Cosme & Gilquin's (2008: 270) results suggest that postmodifying *with*-PPs tend to have congruent French counterparts (e.g., *its strategic relationship with Syria* corresponding to *sa relation stratégique avec la Syrie*) much more frequently than when they occur in other syntactic functions.[6] Therefore, when exploring the cross-linguistic correspondences of PPs, the comparison cannot be purely formal, but should "concern the mapping between form and function". (Gast 2012) At the same time, "[the] degree of congruence in translation differs across translation directions". (Hasselgård 2021: 159)

For the English translation counterparts of Czech PPs, "the general distribution of equivalents is typically in the proportion of 2:1 (two prepositional equivalents to one non-prepositional)". (Klégr & Malá 2009: 17) The range of prepositional equivalents is quite broad, usually with one dominant preposition both in terms of frequency[7] and the number of senses it can translate (e.g., *in* for *v/ve*). The degree of formal congruence of the translation counterparts appears to be related to the syntactic function of the PPs, with postmodifying and adverbial PPs being translated by congruent counterparts more frequently (65–70% of counterparts) than those functioning as complements.

Postmodification, generally, "displays correspondence [between Czech and English] not only in the syntactic function but also in the form of realization". (Dušková 2015: 136) Where postmodification is realized by a PP in Czech, the English translation was found to be congruent both syntactically and formally in a vast majority of correspondences (see Section 1.2), with most divergent counterparts comprising premodification by a converted noun, e.g., *přiznání k dani – tax return* (Dušková 2015: 142–145). The correspondence between a Czech postmodification by a PP and an English premodification by a converted noun – "a

[6] This may be connected with the fact that in postmodification, the preposition tends to express "the prototypical sense of mutual relation, rather than the more metaphorical senses which are characteristic of *with*". (Cosme & Gilquin 2008: 271)

[7] The frequency of the dominant counterpart appears to be different for each preposition. For instance, 49.7% of instances of the Czech preposition *v/ve* were translated using its dominant counterpart, the English preposition *in*, and 18% of *po* by the dominant preposition *after* in Klégr & Malá's (2009) data.

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formally indeterminate word class acquiring its syntactic function from its position" (Dušková 2015: 159) – reflects the typological distinctions between the two languages, and the analytic and nominal character of English in particular.

The premodifying nouns, as well as adjectives, were also listed by Klégr et al. (2012: 178–180) as possible non-prepositional correspondences of postmodifying PPs comprising the prepositions *v/ve* 'in', *na* 'at' and *s/se* 'with', e.g., *tik ve tváři* 'a tic in face' – *facial tic*, *dopis na rozloučenou* 'a letter for farewell' – *a farewell letter*, *pouzdro s pistolí* 'a holster with pistol' – *pistol holster*. The postmodifying *v/ve*, *na* and *s/se* PPs were also found to be translated by participial postmodifiers, e.g., *křeslo v kombinaci dřeva a kůže* 'a chair in a combination of wood and leather' – *chair made of wood and leather*, *pohled na zelené dálky* 'a view of green distances' – *a distant view clothed in fresh green*, *košík s malým Mojžíšem* 'the basket with little Moses' – *the basket carrying little Moses*.

As the above studies have shown, the translation correspondences where a Czech postmodifying PP has a non-prepositional English counterpart are rare. This appears to be mainly due to two factors: first, the constancy of syntactic function and realization form of postmodification in translation from Czech to English (Dušková 2015), and secondly, the typological differences between the predominantly synthetic Czech language and analytic English. In Czech, the relation between two nouns within the noun phrase is conveyed most frequently by a case marking of the modifying noun (Štícha et al. 2013: 728), cf. *kousek čokolády* 'piece-NOM chocolate-GEN' – *a bar of chocolate*; *procházka lesem* 'walk-NOM forest-INSTR' – *a walk in the forest*). In the English noun phrase, on the other hand, the preferred means of expressing the relation between the head noun and the postmodifier is the preposition, PPs constituting the most frequent form of postmodification (Biber et al. 2021: 601). Alternatively, the modifying function of the noun can be indicated by word order in English, with the modifier preceding the head noun, e.g., *čokoládové tyčinky* ('chocolate-ADJ bars-NOUN') – *chocolate bars*, *lesní cesta* ('forest-ADJ path-NOUN') – *a forest path*. As the examples illustrate, in Czech the corresponding premodifier is expressed by a denominal adjective. In Czech, conversion is a marginal word-formation process, and premodifying nouns are limited to loanwords (e.g., *jazz klub*, *fit centrum*), abbreviations (e.g., *IBM diskety*, *HIV test*) and expressions bordering on apposition (e.g., *chudák Marie* 'poor-NOUN Mary-NOUN') (Veselovská 2017, Karlík 2017). In English, using the noun as a premodifier requires no morphological changes in the noun. Due to the fixed word-order, including the ordering of the components of the phrase, the modifying function of the noun is sufficiently indicated by its position within the noun phrase. This may be further supported by the loss of the grammatical contrast of number, e.g., *child labour*, *trouser leg* (Dušková et al. 2009: 2.22.21).

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2 Material and method

The analysis presented in this chapter focuses on the English translation correspondences of Czech postmodifying PPs with the seven most frequent Czech prepositions – *v* 'in', *na* 'at', *z* 'from', *s* 'with', *o* 'about', *do* 'to', and *k* 'towards' (Cvrček et al. 2010: 333). The queries[8] also included the vocalized forms of the prepositions – *ve*, *ze*, *se*, *ke*, *ku* – which are used when the following word starts with the same or similar consonant as the preposition, or with a consonantal cluster, e.g., *ve vesmíru* 'in space', *ke stromu* 'towards the tree' (Cvrček et al. 2010: 340–341).

The research was conducted in two steps, exploring:

1. the impact of the direction of translation (CS > en, or EN > cs), and the register (fiction and non-fiction) on the choice of the English counterparts of Czech postmodifying PPs (Step 1), and

2. the types of divergent counterparts of a Czech postmodifying PP (Step 2).

Methodologically, the research falls within corpus-based contrastive studies. The phenomenon under study is a rare one. The method adopted therefore aims primarily at identifying, describing and classifying the relevant patterns of translation correspondence, rather than at providing a comprehensive overview of their distribution and frequency.

2.1 Step 1

In order to explore the impact of both the register and the direction of translation, four subcorpora of the corpus InterCorp v14 were compiled: fiction translated from Czech to English and from English to Czech, and non-fiction in both directions of translation. The size of the subcorpora is given in Table 1. All texts available in the respective categories in InterCorp v14 were used, which resulted in incomparable sizes of the fiction and non-fiction subcorpora. This, however, does not seem to pose a major problem for the present study due to the methods adopted (i.e. working with samples in Step 1, and using the corpus as a repository of examples extracted by individual queries in Step 2).

The Czech texts (originals and translations) in the four subcorpora were searched for contiguous sequences of a noun and one of the prepositions *v/ve* 'in', *na* 'at', *z/ze* 'from', *s/se* 'with', *o* 'about', *do* 'to', and *k/ke/ku* 'towards'. The

[8]The basic form of the query was [tag="N.+"][lemma="v|na|z|s|o|do|k"]

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Table 1: The size of the four subcorpora of InterCorp v14 (Step 1 of the research).

Direction of translation	Register	Size of the subcorpus		
		Number of positions	Number of texts	Number of authors
CS(en)	fiction	4,686,413	56	31
	non-fiction	611,658	5	5
cs(EN)	fiction	25,258,196	215	138
	non-fiction	2,567,506	21	17

results were shuffled and checked manually[9] in order to select only postmodifying PPs and discard the chance sequences of a noun and adverbial (3). Cases of comitative coordination (Škrabalová 2011, cf. Čechová et al. 2011: 320), i.e. "a paratactic relationship between noun phrases syntactically framed as hypotaxis using the preposition *s/se*" (Malá 2017: 207), illustrated by (4), were also removed from the dataset.

(3) CS. Šarceceg vyhodila zbytky *kozy z okna* […] (Petra Hůlová, Paměť mojí babičce, fiction)
en. Shartsetseg threw the *goat out the window* […] (transl. by Alex Zucker)

(4) CS. I *Eva s Marií* [Eva with Marie] se usmívají. (Michal Viewegh, Vybíjená, fiction)
en. *Eva and Marie* were also smiling. (transl. by David Short)

In some cases, "it may be difficult to decide whether a prepositional phrase following a noun is a postmodifier of the noun or an adverbial in the clause". (Biber et al. 2021: 109) In (5), the PP *na koncích propletených ramen lustru* ('at the ends of the interlacing branches of a chandelier') could be analyzed as either an adverbial (specifying the location of 'coming on', cf. the English translation) or a postmodifier of the noun *světla* ('lights') (specifying the location of the lights). Such concordance lines were retained, and analyzed as postmodifiers.

[9]The cases of omission of the modifier resulting in a meaning shift, and free translations where the extent of divergence from the source at clausal level made it impossible to identify the counterpart of the postmodifying PP were discarded.

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In noun phrases with multiple postmodification the difference in the relative position of the postmodifying PP was neglected as long as its correspondence to an English counterpart could be established.[10]

(5) CS. Náhle se pode mnou rozsvítila světla *na koncích propletených ramen lustru*, visícího na dlouhém laně, které bylo upevněné k lucerně kupole hned vedle mé hlavy; […] (Michal Ajvaz, Druhé město, fiction)
en. Suddenly lights came on *at the ends of the interlacing branches of a chandelier* hanging from a long cable attached to the turret of the cupola right next to my head; […] (transl. by Gerald Turner)

From the Czech texts in each subcorpus, relevant clauses comprising postmodifying PPs were excerpted together with their English translation counterparts until a sample of 250 clauses from each subcorpus was compiled. The representation of the individual prepositions in the 1,000-clause sample is given in Table 2. The relative frequency and the types of counterparts will be discussed in Section 3.

Table 2: The representation of the 7 most frequent Czech prepositions in the sample of fiction and non-fiction text, Czech originals, and translations from English.

Preposition	Fiction		Non-fiction		Total
	CS(en)	cs(EN)	CS(en)	cs(EN)	
v ‘in’	54	62	68	64	248
do ‘to’	6	8	9	4	27
k ‘towards’	14	14	33	12	73
na ‘at’	54	59	50	51	214
o ‘about’	24	16	28	34	102
s ‘with’	56	38	23	45	162
z ‘from’	42	53	39	40	174
Total	250	250	250	250	1,000

[10]Examples of multiple postmodification, e.g., EN. přezkoumat poslední *reakce Douga-8612 na tuto zkušenost* [reactions by Doug-8612 to this experience] (Philip Zimbardo, The Lucifer Effect, fiction) – cs. *review the final reactions to this experience by Doug-8612* (transl. by Radek Kašpar and Marika Králíková), reveal the close relationship between the Czech head noun and postmodifying genitive (*Douga-8612*), which is reflected in its obligatory position immediately after the noun (cf. Dušková et al. 2009: 14.13.2).

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2.2 Step 2

As the divergent counterparts of postmodifying PPs are quite rare in both directions of translation, it was obvious that more data was needed to explore and describe the types of non-prepositional counterparts in detail. A larger, more recent version 15 of the corpus InterCorp[11] was used, and more specific queries based on the findings of Step 1 were formulated to identify further instances of the various types of counterparts (e.g., a sequence of a noun and one of the prepositions in question in the Czech text corresponding to a sentence comprising two adjacent nouns in English), relying on the corpus as an "example bank". (Lüdeling & Kytö 2008) Since this step focussed on the qualitative description rather than on the frequency of the types of counterparts, the direction of translation and the register were disregarded. The results will be discussed in Section **??**.

3 The counterparts of Czech postmodifying prepositional phrases (Step 1)

3.1 Congruent and divergent counterparts

The overall majority of counterparts of the Czech postmodifying PPs (76.6 %, 766 instances) were formally congruent. The highest degree of divergence could be observed in Czech translations of English fiction; in non-fiction, generally, formally divergent counterparts were rarer than in fiction[12] (Figure 1), perhaps reflecting the higher degree of standardization of non-fiction texts (cf. Kufnerová et al. 1994:16).

Figure 1: : Formally congruent (PP) and divergent (non-PP) correspondences of Czech postmodifying PPs (samples of 250 occurrences for each sub-corpus)

Most of the formally congruent counterparts (92.2 %; i.e. 706 out of the 766 instances) also performed the same, postmodifying syntactic function as the Czech PP (6). Again, the degree of syntactic divergence was highest between English

[11]Two subcorpora of InterCorp v15 were compiled and used, one comprising Czech original fiction and non-fiction texts and their English translations (5,612,530 tokens, CS(en)), the other English fiction and non-fiction source texts and their Czech translations (29,975,445 tokens, cs(EN)).

[12]In either direction of translation, the differences in the representation of non-prepositional counterparts between fiction and non-fiction are not statistically significant at the level of $p < 0.05$ (log-likelihood, calculated using llwizard available at https://ucrel.lancs.ac.uk/llwizard.html).

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original fiction texts and their Czech translations. In both text-types and directions of translation, most of the syntactically non-congruent counterparts (6.4%, 49 instances) consisted in the Czech head noun corresponding to an English verb or predicative adjective. The English PP functioned as the adverbial or complement of the verb (often non-finite) or adjective (7). A minor type of syntactically divergent prepositional counterparts was represented by complex prepositions (1.4%, 11 instances), e.g., *v závislosti na pojetí jazyka* 'in dependence on the conception of language' – *according to conceptions of language*.

(6) CS. Jeho *pohled na funkce* [view of the functions] jazyka […] založil nový směr ve filozofii jazyka a lingvistice vůbec. (František Čermák, Základy lingvistické metodologie, non-fiction)
en. His *view of the function* of language […] established a new trend in the philosophy of language and linguistics generally […]. (transl. by Patrick Corness)

(7) cs. Obstoupily ho kolem dokola a zahrnuly ho *nářky na své bolesti* [complaints about their aches] a potíže a nejrůznější ženské neduhy. (transl. by Zuzana Mayerová)
EN. They clustered around him, *complaining about their aches* and pains and their various female ills. (Isaac Bashevis Singer, The Family Moskat, fiction)

The following sections will focus merely on correspondences between Czech postmodifiers realized by PPs and their formally divergent English counterparts (both in non-translated and in translated texts), i.e. English counterparts whose realization form is not a PP, and which may perform a variety of syntactic functions.

3.2 Classification of formally divergent translation correspondences of Czech postmodifying prepositional phrases

The non-prepositional translation correspondences of the Czech postmodifying PPs (both CS and cs) can be classified into three groups according to the syntactic function of the English counterpart:

- Premodifiers, e.g., CS. dárek k *narozeninám* 'a present for birthday' – en. a *birthday* present;
- non-prepositional postmodifiers, e.g., CS. jednání *v Moskvě* 'talks in Moscow' – en. the talks *held in Moscow*;

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- complements of a verb which corresponds to the head noun of the Czech noun phrase, e.g., CS. v pohádce […] je *zmínka o svatbě královského sourozeneckého páru* 'a mention of a royal sibling-marriage' – en. the tale […] *mentions a royal sibling-marriage*).

Apart from these types, two further classes were distinguished which involve indirect translation techniques – reduction and merged counterparts.

- Reduction consists in the omission of the counterpart of the Czech head noun, with the postmodifier taking over its role, e.g., cs. potřísnil mu ji blátem a *šťávou z ostružin* 'juice from blackberries' – EN. stain it with mud and *blackberries*.
- What is referred to as a merged counterpart here is a blend of the noun and postmodifier, usually a single word, retaining the meaning of the source noun phrase, e.g., CS. *chuť k jídlu* – en. appetite.

The representation of the individual classes of counterparts in the 1,000-sentence sample is given in Table 3.

Table 3: Translation correspondences of Czech postmodifying PPs.

Type of counterpart	Fiction		Non-fiction		Total
	CS(en)	cs(EN)	CS(en)	cs(EN)	
premodifier	23	46	10	28	107
postmodifier	25	14	22	14	75
verb complement	4	8	13	4	29
reduction	2	4	1	1	8
merge	4	5	1	5	15
congruent (PP)	192	173	203	198	766
Total	250	250	250	250	1,000

As shown in Figure 2, in the translations from English to Czech (cs(EN)), the divergence was most often (74 instances) due to the Czech postmodifying PP corresponding to an English premodifier. Most frequently (67 instances), this involved English nouns used as premodifiers, e.g., EN. an old *hardback* book – cs. stará kniha *v pevné vazbě* 'an old book in hard back', EN. a *zero sum* game – cs. hra *s nulovým součtem* 'a game with zero sum'. The semantic relationship

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between the modifying noun (phrase) and the head is not indicated in any way in English. In the Czech translation by a postmodifying PP, the preposition makes the relationship more explicit.

In translations from Czech to English, on the other hand, the most frequent divergent correspondences of the postmodifying PPs were postmodifying clauses (47 instances), finite in fiction, e.g., CS. pusté Malostranské náměstí *se zaparkovanými tmavými automobily* 'deserted Malá Strana Square with parked dark cars' – en. the deserted Malá Strana Square *where dark cars were parked*), and non-finite in non-fiction, e.g. CS. snahy *o formalizaci teorie* 'attempts at formalization of theory' – en. attempts *to formalise theory*. This type of correspondence tends to make the English translation more explicit than the source, with the predicate of the dependent clause specifying the relationship between the two noun phrases more precisely than the preposition, e.g., CS. některé požadavky *na práci překladatele* 'some demands on the work of the translator'– en. some of the demands *imposed on the translator's work*.

Figure 2: : The ratio (percentage) of classes of formally divergent correspondences across the registers and directions of translation.

The following section gives a more detailed description of the English non-prepositional counterparts of Czech postmodifying prepositional phrases. A larger corpus and a set of more specific queries (cf. 2.2) drawing on the above findings (3.2) were used to search for further instances of the types of correspondences outlined above.

4 Non-prepositional English correspondences of Czech postmodifying prepositional phrases (Step 2)

4.1 Premodification

Czech postmodifying PPs were found to correspond to several formal subtypes of English premodifiers:

4.1.1 Premodification by a noun in English

The English noun was the most frequent realization form of the premodifier corresponding to the Czech postmodifying PP. The premodifying function of the noun is signalled by its position; the partial conversion of the noun into a syntactic adjective is often accompanied by the loss of the contrast of number

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(Dušková et al. 2009: 2.22.21), e.g., cs. nejbohatší obchodníci *s kořením, ořechy, zrny a lístky v Kóčínu* 'the wealthiest merchants with spices, nuts, beans and leaves in Cochin' – EN. the wealthiest *spice, nut, bean and leaf* merchants in Cochin.

This type of correspondence highlights some reasons why the prepositional form of postmodification may have been preferred to the more common adjectival premodification in Czech:

- The adjectival premodifier may be blocked due to polysemy. For instance, what prevents the use of the adjective *deskový* corresponding to *record* in cs. obchod *s deskami* 'store with records' – EN. *a record* store is the polysemy of the noun *deska* in Czech. The adjective *deskový* is derived from the noun when referring to a board or plate, e.g., *desková* hra ('board game'), *desková* malba ('panel painting'), *desková* tektonika ('plate tectonics'), but not when denoting a gramophone record.

- Another hindrance on using a premodifying adjective in Czech seems to be collocation. While the adjective *datlový* ('date') collocates with several products made of dates, e.g., *datlové* víno ('date wine'), *datlový* koláček ('date cookie'), *datlová* šťáva ('date juice') and occurs also in *datlová* palma ('date palm'), it does not seem to be used with date flour, e.g., CS. moučka *z datlí* rozmíchaná v pivu – en. *date* flour dissolved in beer.

- Another reason why a Czech postmodifying PP may be preferred to adjectival premodification consists in a possible shift in reference. While *předsíňové* zrcadlo ('the front-hall-ADJ mirror-NOUN'') generally refers to a type of mirror,[13] zrcadlo *v předsíni* ('the mirror in the front hall') has a specific reference, describing a particular item of furniture.

- The Czech postmodifying PP often corresponds to the English premodification by a proper noun, usually a place name, e.g., cs. v zapadlém a přísně tajném komplexu *ve Virginii* 'in the remote and classified complex in Virginia' – EN. in the remote, classified *Virginia* medical complex, and by compound nouns, often string compounds, e.g., CS. vídeňský spisovatel *z přelomu století* 'the Viennese writer from the turn of the century' – en.

[13] Cf. the following example from the 'House and Living' magazine: *V předsíňovém zrcadle se potřebujeme vidět od hlavy k patě.* 'We need to see ourselves from head to toe in the front hall mirror'. (Dům a bydlení, 38/2017, Křen, Michal et al., 2022. *Corpus SYN*, v. 11 from 14. 12. 2022. Institute of the Czech National Corpus, Charles University, Prague. Available from: https://www.korpus.cz)

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the turn-of-the-century Viennese writer, cs. zprávy *z bojišť* 'news from the battlefields' – EN. *down-in-the-trenches* war stories.

A variety of semantic relations may obtain between the head noun and the Czech postmodifying PP corresponding to the English converted noun (the classification is based on Hasselgård 2021: 150–151, cf. also Malá 2017: 210–211), for instance:

- temporal: cs. pohřebišti *ze starší doby železné* 'a burial from Late Iron Age' – EN. a *Late Iron Age* burial, cs. nějakého strašidelného obchodního centra *z přelomu století* 'some creepy shopping mall from the turn of the century' – EN. some creepy *turn-of-the-century* shopping mall;
- locative: CS. na zvlhlém koberci *v obývacím pokoji* 'on the wet carpet in the living room' – en. on their wet *living-room carpet*, CS. slečny *v baru* 'girls in the bar' – en. the *bar* girls, CS. jedné posluchárny *v druhém patře* 'one lecture room on the third floor' – en. one of the *third-floor* lecture rooms;
- focus: the modifier "specifies some aspect" of the head noun, for instance,
  - the purpose, e.g., cs. pohádka *na dobrou noc* 'story for good night' – EN. *bed-time* stories, CS. příspěvků *v nezaměstnanosti* 'benefits in unemployment' – en. *unemployment* benefits, CS. nožíku *na rozřezávání korespondence* 'a knife for opening letters' – en. *a letter* opener, cs. všechny její seznamy *na nákup a vyřízení potřebných věcí* 'all her lists for shopping and doing necessary things' – EN. all her *shopping* lists and *things-to-do* lists, cs. soutěž *o nejlepší pojmenování letadla* 'the competition for the best name for a plane' – EN. *the best-name-for-a-plane* competition;
  - content or composition, e.g., CS. kelímek *s kafem* 'the cup with coffee' – en. his *coffee* cup, CS. krvavé paštiky *z koňských jater* 'blood-streaked pâté form horse liver' – en. blood-streaked *horse liver* pâté, cs. salát *z řas a chaluh* 'the salad from seaweed' – EN. the *seaweed* salad;
  - occupation, e.g., CS. inteligentního obchodníka *s chmelem* 'the intelligent trader in hops' – en. the intelligent *hops* trader, CS. továrník *na umělé zuby* 'the manufacturer for false teeth' – en. the *false-teeth* manufacturer;
- attribute: the modifier "specifies a property of the NP head", for instance,

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- material, e.g., CS. tajemné tramvaji *ze zeleného mramoru* 'the mysterious streetcar from green marble' – en. the mysterious *green marble* streetcar, CS. dveře jídelny *z mléčného skla* 'the door of the dining-room from frosted glass' – en. *frosted-glass* doors into the dining-room;
- cause, e.g., cs. nová alergie *na sójové boby* 'the new allergy to soya beans' – EN. the new *soya-bean* allergy;
- size, e.g. cs. Východního sálu *o rozloze téměř sto metrů čtverečních* 'the East Room of the size almost one hundred square meters'– EN. the *thirty-two-hundred-square-foot* East Room.

4.1.2 Premodification / determination by a genitive

The correspondence between Czech postmodifying PPs and English nouns in the genitive case may illustrate another difference between the structure of noun phrases in both languages. The genitive serves either as a modifier or a determiner in the English noun phrase, and expresses a relatively broad range of semantic relations to the head noun. These relations are indicated by the prepositions in Czech (albeit loosely due to the polysemy of the prepositions), e.g., the locative determinative genitive corresponding to *ve*-postmodification in Czech, cs. hospodářská činnost *ve městě* 'economic activity in the city' – EN. the *city's* economic activity, and a temporal modifying genitive, cs. úrok *na tři měsíce* 'interest for three months' – EN. *three month's* interest.

4.1.3 Premodification by a denominal *-ed*

adjective

The translation pair CS. plášť *s kapucí* 'raincoat with hood' – en. *hooded* raincoat is similar to the English premodification by a converted noun. Here, however, the modifying function is indicated, apart from the position, by the denominal adjectival suffix *-ed*. The adjectives are often derived from compound nouns, e.g., CS. jakýsi podivný anděl *s psí hlavou* 'a curious angel with a dog head' – en. a curious *dog-headed* angel.

With animate head nouns, this type of modification typically describes the referent's clothes or body-parts, e.g., cs. Bragg *se svým bílým knírem* 'Bragg with his white moustache' – EN. the *white mustached* figure of Bragg, CS. žena *s tlustými lýtky* 'a woman with thick calves' – en. a *heavy-thighed* [sic] woman, cs. pány *v buřinkách* 'men in bowler hats' – EN. the *bowler-hatted* gentlemen, cs. lidé *v maskách* 'people in masks' – EN. the *masked* individuals, CS. barmana *v bílém*

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saku 'a barman in a white jacket' – en. a *white-jacketed* barman. The referents of inanimate nouns are characterized by their shape or material, e.g., CS. vázy *ve tvaru velblouda* 'vases in the shape of a camel' – en. *camel-shaped* vases, CS. nožem *se zvláště širokým ostřím* 'a knife with an especially broad blade' – en. a special *broad-bladed* knife, cs. klobouk *s pérem* 'a hat with a feather' – EN. his *feathered* hat, cs. psí bouda *s plochou střechou* 'a dog kennel with a flat roof' – EN. a *flat-roofed* dog kennel, CS. židle *s nízkými opěradly* 'chairs with low backs' – en. *low-backed* chairs. As the examples illustrate, the most frequent prepositions in this type of correspondence are *s/se* 'with' and *v/ve* 'in'.

4.1.4 Premodification by a past participle

Examples like CS. první pytel *s pískem* 'the first bag with sand' – en. the first *sand-filled* bag illustrate premodification by a compound which consists of a past participle and a noun representing the object or adverbial modifier of the verb (cf. bags *filled with sand*). The formation of a similar Czech type of compounds – deverbal adjectives – is more limited, and the transposition of the verbal action into a quality is expressed formally by an adjectival suffix (Bozděchová 1994: 133ff). The postmodifying PP appears to be a more flexible solution in Czech.

The participles often express a rather general relational meaning, corresponding to that indicated by the preposition, e.g., CS. pahýlem *ve fusekli* 'his arm in a sock' – en. his *stocking-covered* arm, cs. koženým opaskem *s náboji* 'the leather belt with bullets' – EN. the *bullet-laden* holster belt, cs. průmyslu *na páru* 'industry at steam' – EN. *steam-driven* industry. Compounds with more specific meanings correspond to complex noun phrases complementing the Czech preposition, e.g., cs. prsty *s okousanými nehty* 'fingers with bitten nails' – EN. *nail-bitten* fingers, CS. dálnic *se zlatou vozovkou* 'freeways with golden paving' – en. *gold-paved* freeways.

4.1.5 Other adjectival premodifiers

Apart from the above recurrent counterparts, the Czech postmodifying PPs were also found to correspond to adjectival premodifiers. In translation pairs like CS. měsíc *v úplňku* 'moon in full' – en. *full* moon, the modulation appears to be due to the absence of a corresponding Czech adjective and the fixed character of the naming unit in both languages (cf. also cs. život *ve volné přírodě* 'life in wild nature'– EN. *wild* life). Other instances seem to have been rather context-dependent choices, cf. the different translations of cs. tuhle příšernou cestu *do Mexika* 'this horrible trip to Mexico' – EN. this horrible *Mexican* trip, and CS. svý cestě *do Evropy* 'his trip to Europe' – en. his trip *to Europe*.

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4.2 Postmodification

Czech postmodifying PPs can correspond to non-prepositional forms of postmodification in English. In most cases, the English postmodifier was a clausal one. The majority of these clauses were participial, often with semantically weak posture and relational verbs, which function as linking elements, similar to but more explicit than the corresponding Czech prepositions, e.g., CS. sochy *s lampama* 'statues with lamps' – en. the statues *bearing lamps*, CS. vory *s bronzovými lvy* 'rafts with bronze lions' – en. rafts *carrying bronze lions*, CS. pecí *z žáruvzdorných cihel* 'ovens from fire-resistant bricks' – en. ovens *built of fire-resistant brick*, cs. rovných červených kachlů *ve staré římské konstrukci* 'flat red tiles in the ancient Roman structure' – EN. flat red tiles *placed in the ancient Roman structure*.

The same type of predicates was attested in finite postmodifying clauses, e.g., CS. ukazováčkem *s prstenem ze starého zlata* 'an index finger with a ring from antique gold' – en. an index finger *that wore an antique gold ring*, CS. obrovský rudý transparent *s nápisem* […] 'a huge red banner with an inscription' – en. a huge red banner *that read* […].

Another type of non-finite postmodifying clauses were infinitival clauses, which usually complement a de-verbal noun, e.g., CS. snahy *o formalizaci teorie* 'attempts at the formalization of theory' – en. attempts *to formalise theory*.

Other forms of postmodifiers – adverbs, adjectives and appositive nouns – were represented marginally, e.g., cs. trávy *na druhém břehu* 'the grass on the opposite bank' – EN. the grass *beyond*, cs. zálivu Guantánamo *na Kubě* 'bay Guantánamo in Cuba' – EN. *Guantánamo Bay, Cuba*.

4.3 Verb complements

Example (8) illustrates the Czech noun phrase with a deverbal head-noun *porady* ('consultation') complemented by a PP *s Hlavounem*. The English counterpart relies on a non-finite verb form, the gerund *consulting*, complemented by the object *Bigwig*. The form of the complement is based on the valency of the de-verbal noun and the verb, respectively.

(8) cs. Rozhodl se zamířit do lesa bez *porady s Hlavounem* [without a consultation with Bigwig][…] (transl. by Jitka Minaříková)
EN. He decided to go straight into the wood without *consulting Bigwig* […] (Richard Adams, Watership down, fiction)

Czech de-verbal nouns with prepositional complements typically correspond to non-finite or finite clauses. The non-finite counterparts can be exemplified

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by further gerund constructions, e.g., CS. *zapomenutí na vláčné pohyby ruky* 'forgetting-NOUN about the supple movements of the hand' – en. *forgetting the supple movements of the hand*, CS. *sňatku s lenochem* 'marriage with a loafer' – en. *marrying a loafer*, participial clauses, e.g., CS. *satiry na jejich ideologii, puritanismus* 'satire of their ideology, puritanism' – en. *satirising their puritan ideology*, or infinitive clauses, e.g., cs. *učinit ústupek ze své osobní integrity* 'make a compromise from his personal integrity' – EN. *to compromise his personal integrity*, cs. *výpravu do zahrady* 'a raid on a garden'– EN. *to raid a garden*. The correspondences CS. *se zřetelem k frekvenci* 'with consideration to frequency' – en. *considering the frequency*, and CS. *v návaznosti na K. Buhlera* 'in connection with K. Buhler' – en. *following K. Buhler* may illustrate the recategorization of the 'preposition + noun + preposition' construction into a secondary preposition in Czech, mirrored by a de-participial preposition in English.

The translation by a finite clause requires a greater extent of restructuring (9), which also involves filling the position of the subject in English. The *s*-PP can correspond to the verb *have* expressing a general meaning of belonging. Example (10) illustrates the use of the semantically weak verb *have* with an inanimate locative subject (Malá & Brůhová 2019).

(9) CS. Přímou vládu *v Čechách a na Moravě* ['direct rule in Bohemia and Moravia] císař nikdy nevykonával […] (Jaroslav Pánek et al., Dějiny českých zemí, non-fiction)
en. The Emperor never directly ruled *Bohemia and Moravia* […] (transl. by Petra Key et al.)

(10) CS. Po celé délce pokoje se táhl tmavý stůl, kolem něho stály v pravidelných rozestupech vysoké židle *s černým koženým polstrováním* ['tall chairs with black leather upholstery'], přitlučeným k dřevu mohutnými kovovými cvočky. (Michal Ajvaz, Prázdné ulice, fiction)
en. Running the whole length of the room was a dark table with tall chairs at regular intervals around it. These chairs *had black-leather upholstery* attached to the seat with mighty metal studs. (transl. by Andrew Oakland)

4.4 Reduction and merger

The English counterparts resulting from the last two types of correspondence – (syntactic) reduction and merger – look similar; each of them, however, is a result of a different process. Reduction consists in the English version lacking

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the counterpart of the Czech head noun, without a substantial difference in the meaning of the noun phrase, e.g., cs. vyrazily *lidem v prvních řadách* dech 'people in the front rows' – EN. drew a gasp from *the front row*, cs. nic tak radikálního jako *život v Atlantě* 'life in Atlanta' – EN. nothing as radical as *Atlanta*, cs. *ryba na předloze* 'the fish in the original' – EN. *the original*. The meaning of the head noun is generally derivable from the context; what is retained is the rhematic part of the noun phrase, carrying the most important, focal piece of information within the noun phrase.

The head noun and the prepositional postmodfier can be merged in two ways: the first, illustrated by the translation pairs cs. *chuť k jídlu* 'appetite for food'– EN. *appetite*, CS. *do rýhy ve výstřihu* 'down the line in her cleavage' – en. *down her cleavage*, or cs. *vyhození mince do vzduchu* 'the flip of a coin in the air' – EN. *the flip of a coin*, involves the same idea in both languages expressed in English using a different, often univerbal, expression. The second type most often resembles the PP corresponding to premodification by a noun (4.1.1). Here the nominal modifier becomes integrated with the head noun as the initial stem of a compound, e.g., cs. *truhlík na uhlí* 'the scuttle for coal' – EN. *the coal-scuttle*, cs. *hodiny s vestavěným rádiem* 'a clock with inbuilt radio' – EN. *a clock-radio*, cs. *pastičky na myši* 'traps for mice'– EN. *mousetraps*, cs. *police s knihami* 'shelves with books' – EN. *bookshelves*. Alternatively, the English noun can be formed by derivation, adding a suffix that denotes a person, similarly to the Czech head noun, e.g., cs. *obchodníci s látkami* 'traders in cloth' – EN. *clothiers*.

5 Concluding remarks

The Czech postmodification realized by a PP rarely has a divergent translation counterpart. Non-prepositional counterparts were attested in 23.4% of (1,000) postmodifying PPs. The highest incidence of formally non-congruent correspondences was observed in Czech translations of English fiction; in non-fiction, generally, formally divergent counterparts were less frequent than in fiction. The majority of formally congruent counterparts (92.2 %) also performed the same, postmodifying syntactic function as the Czech PPs. The extent of syntactic divergence was largest between English original fiction texts and their Czech translations (cs(EN)).

The degree of cross-linguistic correspondence and the preference for a particular type of counterpart seem to be influenced by the direction of translation here. The proportion of non-congruent counterparts was larger in the cs(EN) direction both in fiction and non-fiction texts. In this direction, the most prominent

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divergent EN sources of Czech postmodifying PPs were premodifiers realized by nouns, a construction which has no direct congruent counterpart in the Czech language, e.g., EN. *cool stream water* translated as cs. *voda z chladného pramene* 'water from cool stream'. In the opposite, Czech > English direction (CS(en)), the translators appeared to rely more frequently on postmodifying clauses as the counterparts of Czech postmodifying PPs.

One of the goals of the present study was to try to shed some light on the motivation for the translation divergence. Most of the systematic differences seem to be ascribable to the predominantly analytic character of the English language, compared to predominantly synthetic Czech. English has available to it several means of expressing modification that have no formally congruent counterparts in Czech. These include the most frequent divergent counterpart of Czech postmodifying PPs – premodification by a noun. Recurrent 'noun + noun' combinations tend to form compounds. The process, which was classified here as merging the head noun and the modifier, can alternatively be viewed as a scale, with premodifying nouns at one end, and compounds at the other. The transposition between the English premodifying noun (phrase) and the Czech postmodifying PP is fairly straightforward. The relationship between the head of the noun phrase and the modifier is retained, no shift from one word-class to another is needed, nor does the syntactic transposition pose major restrictions on the complexity of the modifying noun phrase. The Czech preposition renders the relationship between the head and the modifier more explicit than the juxtaposition in English.

Another feature typical of English (in comparison with Czech) is the abundance of non-finite verb forms. Participles, gerunds and infinitives may act as counterparts of either the postmodifying PP or of the head of the noun phrase. The postmodifying function is typically performed by semantically weak verbs in the present or past participial form, which easily recategorize into secondary prepositions. In this respect, the participial postmodifiers resemble the Czech primary prepositions – both serve as elements that link the head noun to the noun phrase in postmodification, conveying quite broad meanings, such as composition, material, or accompanying circumstance. Czech, as a predominantly synthetic language, generally relies heavily on verb inflection. Participial verb forms are therefore rare. The construction structurally closest to English postmodifying participles is the Czech deverbal adjective, which has to agree with the head noun in gender, number and case, expressed by the adjectival suffix. The Czech postmodifying PP, requiring no formal concord with the head noun, therefore appears to be a more felicitous solution here as long as the simple primary preposition is capable of capturing the relationship between the nouns indicated by the English participle.

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The use of the modifying or determinative genitive as the counterpart of the Czech postmodifying PP highlights the differences in the structure of the noun phrase between the two languages. At the same time, in the inflectionally poor English, the genitive is capable of expressing a relatively broad range of meanings, and it combines with both animate and inanimate nouns. In Czech, the corresponding semantic relations between the head noun and the modifier may be expressed more explicitly by the preposition.

The choice of a particular type of counterpart was shown to impact on how explicitly the semantic relationship between the head noun and the modifier is expressed. Premodification by a noun, with syntactic relations indicated merely by word-order, leaves the semantic relationship implicit in English. The modifying function of denominal adjectives and participial premodifiers is signalled, apart from the position, morphologically, but there is no explicit indicator of the semantic relation between the modifier and the head. In postmodifying clauses, it is often the semantically weak (non-finite) verb that serves as a more explicit equivalent of the Czech preposition, linking the head noun to another noun phrase and expressing the semantic relation between them. As shown in Section 3.2, the universal trend towards increased explicitness in translation (cf., e.g., Bernardini & Russo 2018), seems to assert itself in both directions of translation. In the English > Czech direction, the Czech preposition marks the semantic relation between the head noun and the modifying noun phrase explicitly, compared to the most frequent English source construction, premodification by the noun juxtaposed to the head. In the opposite, Czech > English direction, the translators were found to rely most often on English postmodifying clauses, whose predicates indicate the semantic relation between the head noun and the modifying noun phrase more explicitly than the corresponding Czech preposition.

The above observations may suggest some reasons for the divergent correspondences. They are, however, tendencies rather than rigorously observed rules. As examples (11) and (12) below illustrate, the choice between the prepositional and non-prepositional English counterparts of the Czech postmodifying PPs may be context-dependent, leaving the decision open for the translator.

(11) CS. Na druhé straně chodby stály ve výklencích sochy *z bílého mramoru* [statues of white marble]; […] (Michal Ajvaz, Zlatý věk, fiction)
en. In alcoves on the other side of the corridor there are statues *of white marble*; […] (transl. by Andrew Oakland)

(12) CS. […] přesně na tom místě […] stál teď pomník *z černého mramoru* [a headstone of black marble] s docela jiným pozlaceným jménem. (Milan Kundera, Směšné lásky, fiction)

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> en. […] precisely on that spot […] now stood *a black marble* headstone with a quite different name in gilt. (transl. by Suzanne Rappaport)

Even though the present chapter focussed on a minor type of translation correspondence between Czech and English, it is hoped to have catalogued the recurrent patterns and suggested some motivation for the choice of a particular counterpart. To give a comprehensive picture, however, the correspondences between English PPs and their Czech translations should also be explored in detail both in cs(EN) and CS(en). More research is needed to situate the results within the broader area of translation correspondences between Czech and English at the level of the noun phrase in fiction and non-fiction.

Drawing on parallel translation corpora, this chapter also highlights the importance of sufficient reliable bi-directional data to explore infrequent phenomena.

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Chapter 6

Using diachronic corpora to translate emerging or new specialized fields

Cuchet

This chapter discusses the use of short diachronic corpora to provide translators with conceptual and terminological information considered useful when translating documents dealing with emerging or new medical specialties. While the link between specialized translation and corpora is now well established, the use of diachronic corpora, and in particular short diachronic corpora (i.e. covering only a few years) for translation purposes remains largely uncharted territory. The work presented in this chapter is based on the assumption that the use of diachronic information will provide translators with the cognitive and terminological information needed to understand the conceptual organization of complex specialized fields, particularly when it comes to reporting scientific advances or innovations through translation. The chapter is based on the study of a specialized corpus, covering a short period of time (24 years, from 1993 to 2017), in the field of nanomedicine applied to cancer treatment. This corpus-based study serves as a starting point for assessing which types of diachronic information are most useful and which are not from a translation point of view. The final part of the chapter will be devoted to a discussion of the advantages and limitations of such an approach and how it can be implemented in the context of translator training.

1 Diachronic corpora and specialized translation: A new path to be invented

The link between corpus linguistics and specialized translation has been extensively documented both in empirical research and in professional practice, and the value of using corpora for specialized translation is well established. Over

Cuchet. 2026. Using diachronic corpora to translate emerging or new specialized fields. In Tomáš Svoboda, Olga Nádvorníková & Věra Kloudová (eds.), *Advancing interdisciplinarity in empirical translation research*, 145–167. Berlin: Language Science Press. DOI: ??

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the last twenty years or so, numerous studies (including those published by Aston 1999; Baker 1999; Varantola 2002; Kübler & Aston 2010; Kübler 2011 or Loock 2016) have demonstrated the benefits of using electronic corpora to improve the quality of translations from all points of view. Despite the difficulties associated with creating and exploring ad hoc corpora for translators working under constant time pressure, the use of corpora nonetheless enables translators to make more accurate choices of translation equivalents, to adapt the phraseology they use and to choose the appropriate syntax and style. Other studies have also highlighted the benefits of using electronic corpora for translator training (Bowker 2003; Rodrígues-Inés & Hurtado Albir 2012; or Kübler et al. 2018). For Bordet:

La réflexion sur l'interaction entre constitution de corpus et décision de traduction est par conséquent un terrain de recherche crucial pour la formation de traducteurs aptes à développer et à défendre le métier de traducteur spécialisé.[1] (Bordet 2014: 13)

However, in the vast majority of cases, the possible benefits of using translation corpora have only been explored from a synchronic perspective; very few studies consider the possibility of using diachronic corpora to help translators better understand highly specialized concepts and domains. Only two articles, published in 1999 and 2006 respectively, explore the use of diachronic information in the translation of medical texts and assess the benefits that translators can derive from the diachronic perspective. In the first article, Vandaele (1999) carried out a diachronic study of the meaning of the English terms 'hypnosis' and 'narcosis' in the field of pharmacology, and showed how the very complex semantic evolution that characterizes these two terms is a source of numerous translation errors in French. In the introduction to her article, she notes that:

Il faut parfois, pour comprendre, retracer l'histoire d'un domaine et s'apercevoir que derrière l'imprécision terminologique se cachent problématiques de recherche, controverses, évolution de connaissances.[2] (Vandaele 1999: 315)

Gentile (2006), a professional translator and a teacher, also stresses, in the context of translating specialized texts in the field of psychoanalysis, that the diachronic dimension is an essential vector of knowledge for the translator:

Le point de vue diachronique est essentiel […] non seulement pour la recherche de l'équivalent en traduction, mais pour la compréhension des concepts d'une

[1]Reflecting on the interaction between corpus building and translation decisions is therefore a crucial area of research for training translators capable of developing and defending the profession of specialized translator. (Bordet 2014: 13)

[2]In order to understand, it is sometimes necessary to retrace the history of a field and realize that behind the imprecision of terminology lie research problems, controversies and the evolution of knowledge. (Vandaele 1999: 315)

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discipline.[3] (Gentile 2006: 128)

Other research works show the value of using diachronic corpora for pedagogical purposes, but they are mainly concerned with the teaching of language history: this is the case, for example, of the various contributions that make up the work by Kawagachi et al. (2011), which emphazises the use of electronic corpora to raise students' awareness of the grammatical, lexicological and syntactic features of ancient language states. In the context of learning the history of English, Heremans & Cuyckens (2012) have developed a series of exercises (together these exercises form a working method called 'Diachronex') using large diachronic corpora of general English, in order to gradually familiarize students with the syntactic, grammatical, lexical but also semantic changes that have occurred in English over time. However, no research work has yet investigated the benefits of using corpus-based diachronic information for translating highly specialized texts and documents.

Moreover, the research work cited above is overwhelmingly based on the observation of linguistic phenomena occurring over a long period of time which might be described as 'historical' (spanning many centuries). This chapter, on the other hand, builds on Picton's work (2011) on diachronic terminology and explores the possibility of extracting useful information from changes that occur over a short period of time (referred to as 'short diachrony'), i.e. over just a few decades, a methodological choice that allows for the minute understanding and description of changes linked to scientific progress and innovation. These changes which are linked to the emergence of new knowledge in a field can be a source of difficulty for translators, especially when they occur in highly scientific fields, with highly technical terminology, such as medicine.

1.1 Medical translation, an ideal field of observation for the diachronic dimension

Medical translation is a subject of particular interest, having given rise to a substantial corpus of research literature, in which reference is often made to its 'specificity' (for example Popineau 2016: 78). For Lee-Jahnke (2005), medical translation represents a distinctive exercise given that it conveys scientific content directly related to human beings, and because, historically, the medical field was one of the very first fields in which texts were translated:

Medicine has always held a special position because it deals with human beings, i.e., with persons – their personality, their body and their life. This is prob-

[3] The diachronic point of view is essential [...] not only for finding the equivalent in translation, but also for understanding the concepts of a discipline. (Gentile 2006: 128)

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ably the reason why medical translation belongs to one of the oldest fields of translation. (Lee-Jahnke 2005: 81)

Lee-Jahnke also claims that the large number of texts and documents from diverse medical specialties that require translation means that this area occupies a major place in the professional translation market:

Medical translation has always been of major importance within the field of translation, because a large number of texts are translated and hence this area represents a large share of the market. (Lee-Jahnke 2005: 81)

For Jammal (1999), the specificity of medical translation is rather due to the complexity and technicality of the concepts described:

Pour notre part, nous croyons que la traduction médicale diffère des autres champs de traduction sous certains aspects. Ainsi, prenons la phase de décodage. [...] Il nous semble que, de tous les domaines, le domaine médical est peut-être celui qui est le plus difficile à appréhender. On conviendra, par exemple, qu'il est plus facile de comprendre les tenants et aboutissants du traité de Maastricht, ou la relation existant entre l'inflation et le chômage, que ce qui se passe au cours de la biosynthèse du cholestérol et des troubles qui peuvent causer l'hypercholestérolémie[4]. (Jammal 1999: 217)

The complexity of medical concepts is also accompanied by a well-known and well-documented lexical complexity, with the presence of a large number of terms based on Greek and/or Latin formants (e.g. 'column chromatography', 'cytotoxicity', etc.) for instance, and the frequent use of eponymous terms. For Rouleau (2003), there are also the numerous cases of synonymy (e.g. 'motor apraxia' and 'kinetic apraxia', in the field of motor disorders), and the many obstacles posed by what he calls affixal *faux amis* (e.g. medical adjectives ending in -al in English, which do not always translate into an -al/-ale adjective in French, such as 'dental' / 'dentaire' or 'carpal' / 'carpien').

The most frequently described difficulties in medical translation also include the different types of variation that occur in medical texts, whether morphological ('lipocomplex' / 'lipoplex'), graphic ('nanovesicle' / 'nano-vesicle') or lexical ('small interference RNA' / 'small interfering RNA').

At the lexical level, translation pitfalls also arise from the large number of technical, often compound terms, that may be unfamiliar to the translator ('su-

[4]For our part, we believe that medical translation differs from other fields of translation in certain respects. Let's take the decoding phase. [...] It seems to us that, of all fields, the medical field is perhaps the most difficult to understand. We would agree, for example, that it is easier to understand the ins and outs of the Maastricht Treaty, or the relationship between inflation and unemployment, than what happens during the biosynthesis of cholesterol and the disorders that can cause hypercholesterolaemia. (Jammal 1999: 217)

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percritically dried insulin-loaded microparticles', 'immunotherapy based on immune checkpoint blockade', 'CTL-mediated apoptosis', etc.). In the medical field, Maniez has studied complex noun groups (in particular adjectival nouns in 2001 and nominal premodification in 2007), and collocations (2002). In his 2007 article, Maniez stresses the need to focus on complex lexical patterns in translator training, as this type of structure is very common in medical English (2007: 09), and that decoding them frequently presents challenges for the translator. Using a corpus of medical English in cardiology, he shows that complex lexical patterns (such as adjective + noun 1 + noun 2) can pose decoding problems, as is the case with the terms 'atrial septal defect' (a defect of the atrial septum), or 'jugular venous pressure' (pressure of the jugular veins). The potential for misunderstanding is heightened in the case of complex coordinated structures such as in the term 'increased nucleic acid and protein synthesis and glycolysis', since the combination of nominal premodification and coordination can be very ambiguous for the translator to understand (does 'nucleic' qualify 'acid', both 'acid' and 'protein', or 'synthesis'?)

Finally, the frequent use of non-transparent, technical acronyms by experts, as shown in the context below extracted from the corpus, represents another salient feature of the medical language[5] and is among one of the most challenging for translators to navigate:

(1) EMA has been required the full physicochemical characterization data of NMc in terms of zeta potential, particle size, size distribution, spectroscopic investigation (NMR, FTIR, UV, LC-MS), degradation products, impurity profiling, and solid-state characterization by DSC, DTA, TGA, solid state NMR and XRD. [2015][6]

1.2 Translating innovation: An additional difficulty

In addition to the obstacles mentioned above, medical translators face specific difficulties when translating texts that contain new concepts or deal with emerging specialties.

Following Kuhn (1970), we consider that innovative or emerging specialties refer to previously unknown scientific fields (such as xenotransplantation in

[5] A number of medical experts point out that the abundance of non-transparent acronyms in the medical field creates major communication problems, including for experts in the field. For example, in the editorial of the journal *Diagnostic and Interventional Imaging*, Soyer (218: 590) claims that "too many acronyms/initialisms make the reading difficult and annoying".

[6] Contexts are followed, in square brackets, by the year from which they are taken in the corpus.

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medicine, for example) or to fields that already exist, but which are undergoing a significant period of scientific progress and innovation, such as in pharmacology with the development of nanodrugs.

Our assumption here is that translating the terminology of emerging specialties poses a number of specific problems because the translator has to deal with terms that may not be fully stabilized in the lexicon, and that may evolve in a very short period of time as the new concept they denote is discussed, clarified or negotiated by experts (a phenomenon of instability already described by Guilbert, for aviation, in 1965). In addition, new specialties that emerge are often characterized by accelerated hybridization, which means that scientists increasingly use concepts, practices and tools from a wide range of sometimes very different fields (for example, in the medical field, the new concept of augmented medicine is based on robotics, artificial intelligence and 3D printing).

The development of nanomedicine for cancer treatment also reflects this growing hybridization of medicine and science in general, as nanodrugs have been developed using engineering techniques, indicating a shift away from the more traditional fields with which pharmacology was previously associated, such as chemistry and anatomy. A comparison of the keywords extracted from the earliest articles in the corpus (1993–1995) with those extracted from the most recent articles (2015–2017) clearly reflects this estrangement from pharmacology's traditional scientific base. Terms from chemistry such as 'sodium carboxymethylcellulose' or 'neoglycoproteins', for instance, which were prevalently used by authors in the first part of the corpus, are no longer present in the most recent articles. Instead, terms such as 'regenerative engineering' or 'Interpenetrative Network, IPN' have emerged.

This hybridization can lead to increased complexity for translators, who have to be able to understand and translate scientific terminology from a diverse range of disciplines. They are seldom helped in this by the lexicographical resources and terminology databases available online, which, apart from the fact that there are few of them dealing with such highly specialized subjects, are also often ill-suited to the task of translating novelty: the information they provide is often not updated quickly enough to accurately reflect the emergence of new terms, forcing translators to turn to other types of resources.

We therefore believe that diachronic corpora and the information they contain represent an alternative source of information for translators, and that they are not only likely to provide useful information about new terms, but can also be a means of providing them with minimal cognitive knowledge that is particularly useful for understanding how and why terms are formed in the context of scientific emergence.

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2 The use of corpus-based diachronic information in translation

2.1 Providing translators with minimal cognitive knowledge

Various methodological approaches have been proposed to train students to deal with medical translation issues. These approaches can be based directly on the translation task itself: Popineau (2016), for example, inspired by Nord's functionalist theory, uses the central notion of functions to help translation students. The translation task is divided into several functions (referential function, expressive function, etc.), each of which represents a precise stage that the translator must complete when moving from one language to another.

Other authors suggest intervening upstream of the translation task, in particular through targeted documentary research, the aim of which is to provide translators with a conceptual knowledge base that they can continue to enrich as the translation progresses. Jammal (1999) for instance, proposes a documentary research method based on what he calls a 'sequential order' which for the translator consists in searching for information according to a medical logic. For example, in the context of translating documents relating to a particular pathology, it would be appropriate to look first for information on the anatomy or part of the anatomy concerned, then on the physiology of the cells, organ or organs affected, then on the clinical manifestations of this pathology, then on the pharmacological aspect of possible treatments, etc. This method, which enables the trainee translator to acquire essential knowledge that he or she will later use in translating, does however require a prior knowledge of medical logic and the basic principles that underpin it in terms of semiology, cell biology and pharmacology, which perhaps not all medical translators may possess.

Vandaele (2001), aware of the conceptual complexity that characterizes the medical field and of the difficulties that translators may encounter in grasping this conceptual complexity, goes a little further upstream from the documentary stage and suggests translators be provided with a minimum of cognitive knowledge that will enable them to carry out an appropriate documentary search and to target the selected information more effectively. This approach is based on the observation that:

> Un spécialiste, pour mettre à jour ses connaissances [...] rattache les nouveaux concepts à un bagage cognitif déjà organisé. Il en est de même, d'ailleurs, pour un traducteur d'expérience spécialisé dans un domaine particulier. Qu'en est-il lorsqu'un apprenti-traducteur ou un traducteur

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> a devant lui un texte traitant d'un domaine de connaissance qui lui est étranger ? [...] Faute d'*ancrage* à un bagage cognitif minimal préexistant, le sujet éprouve la plus grande difficulté à saisir le fil logique qui pourrait lui permettre de répondre efficacement aux questions qu'il se pose sur le texte.[7] (Vandaele 2001: 16–17)

In the same vein, Sager (1990) claims that it is

> important to realise that terms can only be used if the user already possesses the configuration of knowledge which determines the role of the term in a structured system. (Sager 1990: 75)

Providing the translator with a minimal configuration of knowledge seems to us particularly relevant in the context of scientific progress and emergence, and in increasingly hybrid scientific fields such as nanotechnologies applied to cancer treatment.

2.2 'Knowledge-rich' diachronic information

Our research assumption, that knowledge-rich diachronic information provides translators with the cognitive knowledge and information needed to translate emerging medical fields was tested using a diachronic corpus, covering a short period of time (24 years, between 1993 and 2017).[8] The corpus of approximately 1,650,000 words contains scientific articles, book chapters and doctoral theses written in English by experts in the field of nanomedicine. Over the 24 years of the corpus, research published by 256 different authors has been collected.

The corpus was analyzed using two free software packages, Termostat web 3.0[9] (an automatic term acquisition tool) and Lancsbox[10] (a new-generation software package for analysing linguistic and corpus data). The analyses carried out

[7] Specialists, in order to update their knowledge [...] attach new concepts to an already organized cognitive baggage. The same applies, moreover, to an experienced translator specializing in a particular field. But what happens when a translator or an apprentice translator is faced with a text dealing with an unfamiliar area of knowledge? [...] Without being anchored to a pre-existing minimal cognitive baggage, the subject finds it extremely difficult to grasp the logical thread that might enable him or her to effectively answer the questions he or she has about the text. (Vandaele 2001: 16–17)

[8] The results presented in this chapter are all drawn from a unilingual corpus of medical English, in the field of nanomedicine. A diachronic corpus of comparable size and nature in medical French has also been compiled for a larger research project, but it is not used in this chapter.

[9] http://termostat.ling.umontreal.ca

[10] http://corpora.lancs.ac.uk/lancsbox

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Table 1: Number of texts and words per chronologic segment.

Time period	Number of texts	Number of words
1993–1994	20	127,154
1995–1997	25	144,540
1998–2000	27	171,549
2001–2003	24	131,816
2004–2006	33	213,163
2007–2009	24	131,695
2010–2012	33	211,102
2013–2015	35	236,690
2016–2017	35	285,947
total	476	1,653,656

on the corpus were deliberately kept simple (based on statistical comparisons, extraction of contexts and definitions, study of changes in term constructions, etc.) in order to be easily reproducible by translators, who very often do not have the time or training to carry out complex and technical corpus-based studies.

Our exploration of the corpus focused on the extraction of diachronic information, which is likely to provide translators with elements of knowledge about new terms, as well as about the evolution of emerging concepts, thus equipping them with a minimal cognitive knowledge, as mentioned by Vandaele (2001).

We will refer to this useful diachronic information as 'knowledge-rich' information, using and adapting the concept of 'knowledge-rich contexts' described first by Meier in 2001 and then by Barrière in 2004. Knowledge-rich contexts are defined as sets of sentences that, more than others, contain knowledge-bearing information that can be organized in databases:

In large text corpora, Knowledge-Rich Contexts (KRCs) are a subset of sentences containing information that would be useful to a human for building a knowledge base. (Barrière 2004: 188)

Knowledge-rich information is therefore considered to contain diachronic information that is useful to the translator for understanding and translating emerging specialties.

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3 Benefits derived from the diachronic perspective

A certain amount of lexical information, when studied in the light of diachrony, is likely to constitute knowledge-rich information for the translator. This includes the study of term variations, of changes in the construction of complex terms, and the study of terminological and metaphorical families. Knowledge-rich information is also provided through the study of terms that are part of generational renewal.

3.1 Term variation

The first type of knowledge-rich diachronic information that we have extracted from the corpus relates to the variation in the form of terms over time. Indeed, describing these different variations in the corpus is a useful way of showing the translator how the new terms in the domain have been formed. In our case, corpus exploration provides valuable information about the form of some of the terms ending in -plex, extracted from the later part of the corpus (2010 and later). For example, the data (see table below) show that the terms 'polycomplex', 'lipocomplex' and 'nanocomplex' have been reduced over time to 'polyplex', 'lipoplex' and 'nanoplex'. These observations are consistent with Humbley's study (2008) of specialized terms in the field of e-commerce, in which he shows that some new terms are rapidly truncated either as initialisms or as blends.

Table 2: Changes in observed frequencies in two time periods (P1 and P2) of the corpus for three terms ending in '-plex/es'.

Sub-corpus	P1 (2001–2009)		P2 (2010–2017)				
size (No of words)	476,674		733,739				
Frequency	abs.fq	ipm	abs.fq	ipm	DIN	RR	95% CI
Polycomplex/es	521	1,092.99	85	115.85	-80.83	2.18	2.115–2.256
Polyplex/es	106	222.37	590	804.10	56.67	0.39	0.324–0.461
Lipocomplex/es	464	973.41	22	29.98	-94.02	2.43	2.379–2.473
Lipoplex/es	3	6.29	304	414.32	97.01	0.03	0.008–0.076
Nanocomplex/es	188	394.40	5	6.81	-96.60	2.47	2.418–2.532
Nanoplex/es	0	0,00	431	587.40	100	0	0.0–NaN

DIN = difference index (see Fidler & Cvrček 2015), RR = risk ratio, CI = confidence interval at $p < 0.05$.

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The data extracted from the corpus also shows a certain amount of synonymic variation around new terms that emerge in the corpus. For example, the acronym 'siRNA' may lack transparency for translators, so studying the variations of its extended form over the 24 years of the corpus can help translators to better understand the term and how it was formed. Depending on the author, the term 'siRNA' in the first part of the corpus corresponds to either 'small interfering RNA' or 'small interference RNA', as shown in the extracts below.

(2) Using the amine groups of CMCS, polyethylene glycol (PEG) residues were coupled to form a protective structure that entrapped a CaP core containing *small interfering (si)RNAs.* [2008]

(3) Especially in the field of DNA-based and *small interference RNA (siRNA)*-based therapeutic attempts such as gene therapy, viral vectors were considered the ideal targeting shell. [2010]

(4) In several studies, *small interfering RNA (siRNA)* has been shown to allow specific gene silencing in human cells. [2011]

However, over time, and especially since 2010 in the corpus, the extended form 'small interfering RNA', which may seem less explicit and transparent from a scientific point of view than 'small interference RNA' has become the dominant term used by experts, as Table 3 shows.

Table 3: Changes in observed frequencies for the two variants 'small-interference RNA' and 'small-interfering RNA'.

Sub-corpus	P1 (2001–2009)		P2 (2010–2017)				
No of words	476,674		733,739				
	abs.fq	ipm	abs.fq	ipm	DIN	RR	CI
Small-interference RNA	231	484.61	35	47.70	-82.08	2.21	2.105–2.312
Small-interfering RNA	102	213.98	519	707.34	53.55	0.42	0.349–0.498

DIN = difference index (see Fidler & Cvrček 2015), RR = risk ratio,
CI = confidence interval at $p < 0.05$.

3.2 Premodified constructions

Another point of interest is the observation of changes in the construction of some key terms in the corpus. The number of doubly premodifed constructions

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such as '-loaded', '-based' and '-laden' increases in the data from 2009 onwards (with a few sporadic occurrences, however, between 2006 and 2009) in connection with terms such as 'liposome' or 'gel':

(5) For more widely therapeutic application of these *protein-loaded gels*, injectable dosage or new pharmaceutical forms a required. [2009]

(6) The purpose of this study is to evaluate the retinal toxic response induced by *ganciclovir (GCV)-loaded gel* in Wistar rats' eyes. [2013]

(7) The mechanical and rheological properties of *poloxamer-based gels* are closely correlated and directly affect the gel retention time within the eye. [2015]

(8) In vitro studies of that preparation exhibited that *Span 60-based gels* can improve the ocular bioavailability of diclofenac sodium for the prolonged period of time and there were no ocular irritation effects. [2013]

(9) These experiments were performed to determine the equilibrium release time for the *microemulsion-laden gels*. [2011]

(10) Results also show that the transport of CyA through the *surfactant-laden gels* is controlled by diffusion through the gel. [2012]

The total number of doubly premodified terms increases sharply in the corpus, with 19 occurrences of such constructions in the period 1993–2000 and 893 in the period 2010–2017.

Table 4: Changes in observed frequencies for doubly premodified terms, between the very first and the very last time period of the corpus.

Sub-corpus No of words	P1 (1993–2000) 443,243		P2 (2010–2017) 733,739				
	abs.fq	ipm	abs.fq	ipm	DIN	RR	95% CI
Doubly premodified terms	19	42.87	893	1,217.05	93.20	0.06	0.035–0.086

DIN = difference index (see Fidler & Cvrček 2015), RR = risk ratio, CI = confidence interval at $p < 0.05$.

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It is important that translators are made aware of the increase in this type of premodified complex matrices, as they are difficult to understand and translate. The use of diachronic information here is useful in informing translators about the progress that has been achieved in the field over time: gels in the corpus data appear to have moved from simple, often bioadhesive gels to more complex, drug-loaded, more effective gel formulations. Similarly, liposomes have moved from simple bilayer liposomes to substance-loaded or coated liposomes in the space of 24 years.

(11) *Stable liposomes* were rendered pH-sensitive by complexation to a polymer that undergoes marked temperature-and pH-dependent water solubility changes. [1993]

(12) TBS buffered solutions with 4, 2, 1 and 0.5 mg ml^{-1} empty *bilayer liposomes*, 3.5 μm diameter, and 35, 70, 150, 200, 250 ppm CHT were prepared. [1995]

(13) *Biopolymer-coated liposomes* (3 mM and 6 mM) were prepared by the thin film method and hydrated with phosphate buffer (PB) or glycerol phosphate buffer (G-PB). [2016]

(14) *Chondroitin sulfate (CS)-stabilized liposomes* are effective carriers of kartogenin. [2017]

(15) We investigated the interactions of empty liposomes and *KGN-loaded liposomes* with both CS derivatives using various physicochemical techniques. [2017]

The contexts extracted from the corpus show that the increasing pre-modification of the term 'liposome' over the course of 24 years is also accompanied by an increase in the use of the adjectives 'conventional', 'traditional' and 'classic' in contexts where the two generations of liposomes are compared: liposomes become 'conventional' or 'classic' when they are replaced by the next generation.

(16) Thus a single or several breaches of lipid membranes will not lead to a total drug release from the internal aqueous contents, which is not like *conventional liposomes*. [2007]

(17) Therefore, results of these experiments pointed out a higher deformability of PEVs than the *conventional liposomes* as shown by Table 3 where deformability index (DI) for the studied formulations is also reported. [2010]

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(18) For instance, fluorescently-labeled bacitracin, a polypeptide antibiotic, was delivered from ethosomes to a depth of 200 μm in dermatomed human cadaver skin, whereas its delivery from *classic liposomes* was negligible [2013]

The analysis of the changes that occur in complex noun groups, especially in premodified groups, therefore fits perfectly into the network of diachronic information described in this work, which is likely to be useful to the translator.

3.3 Terminological families

Observing terminological families (e.g. families of specialized terms which share the same form) and how they evolve over time can prove useful in increasing the translator's cognitive knowledge and provide important elements of understanding of the field.

For example, translators may be surprized to find that in texts on nanomedicine, some of the terms they need to translate are actually based on the prefix 'micro', which refers to a completely different scale and a completely different generation of drugs as explained in the context below: nanodrugs are part of a new generation of drugs that are smaller than one nanometer in size and represent a scientific breakthrough compared to the microscopic formulations used previously.

(19) Three distinct but overlapping eras of increasing complexity can be defined: The first one, starting as early as in the mid-1960s, was termed the MACRO era where polymers were used to form macroscopic devices such as slabs, films, rods, rings or spheres etc. to accommodate drugs for controlled release. […] A second era in the history of drug delivery, beginning in the early 1970s, is the MICRO era, closely linked to the use of biodegradable polymers, e.g., to manufacture drug loaded microparticles. […] The NANO era, finally, is the third and currently still the most dynamic era within the development of polymeric delivery systems. [2010]

From a linguistic point of view, we would therefore expect to find more terms based on 'nano' in the most recent part of the corpus. In fact, if the frequency of the prefix 'nano' indeed increases in the corpus data, as shown in the Table 5, the number of terms based on the prefix 'micro' does not decrease, which is much more unexpected.

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Table 5: Changes in observed frequencies for terms based on *micro* and *nano*, with or without hyphen, between the very first and the very last time period of the corpus

Sub-corpus	P1 (1993–2000)		P2 (2010–2017)				
No of words	443,243		733,739				
	abs.fq	ipm	abs.fq	ipm	DIN	RR	95% CI
micro	516	1,164.15	2,108	2,872.96	42.33	0.52	0.483–0.564
nano	168	379.02	3,722	5,072.65	86.10	0.11	0.099–0.133

DIN = difference index (see Fidler & Cvrček 2015), RR = risk ratio, CI = confidence interval at $p < 0.05$.

Taking a closer look at the contexts extracted from the corpus we can see that a new family of terms, based on 'micro', appears in the last part of the data, together with the new concept of regenerative medicine.[11]

(20) [...] the Tian group from Boston University recently investigated how physical and chemical signals affect vascularization of directly seeded capillary-scale channels to apply those signals to create *microvessels* that mimic human capillaries in size and function. [2015–2017]

(21) Such inflammatory *microenvironments* can drastically limit the integration of biomaterials with surrounding native tissue. [2015–2017]

This new field of regenerative medicine implies that micro-assemblages or ensembles are built in micro-environments. This is the reason why the number of occurrences of terms in 'micro' increases, rather than decreases in the second part of the corpus. In fact, it is a new conceptual and terminological family of terms that is based on 'micro'. The diachronic perspective could therefore help the translator to understand that there are in fact two families of terms based on 'micro' in the corpus; a first generation of terms describing drugs that are gradually being replaced by nano-sized formulations, and a second generation of terms which are newly created in relation to the emerging concept of regenerative medicine.

[11] According to Altyar et al., "With the evolution of interdisciplinary medical sciences in the last decades, a new branch of medicine occurred, namely regenerative medicine (RM). It applies engineering and life sciences principles to heal or replace damaged tissues, promote organ renewal, restore lost functions, and continually bridge permanently emerging damages". (Altyar et al. 2023: 114)

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3.4 Metaphorical families

Another type of families, metaphorical families, are worth studying from a diachronic perspective, as it is evident in our corpus that new terms tend to be organized into sets of well-identified metaphors. As Fries (2016) points out in her book, the terms describing nanotechnologies in their various application areas often have a metaphorical basis and most of them, more specifically, are organized around the metaphor of civil engineering ('quantum well', 'tunneling effect').

The data extracted from the second half of our corpus also confirms the emergence of two major metaphorical families organized around the notions of 'network' and 'complex assembly'.

Table 6: A first metaphorical family organized around the notion of 'network'.

Network
IPN (interpenetrating network)
microfluidic network
functional vessel network

Table 7: A second metaphorical family organized around the notion of 'complex assembly'.

Complex assembly
siRNA transfection complex plasmid DNA polyplex
DNA duplex
self-limiting platform
fibrin scaffold
tissue-like architecture

This type of diachronic information could help the translator to understand that it is useful to identify the appearance of these sets of new metaphorical terms, in order to translate them as part of a coherent whole rather than as isolated elements.

3.5 Terminological generations

Finally, to conclude this section on the benefits of using knowledge-rich diachronic information for translation, we would like to show that studying a

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corpus over time also makes it possible to identify generational families, i.e. families of terms that are specific to one scientific stage and that are replaced by others as experts move on to another stage. This is the case for the terms based on 'nano' and 'micro' already mentioned, but it is also the case for other terms whose replacement in the lexicon is much more rapid, sometimes within the space of a year, which can be a source of difficulty for the translator.

For instance, as shown in the table below, the term 'solid lipid nanoparticle' is gradually replaced in the corpus, by the term 'nanostructured lipid carrier', which refers to a more efficient and more complex generation of lipid carriers. Similarly, the first 'carbon nanotube' which is mentioned for the first time in the 2005–2008 period of the corpus, is replaced by a new generation of 'single-walled carbon nanotubes' (2009–2012), then by another new generation of 'multi-walled carbon nanotubes' (2013–2017), and finally by the latest generation of 'f-engineered carbon nanotubes', mentioned in the last year of the corpus.

Table 8: Two examples of generational changes.

	1993–2000	2001–2004	2005–2008	2009–2012	2013–2017
solid lipid nanoparticle	344	547	512	275	31
nanostructured lipid carrier			52	425	1,502
carbon nanotube			7	57	43
single-walled carbon nanotube				5	67
multi-walled carbon nanotube					38
Year 2017: f-engineered carbon nanotube					12

While some of the results extracted from the corpus form part of a knowledge-rich information network, other diachronic results seem to be less useful for the translator, either because they do not really provide 'valuable' terminological and/or cognitive information, or because they do not allow for an adequate solution to some of the difficulties that are typical of medical translation, such as for instance spelling variations, or the presence of opaque acronyms.

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4 Limitations of the diachronic perspective

4.1 Definitions

Definitions fall into this category of diachronic information that is not essential for specialized translation. The way in which the structure of the definitions evolves over time in the corpus is very interesting, because, as can be seen from the examples below, most of them are based on a comparison between what is scientifically new and what is considered 'old', thus showing the transition from one generation of particles to another.

(22) Suitable small particle systems include microspheres and microcapsules as well as nanoparticles and nanocapsules. *Microparticles and microcapsules have a particle size above 1 µm.* Microspheres are monolithic particles possessing a porous or solid polymer matrix, whereas microcapsules consist of a polymeric membrane surrounding a solid or liquid drug reservoir. *Nanoparticles have a particle size in the nanometer size range below 1 µm [4].* [1995–1996]

1. This information, although very interesting, does not seem to be sufficiently valuable or 'knowledge-rich' for the translator. Moreover, it would clearly take too much time for the translator to extract a relevant number of definitions from different chronological periods of the corpus and to read them in detail in order to find useful information about the succession of generations in the field.

4.2 Spelling variations

Similarly, no diachronic information extracted from the corpus seems to be sufficiently useful to help the translator understand and translate appropriately the spelling variations of terms constructed with 'nano' (for example, the term 'nanovesicle' can also be spelled 'nano-vesicle' or 'nano vesicle'), as the three variants are used very irregularly in the corpus data and do not seem to follow any clear trend over time.

4.3 Acronyms

The very irregular use of acronyms over time in the corpus is not easy to organise into useful information for the translator; the only real information that can be extracted from the data is that, in the second part of the corpus, acronyms are

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Table 9: Changes in spelling variation of the term 'nanovesicle' over three time periods of the corpus.

Sub-corpus	P1 (2001–2003)		P2 (2007–2009)			P3 (2013–2017)				
No of words	131,816		131,695			522,637				
	abs.fq	ipm	abs.fq	ipm	DIN (P1-P2)	abs.fq	ipm	DIN (P2-P3)	RR	95% CI
Nano vesicle	272	2,063.48	362	2,748.78	14.24	321	614.19	-63.47	2.64	2.485–2.831
Nano-vesicle	26	197.24	198	1,503.47	76.81	206	394.16	-58.46	2.44	2.206–2.692
Nano vesicle	89	675.18	61	463.19	-18.62	191	365.45	-11.80	1.20	0.967–1.497

DIN = difference index (see Fidler & Cvrček 2015), RR = risk ratio, CI = confidence interval at $p < 0.05$.

increasingly used in matrices where they are attached to other acronyms,[12] thus increasing the number of complex and opaque terms, that diachronic information fails to explain.

Again, the use of diachronic information is not sufficiently relevant to help the translator better understand the terminological specificities of a specialized field such as nanomedicine.

5 Future perspectives: The use of diachronic corpora in translator training

The aim of this chapter has been to highlight the use of diachrony in the translation of specialized fields that are subject to constant scientific progress, such as nanomedicine. We believe that the diachronic perspective offers a highly productive means of enabling professional translators to grasp newly emerging concepts and their terminology. By studying a specialized corpus spanning 24 years, we have been able to show that a certain amount of diachronic information is indeed likely to help translators better understand recent developments in a field. We have called this type of information 'knowledge-rich information'. In our corpus, knowledge-rich information can be found when looking at changes in the structure of complex terms over time (especially in doubly premodified terms, but also in terms based on -plex/plexes which are shortened over time). It can also be found in changes in families of terms and metaphors over time.

In our opinion, the advantages of using the diachronic dimension in translation are real, but the compilation and exploration of a diachronic corpus is a

[12] As this extract from the corpus shows: "P(MAA-g PEG-co MMA) underwent the largest size increase upon increasing pH, while P(MMA-g-PEG-co-TBMA) underwent the smallest size increase". [2017]

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time-consuming solution for a professional translator, even for a corpus covering a short period of time, as we have seen here, and even if the queries for exploring the corpus are kept simple.

A possible future perspective therefore is to integrate the diachronic perspective into translation training, especially at Master's level, when students specialize in translating much more technical documents that require minimal cognitive knowledge in order to be correctly understood and handled. To this end, we propose to enrich the research presented here by designing practical exercises (see Appendix to the chapter) following the work of Heremans & Cuyckens' (2012) work on Diachronex. The aim of these exercises would be to use corpus-based information to help trainee translators understand as quickly as possible how the field is organized, what the main concepts and terms are and how they have evolved over time.

A further advantage of such exercises would be to give trainee translators a more general knowledge of the dynamics of language evolution and to make them aware of the fact that specialized lexicon evolves, sometimes in a short or even a very short space of time when they go through periods of innovation. These exercises would thus contribute to the development of an integrated and more comprehensive approach to translation training.

Appendix A Appendix

Here are two examples of possible exercises based on the use of a pilot diachronic corpus designed for classroom use:

Exercise 1 – Finding evidence of changes in the corpus, for example by looking at changes in a list of frequently associated terms.

The terms below are listed in decreasing order for each time period, with the term at the top of the list being the term most frequently used in association with "release". How has the release of an active substance evolved over the last five years?

One answer, according to the following table, would be that the position of the adjectives 'controlled' and 'slow' changes over time, and that their downgrading towards the end of the list in the period 2009–2017 indicates that they are less used, unlike terms such as 'faster' or 'burst', whose position moves towards the top of the list, indicating that drug release systems have become faster, for example.

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1993–2000	2001–2008	2009–2017
drug	drug	drug
controlled	pulsatile	***burst***
slow	**controlled**	sustained
linear	sustained	**faster**
continuous	***burst***	localized
zero-order	on-off	**controlled**
in vitro	**slow**	triggered
pulsatile	transient	quick
immediate	in vitro	controllable
prolonged	**faster**	**slow**

Exercise 2 – Finding out information about changes in the structure of complex terms over time.

Identify changes in the structure of the term 'drug delivery system' in the following list of contexts, from oldest to most recent.

One answer might be that the structure of the terms becomes more complex over time, perhaps indicating that drug delivery systems become more complex over time. It is also worth noting that the full term is increasingly used in conjunction with its acronym and that in two recent examples, only the acronym is used to refer to drug delivery systems.

- A new technique for the production of this material, based on the foaming of suspensions and in situ polymerization (gelcasting method), resulted in a material whose characteristics are likely to make it useful as an *implantable drug delivery system.* [period 2000–2002]
- A conceptually novel *periodontal drug delivery system (DDS)* is described that is intended for treatment of microbial infections associated with periodontitis. [period 2000–2002]
- Chitin/PLGA blend microspheres as a *biodegradable drug-delivery system*: phase-separation, degradation and release behavior. [period 2000–2002]
- The global aim of this research project was to develop a *self-nanoemulsifying drug delivery system (SNEDDS)* for non-invasive delivery of protein drugs. [period 2008–2010]
- To overcome low oral bioavailability of poorly water-soluble drugs, *self-emulsifying drug delivery systems (SEDDS)* have been noted as a promising strategy. [period 2008–2010]

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- polypeptide-block-poly(vinylpyrrolidone) as prospective *self-assembled drug-delivery systems* have attracted growing interest because of their unique morphological properties. [period 2008–2010]
- *Supersaturatable self-emulsifying drug delivery systems (S-SEDDS*) were explored to improve the oral absorption of AMG 517, a poorly water-soluble drug candidate. [period 2008–2010]
- *Carboxymethyl cellulose (CMC)-loaded Co-Cu doped manganese ferrite drug delivery systems (DDSs)* as a new dual-modal simultaneous contrast agent for magnetic resonance imaging. [Period 2015–2017]
- The new *supramolecular organic frameworks (SOF) drug delivery systems (sof-DDSs)* can adsorb dianionic pemetrexed (PMX), a clinically used chemotherapeutic agent instantaneously upon dissolving in water, which is driven by both electrostatic attraction and hydrophobicity. [Period 2015–2017]
- Three *silica encapsulated lipid-based drug delivery systems (LBDDSs)* were developed and investigated: (a) *a self-nanoemulsifying drug delivery system (SNEDDS)*, (b) a solid *SNEDDS formulation*, and (c) silica–lipid hybrid (SLH) microparticles. [Period 2015–2017]
- To finish, the field of encapsulation in nanoparticles is broached in order to put forward recent advances in innovative *skin drug delivery systems (ISDDSs)* of antipsoriatic active agents for a better efficacy, safety and compliance. [Period 2015–2017]
- Ideally, a community of *biohybrid bacteria-based DDS* can be engineered that are wirelessly controlled, a strategy that holds promise for increasing the efficiency and reducing the operational cost for targeted drug delivery. [Period 2015–2017]
- Herein, we developed a novel formulation, *non-aqueous self-double-emulsifying DDS (SDEDDS)* by formulating mixtures of hydrophilic surfactants and oil-in-oil (O/O) emulsions, which could improve *trans*-resveratrol solubility through formulations optimization. [Period 2015–2017]

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Chapter 7

Giampieri

Plain English is language that can be understood by the layperson. Over the decades, scholars and linguists have advocated the drafting of plain English texts in the legal field (Tiersma 1999; Tiersma & Solan 2012; Williams 2023). Nonetheless, many legal documents are still drawn up in the most intricate form, which is referred to as legalese. Even contracts addressed to non-native speakers of English (such as employment agreements) tend to be drafted in an archaic style. In this respect, there seems to be a significant difference between employment agreements and employment offer letters. This chapter is aimed at bringing to the surface the nuances between these two types of contracts. In particular, it unveils whether legalese features are used in both types of documents and if their drafting style is formal or informal. To this purpose, two corpora are constructed and analyzed: one composed of employment agreements, and the other of employment offer letters. This analysis highlights recurrent phrases, formulaic expressions, and legal terminology in such documents.

The chapter findings highlight that there seems to be a more pervasive use of plain English language in employment offer letters, whereas employment contracts tend to be drafted in legalese. The chapter presents implications of corpus-based analyses in translation studies. More precisely, it shows how corpus-driven activities can be adopted in the translation process. Its rigorous methods and principles may, in fact, be replicated in other fields, thus fostering interdisciplinarity in translation research.

1 Introduction and research questions

This chapter is aimed at exploring the language of employment contracts and employment offer letters by unveiling differences in legal language and style. Exploring the linguistic similarities or differences in these two legal documents is relevant since they are recurrent in national and international contexts.

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This chapter also wishes to scrutinize if and how corpora can be useful to carry out intralingual translations (Jakobson 1959), i.e., whether they help translators produce texts which are plainer and/or more accessible.

The following questions are posed: 1) To what extent do these documents feature elements of *legalese* or plain English?; 2) Are employment contracts and employment offer letters addressed to different categories of employees?, and 3) Do they differ in terms of content and purposes?

This chapter also discusses implications in the field of translation studies. Corpora, in fact, are useful when making hypotheses about language patterns in a second language. For this reason, further questions that this chapter wishes to address are the following: 4) Can corpora help discover the nuances of a second language for translation purposes?, and 5) Can corpora help translators deliver intralingual translations?

The added value of the study carried out in this chapter lies in the fact that it shows a valid corpus-based approach to understand and explore the language of employment legally binding documents. This contribution carries out detailed corpus-based analyses at both word and phrase level. Therefore, it shows possible translation strategies and tools which can be replicated in other academic studies and research. The novelty and usefulness of this approach are the rigorous methods applied and the way corpus consultation is carried out to bring to the fore language nuances and (dis)similarities between the two types of documents. Corpora provide linguistic resources that help users observe and analyse language change over time or in different documents or genres, also within the same domain. In this way, they showcase the linguistic diversity which translators can tap into when expressing terms and concepts in a target language.

2 Theoretical background and literature review

Legal language is considered pedantic and archaic (Bhatia 1997; Williams 2004, Williams 2011 and Williams 2023). Scholars claim that the "language of the law" (Mellinkoff 2004) is intricate for reasons of inclusiveness and preciseness (Coulthard & Johnson 2007: 40). Others argue that lawyers and judges do not have any particular communication intent; they use a complicate jargon only to remark their "legal fraternity". (Tiersma 1999: 52)

2.1 Legalese

Many legal documents are characterized by *legalese* features which make it particularly abstruse, especially to the layperson (Tiersma 1999: 200; Tiersma

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& Solan 2012). *Legalese* features have been analyzed and discussed by many scholars over the years (Bhatia 1997; Tiersma 1999; Williams 2004, Williams 2011, Williams 2015 and Williams 2023; Giampieri 2021). Polysemy and near-synonyms, for example, characterize legal documents. Polysemous words are terms which have several meanings (Alcaraz Varó & Hughes 2002: 35ff), whereas near-synonyms are different words with very similar meanings (Stubbs 1996: 109; Phillips 2003: 154). An example of polysemy is "consideration" which is a synonym of "payment" in legal discourse (Scott 2017: 45), whereas in non-legal contexts, it means "reflection", or "pondering". The word "term" is also polysemic, as it can mean "duration", "condition", or even "word", in certain circumstances. Hence, "term", "duration" and "period" are near-synonyms in many legal contexts. Complex prepositions are another example of how legal language can be difficult to understand. Scholars refer to many instances in this respect, such as "by virtue of", "for the purpose of", "other than", or "pursuant to[1] " (Bhatia 2010: 26–29; Coulthard & Johnson 2010: 10; Mellinkoff 2004). As can be guessed, complex prepositions tend to make texts pedantic. Deixis also characterizes legal discourse, as readers have to move their attention back and forward in the text in order to understand what words refer to. Examples in this case are adverbial constructions such as "heretofore", "thereon", "thereto", but also "aforementioned", "foregoing", etc. (Abate 1998: 14–16). Ambiguity, amongst others, is achieved by resorting to "shall" as a modal verb to either express future actions or obligations, instead of using "will" or "must", respectively (Tiersma 1999; Tiersma & Solan 2012; Williams 2015 and Williams 2023). Passive forms are very frequent and are used to "obscure the actors" (Tiersma 1999: 75). Nominalization is another feature that hallmarks legal documents and conceals actors, or phrase subjects. With nominalization, verb and noun combinations are preferred over straightforward verbs (as in "make an agreement", instead of "agree") (Bhatia 2010: 26–29; Tiersma 1999: 75; Williams 2011; Coulthard & Johnson 2010: 10). Also, in legal texts sentences are very long for reasons of all-inclusiveness (Williams 2004 and Williams 2015). This gives way to syntactic discontinuity and embeddings (Tiersma 1999: 56–57; Williams 2015), which make the understanding of a passage particularly arduous. Finally, legal language tends to be sexist, given the predominance of male pronouns and male possessive adjectives (Griffith 1988: 135).

[1]See also the list on http://www.plainenglish.co.uk/the-a-z-of-alternative-words.html and https://www.plainlanguage.gov/guidelines/words/use-simple-words-phrases/.

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2.2 Plain English

In view of the "opacity" (Orts 2015) of the language of the law, over the years the Plain English movement has tried to make legal language simpler and more accessible to the layperson (see Tiersma 1999; Bhatia 2010; Tiersma & Solan 2012; Williams 2023). Its advocates have proposed many strategies to simplify legal jargon by eliminating archaic expressions, reducing sentence length, using non-sexist language, removing unnecessary words, etc. (see Bhatia 2010). Some institutions and drafters have embraced the principles of plain English in legal writing. For example, the Commission (2023) developed at EU level is an innovative reference document. In the same way, the UK legislation has made huge strides forward and British drafters have produced more comprehensible acts and statutes (Williams 2023).

2.3 Legal language and corpora

Corpora are used to carry out language investigations (Resi 2024) and to foster second language learning, especially in technical fields (Cuchet & Dury 2024). Given its specificity and technicalities, legal language has been studied and analyzed by using and consulting corpora (Vigier & Sánchez Ramos 2017; Biel 2018; Vigier Moreno 2019; Giampieri 2024). Corpora help find and notice word usages in context, collocations and lexical phrases, which abound in legal texts. Legal language is so formulaic and repetitive that legal corpora need not be large to study and analyse its features (Bhatia et al. 2004: 207; Giampieri 2025).

With regard to legal corpora, scholars posit that the ones publicly accessible are generally composed of court decisions and/or legislation (Biel 2018: 29; Prieto Ramos & Guzmán 2021: 255). This means that no corpora on corporate or private legal documents are available due to confidentiality or privacy reasons (Vigier & Sánchez Ramos 2017: 261; Biel 2018). In this regard, however, the work by Giampieri (2024) is innovative as corpora of terms of service are now publicly available. Generally, legal corpora on corporate documents tend to be compiled for specific research projects and are kept private (Vogel et al. 2017: 12). Hence, studying the legal language of private legal texts must be made on a case-by-case basis by creating DIY (do-it-yourself) corpora (Zanettin 2002). Zanettin (2002: 244) suggests building a DIY corpus with 10–50 documents (5,000–40,000 words).

Corpora can be composed either automatically or manually. Automatic (or semi-automatic) corpus compilation is possible by using software solutions such as BootCaT (Baroni & Bernardini 2004) or specific platforms, such as the Sketch Engine (Kilgarriff et al. 2004) (see, for example, the case studies by Resi 2024 and

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Giampieri 2024 and 2025). Manual corpus creation is more time-consuming, but it is focused and targeted, as each document composing the corpus is carefully selected according to a specific project.

Corpora are consulted by translators as they help notice specific language patterns (Zanettin 2002; Vigier & Sánchez Ramos 2017; Giampieri 2024 and and 2025). They may also be exploited to assist translators in simplifying texts in intralingual translation processes (Stodden et al. 2023).

3 Methodology and data collection

In order to investigate the legal language and style of employment contracts and employment offer letters, as well as unveil possible implications in translation studies, two specific corpora are built: a corpus of employment agreements and a corpus of employment offer letters. The corpora are created manually in order to make sure that the documents retrieved were drawn up by native speakers of English. Each document was, in fact, read carefully before considering it for the purposes of this analysis.

As for the employment agreement corpus, the following Google search string was written to retrieve employment contracts: *"employment agreement|contract" site:.onecle.com "governed by the laws of the State of"*. On the basis of this search string, it was possible to retrieve either "employment agreement" or "employment contract" in the onecle.com domain. Onecle.com is a database of US and non-US contracts. The string "governed by the laws of the State of" helped narrow down the results to US agreements. This is, in fact, a formulaic expression written at the end of North-American agreements stating the applicable law. The parties to the contract were also taken into consideration. To this aim, each webpage was read carefully to make sure that at least the drafting party was based in the US and, hence, was using native-speaking language. At the end of the process, 21 files were retrieved (3,812 word types; 104,633 tokens).

The Google search string of employment offer letters was the following: *"employment offer letter" site:.onecle.com.* In this way, employment offer letters were downloaded from the onecle.com domain. As for the previous search, letters were carefully checked to verify that they had been written by native speakers of English. In this regard, only letters of employment issued by US companies were downloaded and saved. This process helped retrieve 31 files (2,650 word types; 46,235 tokens). As can be seen, despite being composed of more files, the second corpus shows fewer word types and tokens. This is due to the fact that employment offer letters are shorter than standard employment agreements. This peculiarity was noticed while composing the second corpus.

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The corpora were analyzed (one at a time) by using the AntConc (Anthony 2024) software solution.

AntConc is an offline concordancer which lists concordances, collocations, KWiC (key words in context), and is equipped with many other functions. For the purposes of this study, the main search queries are used, such as concordancing and collocate listing.

This chapter carries out detailed analyses by following a methodological and rigorous approach. Hence, its methods and principles can be replicated in over fields of study, thus fostering interdisciplinarity.

3.1 Data collection and analysis

The documents composing the corpora were analyzed firstly by following a general approach, then by adopting a more analytical perspective. In the first case, the texts of each corpus were skimmed in order to notice evident differences in their layout and general content. Afterwards, a detailed corpus-based analysis was carried out with the aim of revealing relevant language patterns. More precisely, *legalese* features were searched for in both corpora, such as the modal auxiliary "shall", passive constructions, deictic elements (such as "hereon" or "hereof"), polysemous words (e.g., "term" and "consideration"), etc.

3.2 Feature selection

The *legalese* features that were searched for in both corpora were chosen on the basis of literature findings. The use of "shall", for example, has been criticized by many scholars (Tiersma 1999; Bhatia 2010; Williams 2023) as it is an ambiguous auxiliary which can refer to both future actions and obligations (Bhatia 2010: 28; Williams 2023: 28–32). Sample phrases are "we shall see" and "shall be construed in accordance with" (Williams 2023: 32–33). Passive forms are also used extensively in many legal documents to avoid mentioning an agent and to ensure the formality of statements (Williams 2023: 32). An example is the sentence "the employment relationship shall be terminated upon the expiration of the legal term of notice". (Williams 2023: 32) Deictic elements such as "hereon", "hereto", "hereof", "thereon" etc. (Abate 1998: 15; Bhatia 2010: 26–28) are used to refer back and forward to parts of a document; in this way, they hinder smooth reading (e.g., "the parties hereto", Bhatia 2010: 16). Polysemous words such as "consideration" and "term" have been pointed out by the literature and criticized (Scott 2017), as their meaning in daily usage is very different from the one they assume in legal contexts. For example "consideration" means "payment" in contracts (Scott 2017:

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45). Finally, simplified versions of complex prepositions such as "pursuant to", "other than", "for the purposes of", and "by virtue of" have been proposed by the literature over the years (Bhatia 2010; Commission 2014). For instance, Bhatia (2010) suggests using "under" or "according to" instead of "pursuant to" (Bhatia 2010: 28), whereas the Clear English guide (2014) replaces "for the purpose of" with "to" (Clear English guide 2014: 2).

4 Analysis

This section reports and discusses the corpus analysis of both corpora with a view to exploring the legal language and style of the employment agreements and employment offer letters.

4.1 General observations

By scrutinizing the two types of documents from a general perspective, it is possible to notice some relevant differences.

The employment agreements composing the corpus, for example, are generally organized into clauses and start with premises. They state the duration of the contract; the duties and obligations of the parties; the compensation of the employee; the benefits; the terminating events; non-competition and non-disclosure clauses; dispute resolutions, and the governing law and jurisdiction. Standard "boilerplate" clauses (Cordero-Moss 2011) are also mentioned, such as severability, entire agreement, waivers, etc.

Conversely, employment offer letters are not divided into clauses and have no recitals or premises at the beginning. Being letters, they address the reader/recipient by a "Dear" followed by the receiver's first name (hence, their tone is quite informal since the very beginning). Texts tend to be organized into bullet points, reporting the employee's start date, compensation, duties and benefits. Some refer to arbitration as a dispute resolution, whereas others offer stock options. Many of them express the hope for the recipient to accept the offer (e.g., "I hope that you elect to accept this offer of employment") and highlight the sender's availability in case of questions or doubts (e.g., "please feel free to call me if you have any questions at all").

Given the general overview above, it can be noticed that employment agreements use a more formal tone and far more *legalese* features than employment offer letters. As mentioned, many employment agreements start with premises (or recitals), whereas employment offer letters do not. Also, they feature opening

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formulae such as "Now, therefore, in consideration of the mutual covenants hereinafter set forth and for other good and valuable consideration, the parties hereto agree as follows:", which the employment letters do not have. On the contrary, as already pointed out, letters of employment start with "Dear" followed by the employee's first name and then they continue with phrases such as "we are / I am pleased to offer you the position of", that make the reader feel at ease. Hence, there seems to be less detachment between the two parties. At the same time, employment agreements end with a closing formula, such as "In witness whereof, the parties have executed this Agreement as of", which employment letters do not have. These, in fact, end in a much simpler way, with closing sentences such as "If you have any questions regarding this employment offer, please call (…)", followed by the statement "I have read this offer and I understand and accept its terms", or simply "agreed and acknowledged". Furthermore, by opening and reading any employment offer letter, the use of the second-person pronoun "you" comes to the fore, as well as second-person adjectives ("your"). This makes the tone less formal and more direct (see also Williams 2023). On the contrary, employment agreements use third-person references, such as "Employee" or "Executive", and tend to use male pronouns and male possessive adjectives ("he", "him", "his").

The findings above can be considered as relevant when translators need to simplify language and/or make it more accessible to laypeople (see Stodden et al. 2023). As a matter of fact, such observations reveal how intralingual translation strategies can be followed.

In light of these initial findings, it would be tempting to speculate that employment agreements are addressed to high-ranked managers, whereas employment offer letters to lower positions. This assumption is confuted by the fact that positions of Executive or Directors, for example, are offered in both types of documents. For instance, "Executive shall serve as executive vice president" is a phrase found in some employment agreements, and "offer for the position of Executive vice president" is mentioned in several employment offer letters.

4.2 Corpus analysis of legalese patterns

An in-depth corpus analysis of the documents composing the two corpora is now carried out. The investigation compares the language used in the employment agreement (EA) corpus and in the employment offer letter (EOL) corpus, respectively. Table 1 below shows some small-scale quantitative data related to a comparative analysis of some *legalese* features that is featured in both corpora.

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As mentioned, the *legalese* patterns are chosen on the basis of literature findings (see Section 3.2 on *legalese* feature selection). The occurrences of *legalese* patterns are reported in entire figures and in percentage values. The percentage values are calculated against the number of word types of each corpus (i.e., 3,812 in the EA corpus and 2,650 in the EOL corpus).

Table 1: A small-scale comparative quantitative analysis of legalese features in the employment agreement (EA) corpus and in the employment offer letter (EOL) corpus.

	EA		EOL	
Legalese features	Count	%	Count	%
Shall	1,559	40.9%	206	7.8%
Passive forms	1,082	28.4%	552	19.7%
Deixis (e.g., "hereon", "hereof", "thereon"...)	946	24.8%	127	4.8%
Polysemous words ("term")	250	6.6%	43	1.6%
Polysemous words ("consideration")	54	1.4%	4	0.1%
Complex prepositions ("pursuant to")	139	3.6%	33	1.2%
Complex prepositions ("other than")	97	2.5%	23	0.8%
Complex prepositions ("for the purpose(s) of")	22	0.57%	1	0.03%
Complex prepositions ("by virtue of")	4	0.1%	0	-

Comments to the figures of Table 1 are reported on the next pages. The comments help understand the language used in both corpora. They also supplement the quantitative data of Table 1 by providing qualitative insights (i.e., sample phrases), and by presenting and discussing language patterning from the two corpora.

As far as "shall" is concerned, it is evident that the EA corpus makes a far larger use of this controversial modal verb (40.9%). On the contrary, in the EOL corpus it is used more sparingly (7.8%). Sample paragraphs sourced from the EA corpus are the following: "Employee shall devote all working time, attention, knowledge, and skills solely to the Company" and "Employer shall have no right to terminate under this Section". Examples from the EOL corpus are the following sentences: "you shall (a) devote your entire working time for or at the direction of the Company" and "you shall not have the ability to terminate this Agreement". Table 2 below reports these and other sentences.

Although "shall" is used in both corpora, it is observable that both EAs and EOLs use "will" to a different extent. The EOL corpus mentions it more systematically than the EA corpus (789 hits, 29.8%, against 415, 10.9%, in the EA corpus).

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Table 2: Sample phrases with "shall".

No.	Sample phrase (EA)	Sample phrase (EOL)
1	*Employee* shall devote all working time, attention, knowledge, and skills *solely* to the Company.	*You* shall (a) devote your entire working time for or at the direction of the Company.
2	*Employer* shall have no right to terminate under this Section.	*You* shall not have the ability to terminate this Agreement
3	*Employer* shall reimburse *Executive* for all traveling, hotel, entertainment and other out-of-pocket expenses	*The Company* shall promptly reimburse *you* for all actual and reasonable business expenses.

The modal "must" also appears, but not considerably (34 hits, 1.3%, in the EOL against 33 hits, 0.8%, in the EA corpus).

In addition, the EA phrases reported in Table 2 above feature the words "Employee", "Executive" and "Employer", whereas the EOL statements use "you" and "The Company". This aspect was already discussed and noticed above. Such appellatives highlight the fact that letters of employment tend to me more direct. Moreover, the EA corpus contains the formal term "solely" (line one), whose plain equivalent is "exclusively", or "only".[2] In this respect, "solely" is used 17 times in the EA corpus (0.45%), whereas only once in the EOL corpus (0.04%).

Passive forms are not particularly easy to source via corpus analysis, given that the corpora are untagged. However, targeted searches can be carried out by writing the search string "be *ed". By doing so, it is possible to retrieve passive forms composed of the bare infinitive "be" followed by a regular past participle. This kind of passive form is particularly frequent in the EA corpus (804 counts, 21%, against 347, 13%, in the EOL corpus). A similar search string, i.e., "is *ed", confirms instances of passive constructs in the EA corpus (219 hits, 5.7%, against 101, 3.8%, in the EOL corpus). The search for "are *ed" shows slightly more hits in the EOL corpus (74, 2.8%, against 59, 1.5%, in the EA corpus). Table 3 exhibits corpus-sourced passive constructions.

Table 3 above highlights that both corpora make use of passive forms. However, EOL clauses are less formal as they tend to feature plainer words and expressions. For example, instead of "relocate" (line one), the clause in the EOL corpus

[2]See https://www.collinsdictionary.com/it/dizionario/inglese-sinonimi/solely.

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Table 3: Sample phrases with passive forms.

No.	Sample phrase (EA)	Sample phrase (EOL)
1	Executive shall (…) and he shall *not be required to relocate* his residence or principal office to any place outside (...) County.	*You* will not *be required to move* to the San Francisco Bay Area.
2	*It is agreed* by the parties *hereto* that...	*It is intended* that...
3	Any prior agreements or understandings with respect to Employee's employment by Employer, *are canceled* as of the Effective Date.	Any contrary representations which may have been made to you *are superseded* by this offer.

contains "move to". Also, the first clause of the EA corpus uses sexist language ("*he* shall"), whereas the corresponding clause of the EOL corpus displays "you".

The search string "is *ed" (second line of Table 3 above) reveals interesting patterns. For instance, the EAP corpus contains formal expressions such as "it is agreed". This construct does not appear in the EOL corpus. Deixis also comes to the fore; in particular, with the adverbial expression "hereto" (line two of Table 3).

As already mentioned, only the past participle of regular verbs is investigated. Nonetheless, it is evident that this type of passive form is more recurrent in the EA corpus (1,082 counts, 28.4%, against 522 occurrences in the EOL corpus, corresponding to 19.7%).

As noticed, deixis appears in some EA clauses. If "*here*" is searched for in both corpora, 946 hits are retrieved from the EA corpus (24.8%), whereas only 127 (4.8%) are found in the EOL corpus. In this regard, it should be highlighted that the "*here*" search provides also unrelated elements, such as "therefore", "where" and "elsewhere" in the EOL corpus. Since "therefore" is a linking word and "where/elsewhere" are place adverbs, they are ruled out. The deixis elements obtained from both corpora are the following: "therein", "thereof", "thereto", "thereafter", "hereto", "hereof", "hereby", "hereunder", etc. Table 4 illustrates some examples.

Expressions such as "thereof", "hereof", "hereunder" (line 1) and "whereas" (line 2) or "thereafter" (line 3) make the reading and understanding of legal texts less smooth (Williams 2023). As regards "hereby" (line 4), it could be argued

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Table 4: Sample phrases with deixis.

No.	Sample phrase (EA)	Sample phrase (EOL)
1	Executive may sit on the boards of directors of corporations and committees *thereof* without violating his obligations *hereunder*.	With respect to the subject matter *hereof*.
2	*WHEREAS*, the Company and Employee entered into an Employment Agreement dated as of July 22, 1996 (as amended to the date *hereof*, the "Employment Agreement").	The Company consummates a change of control transaction, *whereby* fifty percent (50%) or more of the voting stock of the Company changes ownership.
3	*Thereafter*, this Agreement terminates and the Executive shall receive those benefits.	*Thereafter*, you will accrue vacation at the regular (...) vacation accrual rate.
4	The parties do *hereby* agree as follow: [*sic.*].	You *hereby* agree that the Confidentiality Agreement will govern and be applicable to your prior employment.

that it is slightly more comprehensible or, at least, more frequent and accepted (Williams 2023: 74). This adverb is featured in both corpora, although with different percentage values (84 hits in the EA corpus, 2.2%, and 8 hits in the EOL corpus, 0.3%).

The polysemous word "term" appears 250 times (6.6%) in the EA corpus, and only 43 times (1.6%) in the EOL corpus. With regard to the word "consideration", the EA corpus produces more hits (1.4% against 0.1% of the EOL corpus). Sample paragraphs or phrases with such polysemous terms are reported in Table 5.

The examples reported in Table 5 above show that polysemy chaaracterize both corpora, although the EOL corpus produces fewer occurrences.

With regard to the polysemous word "term", it is evident that line one of Table 5 contains instances where it means "duration" in both corpora, whereas it is a synonym of "word" in both the examples of line two. Finally, line three shows that "term" is a replacement for "condition".

Also, the expression "in consideration" is polysemous in the EA corpus, as it

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Table 5: Sample phrases with "term" and "consideration".

No.	Sample phrase (EA)	Sample phrase (EOL)
1	The Original *Term* and any 12 month period commencing immediately after the end of any *Term*.	During the *term* of your employment.
2	The *term* "Affiliate" shall include, but shall not be limited to...	The Severance Plan defines the *term* "Cause" as a participant's: Material neglect...
3	(A) Upon all of the *terms* and conditions described in this Agreement. (B) No waiver by any party of any *term* or condition of this Appendix A	Under the *terms* and conditions described above.
4	As a condition of my employment with (...) its subsidiaries, affiliates, successors or assigns (together the "Company"), and *in consideration of* my employment with the Company.	Like all Company employees, you were required, as a condition to and *in consideration of* your employment with the Company, to sign the Company's standard Proprietary Information.
5	In exchange for the severance benefits described in Section 2.3(a)(i) of my employment contract with the Company dated (...) ("Employment Contract"), which I acknowledge is sufficient *consideration* to support this Agreement,	You also will be eligible for *consideration* to receive long term incentive equity awards

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signifies "as a counter-performance" (line four). In the EOL corpus, by contrast, this expression means "considering"; hence, it is not a controversial equivalent of "as a form of payment" (see line four). Conversely, in the fifth line, both terms are polysemous.

Complex prepositions hallmark the two corpora. If the expression "for the purpose* of" is queried in the EA corpus, 22 hits (0.57%) are found, whereas only one occurrence is retrieved from the EOL corpus (0.03%). The expression "pursuant to" is used 139 times in the EA corpus (3.6%) against 33 (1.2%) in the EOL corpus. The phrase "other than", meaning "except", shows 97 occurrences in the EA corpus (2.5%), but only 23 (0.8%) in the EOL corpus. Finally, the prepositional phrase "by virtue of" produces 4 hits (0.1%) in the EA corpus, but no results in the EOL (see Table 6 below for phrases in context).

Table 6: Sample phrases with complex prepositions.

No.	Sample phrase (EA)	Sample phrase (EOL)
1	*Pursuant to* the terms hereof, *for the purposes of* entering judgment with respect to such award.	*For the purpose of* calculating any benefits.
2	Employment is terminated by the Company without Cause *pursuant to* paragraph 16(a)(iv) below.	Severance payments made *pursuant to* Section 9 or Section 10.
3	Any attempted assignment, transfer, conveyance or other disposition (*other than* as *aforesaid*) of any interest.	You are not eligible for any severance benefits *other than* those set forth in such agreement.
4	The Employee will not copy or duplicate any of the *aforementioned* documents.	The occurrence of one of the *aforementioned* events.
5	*WHEREAS, by virtue of his* employment with Employer, Executive will become acquainted with certain confidential information.	-

As can be observed in line one and three of Table 6, several complex prepositions appear in the same sentence of EAs (i.e., "pursuant to" and "for the purposes of" in line one, and "other than" and "aforesaid" in line three). Line four also

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features "aforementioned" in both corpora. In this respect, there are nine occurrences of "aforementioned" and "aforesaid" in the EA corpus (0.2%), whereas only one instance of "aforementioned" in the EOL corpus (0.04%) (line five). Finally, in line five, other *legalese* traits come to the fore, such as sexist language ("his"), and the term "whereas", which is an expression setting the contract premises or recitals.

In addition to the features explored above, there are other *legalese* peculiarities in the two corpora which are not specifically quantifiable and, for this reason, are not reported in Table 1. These elements are relevant for text comprehensibility and are, for example, repetitions and use of near synonyms; nominalization; long sentences, and syntactic discontinuities. Repetitions and near-synonyms, for example, characterize both employment agreements and employment offer letters. Table 7 below reports some sample phrases.

The clauses shown in line 1 of Table 7 above concern the prohibition for the employee to use trademarks or copyrighted materials during or after employment. To this aim, all the company's proprietary information is listed. Line two focuses on the people working with or for the employer, while line three refers to the employer's policies and customs. Although near-synonyms characterize both corpora, the lists mentioned in employment agreements is longer and more inclusive.

As discussed above, another interesting *legalese* trait is nominalization. Both types of documents feature nominalization. For example, "make" is often followed by an object, such as "make an election" (instead of "elect" or, better, "choose"), "make payment" (instead of "pay"), "make changes" (instead of "change"), etc. These and other examples are reported in Table 8 below.

In line two (B) of Table 8, the plain expression "you will need to present documentation" can replace the nominalized phrase "upon presentation of such documentation", which appears in both corpora. In the same way, the expression "based on the achievement of" could be simplified by writing "based on / subject to achieving" (line three, B).

Also long sentences (and syntactic discontinuity) characterize legal documents. In the current analysis, sentences are considered long when they contain more than 50 words (see Williams 2023: 67–68). Williams (2023), in fact, defines as long the sentences which exceed 45 words (Williams 2023: 67). The corpus of employment agreements produces more instances in this respect. One of them is reported in Table 9 below.

Both the clauses of Table 9 are entitled "change of control". The clause of the EA corpus is composed of 116 words; hence, it is extremely long and complex. Syntactic discontinuities are underlined and there are many *legalese* features, such as

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Table 7: Sample phrases with repetitions and near-synonyms.

No.	Sample phrase (EA)	Sample phrase (EOL)
1	(A) Computer programs, systems and/or software; ideas, inventions, trademarks, business information, know-how, processes, improvements, designs, redesigns, discoveries and developments. (B) Trade or business secrets, including business opportunities, marketing or business diversification plans, business development and bidding techniques, methods and processes, financial data and the like (collectively, the "Protected Information").	(A) Trade secrets, confidential and proprietary information. (B) Trade secrets, confidential business or technical information or know-how not generally known to the public.
2	Employer, its Affiliates, and all of their agents, employees, officers and directors.	[*Company name*] and its affiliates and subsidiaries.
3	(A) Company policies, welfare or benefit plans, or other employment practices. (B) Plans, policies and arrangements.	The Company's practices and policies.

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Table 8: Sample phrases with nominalization.

No.	Sample phrase (EA)	Sample phrase (EOL)
1	(A) You shall advise the Company in writing whether or not you have *made an election.* (B) The Corporation shall have no further obligation to *make any payment.* (C) The Corporation shall not *make any changes* in such plans or programs.	(A) We neither need nor desire to *make any unauthorized use* of any intellectual property. (B) You will not need to *make any cash payments* to cover such tax withholding. (C) Please *make arrangements* with me to complete the necessary eligibility to work.
2	Upon *presentation* of such documentation thereof.	(A) Upon *presentation* of appropriate documentation in accordance with the Company's expense reimbursement policies. (B) You will need to present documentation.
3	(A) Based on the *achievement* of performance goals. (B) Based on achieving a combination of revenue, profit and other business performance targets.	(A) Based upon the *achievement* of performance targets. (B) Subject to [*Company name*] achieving certain stated performance goals.

"hereof", "other than", "subsequent to", and "shall". The clause sourced from the EOL corpus is even longer (126 words) and the one reported in Table 9 is only an extract of the entire clause. Other *legalese* patterns are as follows: "pursuant to", "herein", nominalization ("the consummation of"), and "sole". Although longer, the EOL clause contains fewer instances of syntactic discontinuity. Furthermore, the organization of the text is more straightforward. Sequencing and indexing, in fact, make the clause more comprehensible. Also, the tone is less formal than the one of the corresponding EA clause. The clause reported in Table 9 above is only an example of many similar (long) sentences of the clauses composing the EA and EOL corpora.

The overall tone of both employment agreements and employment offer letters is now scrutinized. Table 10 below lists a few sample sentences and phrases,

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Table 9: Long sentences and syntactic discontinuities.

No.	Sample phrase (EA)	Sample phrase (EOL)
1	Individuals who, as of the date *hereof*, constitute the Board (the "Incumbent Board") cease for any reason to constitute at least a majority of the Board; provided, however, that any individual (*other than* an individual whose initial assumption of office occurs as a result of an actual or threatened solicitation of proxies or consents by or on behalf of a Person other than the Board) who becomes a director *subsequent to* the date *hereof* whose election or nomination for election by [*Company name*]'s shareholders was approved by a vote of at least a majority of the directors then comprising the Incumbent Board shall be considered as though such individual were a member of the Incumbent Board.	In the event that: (i) the Company consummates a change of control transaction, whereby fifty percent (50%) or more of the voting stock of the Company changes ownership *pursuant to* such transaction (a "Change of Control"); and (ii) within twelve (12) months after the *consummation* of a Change of Control, your employment with the Company is either (a) terminated by the Company or successor entity without Cause (as defined in Section 9 *herein*), or (b) terminated by you for Good Reason (as defined in and in accordance with the paragraph below); and (iii) you sign, date, and deliver to the Company the Release and allow it to become effective; then you will receive the following as your *sole* severance benefits (the "Change of Control Severance Benefits"):

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where it is evident that the register used in employment letters is far more direct and, to some extent, plainer.

Table 10: Tone.

No.	Sample phrase (EA)	Sample phrase (EOL)
1	*The Executive shall be eligible to* receive a target annual cash bonus of forty-five percent (45%) of his base salary under The New York Times Company 1991 Executive Cash Bonus Plan (the "Executive Cash Bonus Plan").	(A) *You will be eligible* for an annual bonus of up to 20% of your base salary. (B) *You may be eligible* to receive an annual discretionary bonus.
2	*In the event that Executive accepts employment* with, or provides any services to (whether as a partner, consultant, joint venturer or otherwise), any person or entity which offers products or services that are *competitive with any products or services offered by the Company* or with any products or services that Executive is aware the Company intends to offer, Executive shall be deemed to have voluntarily *terminated employment* with the Company	*You must not* engage in any work, paid or unpaid, that creates an actual conflict of interest with the Company.

The first line of Table 10 focuses on an annual bonus that the employee may receive. The tone used in employment letters is more informal and direct (e.g., "you will be eligible for an annual bonus" against "the executive shall be eligible to receive a target annual cash bonus").

The second line sets the prohibition for the employee to engage in any activity which could compete with the employer's. Also in this case, the sentence of the letter of employment is much shorter, more direct and to the point.

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5 Discussion

The analysis carried out in the sections above showed a recurrent and systematic use of *legalese* in employment agreements. By contrast, employment offer letters are characterized by *legalese* to a lesser extent. There are long sentences, complex prepositions, passive forms and nominalization, but *legalese* features are less pervasive, texts are smoother and, hence, more understandable.

For example, despite the persistence of "shall" in many letters of employment, there is also a recurrent use of "will" and "must" (see Table 1 and 2). Passive forms and polysemous words characterize both corpora, but the EOL corpus produces fewer instances. Complex prepositions hallmark employment agreements and employment letters, but those in employment offer letters are fewer than the prepositions found in employment agreements (see Table 1, where "other than" shows a frequency percentage value of 2.5% in the EA corpus, but only 0.8% in the EOL corpus). Phrases with near-synonyms and repetitive words are mentioned in both corpora, but in letters of employment they are far shorter (see Table 7, for example). Sentences are long in the two corpora, but those in employment offer letters tend to be organized in bullet points and/or by sequencing. These writing strategies make EOLs more comprehensible and easier to read (see Table 9). In addition, by considering the overall tone of employment offer letters, it is evident that they are more direct, if not almost colloquial, in comparison with employment agreements.

Such considerations can be taken into account when delivering intralingual translations and when addressing target texts which, to some extent, must be drawn up in plainer language.

6 Conclusions and future research

This chapter was aimed at exploring the legal language used in employment agreements and employment offer letters. In particular, it wished to bring *legalese* features to the surface, analyse them and investigate to what extent employment agreements and employment offer letters make use of *legalese* patterning in native speaking contexts. In this way, it also investigated possible corpus-assisted applications in (intralingual) translation studies.

This chapter contributed to highlighting the fact that employment agreements are drafted in a formal and archaic style, despite being addressed to both native and non-native speakers of English. They feature *legalese* patterns, which are used systematically throughout the documents. By contrast, in employment of-

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fer letters, alternative (and plainer) linguistic solutions are proposed. Long sentences, syntactic discontinuities, nominalization, or complex prepositions are still used by the drafters of employment contracts and letters of employment, but the pedantic aspect of the language of the law seems to be less pervasive. As a matter of fact, plainer linguistic solutions are proposed in employment offer letters, such as shorter all-comprehensive lists of near-synonyms; a reduced recurrence of "shall" in favour of a more systematic use of "will"; fewer deixis patterns and simpler passive forms. Also, despite the challenging sentence length, letters of employment are organized in bullet points or in sequences, thus allowing readers to follow the text content easily.

Differently from what could be expected, the findings of this chapter highlight that employment agreements and employment offer letters do not differ in terms of the duties or positions offered to the recipient, as they are both addressed to a variety of employees, including executives. Hence, the reason(s) why companies decide to resort to one type of document instead of the other might be further explored by researchers.

There are many implications in the field of translation studies. Users can explore either comparable or parallel corpora (Fantinuoli & Zanettin 2015; Resi 2024) in order to understand and verify language nuances and make informed decisions as to the best translation candidates in a given context. Translators could also analyse collocations in different languages and retrieve topical vocabulary, as well as multi-words and lexical phrases. For example, as the language of the law is particularly characterized by formulae and fixed language expressions (Giampieri 2025), translators could explore such patterns thanks to corpus consultation. Most of the corpus platforms and corpus software solutions available today contain features that allow users to carry out language investigations (Cuchet & Dury 2024; Resi 2024). At their disposal, translators have collocation listing options, as well as key-word-in-context functions and lemmatization.

By creating a similar corpus composed of Italian *contratti di impiego* (back-translation: "employment contract"), it is possible to notice words and expressions which translate most of the terms analyzed above. Examples of such phrases are *premesso che* (back-translation: "whereas"); *orario di lavoro* (back-translation: "working time"); *traslocare* or *trasferirsi* (back-translations: "move to" or "transfer to"); *recedere dal contratto di impiego* (back-translation: "terminate employment"); *avere diritto a* (back-translation: "be entitled to"); *rimborsare spese* (back-translation "to reimburse expenses"), just to mention a few. An Italian corpus could be composed by searching for templates or drafts of *contratto di impiego* in Italian domains (search query: *"modello contratto di impiego"*

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site:.it). In this way, sample contracts are downloaded and their language and terminology can be analyzed by using the AntConc software solution.

The limit of this study lies in the corpora taken into consideration. An analysis of the contracts drawn up in English as a *lingua franca*, for example, could have yielded more comprehensive, or perhaps different, results. In this respect, it could be interesting to explore and verify to what extent employment offer letters and/or employment agreements are used in non-native speaking contexts and which type of language (i.e., plainer or more intricate) and register or tone are applied. This is something that future research might wish to explore. Also, analysing the drafting style of other legal documents written in English as a first or second language could be another avenue to explore.

The implications of this study are manifold, as this chapter provides evidence that corpus consultation helps deliver intralingual translations (Jakobson 1959). In other words, corpora can be used in the translation process to deliver target texts drafted in a plainer legal style. Hence, corpus consultation could be fostered for the benefit of both students in Translation Studies and professional translators. Further analyses could explore these aspects and propose the application of *ad hoc* corpora in (intralingual) translator training and/or (intralingual) translation practice. The principles and methodologies followed in this chapter can be replicated in other fields of study, thus fostering interdisciplinarity in empirical translation research.

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Chapter 8

Translating Rick & Morty's audiovisual novum parody: strategies and effects

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Since 2013, Dan Harmon and Justin Roiland's postmodernist TV show *Rick and Morty* has staged a science-fictional world filled with dark and nonsensical humour and a tinge of nihilistic philosophy. The chapter offers an analysis of *Rick & Morty*'s invented terms that parody the genre, and their French translation in the dubbed version of the TV show. The authors use all the genre's tools, including linguistic creativity, to ground their original world, and to generate humour. Using contrastive linguistic tools, such as semantic and morphosyntactic analysis, as well as translation studies theories and approaches, especially the notion of Nida's dynamic equivalence, functionalist approaches to translation and previous research on translating humour, the chapter aims for a better understanding of the translation of parodic elements in a science-fictional context, especially in relation to the visual content.

The chapter offers a model to evaluate the generic and humour fluency of the target terms using linguistic criteria while taking into consideration all the constraints in relation to translating audiovisual material as well as the relationship between text and image. Translation strategies may create distinct effects on the target audience as fictive terms are necessarily connected to the visual and narrative content. The analysis aims to provide a refined vision of the priorities and restrictions in play during the translation process of novum parody. Thus the chapter highlights the close relationship between text, image and translation choices. Translators prioritize one of the aspects of the invented terms (humour, parody or genre) according to the audiovisual context. The chapter also offers to validate the proposed model by analysing another audiovisual work, *Futurama*, in order to confirm the hypotheses.

Alice Ray. 2026. Translating Rick & Morty's audiovisual novum parody: strategies and effects. In Tomáš Svoboda, Olga Nádvorníková & Věra Kloudová (eds.), *Advancing interdisciplinarity in empirical translation research*, 191–229. Berlin: Language Science Press. DOI: ??

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1 Introduction

It is common to hear that humour is untranslatable or that it gets lost in translation. However, many researchers have demonstrated that it is not always the case (Low 2011; Hofstadter 1998; Delabastita 1997 and Delabastita 1997; Gottlieb 1997). Every language element can be adapted and adjusted according to the context. In fact, some adaptations and adjustments may be required to achieve dynamic equivalence, that is to generate the same *effect* on the target audience, or at least to fulfil its function in the target culture. In fact, "translatability does not require that the TT [target texts] use the same linguistic structure, but merely that it delivers, broadly speaking, the same joke". (Low 2011: 60) Besides, what is funny or not is dependent both on culture and on individuals, making the translation of humour all the more complex. Of course, language or culture-specific humour demands more creativity from the translator as the solution may not be straightforward and may require the use of transcreation to "suit the characteristics of an intended audience". (Díaz-Millón & Olvera-Lobo 2021)

Zabalbeascoa (1996) established a translation model, specifically regarding TV comedy, based on priorities and restrictions, with the priorities "as a means of expressing the intended goals for a given translation task and the restrictions are the obstacles and problems that help justify one's choice of priorities and, ultimately, the solutions adopted in the translation". (p. 243) Of course, when translating (dubbing) TV comedy, the difficulty of transferring humour is coupled with the inherent constraints (restrictions) of the media: lip synchronization and isochrony. In our case, *Rick & Morty* being an animated TV show, the degree of tolerance for differences in lip synchronization is higher than a comedy with live actors, as the lip motion is recreated even in the source language and may not be completely accurate in any case. However, translators may still be dealing with synchronization constraints, especially isochrony: the characters – animated or not – cannot be seen talking when they are not and vice versa, or otherwise the illusion is shattered, and the audience cannot experience the show properly. Another restriction that may occur during the translation of TV comedy is the differences that may emerge "in the background knowledge of the original and prospective audiences;[1] differences in cultural and moral values, customs and traditions; differences in conventional themes and techniques of joke-telling" (Zabalbeascoa 1996: 248) and the relationship between the visual content and the dialogue can also be a source of challenges as it can limit creative solutions. However, all the restrictions can be overcome with creativity, by

[1] *Rick & Morty* offers a perfect example in its last season 6, when the authors made the character of Joseph Campbell appear: this is a reference that the target audience may not have.

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adjusting the humour to the context and the target audience, as "[w]hat matters in this case is the perception of the source text's humour as a basis for the decision to make the translation a humorous text" (p. 247), the goal being to achieve "popularity and [create] a faithful following". (p. 245)

One of the specificities of science fiction is its use of language to stage new concepts and objects that are not real and create the cognitive estrangement of the genre as defined by Suvin: "SF is, then, a literary genre whose necessary and sufficient conditions are the presence and interaction of estrangement and cognition, and whose main formal device is an imaginative framework alternative to the author's empirical environment" (Suvin & Canavan 2016: 20, emphasis removed) (see Spiegel 2008 for a re-evaluation of the cognitive estrangement). The idea behind the concept of cognitive estrangement implies that the genre naturalises alien elements (novums) by giving them plausibility, a sense of realism. The novums, the "(novelty, innovation) validated by cognitive logic" (Suvin & Canavan 2016: 79), create a new (or partially new) paradigm that helps to rationally embroider a new world. One of the tools of sci-fi writers is fictive terms, that is, all the neologies used to name novums and create a lexicon that *could possibly be* and is in fact already used as familiar linguistic artefact in the fictive world. Fictive terms often describe pseudo-scientific or pseudo-technological novums and most of them are rooted in the audience's knowledge of the world:

> Fictive neologies have a paradoxical function. They conjure up a sense of the inevitability of a new thing, or of a new discourse in which the neologism is embedded. If there is a word for it, it must already exist. And so the future must also already exist. The more convincing it is in creating an illusion of projected historical reality, the more successful the neology. Yet fictive neology also displays that it is fiction. Because the future cannot exist yet, we know that the neology is a playful, poetic conjuring device, suggesting that any imaginable future is always a poetic construction. (Csicsery-Ronay Jr. 2008: 13)

The translation of science fiction has been the object of previous research (Gouanvic 1997 and Gouanvic 1999; Wozniak 2014; Ray 2019b and Ray 2019a; Campbell 2021 and others) and Suvi Korpi has established the concept of "generic fluency": "related to the quality of reception rather than textual fluency. A translation of a SF text is suggested to be regarded as generically *fluent* when it preserves the voice of the genre, the fictive neologies". (2021: 91) Although neologies may not be the main world-building tool in some sci-fi works, when fictive terms are present in the original source text, they are of great importance as they are

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used to anchor the new world into pseudoscientific plausibility and credibility and "the translators play an important role in preserving the genre-typical elements as the voice of the genre". (Korpi 2017: 88)

The science fiction genre has proven to be a place for humour, whether it is occasional or the global tone of the work. Science fictional humour has not been the subject of systematic research, even though some scholars wrote about specific humorous works of science fiction (Rosinsky 1982; Traub 2018; Koltun 2018; Candelaria 2018) and Dordain's PhD dissertation (2014) attempted to define and understand science fictional humour. The same can be said about translating humorous sci-fi works.

Considering what is known about the translation of humour and the specificities of science fiction, it becomes clear that translating science fictional humour may have its own constraints, especially regarding humorous sci-fi language-related elements. *Rick & Morty* is a perfect example of a work of humorous sci-fi and uses many fictive terms as a world-building tool. The aim of the chapter is to understand how language is used to create parody in the TV show, and the way these parodic fictive terms are dubbed into French, whether they retain their generic features and their parodic nature, or if the constraints (linguistics or technical) may have changed the priorities to retain only one of the fictive terms' features (or none). Another research topic here is if the chosen priorities consider the visual context of the term during its translation. If fictive terms are hybrids by nature, humorous fictive terms may contain oppositional features, as they must remain cognitively estranged with lexical plausibility as well as a tool for parody and humour. Taking an interdisciplinary approach, the chapter's research uses both the conceptual tools of Translation Studies and of Linguistics.

First, *Rick & Morty* will be introduced to define the difficulties arising from the very nature of the show, and the methodology used to conduct the research will be explained. Then, I will categorise the linguistic creativity employed in the TV show as *Rick & Morty* displays many different uses of linguistic creativity, but only one will be analyzed in the chapter, "novum parody", that is the humorous fictive terms that parody the science fiction genre. Then, the parodic strategies of those terms will be analyzed to understand the way they were created and to linguistically explain why they can be considered as "novum parody" and "humorous". The following sections will focus on their French translation, especially the translation strategies regarding generic fluency (see above, Korpi 2017 and Korpi 2021) and what I will call humour fluency (i.e. the retention of the humorous features of the terms). I will also focus on the translation effects (i.e. the potential reception of the terms in the target culture) regarding the concept

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of priorities and the visual component of the text. In the third section, the categorizations established thanks to the linguistic creativity of *Rick & Morty* will be tested on another humorous science fiction show, *Futurama*. Finally, the chapter's analyses will provide a basis for concluding our findings and determining the treatment of the relationship between parody, genre and humour in an audiovisual work. A list of the used abbreviations in this chapter can be found at the end.

2 Corpus and methodology

2.1 Rick & Morty

Rick & Morty is an animated TV show created by Justin Roiland and Dan Harmon. It first aired in 2013 and it has run for 6 seasons. The audience follows Rick, a mad scientist who tries to seek comfort and solace at the bottom of the glass and Morty, his grandson, who is as insecure and naïve as his grandfather is nihilistic (Abesamis & Yuen 2019). Together, they go on adventures through space and the multiverse thanks to a spaceship and a portal gun (a device that opens portals through space). The show is, I think, one of these perfect examples of postmodernist humour using satire and absurdism to highlight the grotesque nature of our societies and question our social constructs and beliefs: "*Rick and Morty* is full of rich references to existential and absurdist philosophy, which are consistently used to satirise and belittle traditional social constructs such as the institutions of marriage, family, religion, and government". (Koltun 2018: 101) The first sentence Genovesi writes on the show in his article is to state that "*Rick & Morty* is not an easy show to watch. It is cynical and loud, has very dark humour, and systematically deconstructs everything in which Western society used to believe". (Genovesi 2020: 2) Rick is an absolute genius, aware of the presumable meaninglessness of life and of the universe. For him, nothing matters, and there is no purpose to anything:

> Indeed, the very meaning of the consequences of human actions and decisions are meaningless since there are other universes in which alternative actions and decisions are taken and to where it is always possible to flee. Even if we decide to do some good, we know that in some alternative reality our counterpart is making the bad decision and causing harm instead. There is no way of avoiding the emergence of suffering and injustice in the multiverse. (Genovesi 2020: 12)

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Watching the show, the audience faces a double dichotomy: it is simultaneously "dark and hilarious; profound and stupid". (Miranda 2017) We laugh with Rick at the absurdity of life itself so that we don't cry, just like he drinks to forget he knows that nothing makes sense. The TV show was adapted into French by Xavier Hussenet, Thibault Codevelle et Soriya Leang.

To stage a postmodernist nihilistic vision of human (and extraterrestrial) life, the authors of *Rick & Morty* have chosen to parody and explore science fiction tropes. Indeed, they offer the perfect tools to highlight cultural relativism, universe meaninglessness, and the dismay of human lives by creating satire through alien environments and absurdist situations[2] that still display cognitive estrangement, thus enhancing the relationship between the show and the audience's reality:[3]

> It explores, and often trivialises, traditional science fiction tropes using clever and dark humour, with the fundamental example being the perverse spin on *Back to the Future's* Doc and Marty character dynamic. (Koltun 2018: 101)

The authors use the specificities of the genre to create humorous situations, especially with the concept of novum. Of course, some novums are quite far-fetched in the show, but as long as they are (or can be) rationally explained in the science-fictional world, they are plausible and may enter the xeno-encyclopedia of the genre.

> Humour interacts in a very specific way with some science fictional features. The process gives rise to a distinctive form of humour: science fictional humour. It draws on the creative power of science fiction to offer innovative and original inventions designed to induce laughter. (Dordain 2014: 494, my translation)

The TV show abounds with absurd concepts and objects that are named by the author to highlight their absurdist and humorous nature ("The Machine of Unspeakable Doom", "the real fake doors", "plumbus", to name but a few of them).

> From the point of view of humour, the xeno-encyclopedia allows authors to conceive creatures, concepts and places that precisely meet their needs

[2]In the episode "Anatomy Park", Morty is shrunk to save an amusement park built inside the body of a man.

[3]Even though the situations are absurd, they are supported by novums and pseudo-scientific and pseudo-technological explanations. They may seem plausible.

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> without bothering about realism, or even verisimilitude, so that they can produce humorous scenes. (Dordain 2014: 374, my translation)

All forms of humour constantly overlap during episodes, creating very dense and sometimes hard-to-follow stories. However, the substantiality of the show and the fact that it needs to follow the codes of the science fiction genre create a real challenge for translators who must also follow the technical rules of dubbing: isochrony, lip synchronization, and characters' language specificities. The translation of audiovisual comedies, whether via subtitling or dubbing, has been previously studied in numerous articles and books (Zabalbeascoa 1996; Attardo 2002; Martínez-Sierra 2006; Bucaria 2008; Jankowska 2009). The one element that clearly emerges from the research is the fact that humour is culturally bound and its translation may be a real challenge in that respect, especially when coupled with technical constraints and no possibility of footnotes:

> The translation of cultural elements becomes even more complicated when a movie [or any audiovisual works for that matter] has to convey entertaining factors and a translator has to face various humorous phrases, wordplay, allusions, irony, or idiosyncratic metaphors. It follows that humor contains a wide array of linguistically and culturally based expressions, which make a translator's work difficult because some elements seem to be hardly translatable. (Burczynska 2012)

Some researchers have already offered models for understanding translation strategies regarding humour and helping translators dealing with humour. Zabalbeascoa (1996) offers a model of translation based on priorities (the intended goals in the translation) and restrictions (the obstacles and problems justifying the choice of priorities). All depend on the type of audiovisual work, of course, but concerning TV comedy (animated or not), the priorities are clearly to entertain, to be funny, and then achieve popularity (Zabalbeascoa 1996: 245). His model classify jokes from the translator's perspective: the international joke; the national-culture-and-institutions jokes; the national-sense-of-humour-joke; the language-dependent jokes; the visual jokes and the complex joke. Martínez-Sierra proposes to complete the model with another classification: Community-and-Institution; Community-Sense-of-Humor; Linguistic; Visual; Graphic; Paralinguistic; Sound; Non-Marked (Martínez-Sierra 2006). The strategies deployed to translate the jokes will depend on both priorities and restrictions but also on the type of jokes.

Attardo (2002) also offers a metric to approach humour translation using the six Knowledge Resources of the General Theory of Verbal Humour: Language,

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Narrative Strategy; Target; Situation; Logical Mechanism; Script Opposition. All the Resources are classified in a linear scale and "the degree of perceived difference between jokes increases linearly with the height of the Knowledge Resource in which the two jokes differ". (Attardo 2002: 183)

Thus, according to the humour displayed in the source text, translators may have to find creative solutions and sometimes even completely adapt the joke to achieve the same effect of the audience (or not, according to the given priority – see Zabalbeascoa 1996). There also remains the question of censorship and acceptability that can be linked with humour, as it is culturally bound, and each culture has its own boundaries. Bucaria (2008) is interested in the ways dark humour crosses cultures in audiovisual translations (English and Italian languages) and concludes that dark humour is mostly transferred in Italian, and when omitted, it is more probably because of linguistic or technical constraints than censorship or acceptability. That may not be the case in every culture or/and for every type of humour.[4] *Rick & Morty* offers a new perspective on the question as humour is combined with science fiction's cognitive estrangement. The models previously proposed by researchers (especially Attardo 2002 and the use of the General Theory of Verbal Humour in relation to translation) do not consider the specificities of science fiction or science fictional neologies. A new model must be developed to understand the intricacies of the fictive terms and the translation challenges they create.

2.2 Methodology

To understand how *Rick & Morty*'s fictive terms are transferred into French and to analyse whether it is possible to retain both the science fictional and humorous features of the terms, all the science-fiction language-related elements were extracted from seasons 1 to 4 in the source text as well as their dubbed French translations.[5] According to their context, they were divided into three categories I will detail in the analysis section: fictive terms, referential parody, and novum parody. To answer the research question, the focus of the chapter will be on novum parody as they are the fictive terms with the most humorous traits. 388 terms were identified out of the 1,139 extracted. The analysis I propose in the chapter is both quantitative and qualitative.

[4] In the TV show *Friends*, for example, the character Monica says to Joey and Chandler "Rachel and I will kiss for one minute" and it was dubbed into French as "Rachel et moi on vous embrasse pendant une minute", shifting the classic heterosexual male fantasy of two women kissing each other into a more "traditional" proposal of two women kissing two men.

[5] The analysis of the humorous and science fictional features of the terms also drew on my linguistic skills as a French native speaker.

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After the extraction and categorization of the fictive terms, all the terms falling under the category "novum parody" were analyzed and categorized to answer the question: how are the terms created in the SL? Nine categories were identified and each one of them is based on objective linguistic criteria (morphosyntactic and semantic).

Then, the corresponding target terms were analyzed to clarify whether they retained both the science fictional and humorous features of the source terms.following Korpi's concept of generic fluency and the two strategies identified: retentive strategy and recreative strategy. Then, I analyzed the humorous aspects of the terms in the dubbed version, and three translation strategies were identified: non-parodic, adaptation, and preservation.

Finally, to clearly identify whether novum parody terms are both cognitively estranged and humorous in the French dubbing, four translation effects were established according to the previous strategies and the visual and narrative context of the terms as "A competent translator should be able to recreate the images that the source text tends to generate in the audience". (Elejalde Sáenz 2020: 47) The audiovisual nature of the show implies a strong connection with the visual content, and as in any artistic work, the context of the terms is also crucial to create a particular effect on the audience. I established four different translation effects based on linguistic criteria and the broader context of the terms,[6] they will be defined during our analysis section: omission, humour, cognitive estrangement, and full retention.

The categorizations that are offered here were the results of my own analysis which has been later reviewed by my colleague Mr. Thierry Robin, professor in English Literature and Language, whom I warmly thank for his time and energy.

3 Analysis

3.1 Linguistic creativity

Rick & Morty displays tremendous linguistic creativity not only to consolidate the sci-fi universe, but also to make the audience laugh and to develop the characters. After the extraction of all science fiction language-related elements, three types of linguistic creativity were identified:

- Fictive terms: typical invented terms that can be found in many science fiction works. They describe novums as rooted in our scientific and technological knowledge. Even if the science displayed is not quite accurate

[6]It should be interesting to compare our results with questionnaires on real spectators.

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(or is completely extrapolated), the concepts or objects named with fictive terms have a rational explanation in the science-fictional world. Many fictive terms are coined using pre-existing languages (Latin, Greek, English, French, etc.) and usual word-formation patterns (Gindre 1999). Some others are in fact neosemes; they can already be found in dictionaries, but their meaning is derived to reflect a new reality.

  - "portal gun": N+N describing a new invention allowing characters to go from one place in the universe to another through a portal.
  - "insect": neoseme describing a race of alien.
  - "Gazorpians": N+suffix describing a race of alien with the typical suffix "-ian" used to indicate a relation between a person and a place.

- Referential parody: references to other pop culture works.[7] The authors pay tribute to a specific work or artist within the context of an episode.
  - "to Cronenberg the world up": to turn people into hideous monsters resembling Cronenberg's works with body horror.
  - "to go blade runner": to rebel against humans, as in *Blade Runner* (Scott 1982).
  - "Snake Net": a wordplay with "Sky Net", the deadly artificial intelligence leading the robots in the dreadful war between robots and humans in *The Terminator* (Cameron 1984).

- Novum parody: all the fictive terms of *Rick & Morty* that ground the science fictional world but clearly amplify some tropes and features of the genre to such an extent that they become parodic, or at least humorous. Novum parody terms are created through different linguistic devices, but their humorous features stem from science fiction.
  - "a flumbus": a new and weird object.
  - "to Glorzo something": the infestation of the Glorzos (an alien race that acts as a parasite) on a person or a planet.
  - "mega-fruit": the fruit of the mega-branches. Here, the use of an imprecise prefix for a wonderful new discovery transforms the novum into parody.

[7]The direct references such as quoting the title of the work, or the name of the artists were not included in this category as they were not considered as novums but only references.

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The focus of the analysis will be on novum parody, as they are lexical or phraseological creations that intertwine both science fiction and humour. Indeed, fictive terms do not contain any humorous traits and are mostly used to ground the sci-fi world and create plausible novums. On the contrary, referential parody is made to be funny and pay tribute to many of the classical texts of pop culture. However, if referential parody elements participate in the multi-referentiality of the show, their humorous features mostly lie in the references rather than in linguistic creativity.[8]

To be clearer on what was considered as novum parody, here is the list of the criteria:

- Any new inventions, objects or concepts which exaggerate sci-fi tropes using linguistics (word-formation pattern especially).
- Terms which are not parodic by themselves, but become so through repetition and accumulation:
  - "alien hospital"; "alien towel"; "alien Internet" in season 2, episode 8.
  - "infinite sisters"; "infinite grandkids"; "infinite daughters" repeated in the entire show.

On the contrary, the category "novum parody" does not include:

- All the alien words that are supposed to be from an alien language (*except* when they are used with English words).
- Proper names (toponyms, anthroponyms and alienonyms).
- Titles of other artistic works.
- Terms containing the adjective "sci-fi" as these terms were considered as part of the show's metadiscourse which also deserves its own analysis regarding translation.
- All the insults.[9]

[8]However, some instance of referential parody use linguistic creativity to highlight the tribute: "Isaac Asi-hole" (s3 ep09); "She Clive Barkered herself" (s3 ep05); "Are you seriously Sawing the Vindicators?" (s3 ep04). Referential parody deserves its own analysis regarding translation as the challenge lies on both cultural references and humour, not so much on cognitive estrangement. Maybe in our next adventure.

[9]Insults can be fertile ground for creativity, but they are not specifically designed to ground the sci-fi world, even though some of them use the genre to be even more creative and create new insults only understandable in the science fiction world.

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3.2 Parodic strategies in the original

As the aim of the chapter is to evaluate the balance between the sci-fi aspects of the terms and their humorous nature in their French translation and according to the audiovisual nature of the text, there is a need first to categorise how the authors have created the comedic aspect of novum parody. After analysing all the 388 terms, nine categories were identified through specific linguistic criteria. Some terms can be classified in more than one category but, in order to facilitate the quantitative analysis, only the main category of humour was considered in the analysis.

3.2.1 Linguistic parody

Linguistic parody includes all the terms that highlight the absurdity of the novum through recognisable lexical patterns ("nano-doctor", "holophobic") as well as the terms that are supposed to show language evolution ("the boom-boom"). Some of the terms use non-existing roots that only sound like they belong to a technical or scientific lexicon through known affixes.

- "headward": the humans see giant heads appear in the sky and some of them create a religion. "headward", a transformation of "toward", means to go towards the heads.
- "universe-destroying un-thought-out technology": the accumulation of modifiers creates a term that could resemble a technical term, but the elements are in fact imprecise and non-technical ("un-thought-out"), parodying this kind of technical jargon in science fiction.
- "defraculate": there is no real definition of the term as Rick (or anyone) doesn't explain what a "defraculator" does. The term uses affixes "*de-*" and "*-ate*" that are common in English words, especially in technical terms (*hydrogenate*, *deactivate; denominative; degrade*).

3.2.2 Semantic parody

Semantic parody includes all the terms whose parodic or humorous features are found in the semantic content. It is not as much as the way the terms are morphologically created, but the meaning of the new collocation which is absurd or parodic (exaggerating science fictional tropes). All terms that stage an absurd novum are included in semantic parody as well as those that already exist but are redefined by the TV show through a new absurd meaning.

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- "infinite daughter": the repetition of the modifier "infinite" throughout the TV show highlights the absurd aspect of the collocation. It means that the multiverse is infinite, therefore Rick has an infinite number of his daughter.
- "a perfectly realized, toilet-filled simulation of heaven": the idea behind the term is quite absurd as "toilet-filled simulation" and "heaven" seems two opposite ideas.
- "giant hidden camera": the whole idea of a hidden camera is to be small and undetectable. The "giant" nature of this particular camera gives the object a humorous tone.
- "sound-erasing sneakers": shoes that erase any sound. The novum is humorous because the invention is quite absurd and useless.

3.2.3 Familiarising estrangement

Familiarising estrangement includes the terms that transform familiar collocations or concepts by adding a sci-fi (often exaggerated) element. Concepts and objects sound quite familiar to the audience but are twisted to sound science fictional. It is often used in the description of alien or alien environment; thus, it is far-fetched to assume those human concepts and objects will be used as such in extraterrestrial societies.

- "Morphizer customer service": a Morphizer (a non-parodic fictive term) is an alien machine that gives the ability to shrink or grow. In the episode in which the Morphizer appears, the character using it has some trouble with the controls. One piece of advice is to call the Morphizer customer service. Once again, customer service is a human concept and applying those concepts to alien technology is humorous.
- "Planetary mindset": the idea that someone who is born and lives on a planet has a particular way of thinking is humorous as it turns social theories to a much larger scale between the ones living on a planet and the intergalactic ones who fly around in space.
- "Low-calorie petroleum flakes": the liquid substance that is used to refine products or create gasoline, and that is highly toxic in our human world is in fact used by aliens as an ingredient in their cereals, but do not worry, they are low-calorie!

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3.2.4 Phonetic estrangement

Phonetic estrangement includes all the terms that sound alien but are used alongside English words to give a sense of (alien) exoticism. This is also a parodic tool as the use of phonetically estranged words with English sentences amplify the science fictional trope that English will be used in the entire galaxy as the *lingua franca*, which is quite absurd. The phonetic estranged terms are often created with unusual sounds for the English language, making them humorous.

- "give me the Blimfarx": the audience has no clue as to the meaning of the word "*blimfarx*" when heard the first time as the sentence is pronounced in the background by an alien The randomness of the sentence and the use of a very exotic phonetics makes it humorous. Later, it is understood that "blimfarx" is intergalactic money.
- "To squanch": a very versatile word used as a verb, a noun or even an adjective by a character named Squanchy from Planet Squanch. It can also be used to imitate existing word, such as "ensquanché" for the French "*enchanté*". When used as a verb, "to squanch" can refer to a sexual act, but otherwise the noun and adjective "*squanchy*" can in fact replace any other words:
  - "Rick! Squanchy party, bro!"
  - "I'm squanching in here!"
  - "Hey, squanchers! The party squanches out at my place"
- "plumbus": a device used for many different things. This object appears in a commercial video that explains how plumbuses are made.

3.2.5 Undertechnicalization

When visiting a science-fictional world with a mad scientist as the main protagonist, the audience expects to see many brilliant inventions with advanced technology, whether invented by the scientist or by intelligent beings. The authors play with this expectation by oversimplifying some of the inventions or discoveries. This can be attained using imprecise terms to describe pseudoscientific novums, the familiarization of scientific notions or objects with daily items or the use of the familiar register to describe pseudoscientific novums. In the context, it is a parodic strategy as the expectations of scientific or technical terms to describe the sci-fi word is counteracted by oversimplified terms.

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- "Snake-makery": to turn someone into a snake. The term uses the noun "snake" with a derivative "makery". For an invention that is quite amazing (as far as I know, nobody knows how to turn people into snakes or into anything for that matter), the term does not reflect the complexity of the concept.
- "Regular matter": this is the opposite of dark matter. The adjective "regular" sounds very general and not scientific.
- "Special grappling shoes": it describes shoes that allow its owner to stick on any walls. The adjective "special" is used to name the invention but it is not a very technical term.

The spaceship is also referred to as "the thing" throughout the show: "fly this thing" or "to land the thing" as if it was any familiar object (which it is for Rick and Morty, but not for the audience).

3.2.6 Accumulation

The accumulation strategy involves the use and re-use of invented terms in a short period of time or repeatedly throughout the episodes. The accumulation strategy uses terms that are often parodic in themselves and so belong to another strategy; the fact that they are repeated numerous times adds the strategy of accumulation.

- "Gear wars", "gear college": the terms describe a world in which people are made of gears ("gear people"), so in a world with gear people, everything is made out of gear.
- "Infinite daughter", "infinite grandchildren": as explained above, Rick travels through the multiverse and meets a great number of versions of himself, of Morty and any other characters. Thus, in his point of view, he has infinite daughters and grandchildren as the multiverse offers an infinite version of them.
- "Defraculate", "defraculator", "fraculation": in only a few seconds, all the terms are pronounced, which highlight the absurdity of the inventions that are not defined by the characters:
  - "- Look, I'm not paying 70 smidgens for a broken defraculator.

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- That is multi-phase quantum resonator.
- Well, does it defraculate?
- No.
- Then, it's a broken defraculator.
- Like you would even know dick about fraculation."

3.2.7 Metadiscourse

One of the specificities of *Rick & Morty* is its use of metadiscourse. Repeatedly, characters, and Rick in particular, shows that he knows he is in a TV show.[10] However, the parodic strategy of metadiscourse in our chapter describes specifically the use of the characters as novums, as if they were objects or concepts of the science-fictional world themselves.

- "anti-Rick speech": a speech that defies the very notion of "being Rick". Just like there are anti-capitalist speeches, there are anti-Rick speeches.
- "Rickless": a character, Morty in particular, who doesn't have a Rick by his side.
- "a Summer"; "a Rick"; "Mortys"; "Ricks": as all the characters have infinite versions of themselves through the multiverse, one version is randomized and can be used with an indefinite article or with a plural. The characters seem depersonalized as though they mean nothing in the whole universe which meets the nihilistic tone of the TV show.

3.2.8 Wordplay

The wordplay strategy involves the transformation of an existing idiom or term to create puns. The new idioms fit the science fictional world the audience is exploring while being familiar.

- "Plutanamo Bay": it describes a prison on Pluto. The collocation is a transformation of an existing location in the United States, also known to be a detention camp: Guantanamo Bay.
- "To hit dream bedrock": in the episode, Rick and Morty are chased by a Freddy Krueger character (Craven 1984) in their dreams. At one point, they

[10] "You ruined the Season 4 premiere" (s4 ep01); "break the fifth wall" (s4 ep06); the use of the adjective "sci-fi", "a sci-fi word" (s2, ep06), "sci-fi friends" (s1 ep11).

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have reached the last level of dreams, and they can't escape anymore in someone's else dream (the whole episode is a crossover between Freddy Krueger and *Inception* [Nolan 2010]). This is a transformation of the idiom "to hit rock bottom".

- "Gorgon shit": the authors replace the noun "bull" in the term "bullshit" by "gorgon" which, let's assume, is the equivalent of an alien bull.

3.2.9 Technobabble

In science fiction, technobabble refers to all the pseudo-scientific discourses that characters use to explain complex scientific or technical phenomena. In fact, the speeches often have no scientific background, or they extrapolate altogether, but they sound scientific.[11] In *Rick & Morty*, the authors also use technobabble, but they use it in a humorous way to parody the genre, as it is clear for the audience that it is non-sense.

- "a couple cranks on the scaling knob with a four dilabyte cross fade would fix that in a jiff".
- "isolating a moment in time splitting your probable selves and shunting you into a near-duplicate equally probable reality, transporting you into it at the moment of parallel determination".
- "I put a spatially tessellated void inside a modified temporal field until a planet developed intelligent life".

3.3 Translating novum parody

The translation of novum parody has been examined through two lenses: its science fictional aspect and its humour. For the former, the question is whether the terms retain the cognitive estrangement or if it is neutralized (Korpi 2017: 93). In order to do so, I used the strategies established by Korpi with the concept of generic fluency: retentive and recreative. For the latter, the parodic features of novum parody translation, three strategies have been identified in our corpus: non-parodic, adaptation and preservation.

In the last part of this section, I will focus on the potential effect the translated novum parody terms have on the audience, if the translation meets dynamic

[11] The best examples of technobabble in audiovisual science fiction can be found in *Star Trek* (see Wozniak, 2014).

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equivalence (so the same *effect*) or not. Indeed, as *Rick & Morty* is produced for an audience, there is a need to understand how the translation of novum parody can affect the audience: have the changes in novum parody terms completely changed their perception in the target language? Here the question focuses on the intersemiotic relation between the dialogue (and so novum parody) and the images, which, undoubtedly, impact the translation choices (Torresi 2008; Kourdis & Kukkonen 2015). The analysis of the translation effects is, in the chapter, considered within the frame of linguistic criteria and the extralinguistic context.

3.3.1 Translation strategies

According to Korpi, if a translated science fiction text does not retain the neologies, the text is partially generically unfluent and so "the genre changes in the translation process" (2017: 89). The importance of neologies in defining the genre shall not be debated here, but I believe they are quite important when used by the author, and their translation can change the reception of the whole work, as they participate to the science fictional worldbuilding as well as the specific atmosphere of the work. The concept of "generic fluency" is relevant to understanding how the fictive terms are transferred from one language into another and if the estranged wor(l)ds are (re)constructed the same way in both languages. Moreover, the constraints of dubbing and the link with the visual content may determine the translation choices. With audiovisual work, novums can be contained in the visual context rather than in the dialogues, or be highlighted in both, which can offer a choice to the translator: when the novum can be seen, is the fictive term absolutely useful in the dialogue?

First, novum parody will be analyzed to understand if the cognitive estrangement of the terms is retained or neutralized in the dubbed version of *Rick & Morty*. The first strategy is "retentive": "by choosing a retentive strategy, the translator has chosen to retain the novum's cognitively estranging nature and preserved the neology as it is. Hence the genre remains, from the point of view of neologies, fluent in the translation" (Korpi 2017: 93). The second strategy is "recreation" which consists of "the extrapolation or generalisation of the neology." (Korpi 2017: 93). Thus, recreative strategy describes all the occurrences that eliminate, neutralize, or completely transform the science fictional aspect of the terms. The translator (or the constraints they face) may have chosen to use their creativity to highlight another aspect of the term of compensate later on another term. The first analysis will allow to determine whether the French dubbing is as cognitively estranged as the original version.

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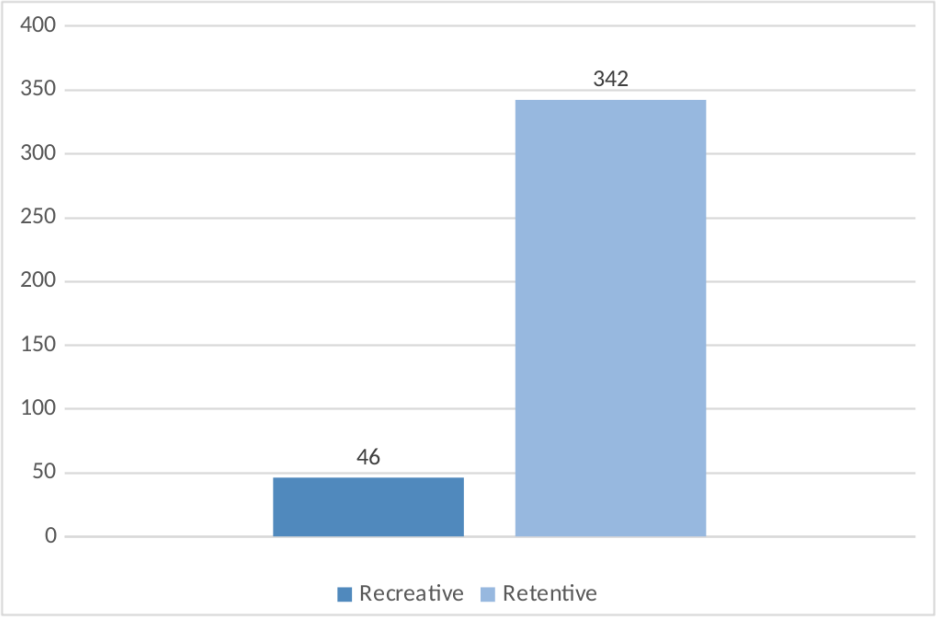


Figure 1: Retentive and recreative translation strategies in Rick & Morty's seasons 1 to 4.

As seen in Figure 1, most of the novum parody terms retain their science fictional nature as 342 (88%) terms employ a retentive strategy while only 46 show a recreative strategy (12%). Here are some examples of novum parody terms with a retentive strategy or a recreative strategy:

(1) SL: "shopping for a brand-new motherworld "
TL: "*partir à la recherche d'un nouveau putain de monde*"
S2 ep10[12]

(2) SL: "alien car battery"
TL: "*batterie*"
S2 ep10

(3) SL: "headism"
TL: "*tétaïsme*"
S2 ep5

[12]To provide a smoother experience, each term given as an example is identified by the season and episode number in which the term is used as follows: S1 ep01.

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(4) SL: *"The Rick-patented Anthology generator"*
TL: "*générateur d'anthologies breveté par Rick"*
S4 ep6

In examples (1) and (2), French dubbing neutralises the cognitive estrangement. In (1), the source term offers a two-layer novum: a. we can shop for a world, even figuratively; b. the idea that we can obtain a new motherworld is absurd. In the French dubbing, the estrangement is erased as the character is only talking about researching a new world for them to settle on, generalising the collocation. The same goes for example (2) with the omission of the adjective "alien" in the French version, so there is no novum or any estrangement. Both examples are terms offering a recreative strategy. The science fictional term or collocation are transformed in the target language and lose its parodic aspects. The cognitive image projected by the terms is completely recreated in French. However, unlike science fiction literature, the cognitive image is used with a visual image that will allow some recreative translation to retain their parodic features.

On the contrary (3) and (4) offer a retentive strategy as "headism" and "The Rick-patented Anthology generator" are dubbed into French with similar cognitively estranged terms. The first one is translated with "*tétaïsme"*, with the same lexical pattern N+suffix and (4) retains all the elements that give the term its science fictional features: "anthology generator"; "*générateur d'anthologies*" and the fact that it is "Rick-patented", "*breveté par Rick*". The linguistic constraints of French language require the use of a verb and a preposition "*breveté par*" but the cognitive estrangement is retained.

It is interesting to note that most of the novum parody terms with a recreative strategy were categorized in "semantic parody" (13 occurrences). Many of them are translated with non-fictive terms: "childproof world"[13] translated with "*monde sûr pour les enfants*" for example, and some others were completely omitted: "turbulent power" or "butt person".

The visual channel can still provide the cognitive estrangement and the novum even though the fictive terms are translated through non-cognitively estranged terms or are completely omitted. The novelty is no more in the language, but it is still on the screen (in the context of the scene, not specifically at the exact moment of the lexical novum in the source language): "butt person" for example

[13]Even though the pattern "something-proof" ("bulletproof") usually marks our safety against the "something", here it is the opposite, a "childproof world" is a world created by Rick for his daughter for her to be completely safe (the ground is bouncy, she can't drown in the rivers, etc.).

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is illustrated right before the lexical novum is pronounced and so the French-speaking audience can still have access to the cognitive novum (see). However, some of the novums are only present in the dialogue and when they are dubbed through omission or recreative strategy, they can be completely missed by the target audience. It is the case, for example with "space-skiing" translated by "*en dévalant les pentes*".



Figure 2: The "butt person", S01 ep08.

AVT is restricted by some technical constraints, especially isochrony in our case, as French actors cannot continue to speak when the characters on the screen have finished talking. As French has an expansion coefficient[14] more important than English, some fictive terms can be a real challenge to translate in respect to time constraints. That can explain some of the recreation strategy which, as will be seen in the next section, does not necessarily mean that the cognitive estrangement is neutralized, only that it is not contained in the neology *per se*. Here, the link with the visual content is the key: even though the local strategy is recreative, the reception of the term can still retain the sci-fi features. However, some terms using a recreative strategy could have been translated with a retentive strategy: "space-skiing" for example, which has been translated without any mention to the "space" nature of the sport. The answer here may be in the restrictions which change the priorities of the translators.

[14] "First, let's be clear: expansion is real. It is acknowledged, even institutionalised and quantified. […] Thus, for English into French translation […], the SFT [Société française des traducteurs] predicts an expansion coefficient of 25%." (Durieux, 1990: 55)

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Once each novum parody term has been categorized according to its translation strategy in regard to generic fluency, the same must be done with its humorous features. Humour fluency has been analyzed according to three parodic translation strategies: first, *non-parodic strategy* consists of erasing all parodic features in the target language (according to the parodic strategies defined above). The term can still be humorous (especially with the visual content), but it is not parodic of the genre anymore. Then, *adaptation strategy* defines terms that are still considered as parodic, but they do not use the same humorous device as in the source language (see 3.2); and finally, *preservation strategy* describes all the terms that use the same parodic strategy in both the source and the target languages.

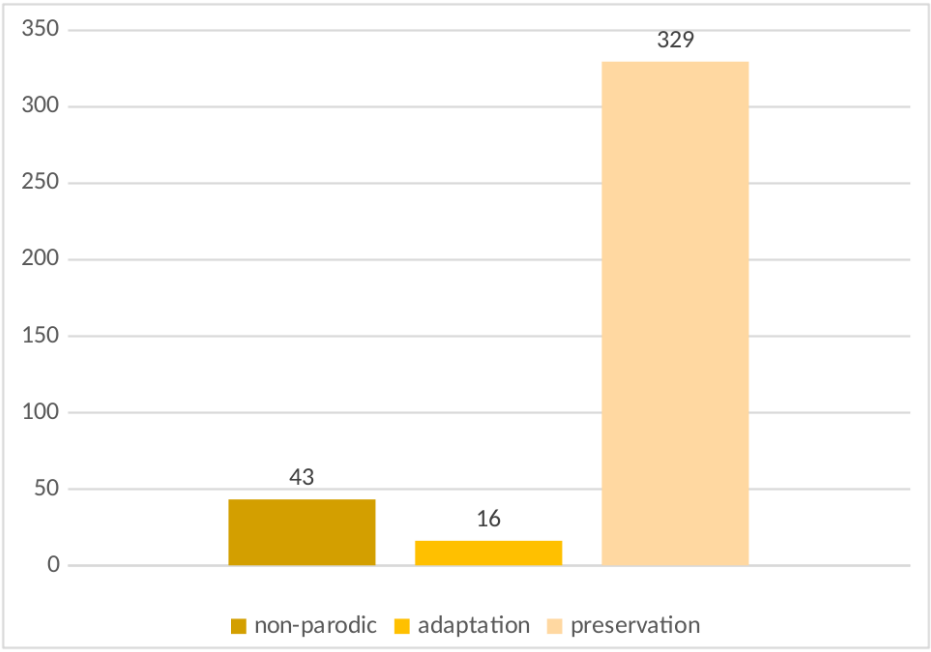


Figure 3: Humour fluency in the translation of Rick & Morty seasons 1 to 4's novum parody.

Most of novum parody has been translated with preservation strategy and retain their humorous nature (85%). However, 11% have been translated through non-parodic terms and 4% have been adapted.

(5) SL: "gear people"
TL: "*créatures à mécanisme*"

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S4 ep10

(6) SL: "unprecedented Rickicidal epidemic"
TL: "*vague de crime sans précédent*"
S1 ep10

(7) SL: "six-chewable figure income"
TL: "*salaire par voie orale à 6 chiffres*"
S3 ep01

(8) SL: "worm thing"
TL: "*vers de terre débile*"
S1 ep1

Example (5) retains the novum as the science fictional concept is preserved in the French translation. However, both the absurd collocation and the accumulation ("gear college", "gear cream", "gearhead") are neutralized. The same can be observed with (6), "unprecedented Rickidical epidemic". The audience can perceive the parodic feature of the term with the invention "Rickidical", ("Rick"+"suicidal" as only Ricks are murdering each other, it is kind of a suicide). In French, the novum parody is translated with "*vague de crime sans précédent*", which is an idiomatic collocation, but does not retain any of the genre or humorous features of the source term.

On the contrary, example (7) is translated with a term that retains both generic fluency and the parodic nature of the source term. "six-chewable figure income" is dubbed with "*salaire par voie orale à 6 chiffres*". Both terms use an idiomatic collocation but transforms it to fit the science fictional world. English adds the adjective "chewable" and French dubbing adds "*par voie orale*", which is quite absurd when attached to an "income" or a "*salaire*".

In example (8), "worm thing" is humorous because of its undertechnicalized nature (the worm thing is an alien entity resembling a worm). It has been translated with "*vers de terre débile*", which fits the category "semantic parody" with the collocation implying that: a. a "*vers de terre*" (worm) can be an alien (just like in the source term) and b. it can be stupid, and therefore the opposite can be true and a "*vers de terre*" can be an intelligent being (in the human sense). Here, the linguistic parody has been adapted with a new humorous strategy, but the dubbed version retains both generic and parodic features of the fictive term.

check agreement. Currently, it is Earth which is débile, due to singular agreement. Or maybe *ver* should be singular

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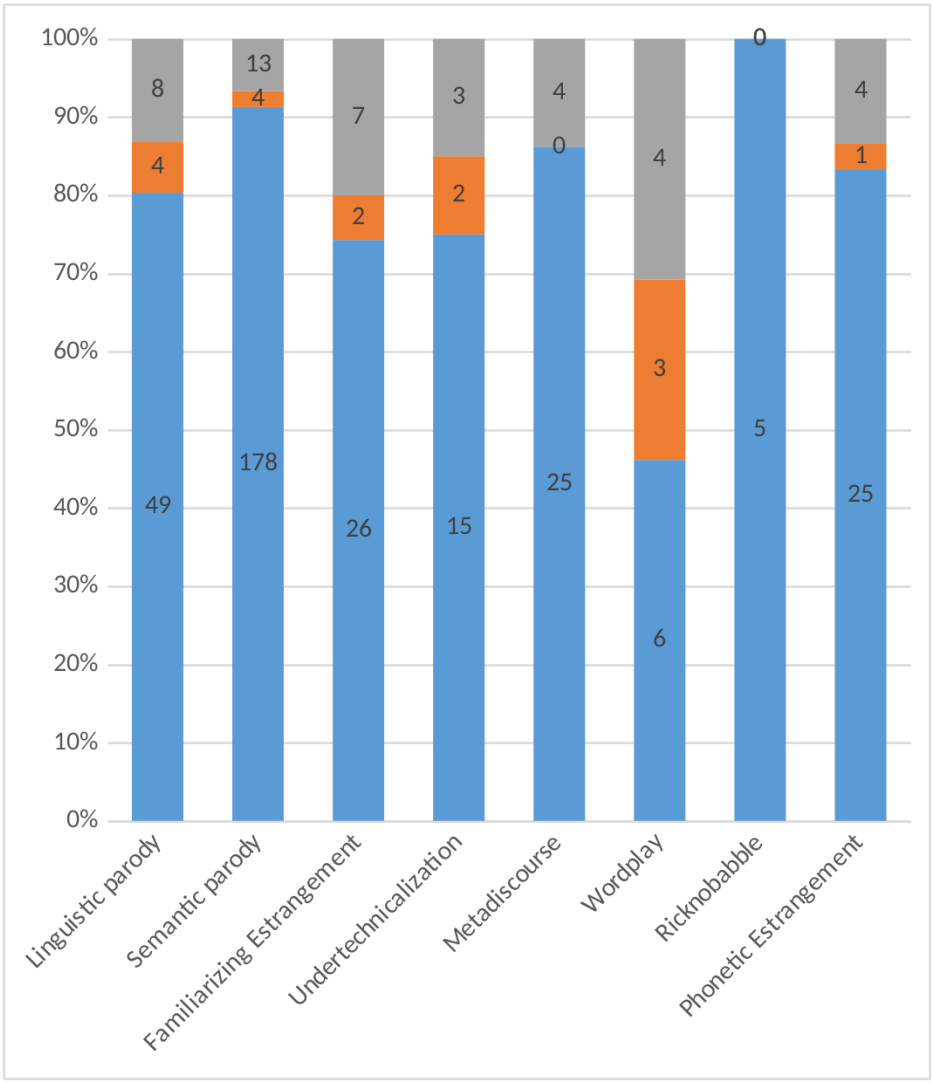


Figure 4: Parodic strategies and parody translation

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It is quite clear that both science fictional and parodic aspects of novum parody are priorities for translators (see) as 329 terms are translated with a novum parody occurrence. Nevertheless, technical and linguistic constraints require the translators to adapt generic and humour fluency. The strong connection between dialogue and visual content is the key to understanding the priorities when both aspects of novum parody cannot be retained in the dubbing process. For example, "regular legged partner" is translated with "*mec qui a des jambes normales*". The translation uses a recreative strategy regarding generic fluency as the term in French is not a lexical creation anymore, it is a description that, without context, doesn't create a new signified (it literally means "a guy with normal legs"). In English, the term uses a linguistic pattern rather technical as a parodic tool. The novum – and hence the parody – is created only through the visual content in French (see Image 2) in which "*mec qui a des jambes normales*" is visually opposed to a man who has the same legs as a baby. The translation still carries humour (not parodical anymore), but the novum relies entirely on the visual aspect of the scene.



Figure 5: Visual context of "regular-legged partner", S01 ep08.

Looking at the percentages, wordplays are mostly translated with adaptation and non-parodic strategies (54%), which can be explained by the fact that "puns [wordplay] pose special problems for translators because, unlike most kinds of verbally expressed humour, they use the specific features of a particular language" (Low 2011: 59).

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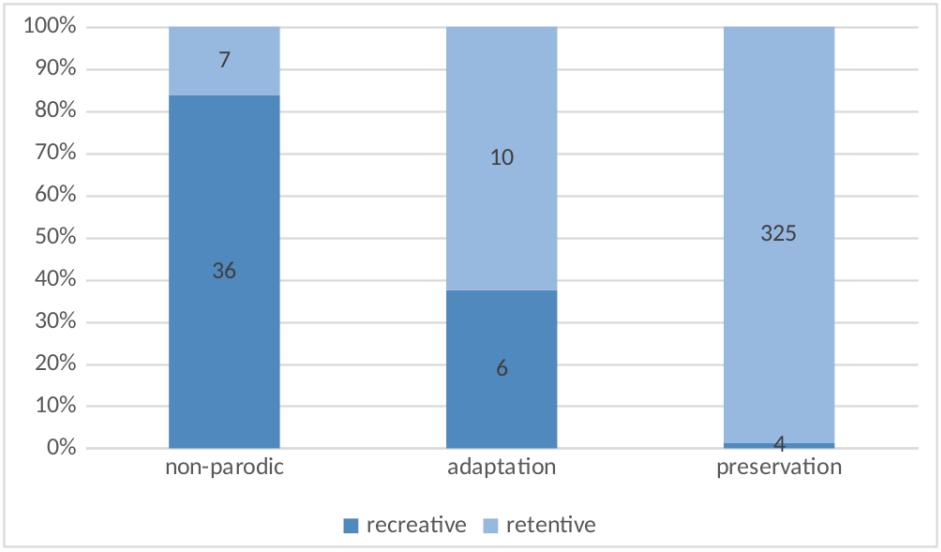


Figure 6: Parodic strategies and parody translation

Furthermore, on the total of undertechnicalized and phonetically estranged novum parody, 15% and 13%, respectively, have been translated with a non-parodic strategy. For undertechnicalization, the terms are either explained ("*changer qui que ce soit en serpent*" for "snake makery") or the sci-fi aspect of the term is neutralized ("*chaussures d'escalade*" for "special grappling shoes"). In the case of phonetic estrangement, in each occurrence the element that created the estrangement is in fact completely omitted in French ("*boîte special ménage*" for "Fleeseeks box" or "*céréales*" for "strawberry nibbles").

Familiarising estrangement also shows many terms translated with a non-parodic equivalent (20%). Most of the terms are in fact translated with non-science-fictional terms. In the French version, they are still familiar terms, but they are not estranged anymore ("*site de rencontre*" for "interstellar dating service"). Here, it seems that the cognitive estrangement relies, in fact, on the visual content and narrative context of the scenes.

(9) SL: "egg-vite"
TL: "*invitation*"
S2 ep10

(10)

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(11) SL: "the bone train"
TL: "*on peut y entrer qu'en passant par derrière*"
S1 ep03

(12) SL: "manipulating the fabric of shit"
TL: "*sacré putain de merdier*"
S4 ep05

In example (9), the wordplay "egg-vite" is neutralized in the French dubbing and translated with "*invitation*" which is a word that already exists and whose meaning is not changed within the context of the TV show. In English, the wordplay works with the visual content as the wedding invitation they receive comes in the form of an egg[15] and the word "egg" has the same number of syllables as "in" and can be associated phonetically with "vite". In French, "*oeuf*" will be difficult to associate with half the word "*invitation*": "*oeuvitation*" omits the last syllable of "*oeuf*" and the wordplay may not be properly understood (even though the visual content is a clue) and "*oeufitation*" is, in fact, not an easy term to pronounce for French speakers. The compound may not be as humorous in French, and the translator has preferred to omit it, as the visual content provides the novum (the form of the wedding invite) and therefore, the cognitive estrangement.

On the contrary, in example (10), the translator has adapted the wordplay. Indeed, "the bone train" plays with the double meaning of "bone" as the part of a skeleton (they are inside a body which has been transformed into an amusement park) and the meaning "to have sexual intercourse". In French, "*en passant par derrière*" is also ambiguous as it can literally mean "to enter from behind" or "have anal sex". Even though the translator has transformed the sentence, they have kept the sexual ambiguity of the wordplay. However, as the French expression does not refer to an object anymore "the [literal] bone train", it is not cognitively estranged. Here, the translator decided that the priority was the wordplay, not the novum (which is visually present on the screen, even though the link between the sentence and the train is not clear in the dubbed dialogue).

The last example (12) was created in the SL with undertechnicalization: "manipulating the fabric of shit". The first part of the term "manipulating the fabric" is quite technical and sounds pseudo-scientific, but "of shit" transforms the fictive term into novum parody. In French, the translation is still funny because of the colloquial and idiomatic language, "*sacré putain de merdier*" ("holy fucking shit"), but it is not parodic anymore as there is no science fictional feature.

[15]The character who marries is a hybrid between a human being and a condor, he lives on Birdworld and is called "condorman".

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However, it is clear that the French dubbing mostly tries to retain the parodic nature of novum parody as the preservation strategy is the most used (85%). Both sci-fi and parodic features of novum parody are priorities in the dubbed French version of the show. Nevertheless, some restrictions may challenge the translators who need to prioritise according to the visual and narrative context (whether the joke is more important than the cognitive estrangement or otherwise, whether the object can be seen on the screen, etc.) and the creative solutions they may come up with. That is why the local strategies which have been analyzed in the section may not be an accurate account of generic and parodic fluency. The parodic fictive terms must be analyzed with their visual and narrative contexts.

3.3.2 Translation effects

First and foremost, *Rick & Morty* is an entertaining TV show and as such, everything, from the fictive terms to novum parody, is produced for the audience. The reception of novum parody is essential to understanding its translation as they are not merely linguistic units but are constitutive of a whole context. Indeed, Korpi notes that

> [n]otably the dichotomy between the retentive and recreative local strategies does not mean that using one or the other would simply guarantee the generic fluency of a text. The results indicate that many of the local strategies can be both retentive and recreative, depending on the context. (Korpi 2017: 94)

The translation of novum parody must also be studied according to the context, narrative and visual, in which the audience will hear them.

Using the previous analysis on generic and humour fluency, this section is dedicated to the analysis and categorization of the translation effects on the audience according to linguistic criteria. Four effects have been identified: *full retention*, *cognitive estrangement*, *humour*, and *omission*. Here the skopos theory established in the functionalist approaches to translation (Reiss & Vermeer 1984; Nord 1997; Vermeer 2012; Nord 2020) will be useful as the function of novum parody is prioritized according to the visual and narrative contexts. The translator needs to understand whether the humorous feature is more essential than the cognitive estrangement, or otherwise, when both can't be retained because of linguistic or technical constraints.

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I will use the concept of Nida's dynamic equivalence: I considered equivalent novum parody terms that retain both generic and humorous fluency, hence creating the same *effect* on the target audience (the novum parody fulfils the same function than in the source text). The functional aspect of the terms is thus taken into consideration just as much as their semantic content. As Nida noted, dynamic equivalence characterises a translation in which "the message of the original text has been so transported into the receptor language that the response of the receptor is essentially like that of the original receptors" (Nida & Taber 1969: 200). However, because the term "equivalence" is deeply rooted in the theoretical background of Translation Studies, I prefer to use the term "full retention" to describe the terms which retain both their cognitive estrangement and their parodic features (340 occurrences), whether the translator used literal translation or chose a creative approach. The novum or the humour can be adapted, modified according to dubbing or linguistic constraints (or even the translator's own creativity), but the target audience is offered the same science fictional humour experience.

- "a frog restaurant" > "un resto pour grenouilles"
- "Alien terror attack" > "*attentat terroriste alien*"
- "this magic stuff" > "*ces bidules magiques*"

The cognitive estrangement effect (25 occurrences) includes all the target terms that translate the science fictional novum into French without any parodic element. The term is no longer funny or absurd, but the audience can still have access to the lexical novum. Therefore, the novum parody term is generically fluent but it is not parodically fluent anymore, it becomes a plain fictive term. Translation effects take into account the visual and narrative contexts of the terms. Even though the term does not retain locally the cognitive estrangement, its use in a specific visual context can create cognitive estrangement.

- "go into god-mode" > "avoir des pouvoirs divins"
- "Plutoquakes" > "tremblements de Pluton"
- "swapped our brains" > "échanger nos processeurs"
- "special grappling shoes" > "chaussures d'escalade" is accessible to the audience in a narrative context where the novum is explained: "grâce à ces petites merveilles de technologie, tu peux grimper sur n'importe quel type

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> de surface, tu peux aller en haut, en bas, en dessous ou tourner à gauche. En clair, tu peux défier la gravité. ” (S01 ep01).

On the contrary, the humour effect (10 occurrences) encompasses all novum parody that does not contain any science fictional features but is still humorous. The audience has access to the comical aspect of the term, but the science fictional concept or object associated with it has been neutralized in the French dubbing. Here, however, the science fictional aspects can be conveyed through the visual channel.

- “gold rickers” > “bande de trous du cul”
- “a quick hand bond” > “une petite session manuelle”
- “regular legged partner” > “mec qui a des jambes normales”
- “interstellar dating device” > “site de rencontre” is used by one of the character who happens to be dating an alien (see). The visual context here highlights the comedy aspect of the situation.



Figure 7: The visual context of the “interstellar dating service”, S03 ep09.

Finally, the omission effect (14 occurrences) describes all novum parody that was not translated as cognitively estranged or humorous, or not translated at

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all. Here the French-speaking audience loses both the science fictional and the humorous nature of the terms which become both unfluent generically and humorously.

- "murder log" > "carnet"
- "working-stiff Ricks" > "employés"
- "peeling skrabquams and slipping nib nibs" > "bosser *comme un esclave*"

3.3.3 Translation strategies and effects

By contrasting both translation strategies with translation effects, some interesting results have emerged. For example, a substantial proportion of non-parodic terms (43) in the target language also show recreative strategy concerning generic fluency (except for seven occurrences) and if 14 show an omission effect, most of them (21) fall under the cognitively estrangement effect.

Concerning the omission effect: the dubbing constraints didn't enable the translator to transfer the lexical novum into French, but the context did not make humour a priority and so novum parody did not transform into a simple joke or a pun either. The priority appears to be in the explanation and expansion rather than estrangement or parody which can be balanced by the visual content, hence the omission of both features.

However, the cognitive estrangement effect can be created even though the translated term uses a recreative strategy: the visual component of *Rick & Morty* allows the translator to omit the linguistic novum as there is its visual manifestation on the screen or a narrative continuity; thus, gaining time to expand another aspect of the dialogue (see example [18], the "special grappling shoes").

The same thing goes with the humour effect: even though some parodic fictive terms are not parodic anymore in the TL, the visual content and the context in which the characters are evolving can still be humorous, but the humour is only conveyed by the situation in which the words are uttered and not by the words themselves (connotation, ambiguity, etc.), see the "interstellar dating service" example.

The results show that if full retention often means retentive and preservation strategies, recreative and adaptation strategies can also lead to full retention, with the context in which the term appears or with an adaptation of the parody strategy to fit linguistic constraints:

- "money bugs" > "la compta"

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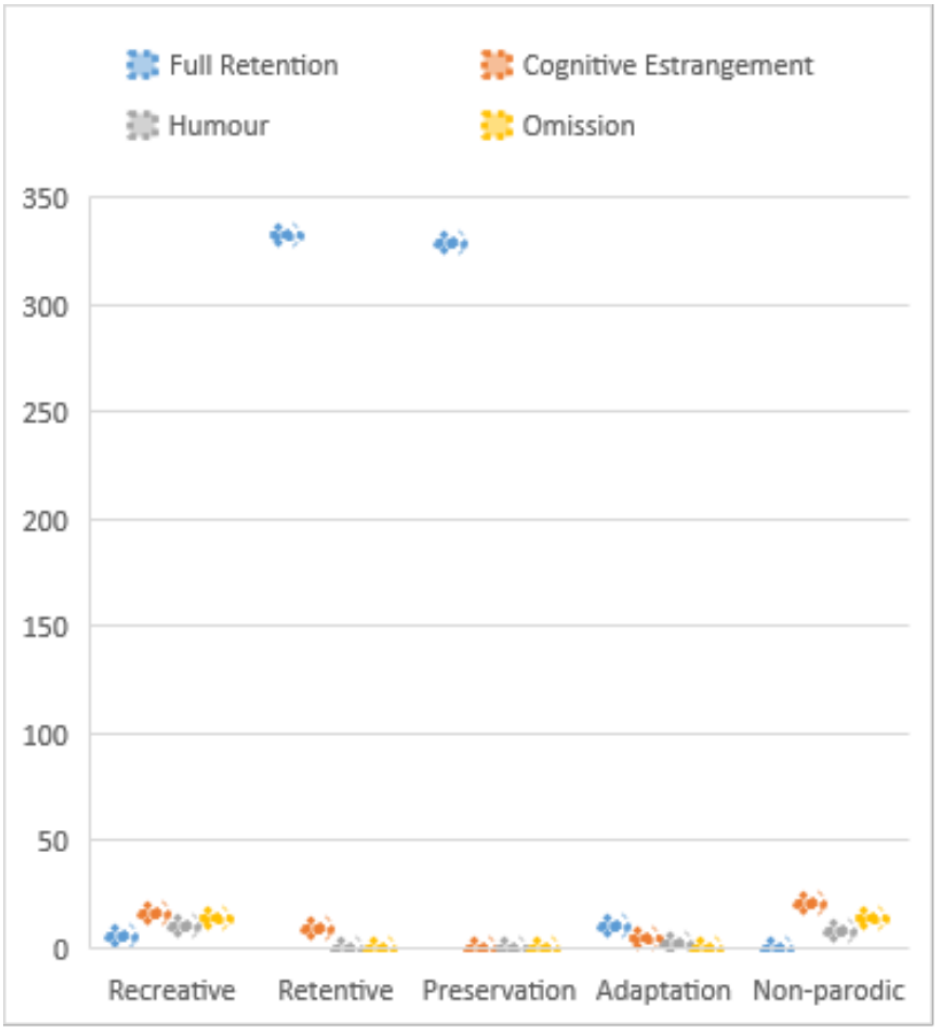


Figure 8: Translation effects and strategies

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- "gearsticles" > "avoir des testicules à la place de la bouche"

shows that the priorities of the translators are both the science fictional nature of the terms as well as their parodic features. However, when the constraints of AVT or the linguistic constraints from English into French are too heavy and do not allow translators to keep both features, they prioritise according to the context, visual and narrative. When humour is mostly conveyed by the term, they keep it, but when the context is funnier and the parody is only the cherry on top, they prioritise the science fictional aspect of the term. Of course, given the nature of novum parody itself, when the term is not cognitively estranged, the parody is either transformed into another form of humour, or completely neutralized. In most cases, both features are retained (87% of full retention effect), even though one of the aspects may only be present in the visual and narrative contexts, and not in the lexical creations themselves.

3.4 Testing the typology: *Futurama*

Futurama is an animated TV show created by Matt Groening. It has run for eight seasons from 1999 to today, and has been produced by Fox, Comedy Central and Hulu. As *Futurama* is also one example of humorous sci-fi, it seemed like a perfect choice to test the model. In brief, *Futurama* tells the adventures of the Planet Express' crew, an interplanetary delivery company in the 31st century, and especially Fry (a human cryogenized from the twentieth century), Leela (a mutant) and Bender (a robot). Although the show isn't as nihilistic as *Rick & Morty*, it does parody the genre by exaggerating some sci-fi tropes and creating absurd situations. The series has had several translators and season 7 was adapted by Cécile Carpentier, Xavier Hussenet, Didier Duclos, Anne Bararuzunza, Flora Seeger, Hélène Castets, Blandine Gaydon, Émilie Barbier, Ludovic Manchette, Aurore Lafage, Thibault Longuet, Audrey Bernière, Cécile Favre et Christian Niemiec.

To test the typology established previously, I extracted all the sci-fi language-related elements from season 7,[16] aired from June 2012 to June 2013. I chose this season as it is the exact same year *Rick & Morty*'s season 1 was released. Then, I extracted the same elements from the dubbed French version of the show and used the previous methodology.

105 novum parody terms were identified[17] ("padded dashboard display"; "disem"; "delay the tentacles", for example). In terms of parodic strategies, I

[16]Sometimes considered as season 10 as some of the seasons were divided in two parts.

[17]Fictive terms and referential parody are also present in *Futurama:* "matter compressor" (S7, ep01) or "deflector shield" (S7, ep03) which is a reference to the TV show *Star Trek*.

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identified six of the previously defined categories (see).[18] However, I haven't found any examples of "phonetic estrangement" or "metadiscourse". For the latter, it can be explained by the differences between both shows. *Rick & Morty* uses dark humour to enhance the nihilistic discourse, and the characters (at least, Rick) are aware that they are in a TV show. In *Futurama*, the characters just live their lives without any knowledge of their fictional status, and the humour is light and parodic, but not nihilistic. It resembles more a science fictional workplace sitcom in many ways.

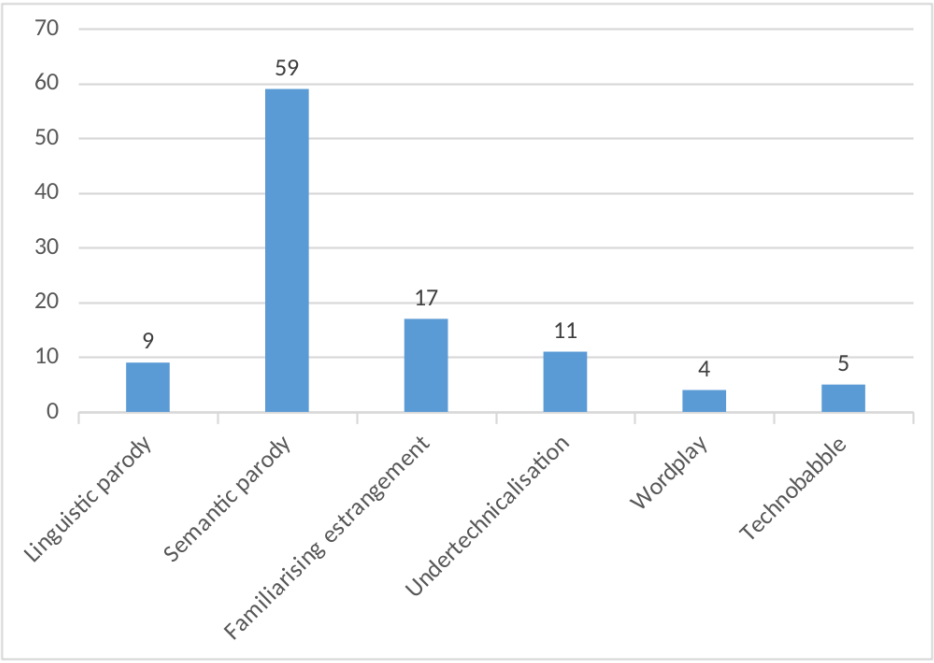


Figure 9: Parodic strategies in Futurama's season 7.

As can be seen in , most of the terms were translated through a retentive strategy (75%) and most of their humorous features were preserved (73%).

Here are two examples of novum parody terms to highlight the notion of priorities and how the relationship between text and image needs to be taken into account to understand translators' choices:SL: "Execubots"

[18]The category "accumulation" was found, but just like for the analysis on *Rick & Morty*'s novum parody terms, this parodic strategy is used in addition to another, and I didn't take it much into account for the test.

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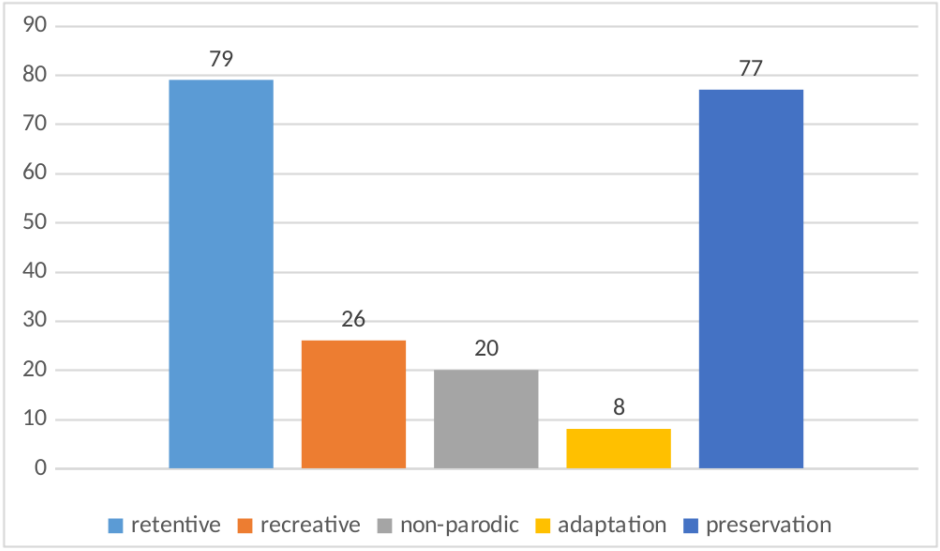


Figure 10: Retentive and recreative translation strategies in Futurama's season 7

(13) TL: "*Robonnets*"
Ep7

(14) SL: "Sewer Mail"
TL: "*C'est le facteur*"
Ep9

In , the term is created through linguistic parody by merging "executives" and "robots" (as they are executive robots) and its science fictional aspect is retained in the French dubbing as "*robonnets*" is also considered as a lexical novum ("*robot*" + "*gros bonnet*"). Here the linguistic parody is preserved in French (even amplified with the use of the familiar term "*gros bonnet*" as a compound's component).

On the contrary, in (14), the novum "Sewer Mail" which implies that there is a whole organized community living in the sewers, is not dubbed as a novum in French. "*c'est le facteur*" is very generic and not specifically science fictional. The familiarizing estrangement of the term is completely omitted, and the term is not linguistically humorous anymore. However, the visual content of the scene highlights the familiarising estrangement and makes "*c'est le facteur*" in French humorous even though the term is not linguistically funny (see).

The translators tried to retain both the science fictional and humorous aspects of the terms. However, the visual content must also be taken into account as it

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Figure 11: "C'est le facteur !".

can change the perception of the translated terms. One non-parodic term can still be humorous as the linguistic and temporal constraints may be counterbalanced through the visual content. Following the methodology, each novum parody term has been analyzed through the lens of the previously defined translation effects (see). Just as in *Rick & Morty*, the translators had both generic and humour fluency as priorities when dubbing *Futurama*'s episodes. However, the change in tone between both TV shows produces some differences: the science fictional features of the terms were sometimes omitted to better preserve the humorous features ("foos" translated with "*petits joueurs*").

In *Rick & Morty*, most of the terms which were created through recreative or/and non-parodic strategies create a cognitive estrangement effect (when not omitted) (see). In *Futurama*, indicates that, when not omitted, the novum parody terms created through recreative and/or non-parodic strategies still create a humour (6 occurrences) or a full retention (3 occurrences) effect. Therefore, humour seems to be prioritized in *Futurama* and can sometimes rely more on the visual content than on the linguistic creations.

4 Conclusion

This chapter deals with several notions such as generic and humour fluencies in an animated audiovisual work. The research question focused on the treatment

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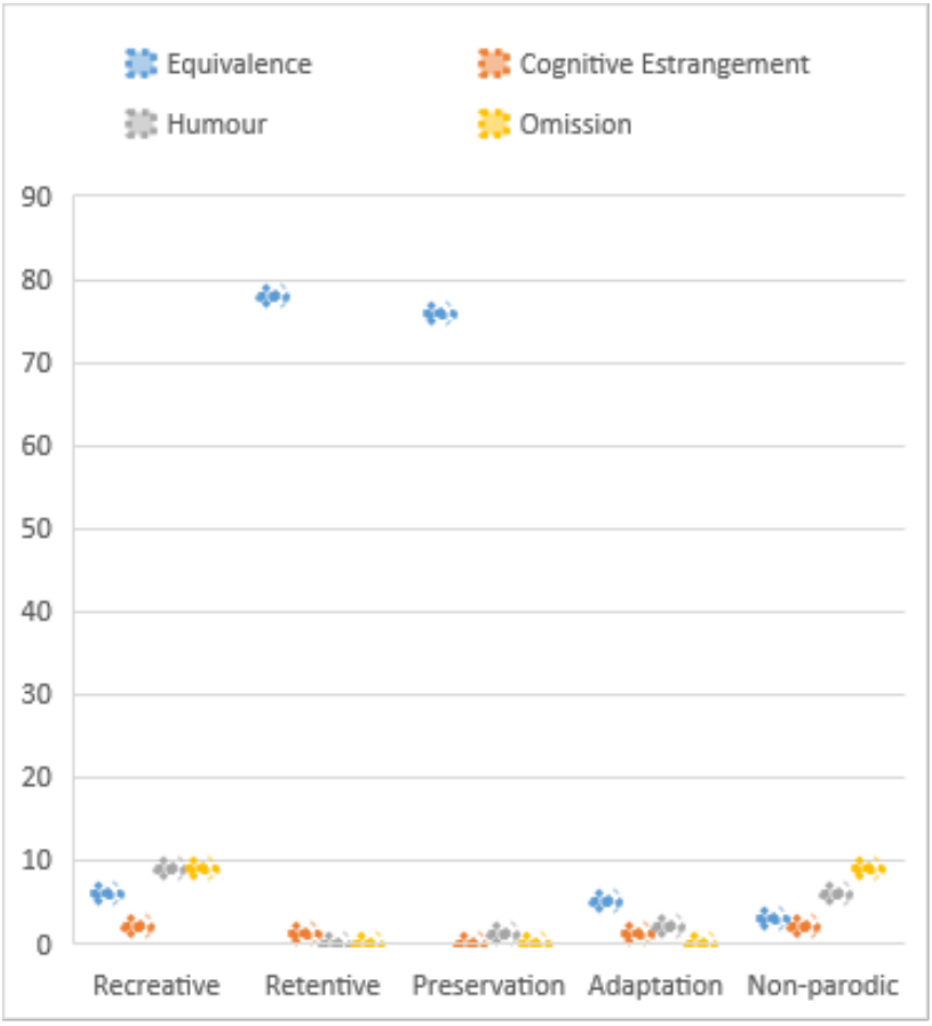


Figure 12: Translation effects and strategies in Futurama's season 7.

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of the different aspects of the science fictional terms that clearly use generic parody to create humour. The audiovisual, linguistic and cultural constraints may prevent translators from offering humorous fictive terms in the target language according to the context. First, different categories of novum parody have been established to understand how the invented terms may convey generic parody in the source text; all the specificities of the language of science fiction is used by the authors to both generate humour and parody. This classification is useful for contrasting the source and target texts and the analysis has revealed that novum parody is considered as such by the translators and that both genre and parodic features are prioritized as much as possible. Nevertheless, when AVT or linguistic constraints prevent any creative solution to retain both generic and humour fluency, the visual and narrative context are used to prioritise either the cognitive estrangement or the humorous aspect of the term, depending on the context or the tone.

The model suggested in the chapter is not flawless, and as any model on humour translation, subjectivity cannot be entirely dismissed. However, the model uses objective linguistic criteria to better understand the way novum parody is handled in AVT. The test on *Futurama* demonstrates its relevance. *Rick & Morty* is a great object of study to further explore, especially in regard to the translation of metadiscourse and cultural references. Further research will be attempted about science fictional humour translation, particularly in audiovisual media such as TV shows or video games, to test and refine the model, so that we can get schwifty in every possible fictive world and in all languages.

Acknowledgements

This chapter is dedicated to my Mamie who didn't watch *Rick & Morty* or read science fiction, but she loved to laugh, and I miss laughing with her.

Abbreviations

sci-fi	science fiction
TL	target language
SL	source language
AVT	audiovisual translation

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Chapter 9

'Default translation': how much free of fault? An exploratory comparison of NMT and human translation

Kamenicka

The chapter centres around an exploratory extension of the concept of 'default translation' proposed by Halverson (2015, 2019), to study and compare the output of professional translators, trainees and NMT systems, for the purpose of cognitive modelling of translation as well as assessing NMT performance in terms of how successfully it imitates translation which is a product of human cognition.

The data is drawn from translations of the same source text by 4 NMT systems (Google Translate, MS Bing, Systran, DeepL), 6 translation trainees and 6 professional translators. The data sets are compared quantitatively, based on a specific metric developed for the purpose of the study, as well as qualitatively, focusing on the solution-type distance between NMT and human translations.

The source text was selected based on specific criteria to suit the default mode of translation and the human translations were obtained in a specific translation regime elicited by the instructions, designed to bring out 'default translation', here understood as a relatively routinized and uninterrupted mode (rather than phase) of translating.

The main findings obtained are, firstly, convincing evidence of DeepL approximating human translations most successfully at collocational level and outperforming the other NMT systems, with a noticeable gap between DeepL and the other systems (and a minimal gap between DeepL and human translations), and, secondly, another solution pattern distinguishing human translations from DeepL, leading to the formulation of the Naturalness Impulse Hypothesis.

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1 Introduction

In the midst of the boom in neural machine translation and its use, closely followed by their reflection in the general media as well as rigorous research in a cluster of disciplines, this chapter aims to explore the differences between products of neural machine translation (NMT) and human translation (HT) from a perspective combining cognitive and quality-related aspects, i.e. to assess how successfully the NMT output imitates translation as a product of human cognition. The study offers insights into some details of how NMT at its best is able to approximate translation by humans as well as attempts to reveal patterns distinguishing human and machine-generated translation at the current stage of development of NMT. The potential use of its findings is seen in (a) the practical area of mapping linguistic and user-oriented characteristics of NMT (both from the perspective of a user and with a view to locating desirables in further improvement of NMT) and, at the theoretical level, (b) reflecting back on understanding human translation better in cognitive and training-oriented terms and contributing towards human translation modelling.

In contrast to studies focusing on evaluating the performance of NMT in specialized registers and with respect to terminologies, the interest in studying the qualitative and quantitative differences between machine-generated and human translation "as such" shifts research attention from situations of predicted difficulty to a more routinized and uninterrupted mode of translation at the human end of the machine-human comparison. Namely, the aim is to see how NMT translation performance can compare to that of humans in translation situations perceived by humans as not particularly challenging. The mode of translating used in the present research design has been inspired by Sandra L. Halverson and the concept of 'default translation' she defined for the purposes of studying cognitive aspects of translating and in reference to research in translation processes (2015, 2019).

In the translation data used, human translation is represented by translations produced by experienced translators as well as students of translation, which offers an opportunity to describe the qualitative and quantitative aspects of the quality of NMT output in comparison with human translation by both experts and trainees, as well as to situate the quality of translation by trainees relative to that achieved by human experts and NMT systems. In the quantitative comparison, a specific metric developed for the purpose of the study was used, combining meaning/coherence disruption count and stylistic/cohesion disruption count, while the qualitative part of the analysis focused on the solution-type distance between NMT and human translations. NMT material for the comparison was ob-

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tained from four NMT systems – Google Translate, MS Bing, Systran, and DeepL. The translations were carried out from English into Czech, i.e. on a language pair showing considerable systemic differences.

2 Default translation

2.1 Default translation according to Halverson

In her 2019 paper, "'Default' translation – a construct for cognitive translation and interpreting studies", Sandra L. Halverson argues that much of empirical research in translation studies has tended to focus on problem solving and "many studies focus on pre-selected areas of predicted difficulty (so-called *areas of interest* or *rich points*)". (2019: 188) The findings of such studies are seen as not very representative since although resolving situations of increased difficulty may take up significant time relative to the whole time spent on translation processes, more research attention should be paid to investigating passages where translators proceed in a more "routine and uninterrupted manner, and where translational choices seem to be made without obvious deliberation" as this type of translating is also representative of what translators usually do and complements the view of how translators perform across different situations and over time (2019: 188).

'Default translation' as Halverson construes it, rooting it in embodied philosophy, which views cognition as embodied, embedded, enacted, and affective (4EA) (2019, 188), is therefore implicitly but very clearly linked with human translation. It is also explicitly defined as an (uninterrupted) *phase* of translation production, without a priori assuming any specific characteristics to be found in the products of these translation phases. Halverson emphasizes that although the introduction of the concept of 'default translation' may be motivated by the interest in studying relationships between the default phases of production and the properties of translated texts and references can be made to 'default translations', it is the phase itself that grounds the construct (2019: 191). She insists that stretches of default translation can be identified in online data only and then provides insights into how this identification can make use of keystroke data and gaze data in process research.

The notion of default translation, in contrast to "challenged translation", has also been used by Carl and Dragsted in "Inside the monitor model: processes of default and challenged translation" (2012), but it is important to note that their motivation for using the notion and their approach to it differ from Halverson's. It is Halverson's default translation that the present study works with and adapts.

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The two approaches are clearly set apart by viewing differently the relationship between default translation and literal translation. According to Halverson (2019: 195), "default translation may involve the selection of translations that bear relatively little formal or semantic similarity to the source, if the two structures are linked often enough in the translational act. Thus, default translation will show evidence of established patterns that originate in shared semantic and/or formal characteristics as well as translational solutions that have become entrenched in spite of their lacking obvious semantic or formal similarities". Carl & Dragsted (2012), in contrast, associate default translation with literal translating. The aim of their study is to provide evidence for their extension of the monitor model (Tirkkonen-Condit, 2005) by comparing process data from copying and translation processes. In this context, consistent with the monitor model, "literal translation is a default rendering procedure, which goes on until it is interrupted by a monitor that alerts about a problem in the outcome. The monitor's function is to trigger off conscious decision-making to solve the problem". (Tirkkonen-Condit 2005: 407–408) Within the monitor model as used by Carl and Dragsted, default translation is identified with literalness and unchallenged, parallel processing of ST and TT while the opposite, challenged mode of translating is equalled with rejecting the first-choice literal solution in favour of a non-literal one.

2.2 Extending the concept

Although Halverson assumes isolation of stretches of default translation based on online processing data, namely saying that "[i]n this discussion, an initial proposal for the isolation of default translation is derived on the basis of existing research" and "stretches of default translation can be identified in online data only" (2019: 198), the mention of the "initial proposal" may be a hint that linking default translation with online data need not be a fixed condition in all research situations. The present study seeks to make a case for extending the concept of default translation beyond the confines of the strict link to specific conditions defined by means of online processing data as a meaningful and justifiable step in certain research settings. Specifically, linking default translating/translation to combinations of processing and performance data – the latter being observational data relating to the performance of translational tasks, such as revision behaviour, use of online or other resources, and other observable behaviour connected with the use of translation technologies – while avoiding grounding it in product data – seems useful for exploratory research. In the present exploratory study, the mode of translating stimulating default translation was elicited by a

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specific choice of the translation task and conditions and monitored through collection of performance data, especially with respect to ensuring relatively uninterrupted production (for more details see Section 3.1).

Some ground for justified extensions of default translation to longer stretches of text based on performance data can be found in Halverson saying that

> default translation involves *the generation of fast and easy translations* which are based *either* on coactivation patterns that have been established through bilingual language use (without translation) or that have been established through prior translation activity and subsequently reused for that purpose more or less frequently, or both. Thus default translation can be found where the translator is translating a particular stretch of language for the first time, if the linguistic material used is familiar and the cross-linguistic relationships relatively unambiguous (2019: 196, first emphasis using italics added)

and that "[t]he 'default translation' construct is being proposed *as a means of investigating routinized translational action by isolating those passages of production that are most likely to contain it*". (2019: 198, emphasis added)

3 Aim and methodology

3.1 The task

To elicit a default mode of translating in human translators, the text for the translation task to be administered to the students of translation and translation professionals was selected so as to meet the criteria of being a non-artificial text for written translation on a non-specialized topic (without specialized vocabulary) and not involving intricacies of style, designed for a general (non-expert) audience. The assignment was based on two passages (with a small overlap between them; of 503 and 521 words respectively) from the transcript of the TED Talk by Bryce Dallas Howard entitled "How to preserve your private life in the age of social media". (Howard Dallas Howard 2022) Neither the speech itself nor its transcript feel overly oral in nature (which might be distracting and requiring some effort detrimental to the default mode of translating directed at resolving the tension between the writing modality and preserving features of orality in the translations).

Prior to collecting the data from the subjects, one pilot was run with an experienced translator/trainer, to fine-tune the instruction-giving and detect potential problems.

The data was collected from all 12 translators based on individual online meetings through Zoom, which consisted of four parts. All translators, professionals

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and trainees alike, were given the same instructions in the first part of the session. They were informed that they would be translating an extract from a relatively easy-to-translate text and since the research was focusing on "what happens in translation" rather than translation quality, they were being asked to deliver a target text in the quality of a draft (a good draft given the relative easiness of the task), not necessarily a top-quality translation in terms of style etc. (to free them of the worry that they would not be allowed to revise their translation). They were instructed they could also think of the task as translating a text for a friend for informative purposes. They would be given the whole text to read before they were to start translating and would translate without online searches or other dictionary searches (and the text had been selected so that this was feasible and not difficult). They were asked to translate at their leisure, without any time limit, from the beginning of the assigned passage till the end, not coming back to translated passages. The pace of their translating should feel comfortable for them – not too slow (they were not to worry about details) and not too fast (there was no time limit).

In case they got stuck at some point, i.e. they found themselves hesitating how to translate a given segment or phrase for noticeably longer than in the rest of the text, they were not to wait to find an ideal solution but leave the phrase out and mark the place in the target text and just continue with the rest of the text. After translating the whole passage, they would be given some time to fill in the gaps, should there be any, so that the translation would not feel unfinished.

To ensure that the translation felt like an easy task, they were given some scaffolding regarding potentially tricky lexis ("chipmunk", "coddle") depending on which of the two passages they were to translate, and a general instruction about translating idioms (not important if they didn't know the exact meaning of the idiom; translate within the context).

They were also told whether they would be translating the first or the second part of the text. They were assured there would be time for them to ask questions, if they had any, before starting to translate.

In the second part of the session, they were given some time to read the whole text. To create an atmosphere of ease, no time limit was given (although the notion was that the informative reading should, ideally, not exceed 10 minutes) and they should tell the researcher when they were ready. No translator needed more time than 10 minutes. They were told that the passage to translate was copied below and set in blue, with paragraphs set apart by a space for their translation, and they would be later writing their translation into the same document. At the end of this part, the translators were offered space to ask questions to make sure the task was understood and they were asked if they would agree to share the

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document on the screen while translating and allow the researcher to record that part of the session solely for checking the process (the recordings would only be stored in the researcher's computer and would not be made accessible to any other parties). All the translators gave their consent.

The third part of the session included the process of translation being recorded and monitored by the researcher. In the fourth part, the translators who needed it were given some time to fill in gaps in their text and a short informal interview was conducted, asking how the task felt to them, whether and where they got stuck (to complement textual and visual data), and whether they wanted to add or discuss anything else. Then they were asked to send the file with the translation to the researcher.

3.2 The translators

Six experienced translators and six translation trainees participated in the research. Each of the two text extracts (referred to here as Source Text A and Source Text B) was translated by three professionals and three trainees and translations, respectively, from four NMT systems – Google Translate, MS Bing, Systran, and DeepL – were obtained (in September 2022). [1] As there was a short overlap between the two source text variants (of one paragraph of 48 words) to ensure the texts would be of as equal length as possible without starting/ending off at a random place, a short part of the text was available in all the 12 human and 4 NMT versions. In all cases, publicly available (unpaid) versions of the systems were used, as this was found to be consistent with the general-topic and unchallenging nature of the text.

All of the professionals had been making their living by translating (and localizing and/or subtitling) for at least seven years at the time of collecting the material (the maximum was 23 years). The trainees were students in their third semester of a two-year Master's degree programme in translation.

3.3 The metrics

To assess the overall quality of the translations quantitatively, a simple two-dimensional metric was designed, to suit the nature of the text selected for the default mode of translating, consisting of the meaning/coherence disruption count

[1] Disclaimer statement: Neither the author nor its employer (department or faculty) have been in any business relationship or partnership with any of the providers of the online translation services.

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and the stylistic/cohesion disruption count. These two counts allowed for scoring all dysfluencies in transferring the meaning of the original.

The meaning/coherence disruption count gives the number of instances where the meaning expressed in the source text has been distorted in such a way that the textual place in the target text is either perceived by a target-only receiver as a disruption of coherence or a significantly different meaning is conveyed (without the receiver necessarily perceiving a disruption of coherence).

The stylistic/cohesion disruption count gives the number of instances which are not meaning/coherence disruptions (correct meaning corresponding to the source text can be constructed) but naturalness of the text with respect to the target language and/or cohesion is noticeably challenged. Meaning/coherence disruptions roughly correspond to the content of the adequacy criterion and stylistic/cohesion disruptions roughly correspond to the content of the fluency criterion commonly used in MT output evaluation. Some examples of application follow:

(1) Meaning/coherence disruption in NMT output:
ST: When I **blanked** at my first piano recital and ran offstage crying, my mom insisted I stay and support my peers.
Bing: Když jsem se na svém prvním klavírním recitálu **vyprázdnila** a s pláčem utekla z pódia, moje máma trvala na tom, abych zůstala a podporovala své vrstevníky. [When I **emptied myself out** at my first piano recital …]

(2) Meaning/coherence disruption in HT:
ST:
Much of this was lost on my mom.
STU2:
Mamka téměř **nic z toho ale vůbec netušila**. [My mom however **had no clue about that**.]

(3) Stylistic/cohesion disruption in NMT output:
ST:
Next up, confidence comes from character, not our looks.
Google: **Dále**, **sebevědomí pochází z charakteru**, ne z našeho vzhledu. [a literal, clumsy translation]

(4) Stylistic/cohesion disruption in HT:
ST:

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> So when my 15-year-old son started a YouTube channel …
> *Systran:* Takže když můj 15letý syn **založil Youtube kanál** … [failure to use a reflexive verb (si založil) + failure to use a more natural common collocation (kanál na YouTube)]

The metric was equally applicable to machine and human translation and distinguishing between dysfluencies of the two types was relatively easy. The strength of the metric is that it accounts for quality of meaning transfer while complementing its assessment with another measure capturing an evaluation of the smoothness and cohesiveness of the translation/s.

4 Results

4.1 Quantitative data and supplementary factors

Table 1: Average times taken to translate the assigned passages.

Subjects	ST A (503 words)	ST B (521 words)
Trainees	29.3 mins	31.7 mins
Professionals	31.0 mins	34.7 mins

The average times needed to complete the translation are given in Table 1. They were recorded only as a supplementary variable though and do not have any explanatory value. It is nevertheless interesting that in both parts, professional translators afforded themselves slightly more time to complete the task. However, both shortest (22 mins and 23 mins) and longest (42 mins and 45 mins) times taken for the task were recorded in the professional group while trainees were a more compact group in terms of time spent on the task.

Table 2 shows that the performance of MS Bing and Systran in this task was nearly equal on both counts. Google Translate outperformed them at the level of meaning/coherence, capturing the core meaning of the text better, but partly at the cost of stylistic smoothness of the text – its stylistic/cohesion count and surface textuality detract from the overall performance quality. As expected, based on both components of the metric, human translators, both professionals and trainees, outperformed all NMT systems significantly and professionals outperformed trainees. The position of DeepL in the comparison is interesting: it stands

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Table 2: Average meaning/coherence and stylistic/cohesion counts for NMT systems and human translators. The counts for human translators have been adjusted to reflect the same length of text.

	Meaning/coherence disruptions	Stylistic/cohesion disruptions
Google	13	28
MS Bing	21	21
Systran	22	21
DeepL	5	15
Trainees	2.7	8.3
Professionals	0.3	4.0

between human translators and the other NMT systems in terms of both counts and it is even closer to translation trainees than to the other most successful NMT system, Google Translate, on account of its meaning/coherence disruption count.

There were also some additional features of translation quality which have not been captured by the metric, mainly the treatment of gender and some technical aspects.

The speaker (the person uttering the text) is Bryce Dallas Howard. The name itself does not clearly indicate the speaker's female gender for a Czech person while the inflected nature of Czech requires the gender of the speaker to be correctly expressed grammatically. This potential issue was identified in the pilot run of the session and a decision was made not to make the gender of the speaker explicit in the instructions given to the translators since attribution of gender to the speaker and some other protagonists of the text/speech was expected to be a major problem for the NMT systems and is not always correctly resolved by translators, especially translation trainees. It was anticipated that the human translators might a) ask about the gender themselves before starting to translate, b) intuit the gender (correctly or incorrectly) or c) in the event that a significant hesitation pause was spotted on the first occasion where the gender needs to be resolved in Czech, the researcher would prompt the translator with the correct female gender. The need to deal with the gender of the speaker (not resolved successfully by any of the NMT engines) as well as that of some other characters mentioned in the extracts produced interesting results but the details cannot be discussed here due to the focus and scope of the chapter. The gender disambiguation issues were not observed to influence the default mode in the rest of the text

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and do not have any impact on the overall results.

Another, technical issue which plays a role in translating from English into Czech and which rendered the NMT output formally flawed (with all four NMT systems) was the failure to convert inverted commas opening the direct speech segments to lower inverted commas which are used in Czech – although including this into the algorithms for language-specific conversion is certainly feasible, it probably was not seen as important enough.

The only NMT system which managed to provide output without any obvious mistake clashing with the norms or written Czech (apart from the dysfluencies of cohesion and coherence) was DeepL. The other systems either missed a compulsory comma or inserted a grammatically incorrect form in a syntactically complex situation; however not all human output was flawless in this respect: the typical mistakes were either missing a compulsory comma or a typo, the frequency being somewhat higher than with the NMT systems. The explanation for this may be that even in a relatively simple text like this one, adherence to the written-language norm may often be what translators fix only in the last stages of editing.

There was yet another small but cumbersome technical issue which plagued the output by the publicly available version of Systran, namely a peculiar formatting of individual words resulting in strange divisions of words ending the lines, which was very difficult, nearly impossible to get rid of and would considerably complicate editing Systran output into a flawless Czech text.

The two components of the metric which was applied in this study as well as the additional factors of output quality as described above suggest that among the NMT systems, the highest translation quality in the task designed to elicit the default mode of translation in human translators was achieved by DeepL, followed by Google Translate. The two remaining NMT systems ranked lower, and the quality of the output by Systran was also hindered by additional technical issues. As will be shown in the Discussion section, the same order of rank in translation quality in English-Czech translation by these NMT systems was observed by Zahradník (2022). The finding is especially interesting given that Zahradník focused on quality in technical texts, specifically in translating specialized terminology across a selection of domains, i.e. in translation far from default translation.

4.2 Qualitative data

A qualitative analysis of the NMT and human translation output in the present study yielded two main findings, which will be presented and discussed in this

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section. Firstly, a thorough line-by-line comparison of the output revealed that even among textual segments (at the phrase rank) which were not marked by coherence dysfluencies or major cohesion dysfluencies, in a number of cases, DeepL imitated solutions chosen by human translators noticeably better than the other NMT systems. The solution types chosen by DeepL actually tended to coincide with the solution types preferred by human translators in general, or those which tended to be selected by professional translators, while the solution types chosen by the remaining NMT systems often differed. In these cases, a gap divided human solutions and solutions by DeepL from other NMT solutions.[2]

The second main finding concerns cases (of the rank of a phrase) where DeepL performed consistently with the performance of other NMT systems while the human translations tended to use a shared solution type. The finding leads to the formulation of a hypothesis concerning the division of attention and effort of human translators in the default mode of translating (and possibly in translation in general) – the Naturalness Impulse Hypothesis, to be explored by further research.

4.2.1 DeepLy approximating humans?

Firstly, evidence of the first observed tendency will be presented by listing the solutions where the said distribution can be observed and they will be commented on.[3]

(5) *In fact, my family has been* in the business of entertainment *for three generations, contending with how to handle putting ourselves out there for the last 70 years.*
Google Moje rodina se ve skutečnosti *zabývá zábavou* …
Bing Ve skutečnosti se moje rodina *zabývá zábavou* …
Systran Ve skutečnosti se moje rodina *věnuje zábavě* …
DeepL Ve skutečnosti se moje rodina pohybuje *v zábavním průmyslu* …
TT1 Vlastně se moje rodina pohybuje *v zábavním průmyslu* …
TT2 V tu dobu moje rodina vlastně *v zábavním průmyslu* …
TT3 Popravdě řečeno, moje rodina působí *v zábavním průmyslu* …

[2]The methodology used, namely monitoring and recording of the text creation process online, provides a guarantee that no NMT engine was used during the translation process; interference from DeepL output can safely be excluded from consideration.

[3]The ST segments are more extensive than the respective translations, to provide some context, and the focus of the comments is on the underlined phrase while other aspects/deficiencies of the translations are mostly left uncommented.

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PRO1 Celá moje rodina je součástí *zábavního průmyslu* …
PRO2 Ve skutečnosti se moje rodina vyskytuje *v šoubyznysu* …
PRO3 Moje rodina po pravdě pracuje *v zábavním průmyslu* …

In Example 1, the ST features the phrase "in the business of entertainment", which is a variation of the more frequent "in the entertainment business". This caused somewhat vague translations by the first three NMT engines, mentioning "entertainment" and a verb signalling personal investment in it, which are however still comprehensible in context. Human translations as well as DeepL provide the Czech equivalent to "entertainment business" ("zábavní průmysl"); one professional translator opted for a near-synonym, "šoubyznys" ("show business").

A very similar distribution of translation solutions occurs in Example 2. The phrase "in the public eye" was mistranslated by the first three NMT systems into a Czech equivalent of "in the eyes of the public", which would be a correct choice had the meaning been "as viewed by the public" but not in this case. The correct solution selected by DeepL coincides with the one by all three professionals and one trainee while two trainees opted for slightly looser translations ("in front of the eyes of the public", "under the surveillance of the public").

(6) … she would need to prepare us for something she herself had never experienced. *Growing up* in the public eye. *And this entailed some extreme parenting tactics. First. No coddling, like ever.*
Google Vyrůstat *v očích veřejnosti.* … [mistranslation]
Bing Vyrůstat *v očích veřejnosti.* [mistranslation]
Systran Vyrůstat *v očích veřejnosti.* [mistranslation]
DeepL Vyrůstání *na očích veřejnosti.*
STU1 Na dospívání *před očima veřejnosti.*
STU2 Dětství a dospívání *pod dohledem veřejnosti.*
STU3 Dospívání *na očích veřejnosti.*
PRO1 Dětství a dospívání *na očích veřejnosti.*
PRO2 Vyrůstání *na očích veřejnosti.*
PRO3 Na vyrůstání *na očích veřejnosti.*

This distribution of solution types was fairly frequent in the data. Without citing the full solutions, let us point to another instance of DeepL using the same correct and natural collocation as all human translators in contrast to the other NMT engines: humans and DeepL translated "fame came with many blessings" as "sláva přinášela/přináší …" (lit. "fame brought/brings …") rather than the dysfluent and literal "sláva přišla/přichází s …". (Google, Bing, Systran)

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(7) *My mom* met *my dad, Ron, when they were 16.*
Google Moje máma *potkala* mého otce Rona, když jim bylo 16.
Bing Moje máma *potkala* mého tátu Rona, když jim bylo 16.
Systran Moje máma *poznala* mého otce Rona, když jim bylo 16.
DeepL Máma se s mým tátou Ronem *seznámila*, když jim bylo 16 let.
TT1 Máma *potkala* tátu, Rona, když jim bylo šestnáct.
TT2 Mamka tátu *potkala*, když jim oběma bylo šestnáct let.
TT3 Moje máma *potkala* tátu, Rona, když jim bylo 16.
PRO1 Máma se s mým tátou Ronem *seznámila*, když jim oběma bylo 16 let.
PRO2 Máma se s mým tátou Ronem *seznámila*, když jí bylo šestnáct.
PRO3 Máma se *seznámila* s mým tátou, Ronem, v šestnácti.

In Example 3, the translations feature solutions for expressing the first encounter of the couple at age 16. All Czech solutions function well and do not attract any attention by lacking in naturalness but while the first three NMT engines and trainees translated the verb with the close Czech equivalent, all three professionals and, significantly, DeepL selected another verb, most frequently used in Czech where the first encounter of a couple-to-be is being referred to.

(8) *After they finally joined forces, after* he actually proposed three times*, and* had kids*, my mom transformed into a full-on warrior woman, ready to protect.*
Google Poté, co konečně spojili své síly, poté, co třikrát požádal o ruku *a měl děti*, …
Bing Poté, co konečně spojili své síly, poté, co ve skutečnosti třikrát požádal o ruku *a měl děti*, …
Systran Poté, co konečně spojili síly, poté, co se třikrát přihlásil *a měl děti*, …
DeepL Poté, co konečně spojili své síly, poté, co *mě* skutečně třikrát požádal o ruku, a *měli děti*, …
STU1 Potom, co konečně spojili síly, táta mámu třikrát požádal o ruku a *měli děti*, …
STU2 Poté, co se konečně dali dohromady, poté, co ji táta požádal o ruku hned třikrát, a *měli děti*, …
STU3 Poté, co konečně spojili síly, co ji táta už potřetí požádal o ruku a co *měli děti*, …
PRO1 Poté, co se konečně dali dohromady, poté, co ji třikrát požádal o roku, a *narodily se jim děti*, …

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PRO2 Když potom, co táta mámu dokonce třikrát požádal o ruku a *měli spolu děti*, konečně spojili síly, …
PRO3 Když nakonec spojili síly, potom, co ji třikrát požádal o ruku, a *měli spolu děti*, …

In Example 4, the phrase whose translations are interesting to focus on expresses that the speaker's mother and father finally had children. It follows after an insertion, appropriately enclosed in commas, specifying that the speaker's father had to overcome his partner's resistance by proposing to her three times, which complicates the attribution of actions to persons for the NMT systems significantly. While human translators with embodied thinking distinguished easily that only the father proposed three times but it was both parents who had kids, which was once paraphrased as "children were born to them" (PRO1), all NMT engines except DeepL – which used the correct majority solution – attributed having kids wrongly to the father only, which in the given context might imply that he had kids with someone else and might be potentially confusing for the addressee.

Although there is not enough space for the commentaries on the examples to discuss other dysfluencies featuring in solutions by NMT systems apart from the phrases in question, let us note another detail in Example 3 which might be illustrative of the differences between output by DeepL and the remaining NMT engines. The grammatical object of the marriage proposal, the mother, "her", is not explicitly mentioned in the ST, which presented a potential difficulty (not for human translators with embodied thinking though). DeepL introduced a mistake here (included in the meaning/coherence disruption count) by attributing the proposal to the speaker instead of her mother. The situation with the other engines is as follows: the verb for the action of proposing has been selected correctly by Google Translate and Bing and incorrectly by Systran (literally "he raised his hand three times") but the grammar of the output by Google Translate and Bing is flawed in not stating the grammatical object of the proposal, which is compulsory in Czech (and this was covered by the stylistic/cohesion disruption count). The mistake by DeepL arose in the context of attempted correctness of the lexicogrammar of the phrase, distinguishing DeepL from the other three NMT systems, this time through a slightly different translation solution patterning.

Examples 5 and 6 show how DeepL approximates situations where the ST uses grammatical means available in English for expressing temporal relations between events beyond a simple temporal sequence. It is the use of past perfect (for which no grammatical equivalent is available in Czech) to stress what the speaker's mother had managed by a certain point (Example 5) and an anticipation

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of the future (“he’d soon film a pilot”), where the contracted form has to be disambiguated to be interpreted as a conditional (Example 6).

(9) And by that point she had flown *solo in an airplane, joined an all-boys gun club, gotten in tons of fist fights and helped her single dad raise her younger sisters.*
Google A v té době *už létala* sólo v letadle, …
Bing A v té době *už letěla* sólo v letadle, …
Systran A v tu chvíli *už letěla* sólo v letadle, …
DeepL A to už *měla za sebou* sólový let v letadle, vstup do chlapeckého střeleckého klubu, spoustu pěstních soubojů a pomáhala svobodnému tátovi vychovávat mladší sestry.
TT1 Do té doby *už* sama *letěla* letadlem, …
TT2 V tom věku už *toho za sebou měla hodně* – …
TT3 V té době *už* sama *letěla* letadlem, …
PRO1 A to už *měla za sebou* sólový let *za kniplem letadla*, …
PRO2 A tou dobou *už* sama *letěla* letadlem, …
PRO3 V té době *už toho zažila a dokázala dost*: …

(10) *It was the early 70s, and my dad was known for playing Opie in the “Andy Griffith Show,” and* he’d soon film *a pilot for a new show called “Happy Days.”*
Google … a *brzy natočil* pilotní díl pro novou show s názvem ”Happy Days”.
Bing … a *brzy natočil* pilotní díl pro novou show s názvem ”Happy Days”.
Systran … a *brzy natočil* pilotní díl pro novou show s názvem “Happy Days“.
DeepL a *brzy měl natáčet* pilotní díl nového seriálu ”Happy Days”.
TT1
Chvíli na to natočil první díl seriálu *Šťastné dny.*
TT2
… přičemž *brzy měl točit* pilotní díl nového pořadu, který se jmenoval „Happy days“ (Šťastné dny).
TT3
… a *čekalo ho natáčení* úvodního dílu seriálu *Šťastné dny.*
PRO1
… a *brzy měl natáčet* pilotní díl nového seriálu „Šťastné dny“.
PRO2

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... a *brzy měl natočit* pilotní díl nového seriálu s názvem „Happy Days".
PRO3
... a *brzy ho čekalo natáčení* pilotního dílu nového seriálu Happy Days.

All solutions in Example 5 are acceptable but DeepL approximates successfully what occurs only in human translations in using a natural-sounding and lexically explicit phrase "mít za sebou", used by one professional and one trainee, similar to another lexical paraphrase used by PRO3 ("she had already experienced and achieved a lot:" followed by a specification of the achievements). The remaining NMT systems, two trainees and one professional, on the other hand, use a solution less explicit about the sequence of events ("already flew/was already flying").

Similarly in Example 6, three NMT systems, except for DeepL, translated the anticipatory conditional by the same phrase causing a cohesive dysfluency but without disturbing meaning in a major way while DeepL coincided with human translations by providing a natural-sounding lexical paraphrase ("was soon to film") identical to solutions by TT2, PRO1 and PRO2 while TT3 and PRO3 selected another similar lexical paraphrase. The solution by TT1 also differed from Google Translate, Bing and Systran, by singling out the event syntactically and not using the literal equivalent to "soon" but translating it more freely as "a while later".

(11) Much of this was lost on my mom. *The only thing she watched on TV was "Star Trek."*
Google Hodně z toho přišla moje máma. [meaning lost]
Bing Hodně z toho se ztratilo na mé matce. [meaning lost]
Systran Hodně z toho se ztratilo na mé mámě. [meaning lost]
DeepL Mámě to moc neříkalo.
TT1 Máma o většině toho vůbec nevěděla.
TT2 Mamka téměř nic z toho ale vůbec netušila.
TT3 Tohle *máma ocenit neuměla.*
PRO1 Tohle všechno *mámě nic neříkalo.*
PRO2 Většinu z toho máma vůbec nevěděla.
PRO3 Mojí *mámě tohle nic neříkalo.*

In Example 7, Google, Bing and Systran translate the phrase in question literally, causing a meaning disruption (correct meaning cannot be retrieved from context), in contrast to a correct and very natural translation by DeepL, very close to human translations by PRO1 and PRO3. TT3 translated the phrase by

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a close paraphrase and TT1, TT2 and PRO2 opted for looser paraphrases centered around the mother not being aware of any of this (where the meaning was sufficiently preserved).

To support the claim that DeepL shows, across a range of situations, significant resemblances to human translations, one last example will be presented, due to the limited space of this chapter.[4]

(12) *Years later, when my son grimaced at* a chipmunk carcass, our cat had left in the driveway, *Cheryl made him clean up the remains.*
Google O několik let později, když se můj syn zašklebil na mršinu veverky, *naše kočka odešla na příjezdové cestě*, Cheryl ho přiměla uklidit zbytky.
Bing O několik let později, když *se můj syn ušklíbl na mrtvolu chipmunku, naše kočka odešla na příjezdovou cestu*, Cheryl ho donutila *vyčistit ostatky.*
Systran O několik let později, *když se můj syn zašklebil na jatečně upraveném těle, naše kočka odešla na příjezdovou cestu*, ho Cherylpřiměla, aby zbytky uklidil.
DeepL Když se můj syn po letech ušklíbl *nad mrtvolkou veverky*, kterou naše kočka nechala na příjezdové cestě, Cheryl ho donutila zbytky uklidit.
TT1 Když se o mnoho let později zatvářil můj syn zhnuseně nad veverčí kostrou, kterou nechala na příjezdové cestě naše kočka, Cheryl ho donutila *zbytky kostřičky* uklidit.
TT2 A když se pak můj syn o mnoho let později zašklebil při pohledu *na mrtvolku čipmanka*, kterého naše kočka ulovila a nechala přede dveřmi, Cheryl ho přiměla, aby ji uklidil.
PRO3 A když se můj vlastní syn o spoustu let později ušklíbl při pohledu na mrtvou krysu, kterou naše kočka nechala ležet před domem, Cheryl mu nařídila, aby *mrtvolku* uklidil.

The sentence mentions "chipmunk", an animal not native to Europe the translation of which might have disrupted the default mode of translating in the human translators, therefore *chipmunk* (together with the verb *coddle*) was mentioned as one of the potentially difficult lexical items as a part of the instructions (to those who translated the respective part of the text); the translators were prompted with the corresponding name of the biological species ("burunduk") and told not

[4]To save space, the less interesting (but reasonably acceptable) translations by TT3, PRO1 and PRO2 have been left out.

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to worry about this and to translate in context (implying that functional equivalents are acceptable solutions). The NMT translations by Google, Bing and Systran of the sentence, where an incorrect extra comma was added in the transcript published on the internet, are illustrative of the role of pre-editing as the redundant comma has caused a number of meaning disruptions (highlighted). DeepL, nevertheless, managed to translate the whole sentence without any disruption at all.

Moreover, three human translators, TT1, TT2 and PRO3, as well as DeepL, each managed to include a diminutive, a unique item in translation from English into Czech: "remnants of the little skeleton", "the little dead body of the chipmunk, "the little dead body" and "over the little dead body of the squirrel", respectively. The use of a diminutive is a very opportune solution, softening the impression of the scene as one would expect in Czech when the dead body of a small animal is being referred to. Since the use of unique items is observed to be a feature endowing translated texts with a feel of naturalness, associating them with non-translated writing in the respective language and increasing the likelihood that they will not be recognized as translations due to translationese (Tirkkonen-Condit 2005), the contextually very appropriate use of the diminutive by DeepL is another strong piece of evidence of its very good performance in approximating human translations.

4.2.2 Human and more than non-interferential

The features shared by the examples to be presented in this section in support of the hypothesis to be formulated, which formed another pattern spotted in the overview of the translations are the following:

- the solutions by NMT systems are all acceptable and all (including output by DeepL) are typologically different from solutions by human translators;
- the solutions by human translators are less literal and often but not necessarily more natural in Czech in terms of frequencies of occurrence of lexically salient expressions and/or collocationally more natural (they feel "vivid" rather than "neutral");
- the units (simple sentences, or distinct parts of complex sentences) are shorter ones of a rather uncomplicated nature.

In example 9, only TT1 and TT3 use a literal translation analogous to those by the NMT engines, improving it by using the feminine form of *actor*, whose

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use is compulsory for female actors in Czech. The other solutions are less literal and more natural to Czech as they express what the present perfect expresses in English by lexical means available in Czech.

(13) And I've been an actor *since my early twenties.*
Google A *hercem jsem* od svých dvaceti let.
Bing A *hercem jsem* od svých dvaceti let.
Systran A já *jsem herec* od svých raných dvaceti let
DeepL A já *jsem hercem* od svých dvaceti let.
TT1 Já *jsem herečkou* od té doby, co mi bylo dvacet a něco.
TT2 A já *jsem se stala herečkou* chvíli poté, co jsem oslavila dvacáté narozeniny. [I became an actress …]
TT3 A já *jsem herečkou* od svých raných dvaceti.
PRO1 A já *se herectvím živím* od svých dvaceti. [I am making my living by acting …]
PRO2 A já *začala hrát* hned po dvacítce. [I started acting …]
PRO3 A já *se jako herečka živím* už zhruba od dvaceti. [I am making my living as an actress …]

All NMT translations in Example 10 are literal, with a small but notable variation by DeepL, which leaves out the initial causal connector (just like T1), in contrast with the human translations which break the (certainly acceptable) literalness by changing the noun phrase into a verbal phrase in consistency with general tendencies in Czech (TT1, PRO2), changing the connector (TT2, TT3, PRO1, PRO2), or a verbal paraphrase accentuating that coddling was out of question (PRO3).

(14) So that meant *zero coddling. Put your comfort away, she'd preach.*
Google Takže to znamenalo nulové mazlení. …
Bing Takže to znamenalo nulové rozmazlování.
Systran Takže to znamenalo nulové hýčkání.
DeepL To znamenalo nulové rozmazlování.
TT1 To znamenalo, že s námi rozhodně nejednala v rukavičkách.
TT2 A to znamenalo žádné rozmazlování.
TT3 Což znamenalo absenci rozmazlování.
PRO1 Což znamenalo nulové rozmazlování.
PRO2 Což znamenalo, *že* se s námi nemazala.
PRO3 Takže rozmazlování nepřipadalo v úvahu.

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The ST in Example 11 features the noun "challenge", the frequency of whose direct equivalent in Czech (*výzva*) was very low before the surge of post-1989 translations from English and which still has a subtle connotation of a stylistic/conceptual importation, which makes the translation situation interesting. All NMT translations are literal and identical, coinciding only with the solution by TT2. All other human translators diverted from the literal version by varying the pronoun ("any time a challenge arose", TT1), using a near-synonym to the noun ("obstacle", TT3), using a verbal phrase instead of a simple verb ("which occurred in front of us/which was an obstacle on our way", PRO1 and PRO2) or linking the challenge with an opportunity in a looser paraphrase ("each new challenge presented an opportunity", PRO3).

(15) Any challenge that arose *was an opportunity to find your sea legs.*
Google Jakákoli výzva, která se objevila, …
Bing Jakákoli výzva, která se objevila, …
Systran Jakákoliv výzva, která se objevila, …
DeepL Jakákoli výzva, která se objevila, …
TT1 Kdykoli se objevila nějaká výzva, …
TT2 Jakákoliv výzva, která se objevila, …
TT3 Jakákoli překážka, která se před námi objevila, …
PRO1 Jakákoli výzva, která se nám postavila do cesty, …
PRO2 Jakákoli výzva, která se nám postavila do cesty, …
PRO3 Každá nová výzva představovala příležitost najít rovnováhu.

The examples could go on, but one more will have to suffice: Example 12 is somewhat different from the preceding ones by virtue of the human translations (except for one, by PRO1) contrasting with the uniform NMT solutions in form, but not differing in naturalness. The collocation "jako malý/malá" (lit. "as little/being little") seems equally likely in the context as "jako dítě" (lit. "as a child") but it seems not to be accidental that human translators opted for the less literal variants, perhaps for the sake of avoiding literality itself. The solution by PRO2 features a somewhat expressive expression "děcko" as a less frequent and more marked equivalent to "kid". The phenomenon might indicate "a fear of interference" where the translator can easily afford investing into a variation on the very literal solution although the latter is not marked by negative interference.

(16) As a kid, *I was terrified of snakes, so my mom got me a pet snake.*
Google Jako dítě jsem se bála hadů, a tak …
Bing Jako dítě jsem se bál hadů, takže …

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Systran Jako dítě jsem se bál hadů, tak ...
DeepL Jako dítě jsem se bála hadů, a tak ...
TT1 Jako malá jsem se bála hadů, tak ...
TT 2 Jako malá jsem se bála hadů a tak ...
TT 3 Jako malá jsem se bála hadů, tak ...
PRO1 Jako dítě jsem se bál hadů, tak ...
PRO2 Jako děcko jsem měla hrůzu z hadů, a tak ...
PRO3 Jako malá jsem se bála hadů, a tak ...

The examples quoted in this section and other similar ones appear interesting in the context of the monitor model (Tirkkonen-Condit 2005), according to which literal translation is the first-choice solution adopted in the translation unless the monitor(ing) function "alerts about a problem in the outcome" and "triggers off conscious decision-making to solve the problem". (Tirkkonen-Condit 2005: 407–408) The examples quoted here representatively – but not exhaustively – however indicate that although the literal or close-to-literal NMT solutions

would easily pass the test, actual human translation solutions – whether by trainees or professionals – elicited under the default mode of translating differ from them systematically even though there seems no obvious reason for the monitor having to be triggered.[5]

Yet another mechanism therefore seems to be operating in these cases, and it is hypothesized here that in the relatively unchallenged default mode of translating, the translator reaches for a non-literal but cognitively relatively easy-to-access solutions more often than the monitor model would suggest, which is, nevertheless, consistent with the concept of "minimax strategy" proposed by Jiří Levý (1967: 1179), which amounts to the translator choosing solutions yielding maximum effect with a minimum effort. The possible reason may be sought in the fluent and unchallenged mode of translating and in the relative simplicity of the segments in which this phenomenon occurred even in comparison with their textual surroundings. Where the translation is particularly unchallenged (compared

[5]Having said this, I do not deny the possible influence of translator/trainee "programming" through instruction in translation training to avoid interference or awareness of translation norms in that respect. This possible influence is viewed here as consistent with entrenchment (see the quotation from Halverson (2019) in 2.1) and the paper distinguishes between avoiding negative interference (triggering the monitor mechanism in a direct sense) and rejecting stylistically neutral more literal solutions in favour of non-literal solutions based on dynamics of translational efforts and processing capacity (see the Naturalness Impulse Hypothesis below). I am indebted to Giuseppe Palumbo for making me aware of the need to clarify how conditioning by training is approached here.

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with the textual surroundings), translators, with some degree of automation and understanding of the translation task as a mediation activity, may be choosing cognitively easily accessible solutions which are freer than the correct but literal ones to specifically avoid literalness and inject the translation with a sense of freshness to forestall an impression of dullness, which might otherwise come with segments where more cognitive effort has to go into other issues. The hypothesis may be referred to as the Naturalness Impulse Hypothesis. It

seems to correspond well to Halverson's notion (already quoted above) that "default translation may involve the selection of translations that bear relatively little formal or semantic similarity to the source, if the two structures are linked often enough in the translational act. Thus, default translation will show evidence of established patterns that originate in shared semantic and/or formal characteristics as well as translational solutions that have become entrenched in spite of their lacking obvious semantic or formal similarities". (2019: 195)

5 Discussion of results

It is not the purpose of this chapter to provide a comprehensive comparison of NMT systems' performance within the multiple contexts which might be relevant (the same language pair vs. other language pairs, different text types and content domains, comparisons along the temporal dimension, etc.), among other things because systematic findings for the English/Czech language pair at the point of publication are rather scarce and – to the author's knowledge – more or less restricted to final theses written in university translation programmes. There is also a considerable gap between publications arising from the MT community and focused mainly on improving MT architectures and evaluating the outcomes of this process of the one hand (such as the WMT report series and other related publications, e.g. Kocmi et al. 2023) and translation studies approaches on the other. It is significant that the latter publication, summarizing the findings of testing the submitted systems on 8 language pairs, most of which were tested in both directionalities, reports that "For most language pairs, *apart from English and Czech*, MT systems produce outputs that cannot be identified as being worse than the manually produced reference translations in a statistically significant way, using our current evaluation methods" (Kocmi et al.: 16; emphasis by the present author), singling English-Czech MT out as involving more relative difficulty than other language pairs tested (which included, for instance, Chinese, Japanese, and Hebrew).

Some notable exceptions for translation within the English–Czech/Slovak language pairs as pairs of analytical vs. inflectional languages are, in this respect,

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Munkova et al. (2021), Munkova et al. (2023) and Kloudová et al. (2023), focusing on MT evaluation in the context of language complexity, proposing an error typology for such contexts, and exploring the methodology for producing optimal reference translations, respectively. At the level of university theses, the Bachelor's final thesis by Kenderová (2019) is an interesting attempt at exploring the grammatico-syntactic transformations and success/failure in Czech-English NMT by Google Translate while Master's theses by Řehořová (2022) and Nováková (2023) apply different human evaluation methods to several kinds of NMT output and NMT output alongside HT, respectively.

The first main finding, namely the superiority of performance by DeepL over other NMT engines in the default translation task and its ability to approximate solutions by human translators better than Google, Bing or Systran (as of September 2022), is perhaps interesting to compare with results obtained in a final Master's thesis by Zahradník (2022),[6] in a setting vastly different from the one in this research. Zahradník performed a translation quality assessment of performance of the same NMT engines on expert and popular science texts, namely research articles, abstracts, popular science and technical documentation, also in translation from English into Czech, focusing on quality in terminology translation. The thesis by Zahradník includes a comprehensive manual evaluation of the translation of 101 term candidates, with the following results: Table 3 indicates that even in translating terminology from English into Czech, a task radically different from default translation, the best results were achieved by DeepL, followed by Google and Bing, and Systran was the least successful of the four NMT engines. In this case, however, the scores had a narrower distribution. A supplementary assessment of grammatical and formal correctness in the expert and popular science texts by Zahradník rendered the same order of rank, the error scores for DeepL, Google, Bing and Systran being 24, 25, 39 and 51, respectively (Zahradník 2022: 67).

The aim and conditions under which the concept of default translation was envisaged by Halverson (2019) have been extended for the purposes of this exploratory study as described in Section 2.2. The research setting for elicitation of the extended default mode of translating in human translators has also been described in detail (in Section 3.1), but some further details could not be provided due to the limited scope of this chapter. Consistent with the observation by Halverson (2019: 204), default translation studied on language pairs with considerable structural differences – such as English and Czech – provides a good

[6] As the reviewer of the thesis, the present author is very familiar with its content and confident of its quality.

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Table 3: A comparison of NMT accuracy achieved in translation of terminology across genres and domains, adapted from Zahradník (2022: 58).

	Google	Bing	Systran	DeepL
Correct translations	53	52	48	55
Total mistranslation	28	29	32	24
Acceptable variants	20	20	21	20
accuracy total	53	52	48	55

opportunity for research to remove some of the noise present in other translation settings and explore specific hypotheses.

The inspiration for extending default translation to longer stretches of text was partly drawn from sight translation (alternatively referred to as sight interpreting), where reading through the document to be sight-translated is a relatively common practice – similar to the pre-translation reading used in the present study. In "Exploring Translation and Interpreting Hybrids. The Case of Sight Translation", Dragsted & Hansen (2009) compared output quality in interpreters' oral translation, translators' written translation and translators' oral translation as an intermediary mode, concluding that the lower output rate in written translation was not justified by the significantly higher quality of the written products (588). The characteristics of sight translation which connect it to the default mode of translating are the relative simplicity of the STs and online processing (also referred to as parallel processing). In the present research design, however, the written mode was elicited as this was the habitual mode for the translators and a switch to another mode of production might have become a factor influencing their production, especially that of the trainees, for whom it might be a stressful factor disturbing the desired default mode of translating. Moreover, although e.g. Oster (2017: 31) assumes in reference to what is "widely accepted" that the main difference between translation and interpreting is the degree of monitoring (and in the model she adopts it is the *only* difference), she also notes that monitoring is a dynamic capacity (29) and that although it is generally stronger in written production than in oral production, the factors influencing it should be explored in detail.

The empirical study on "The influence of self-monitoring on the translation of cognates" by Oster (2017) also highlights one crucial question, which the present study might contribute to answering: namely the question of how the tendencies

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towards language transfer and normalization (Toury 1995: 310–315) operate and the mechanisms by which translators withstand interference. The assumption by Tirkkonen-Condit (2005) of the existence of a 'monitor' or 'monitor function' to trigger repair in case the literal translation does not work well is a very general one and the operation of the actual mechanisms is far from clear. Referring to previous research, Oster mentions motivation, age and time pressure as factors influencing monitoring (29), but it is evident that a full list – had we had one – would be much longer (e.g. attention/tiredness or training certainly play a major role in realistic translation) and the question remains how dynamic actually monitoring is and how to model it. Questions linking the operation of the monitor, literal versus non-literal default translation solutions, immediate response solutions and delayed translation solutions and translation quality are major questions to be addressed by rigorous research.

The three-fold origin of the translations compared within the research presented here has a broad potential; due to the limitations of the scope of presentation, however, the trainee/professional comparison (with respect to the tendencies observed) has had less attention paid to it here than the NMT/human comparison.

In the light of the Naturalness Impulse Hypothesis having been formulated as a result of this research, it seems useful to point out that comparing NMT output with human translation output in a task expected and designed to be easy for the human translators may generally be a setting favourable to spotting patterns characteristic of human translation, as non-human and non-embodied attempts at coping with the task may provide a good background for them to stand out against. It is however important to note that it is not human cognition in translational behaviour *at its most general* that this chapter seeks to characterize: human translations by subjects conditioned by translation training and/or translation norms and NMT translational output are compared, as it is translation by these human subjects that NMT seeks to emulate. Nevertheless, how the Naturalness Impulse Hypothesis concerns or does not concern translating by non-translators with SL and TL knowledge may be interesting to research in the future.

Although the monitor model is used here for reference as a general background, this study does not seek to confirm or develop the model and the sets of translation solutions that have led to the formulation of the Naturalness Impulse Hypothesis are actually at odds with the monitor model. It is probably not accidental that Halverson (2019) avoids reference to it, beginning her argumentation from scratch and counting on a firm empirical basis under carefully controlled conditions. The very existence of instances where a smooth literal translation of

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a segment is rejected in default translation in favour of a less literal variant certainly calls for challenging the assumption that the monitor model might provide a safe default basis for describing and explaining translation processes.

Tirkkonen-Condit herself mentions the factor of *time* in reference to a previous study providing evidence of the operation of a monitoring mechanism ("the monitor needs time for its operation, and that is why the [different mapping condition] sentences in the experiment take a longer time to translate") (2005: 410). With the degree of monitoring being a feature setting written translation and interpreting apart, it is clear that temporal factors are, among other factors, likely to influence monitoring and translational output in human translation in such ways that it is safer not to assume anything a priori about.

Default translation, just like interpreting (and other types of translation, for that matter), is a process taking place in time. In interpreting, the time factor is more palpable than elsewhere but also in sight interpreting or the default mode of translating, which is "easy and fast" (Halverson 2015: 320), the translation pace is likely to play a non-negligible role. In reality, time always plays a role in translation. Effort models for interpreting (simultaneous and consecutive) (Gile 1997/2002, 2009) regard time as a factor limiting processing capacity. Based on the (implicit) assumption that output monitoring is minimal in interpreting, monitoring is not included as a separate source of effort (i.e. complex operation) in interpreting, but it may be justifiably included as a source of effort in modelling translation, including default translation.

As for the limitations of the study beyond the small-scale extent of this exploratory probe, the elicitation of the default mode beyond the control of variables from online processing comes at a certain cost and one of the consequences is that the translation situation does not conform to parameters associated with typical scenarios of professional (or semi-professional, with the trainees) translation. The author of this chapter however believes that although the binary distinction between "professional" and "unprofessional" translation may be useful in some settings, its validity is relative; namely, that translators' professionalism or semi-professionalism in approaching translation – in terms of applying general norms such as withstanding ST/SL interference – does *not* crucially depend on adherence to fixed translation scenarios. The fact that stronger rather than weaker resistance to interference than expected on the basis of models described in literature has been observed, also seems to support the legitimacy of the adoption of the research scenario. Further research may treat the potential objection for instance by adding an editing stage to the default translation to make sure that the effect of the hypothesized Naturalness Impulse Hypothesis does not disappear when the translation product from an easy translation task undergoes

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the standard editing stage.

6 Conclusion

The adoption of an extended understanding of the concept of default translation as coined by Halverson, freed from being defined strictly by online processing data is a step bringing new opportunities as well as involving some risks amounting to limitations of the study. The opportunities resulting from contrasting human performance with the performance of NMT engines in a translation task which is relatively simple from a human perspective has helped to bring out some regularities in the behaviour of humans conditioned by translation training and practice, which are interpretable as a part of active forestalling of interference.

Specifically, the findings of Section 4.2.2 led to the formulation of the Naturalness Impulse Hypothesis, which suggests that on the background of relatively unchallenged translation (in the default mode), particularly unchallenged segments[7] – where monitoring of literal output does not trigger repair activity, which would require extra time/effort – may be inviting the translator to use the available part of the overall processing capacity to actively forestall a 'shining-through' effect in their translation (to conform to the general norm that, in most even if not all types of translation, "a successful translation should not be immediately recognized as a translation by virtue of the language"). These patterns seem to amount to a less conscious if not unconscious component of withstanding interference with some degree of automation; qualifying them as such is based on corresponding strategies not having been – to the author's knowledge – covered by translation norms as reflected on either in descriptive research or in texts written to support translation training. The patterns spotted resulting in the formulation of the Naturalness Impulse Hypothesis, nevertheless, do agree with the minimax strategy (achieving maximum effect with a minimum effort) formulated at a more general level by Levý in 1967 and with Pym's (2015) perspective of the role of risk management in translation.

The limitations of this study, on the other hand, besides the obvious ones having to do with the small-scale research design centred around only one source text (with two subtext variants) and a limited number of participants, include mainly the extension of the default translation condition and its practical execution. The nature of the translation task as described in Section **??** makes it akin to

[7]How to define the borderline between "relatively unchallenged" and "particularly unchallenged" segments in default translation would be, of course, a pending question for research. At this point, only a preliminary formulation of the hypothesis is being offered.

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hybrid forms of translation such as sight translation/interpreting. While this may be viewed as supportive to exploring the *general* characteristics of translational behaviour in human agents acting under and influenced by translational norms operating beyond specific genres or situations, it might be argued that the task at hand with the instructions having been given does not satisfy the conditions habitual in professional translation.

Last but not least, in the context of the current technological turn in translation and boom in NMT and post-editing, it is worth noticing that Halverson herself comments on the relationship between corpus data and/or machine translation on the one hand and default translation on the other: "Corpus frequency or translation selection in statistical machine translation may be effects of default translation, while at the same time corpus frequency may be considered a reflection of community-wide language norms that impact what becomes a default in the first place". (2019: 191) Studying how selecting between literal and non-literal translation solutions in default translation by humans operates is certainly an intriguing task and comparing human output with NMT output may help to highlight how the translator's knowledge "including but not limited to linguistic and metalinguistic knowledge, established routines for action, and conceptualization of the specific task at hand" (Halverson 2019: 191), the latter being as general as possible in default translation, enters as a factor. The Naturalness Impulse Hypothesis may hopefully serve as a small step in that direction.

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Abbreviations

NMT	neural machine translation
HT	human translation
TT	translation trainee
PRO	professional
ST	source text
TT	target text

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Chapter 10

A comparison of human simultaneous interpreting and automatic simultaneous speech translation as a basis for further research

Kloudová

Simultaneous interpreting (SI) is a cognitively difficult task which requires specialized training and continued learning to deliver the best performance. Even with full attention, occasional errors in interpreting are inevitable, and they are likely to raise in longer sessions. Nowadays, with the advances in deep learning, we are witnessing quality increase in automatic simultaneous speech translation (ASST). However, compared to human interpretation, whose output is the spoken word, the ASST model creates a translated transcription of the original.

The presented study assesses the state of the art by analysing translation departures in human SI (professional and trainee interpreters) and ASST systems. Two terminologically rich speeches, delivered by non-native English speakers, were remotely interpreted by two M.A. interpreting students, and one professional interpreter, to Czech. We had the speeches translated also by an in-house speech translation system.

Our quality assessment focuses on content transfer (the analysis does not reflect on other critical aspects as latency or pleasant voice) from the original speech, extending Barik's taxonomy (1971), which identifies omissions, additions, substitutions, and semantic errors, with grammatical errors and formal defects (Kopczyński 1980).

Our findings reveal that trainee interpreters outperformed the professional one in multiple areas, such as avoiding over-reliance on original structure or excessive use of demonstrative pronouns. In comparison, the ASST model translated more literally, with outputs having fewer translation departures than human interpretations. However, focusing on content transfer only abstains from considering all

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aspects of interpreting quality assessment of human SI that are more important to the client. Future research aims to expand our sample size and evaluation criteria, incorporating more evaluators, students, and professionals, to further validate our findings.

1 Introduction

The confluence of translation studies, linguistics, cognitive science, and technology promises a new horizon for the field of translation and interpretation. While traditional simultaneous interpreting (SI) has long been an integral part of multilingual conferences and events, the emergence of Automatic Simultaneous Speech Translation (ASST) platforms offers real-time solutions, often in the form of transcript translation. In today's age, ASST has reached commendable levels of precision.

However, how does the machine's attempt at interpreting stack up against its human counterpart? The evolution of ASST underlines an essential axiom for both professional interpreters and students: the imperativeness of continuous learning. As technology surges ahead, it becomes crucial for interpreters to maintain their edge, updating their skills and embracing technological advancements.

In this chapter we undertake a tri-fold analysis, juxtaposing the efforts of a professional interpreter, interpreting students, and ASST, with a focus on classifying the types of translation departures across these domains. Section 2 lays the foundation by introducing key terminology and discussing previous works in both realms of interpreting and automatic speech translation. Our methodology, presented in Section 3, elucidates our research setting, the metrics for quality assessment, and the qualitative analysis model we have adopted. Section 4 presents an in-depth analysis of two audit speeches, revealing critical insights. The discussion segment in Section 5 extrapolates these findings, contextualizing them within the broader panorama of translation studies and machine translation. We also add our research limitations in Section 6.

This chapter attempts to understand, through a small sample, what the current state of ASST (i.e., transcript translation or live subtitling) is vis-à-vis human interpreting and reflects on the future development in the intersection of technology and interpreting.[1]

[1]Computer Assisted Interpreting (CAI) is also a part of the landscape but we will not cover it in our chapter.

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2 Terminology and related work

2.1 Simultaneous interpreting

Interpreting is a communicative act. An interpreter, hearer and speaker in one person, is a mediator of communication between people who do not speak the same language and come from different cultural backgrounds. When mediating communication, quality is more important than quantity (Kopczyński 1980).

Simultaneous interpreting (SI) is basically the "youngest" type of interpreting. It dates back to the 1920s, when the first interpreting technology IBM-Filene-Finlay was used. The use of simultaneous interpreters was pioneered by the International Labour Organisation (ILO) and on a smaller scale also by the United Nations. In 1935, I. P. Pavlov's speech was simultaneously interpreted at a conference in Leningrad. The most important and famous moment when simultaneous interpreting was used was the 1945 and 1946 Nuremberg Trial with war criminals. It was the first time when simultaneous interpreting was used on such a large scale and with four languages (German, English, French, Russian). Given its nature, simultaneous interpreting is the preferred type of interpreting in the United Nations, UNESCO, NATO, OECD, IMF or in the EU institutions, but of course it can also be found elsewhere in the private and public sectors.

Simultaneous interpreting can be defined in the following way:

Simultaneous interpreting in the booth – interpreting with technical equipment, an interpreter uses headphones, where they can hear the original speech, and a microphone for the transmission of the message to the target language (...) Simultaneous interpreting requires more focused attention, readiness, and is generally more demanding in terms of immediate logical analysis, processing of the received information and its parallel translation into the target language (Čeňková 2008: 21).[2]

The very name "simultaneous" suggests that the interpreter interprets at the same time as the speaker speaks. They maintain a phase shift, also called "decalage", from tenths of a second to full seconds. Interpreting consists of several phases: listening, understanding, analysing, re-expressing (Jones 2002: 6). The peculiarity of SI is that the reproduction phases need to take place before it is possible to listen to the whole interpreted segment. There is also an acoustic difficulty, because the interpreter has to listen and speak at the same time, which is an unnatural process that needs to be trained. The mental difficulty lies in the fact that in simultaneous interpreting the interpreter does not know which direction the speech will take (Jones 2002: 66–67).

[2]The translation into English is our own.

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Daniel Gile's effort model is related to the phases of SI (Gile 2009: 157–190). An interpreter has a certain processing capacity, its saturation leads to interpreting defects. During various phases of interpreting, the interpreter will exert efforts of listening and analysis, short-term memory, production and coordination. The coordination effort ensures that the previous efforts are used efficiently during simultaneous interpreting.

Simultaneous interpreting is a complex activity, in which the interpreter has to use different interpreting strategies, techniques and procedures. They include, "first of all, divided attention, anticipation, the principle of linguistic economy and backward error correction". (Čeňková 2008: 52) Roderick Jones (2002: 83) adds that "the interpreter must always speak coherently and grammatically, and in particular that they must always finish their sentences." According to "Golden Rules of Simultaneous Interpreting", mentioned by Jones, the interpreter should: communicate, use technology, be able to hear the speaker and themselves properly, not try to interpret something they have not heard (or acoustically understood), concentrate as much as possible, not get distracted by single problematic words, cultivate split attention (listen actively to the speaker and monitor critically their own output), use (if possible) short and simple sentences and express some sense with every sentence (Jones 2002: 71–72).

2.1.1 Remote simultaneous interpreting (RSI)

Remote interpreting has been on the interpreting market since the 1970s. According to Braun, "remote interpreting (RI), refers to the use of communication technologies to gain access to an interpreter in another room, building, town, city or country. In this setting, a telephone line or videoconference link is used to connect the interpreter to the primary participants, who are together at one site". (Braun 2015: 352) for Standardization 20178:2017(en) sets the term "distance interpreting" (also "remote interpreting") and defines it as "interpreting of a speaker in a different location from that of the interpreter, enabled by information and communications technology (ICT)".[3]

Because of the rapid growth of "remote interpreting", in January 2019 (updated in March 2020), the International Association of Conference Interpreters (AIIC) issued a recommendation for distance interpreting[4] and also created guidelines

[3] *for Standardization 20178:2017(en) Simultaneous interpreting — Quality and transmission of sound and image input — Requirements*. Available at: https://www.iso.org/obp/ui/#iso:std:iso:20108:ed-1:v1:en [15 May 2021]

[4] *AIIC Guidelines for Distance Interpreting*. Available at:https://aiic.org/document/4418/AIIC%20Guidelines%20for%20Distance%20Interpreting%20(Version%201.0)%20-%20ENG.pdf [15 May 2021]

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for distance interpreting during the Covid-19 pandemic.[5]

The precise form of RI may vary considerably depending on where the speakers, audience and interpreters are located and which participants in the interpreting process are at the same place.[6] In case of remote simultaneous interpreting (RSI), a more complex technological solution is usually needed: one that allows the organisers to put the interpreters into virtual "booths" and enables the audience to switch between different language channels. There are two types of platforms: ones that were made specifically for organizing multilingual interpreted conferences (for example KUDO, Interprefy or Interactio) and ones that focus on video conferencing meetings and video communication in general and interpreting is just a bonus feature (for example ZOOM or Webex; both these platforms have integrated an RSI function). An advantage of professional platforms such as KUDO or Interactio is that they have some advanced features, they, for example, enable relay.[7] However, they are more expensive and harder to control.

The primary benefits of RI are economic, and it also solves organizational issues like the lack of available interpreters in specific locations or unsuitable spaces for interpretation booths (Ziegler & Gigliobianco 2018: 119–120). Initially, RI faced challenges due to poor data transmission; however, while data transmission quality has improved significantly today, some issues persist. As the AIIC Recommendations put it: "[…] different distance interpreting scenarios provide interpreters with varying quantity and quality of sensory input, with differing possibilities for effective teamwork […]".[8] Therefore, one of the main drawbacks of RI is that it complicates collaboration between colleagues, which is particularly crucial in SI.[9] The demand for RI saw a sharp increase during the Covid-19 pandemic due to the widespread societal restrictions it imposed.[10]

[5] *AIIC Covid-19 Distance Interpreting Recommendations*. Available at:https://aiic.org/site/world/about/inside/basic/covid [15 May 2021]

[6] For a more detailed classification of RI using these parameters, see e.g., Braun 2015 or Ziegler & Gigliobianco 2018.

[7] Relay interpreting is the practice of interpreting from one language to another through a third language.

[8] AIIC Recommendations: https://aiic.org/site/world/about/inside/basic/covid, [24 February 2022].

[9] This is confirmed, for example, by a survey of conference interpreters conducted by ESIT in 2021: 83 % of all respondents said that RSI was more difficult for them. The same survey showed that 77 % of respondents regularly help their colleague, but with RSI, 74 % of interpreters had to find another technical solution, because their colleague was in another location and the interpreting platform did not support a direct connection in the (virtual) booth. ESIT Research Project on Remote Simultaneous Interpreting. First overview of survey results. Available at:https://ec.europa.eu/education/knowledge-centreinterpretation/news/esit-research-project-remote-simultaneous-interpreting_en [17 March 2022]

[10] The long-term impact of this shock on the interpreting market is not yet clear, but it is possible

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2.1.2 Strategies used in simultaneous interpreting

Simultaneous interpreting is a highly cognitively demanding process and therefore, the interpreter is to a certain extent at a disadvantage in the communication situation. Interpreters use various strategies to overcome these issues and handicaps. Kalina (1992: 123) defines a strategy followingly: "A strategy is goal-oriented, so that the goal determines the amount and thoroughness of processing. It may be consciously used but may also have become automatic in so far as the processor will not have to make any cognitive decision." In accordance with Kalina, this chapter uses the term strategy which refers to strategic thought processes taking place during text analysis and production (Kalina 1998: 115).

2.1.2.1 Time lag adjustment

Time lag describes the difference between the moment when an interpreter hears a certain segment of a speech and the moment when they reformulate it in the target language. Time lag is also referred to as the Ear-Voice Span, EVS (Gile 2009: 204). Jones (2002: 72–73) points out that an ideal EVS cannot be described in absolute terms – it always depends on the nature of the text and the delivery of the speaker. Lengthening or shortening of the EVS always comes with certain advantages, but also with problems. Shortening the EVS usually relieves the interpreter's working memory and therefore lowers the effort needed in this regard. On the other hand, shorter EVS means worse anticipation, which might cause calques and an excessive dependence on source language sentence structure. Longer EVS may help achieve a more natural and fluent delivery, but it comes with a risk of overwhelming the short-term memory, especially in more complicated parts of the speech that include lists, numbers, or names (Gile 2009: 204–205).

2.1.2.2 Anticipation

Anticipation refers to approximating how the text or the speech is going to continue. Interpreters often rely on anticipation: the first type of anticipation takes place on the linguistic level, for example because of different syntactic structures of the source and target languages. The second type of anticipation is concerned with the content and the ideas of the speech. This type of anticipation is only

that thanks to the rapid technological development, the advantages of R(S)I will outweigh the disadvantages for some clients and the share of remote interpreting will be greater than it was before the pandemic.

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possible thanks to preparation before the interpretation, but also thanks to factors such as a presentation the speaker may show or specific turns of phrase and words that suggest what position the speaker is likely going to take (Gile 2009: 174).

2.1.2.3 Segmentation

Segmentation, also known as chunking (Kalina 1998: 117) or the "salami technique" (Jones 2002: 91), is a crucial strategy for short-term memory management and therefore also for successful interpretation. Dividing compound sentences into short and simple sentences lowers the short-term memory workload, partially because complete meaningful units are easier to remember, but also because it's possible to quickly produce each unit separately without having to listen to the entire complex sentence (Kalina 1998: 117). At the same time, this strategy makes understanding of the target speech easier for listeners (Jones 2002: 93). However, Gile (2009: 205) warns that formulation of short sentences from a complex compound sentence can in some cases increase demands on the processing capacity and thus it is necessary to consider if this strategy is appropriate depending on concrete circumstances.

2.1.2.4 Reformulation

Reformulation is a necessary part of interpreting, because the goal of the interpreter is "to provide a correct translation of the original in a form that sounds as natural and as authentic as possible in the target language […]". (Jones 2002: 82) In general, reformulation can be considered a core strategy that is indispensable in any interpretation, therefore, we must distinguish between basic reformulation needed for correct and idiomatic expression in the target language and "crisis strategy" reformulation, which is for example accompanied by omission of less important information or by generalization, simplification (of the syntactic structure), summarization or by paraphrasing.

2.2 Automatic simultaneous speech translation (ASST)

These days, we witness significant improvements in the field of natural language processing which, undoubtedly, not only reaches unprecedented text processing and generation capabilities (systems such as ChatGPT and other large language models), but also affects speech processing and speech translation. Therefore, in parallel to two interpretation tracks, our intention extends towards scenarios

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where interpreting is carried out by a machine, which results in an alternative third option – automatic simultaneous speech translation (ASST).[11]

ASST models (i.e. fully automatic systems that are capable of generating translation given speech in foreign language) are naturally limited by several constraints. For example, these models lack the world and situation experience in order to select more and less important parts of the message, and they are so far unable to incorporate contextual cues such as nonverbal or visual inputs. This is in contrast to human interpreters who possess the ability to include surrounding context and nuances of the social situation, contributing to a more holistic interpretation. We hope that our study will highlight some specific differences between human interpreting and ASST, serving as a basis for future improvements of ASST systems.

When an ASST system is created, it can be trivially scaled in terms of number of sessions or session duration. If the presentation in the form of subtitles is sufficient for the audience (instead of voice interpreting), using ASST systems might reduce the overall cost. In other words, softening requirements on the final presentation makes the usage of ASST systems more affordable. This can be especially advantageous in cases where many target languages need to be covered, e.g. in the European Parliament or other multinational fora, by multi-target models (Bojar et al. 2021).

The purpose of this section is to very briefly introduce the current approaches to ASST, focusing more on the aspect of their practical use in interpreting settings than on their internal technical design.

2.2.1 Final output form, overall setting and approaches to ASST systems

We differentiate systems based on how they present their outputs to the audience. Depending on the situation, it is preferable that models present their output either as text (subtitling) or as speech (interpreting). Presenting the translation in the form of subtitles or transcripts available on individual participants' devices (Bojar et al. 2021) can be effective if the source speech content is presented exclusively verbally. In such cases, reading subtitles might not be as distractive for the audience as if the speech content was presented also visually. On the other hand,

[11] Automatic simultaneous speech translation is also referred to as simultaneous spoken language translation (simultaneous SLT) or simultaneous speech translation (ST, SST) in the literature. Note that we explicitly avoid the term "automatic interpreting", to highlight the scale of the difference between automatic outputs and the outputs of human interpreters. Machines, so far, indeed only transcribe and translate sentences of the source language, without any attempt at interpreting their meaning, condensing, simplifying or adapting it for the audience. Other possible terms to describe this process are live translation or live subtitling.

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if the speaker makes use of additional visual inputs such as a presentation, the audience might prefer speech output even if it requires headphones. This setup, however, puts more demands on simultaneity – the audience needs the visual information and the translation to be synchronized as much as possible (Javorský et al. 2022).

The final presentation of systems is tightly connected to their design. There are two major categorizations of models that influence the overall translation quality and simultaneity.

The first classification distinguishes how the models are created. The architecture (i.e. the design of the model) can be either a cascade of several components, or one component that translates the input speech directly to the output text or speech, in an "end-to-end" fashion. In the cascaded setting, a typical model consists of an automatic speech recognition (ASR), segmenter (which splits the flow of words into segments – sentences), a machine translation (which translates individual sentences) and an optional text-to-speech module (Müller et al. 2016, Bojar et al. 2021). Unfortunately, this cascaded approach suffers from the inherent problem of error propagation – each component introduces its own errors and these errors compound as the generation progresses (Ruiz & Federico 2014). The advantage is, on the other hand, that there exist many resources for developing individual components and we can thus optimize each component independently, which improves the overall quality. However, in end-to-end settings, the system is designed to process the flow of words continuously, producing subtitles or speech directly, without any intermediate steps (Polák et al. 2022). Such systems can incorporate crucial information about prosody (i.e. the rhythm, intonation, emphasis) and they generate translation with a smaller delay because they do not need to wait for (partial) outputs from previous components. Although the resources for creating these systems might be limited, overall, the performance has recently become superior to the cascaded models for widely-spoken languages.

The second dimension of classification lies in how the output is delivered. There are either re-translating (Niehues et al. 2016, 2018) or incremental machine translation systems (Dalvi et al. 2018). Re-translating models can afford to change output hypotheses based on the growing context. For instance, translating from a language where gender is not expressed for every noun, more context might provide some indications that help the model to guess the grammatical gender. In subtitles, such changes in the output are perceived as flickering – the updates of text that has already been emitted. It also becomes apparent that speech synthesis (i.e. the automatic generation of speech from text) cannot be used in this setting. On the other hand, incremental models are only allowed to append more

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words and cannot alter the history. In this setting, the audience follows the subtitles more conveniently but the systems cannot correct errors they made and, therefore, the overall quality of the translation can be worse compared to the re-translating approach.

2.2.2 ASST systems vs. human interpreting

In this section we discuss how ASST systems are usually evaluated. One option is to evaluate the quality of the final, textual translation using established automatic metrics (Papineni et al. 2002, Popović 2015, Rei et al. 2020). These metrics assess the similarity (in the simplest case the overlap between words) of the reference and the evaluated translation. Using this type of evaluation is fast but often not reliable. Another way is to evaluate the whole subtitled presentation manually, e.g. in an online setting using so-called Continuous Rating (Javorský et al. 2022) – annotators who understand both the source and target language are asked to continuously rate the quality of the subtitles. The advantage is that humans are able to assess nuances which automatic metrics cannot. On the other hand, the agreement between annotators can be poor (subjectivity may play a major role in the assessment) and it becomes difficult to interpret the results in a meaningful way. Nowadays, both automatic and manual types of evaluation are used in so-called shared tasks (Agrawal et al. 2023, Anastasopoulos et al. 2022). Shared tasks involve cooperative efforts where researchers and professionals collaborate to address a mutual challenge by utilizing shared data and assessment criteria.

The first comparisons of interpreters and systems were performed during these shared tasks. Since the systems are capable of presenting only textual output to the audience, the evaluation was performed only in the form of subtitles. Interpreting was thus transcribed, discarding many of its important qualities such as comfortable setting of audience listening, interpreters summarising the message to avoid overburdening listeners with too much of a speedy rendering, etc. Furthermore, the systems were created in a well-resourced domain and input speech was segmented into sentences and the systems were only given short recordings rather than the whole speech audio as it would be the case in a true-to-life scenario. In such a specific and not quite fair comparison, the results of the evaluation on semantic completeness seem to indicate that systems appear superior to interpreters.

Given all that, there is still a significant gap between what we know about errors observable in the translation generated by automatic systems and the errors in interpreting. We believe that knowing the relations or common attributes between interpreting and ASST might shed more light in how to better design and

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evaluate systems. This is a crucial aspect in developing more robust and reliable systems and moving forward in the field.

3 Method

3.1 Research setting and experimental design

Our study is based on data we collected during remote simultaneous interpreting (RSI) sessions in 2021 and 2022. Interpreters of varying proficiency levels (professional and trainee interpreters) were required to perform RSI of several speeches via Zoom. The speeches were available both as audios and videos, so we decided to show the videos to the interpreters: it is more pleasant for the interpreter to see the speaker and to have a general overview of the communication situation. Besides, the speech was sometimes directly referring to the visual information, e.g. to some illustrations or numbers in the slides.

For our analysis, we selected two audit speeches in the form of monologue presentations. Both audit speeches chosen for our experiment were fully prepared: the speakers had created their presentations in advance. However, the interpreters did not receive the full slides of the presentations in advance, they were only given keywords representing the given thematic area (see the section Research limitations). Both original speakers were non-native speakers with an advanced level of English knowledge.

The RSI was carried out in the language combination English-to-Czech (from B language to A language).[12] Our analysis presented in this study concerns the English-to-Czech direction. We hired one professional[13] interpreter for the language combination English-to-Czech (I1) and several trainee interpreters, two of whom (I2, I3) interpreted either of the two speeches for us. (See Table 1 for more details.)

[12]Besides, the RSI was also carried out in the language combination English-to-German.

[13]We are aware of the fact that we are analysing two interpretations by only one professional interpreter. However, it should be explained that the recording of interpreting performance is very problematic in the market. Professional interpreters are very careful, if the client records their performance, it is for an additional fee. Professional interpreters, unlike students, participate in experiments very rarely, they do not want to be analysed and evaluated, because they are aware that interpreting has a so-called suboptimal quality, because it is an activity that takes place "here and now", interpreters are under a lot of pressure, they have to make decisions very quickly, often omitting or generalizing some information. In addition, interpreting serves in the given context, when analysed ex-post, many strategies may appear inadequate, the evaluator has more time and no pressure compared to the interpreter to evaluate his performance and find an optimal solution.

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Table 1: Basic information on the two interpreted speeches.

	topic	length	professional interpreter	trainee interpreter
audit speech 1	Data Analysis with the Municipality Monitoring Tool (Austria)	10:31	I1	I2
audit speech 2	Open Data in the Netherlands	11:27	I1	I3

The professional interpreter (I1) has more than 10 years of interpreting experience. She studied at the Institute of Translation Studies, Faculty of Arts, Charles University (ÚTRL) Czech and English translation and interpreting. The trainee interpreters (I2 and I3) were (at the time of the experiment) MA students of interpreting (with at least two semesters of SI) from ÚTRL (MA Study Programme: Czech and English translation and/or interpreting).

The RSI sessions were recorded both by the interpreters themselves on a computer or mobile phone and also by us via the Zoom platform (as a backup). The recordings of the interpretation (as well as the original speeches) were automatically transcribed using an automatic speech recognition (ASR) system. The resulting transcripts were then manually corrected by specially trained annotators (by the present authors) according to the audio recordings.

We also had the speeches translated by our ASST system ELITR (Bojar et al. 2021), developed and trained at the Charles University, Institute of Formal and Applied Linguistics (ÚFAL). The specific setup used in this study consisted of the speech recognition module Whisper (Radford et al. 2023) operated in continuous and simultaneous mode using Whisper-Streaming (Macháček et al. 2023) with 1 second minimum chunk size. The automatically transcribed English speech was translated into Czech using the CUNI Transformer Marian model from WMT19 (Popel et al. 2019) in re-translation mode implemented in ELITR submission for Macháček, Kratochvíl, et al. 2020 (Macháček, Kratochvil, et al. 2020). Only completed segments (i.e., not gradual updates in the re-translation) were considered in this study. The outputs of this pipeline are also transcripts.

Then, we as the authors of this chapter, analyzed the (corrected) transcripts according to the proposed qualitative model (see Section 3.3), and we also had the recordings available to compare the transcripts with if necessary (in case of

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human SI). We divided the transcripts into segments (sentences in most cases), aligned them and compared the three interpreted versions (I1 vs I2 or I3 vs ASST) to each other and to the original speech. Our goal was to examine each segment in terms of the qualitative analysis model (see below), and to assign to individual segments one (or more) of the translation departure subcategories, if found there.

3.2 Interpreting quality assessment

It is very difficult to define the quality in interpreting, as it, to certain degree, depends on a subjective perception. According to Jesús Baigorri Jalón (2000: 57), quality assessment is (and always has been) inherently subjective: "The audience's evaluation of the interpreter's performance has always depended solely on their subjective criteria [...] These are the only quality assessment criteria that are reliably available to us." Opinions on the quality assessment vary. Therefore, different criteria (presentation, pleasant voice, grammar, hesitations etc.) or different audience groups are used when assessing the quality.

AIIC has certain quality standards for conference interpreting that all members must meet. Karla Déjean Le Féal (1990: 155) said about the quality standards in AIIC: "What clients hear through headphone interpreting should have the same effect on them as on other listeners who listen to the original speaker. The interpreting should have the same cognitive content and should be expressed clearly and exactly in the same type of language. The quality of language production and presentation should be at least at the same level as by the original speaker, or better, considering that the interpreter is a professional speaker, which is not the case for many speakers, who sometimes even have to speak in a language that is not their mother tongue. Our main goal is to satisfy the audience."

Andrzej Kopczyński (1994: 87–88) thinks there are two ways of looking at quality – a linguistic one and a pragmatic one. The linguistic point of view focuses on accuracy of the content and form. The pragmatic point of view depends on the context and is considerably influenced by several variables (e.g., speaker's goals in the communication, speaker's attitude, the form of the message, interpreter's competence, illocutionary force of the message etc.).

According to Daniel Gile (2009: 37), the satisfactory quality expected by someone engaged in the communication does not always go hand in hand with absolute fidelity to the original, linguistic acceptability or the terminology used. The sender and the receiver are experts and professionals in the given communication situation, while the interpreter might not be. And yet, their performance can be considered satisfactory, if they can fulfil the communication goal of the

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speaker. But at the same time, it does not always have to be the interpreter's fault if the communication goal is not reached. The problem might be on the side of the receiver. Nevertheless, it is mostly the receiver who evaluates the interpreter's performance, however subjective their criteria might be.

Miriam Shlesinger (1997: 128) writes that the quality of an interpreter's performance should be judged by three factors: intertextuality, intratextuality and instrumentality. Intertextuality is the comparison of the original and the new message based on their similarities and differences. Intratextuality deals only with the "product" and its acoustic, logical and linguistic characteristics. And instrumentality focuses on the impact of the speaker's speech and its usefulness. The interpreter has to adjust their speech to the type of the event, the type of text, the situation and the audience.

Errors are an indicator for the quality of interpreting. Kopczyński (1980: 68) says that it is important to define what errors and how many of them can occur, so that the interpreting could be still considered acceptable. The borderline can be very subjective. The theoretical handbooks of interpreting say that an interpreter should not make any errors. This situation is considered ideal, but it is difficult to achieve in real life. Ángela Collados Aís confirms this (2007: 196), she agrees with other scholars that completely flawless interpreting is inconceivable. Therefore, it is important that the interpretation is acceptable in terms of context and pragmatics and that the interpreter achieves the goals of the given communication situation.

3.3 Qualitative analysis model

In our study we focus on quality of interpreting in the sense of transferring the content of the original speech. The basis of our analysis is represented by the taxonomy of translation[14] departures designed by Barik (1971) describing omissions, additions, and substitutions in simultaneous interpreting. If the substitutions are at considerable variance with the original version, we may speak of semantic errors (Barik 1971: 199). The reason we chose Barik's model is related to the fact that it focuses on translation departures at the semantic level, which is the goal of our research. Previous research on the quality assessment of SI into Czech has demonstrated its suitability and applicability (cf. Winklerová 2019).

For the purposes of our analysis, we extend Barik's model with additional factors such as adaptations and extensions, especially concerning grammar errors

[14] According to Barik (1971) we regularly employ the terms *translation* or *to translate* instead of *interpretation* or *to interpret* with the understanding that they refer to *oral* translation (= interpretation), not to *written* translation.

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and formal defects (Kopczyński 1980). A similar methodology for the analysis of chuchotage (whispered interpreting) was chosen by Winklerová in her diploma thesis at the Institute of Translation Studies (Charles University, Prague) in 2019.

Similarly to Barik's work, our analysis does not reflect on other critical aspects of interpretation such as latency or other criteria commonly used in interpretation assessment such as pleasant voice or native accent (Bühler 1986, Kurz 1993, Pöchhacker & Zwischenberger 2010) – our aim is to concentrate purely on the content of the original and translated speech.

In Table 2 we describe the qualitative analysis model in more detail. As mentioned above, our analysis model has been inspired by Winklerová (diploma thesis written in Czech, 2019: 69). The categories 1. omissions, 2. additions and 3. substitutions and errors follow Barik's translation departures scheme (Barik 1971: 200–207), whereas 4. formal errors and 5. grammatical errors represent categories introduced by Kopczyński (1980).

However, we slightly adapt these typologies for our purposes by dividing the first three semantics-based categories into three levels according to their severity for the target text, i.e., the extent to which they affect the final content of the interpreted message as a whole.[15] We also change the coding scheme to be able to clearly refer to each category by its initial letter (omissions: O, additions: A, substitutions end errors: S, grammatical errors: G, formal errors: F).

Our model for the qualitative analysis summarizes the main types of departures and errors of the interpretation. We are aware of the fact that the proposed typology had to be restricted, so that some departures and errors are assigned to the same category even though their nature is different. However, a too detailed categorization system would not be appropriate because of poor clarity and the near impossibility of dividing the items into (only theoretically existing) 'correct' types of shifts. We therefore decided on a compromise: a sufficient division of the categories according to the main criterion which is the severity in terms of how the content of the whole is translated (a similar problem in translation is being approached by Multidimensional Quality Metrics, Burchardt 2013).

However, the form, i.e., the way the target text is formulated, is also an inherent part of the transfer of the original content. Therefore, we decided to include the evaluation of formal and grammatical errors in our analysis. Formal errors are associated with both the spoken language (e.g., false starts or unfinished words) and the translation of the text itself (e.g., interference). These shortcomings can

[15] In his original typology, Barik also divides the departures and errors into four (1. omissions and 2. additions) to five (3. substitutions and errors) categories (some involving subcategories), but he also analyses the possible causes and other factors of the errors described.

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also lead to distortion of the content. Grammatical errors include, for example, deficiencies in morphology (inflection) or syntax (word order, information structure / functional sentence perspective).

The evaluation was carried out by two translation theoreticians (co-authors of this chapter) who are also active translators / interpreters. In cases of doubt, they discussed together trying to reach a consensus.

4 Analysis

In this subsection we describe the analysis of two audit speeches, terminologically very demanding monologue presentations of 10:31 and 11:27 minutes in length, delivered by non-native speakers of English. Their topic is audit in Austria (Audit speech 1) and in the Netherlands (Audit speech 2). In both audit speeches we examined each segment of the transcribed source and both interpretations (A and B) and the output from the machine translation system (C) for possible omissions (O), additions (A), substitutions and errors (S), grammatical (G) and formal errors (F).

4.1 Audit speech 1

The topic of the audit speech 1 (10:31 minutes long) is Data Analysis with the Municipality Monitoring Tool (Austria).

The following Table 3 shows the number of words, characters and segments of transcribed speeches (original and interpretations).

The aim of the analysis was to compare an English original with two interpretations and one (real-time) translation into Czech (two human interpretations and one ASST). We can immediately see that there is less volume of text in the human interpretations and that the two interpretations are shorter than the original, also perceptible in the numbers: English original has 87 segments, interpretation A has 64 segments and interpretation B has 67 segments; the ASST model is more dense, even literal and does not use interpreting strategies (e.g. analysis and filtering of information) such as human SI, which is not an advantage in terms of quality because the interpreting techniques are used for a reason (to avoid cognitive overload in listeners).

In the following Table 4, the interpretation with the highest number of departures is indicated in italics.

The lowest incidence was in the category “additions”, only one (A1), by both interpreters A and B. Both interpreters chose the right strategy, they did not

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Table 2: Model for the analysis.

Category	Code	Description of the categories and their subcategories
1. *Omissions*: items in the original which are left out of the translation	O	*O1*: O of a single word or short phrase, irrelevant to the meaning of the whole (skipping O), acceptable in SI *O2*: slightly affects the meaning of the whole (disrupts logical construction of the text), missing names, numbers, more serious than O1 (disjointed speech) *O3*: significantly affects the meaning of the whole, altered meaning of a sentence: regrouping material from different constructions, more serious than O1 and O2
2. *Additions*: material which is added compared to the original message	A	*A1*: irrelevant to the meaning of the whole (qualifier or elaboration A) *A2*: slightly affects the meaning of the whole (disrupts logical construction of the text: new meaning or relationship), explicitation (adding a connective), more serious than A1 *A3*: significantly affects the meaning of the whole, accompanies rephrasing, misinterpretation, more serious than A1 and A2
3. *Substitutions and errors*: material which is substituted by the translator for something said by the speaker (no added or omitted information)	S	*S1*: mild semantic error, a single lexical item (irrelevant to the meaning of the whole), involving also functional departures (generalization and concretization) *S2*: substantial semantic error, slightly affects the meaning of the whole, wrong numbers (disrupts logical construction of the text) *S3*: gross semantic error, significantly affects the meaning of the whole, considerable difference in meaning, nonsense, opposite meaning, meaningless and confused translation
4. *Formal errors*	F	false starts, hesitation, stuttering, unfinished words/sentences, interferences, repetition of a word or longer segment in the same form
5. *Grammatical errors*	G	incorrect inflection, incorrect word order including functional sentence perspective

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Table 3: The number of words, characters and segments in the transcripts (Audit speech 1, Austria).

	source EN	transcript A (professional interpreter)	transcript B (trainee interpreter)	transcript C (ASST)
words	1,625	1,066	710	1,133
characters	8,981	6,677	4,488	7,068
segments	87	64	67	74

Table 4: The number of translation departures detected in the interpretations of Audit speech 1.

	A Professional interpreter	B Trainee interpreters	C ASST
O1	12	**22**	2
O2	3	**13**	2
O3	0	**1**	**1**
A1	**1**	**1**	0
A2	0	0	0
A3	0	0	0
S1	3	**15**	5
S2	**10**	2	9
S3	5	0	**6**
F	**52**	7	34
G	14	2	**24**

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change the meaning of the segment, on the contrary, they tried to make it easier to understand for the audience and used explicitation. The ASST system as a machine is not capable of pragmatic reasoning, hence we did not observe any addition in any segment.

In the category "substitutions", we can see that the total number of departures is almost equal for all renditions (Interpreter A 18, Interpreter B 17, ASST 20), nevertheless, the distribution of departures into the subcategories is different: the student interpreter has the highest number of departures in the subcategory S1 (15). It appears that those departures can be in most cases considered as a well-chosen strategy that was based on good understanding of the original and good active analysis of the original speech; the professional interpreter has 10 departures in the second subcategory (S2) and 5 in the third one (S3); the ASST system has 9 departures in the second one (S2) and 6 in the third one (S3) – those departures are (relatively) serious errors that can distort the message to a great extent. Here, in most cases, it is a matter of translating the original structure too literally and not recognizing which word types are represented in the given syntactic structures. This results in errors of meaning – syntactically correct in the target language, but semantically meaningless (*a brief overview about our big data tool* translated as *stručný přehled o našem velkém datovém nástroji*, which can be glossed as big tool for data, instead of *stručný přehled o nástroji, který používáme na analýzu velkých dat*). Some departures of this type are repeated many times and we had to count every occurrence (*use the data of the Statistics of Austria* translated as *využijeme data Statistického Rakouska*, meaning the data of the statistical Austria, instead of *využijeme data Rakouského statistického úřadu*).

In the category of omissions,[16] there are more departures in the student interpretation: 22 in the first subcategory vs. 12 by the professional, 13 in the second subcategory vs. 3 and 1 in the last one vs. 0. The first subcategory monitors the omissions of insertions, small comments, secondary information – i.e., utterances the omission of which will not affect the meaning of the interpreted speech. In the second subcategory (O2), the student interpreter tends to leave out exact names and numbers, but retains the overall information – for example, by rounding the numbers, which is an acceptable strategy. In contrast, the ASST system has, as expected, a negligible number of omissions, 5 in total (2 in the first subcategory, 2 in the second one and 1 in the third one).

Grammatical (G) and formal (F) errors were surprisingly predominant in the professional interpretation: there were 52 formal and 14 grammatical errors,

[16]We emphasize again that the category labelled "omissions" bears no negative connotations here.

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which is a relatively high number, considered that the student interpretation had only 7 and 2 of them. The ASST system committed 34 formal and 24 grammatical errors, deficiencies at the formal level are especially interferences which can be seen as related to an overly literal translation (the same reason we observed with the professional interpreter); errors at the grammatical level are assessed mainly as incorrect word order (again due to the literalness, the same phenomenon also occurred with the professional interpreter).

In Table **??** below we present the method of analysis of the original and its interpreting – how the transcribed segments were categorized and commented. In column A, there is the English original, then the professional interpretation (column B), the student interpretation (column E) and the ASST transcript (column H), including the applied categorization of departures (columns C, F and I). For the sake of clarity, we add our commentary after the evaluations, explaining what departures / deficiencies occurred (columns D, G and J, in italics). In case there were more than one (sub)category in a segment, we used red and blue colouring.

(1) Example of the annotation interface for the analysis with (sub)categories of translation departures and errors identified.

a. i. In the second step we assess the municipalities with regard to their significance for the audit activities of the Austrian current Austrian Court of Audit.

ii. V druhým hodnotíme ty municipality ohledně relevantnosti pro ten audit.

iii. G, F

iv. There are two categories in this segment. Interpreter A used an incorrect ending in the word “druhým”, the correct ending is “druhém”. The interpreter overuses demonstrative pronouns in the whole speech; in this segment, the pronoun “ty” is used, but could be omitted.

b. i. Ve druhém kroku jsme dané obce ohodnotili podle aktivit, které se tam provádějí.

ii. S1

iii. In this segment, interpreter B mentioned only activities in general, leaving out that they were activities of the Court of Audit. This was a good strategy, because the omission did not change the message.

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c. i. Ve druhém kroku hodnotíme obce s ohledem na jejich význam pro auditní činnost rakouského účetního dvora.

ii.

iii. This passage corresponds to the original, no translation departure.

(2) a. i. The ranking is always the interesting part, the most interesting part of the whole story, because we can compare any municipalities.

ii. Samozřejmě to seřazení do řebříčku je ta nejzajímavější část, protože pak si ty municipality můžeme srovnat.

iii. G, F

iv. Again, there are two deficiencies on the formal and grammatical level: a slip of tongue (it should be "žebříček", not "řebříček"), bad choice of demonstrative pronouns, the pronoun "to" should be omitted and instead of "ty", it would be better to use the expression "daných".

b. i.

ii. O1

iii. Interpreter B omits this passage, but it is clear from the context what is meant, no relevant information is missing. We consider this to be an acceptable strategy.

c. i. Pořadí je vždy ta zajímavá část, ta nejzajímavější část celého příběhu, protože můžete porovnat obce, obce.

ii. F, G

iii. Again, there are two categories in this segment. On the formal level, the same word was repeated twice ("obce"). ASST copies the structure of the original, interpretation is literal, incorrect word order including functional sentence perspective.

(3) a. i. So, from the existing data we think that our sell well indicators, which I will explain later, and we try to make a indicator model and this monitoring is based on a model which is comprised of indicators that are compiled as assessment indicators, broken down with themes.

ii. Tady jsme vyfiltrovali několik indikátorů, to vysvětlím vzápětí, a snažili jsme se tedy vytvořit model pro ty indikátory a tady ten

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model, vychází z ukazovatelí, kterou které vycházejí podle jednotlivých témat.

iii. G, F S3

iv. We have identified three (sub)categories in this segment. A grammatical error in the word "ukazovatelí", the correct form is "ukazovatelů". On the formal level, the same word was repeated twice ("kterou které"), but we have to admit that interpreter A corrected their incorrect relative pronoun form "kterou" to the apropriate one "které". We, still, used the mark S3, because the segment is unintelligible, there is no analysis, the message is lost and the interpreter only repeats the original words.

b. i. Snažíme se vytvořit indikační model. Toto pozorování se zakládá na modelu, který je složen z indikátorů, které hodnotí daný subjekt.

ii. S1

iii. The salami technique, i.e. segmentation of a long sentence to shorter sections, is well used in this segment. The omission of the insertion about a later explanation of the model does not distort the message and is a good example of a filtering strategy.

c. i. Z existujících údajů jsme tedy vyfiltrovali několik ukazatelů, které vysvětlím později, a pokusili jsme se vytvořit model ukazatelů a toto sledování je založeno na modelu, který je složen z ukazatelů, které jsou sestavovány jako ukazatele hodnocení rozdělené podle témat a…

ii. G, S3, F

iii. We have identified three (sub)categories in this segment. A grammatical error consists in incorrect word order including functional sentence perspective. We, still, used the mark S3, because the segment is unintelligible, there is no analysis, the message is lost and the interpreter only repeats the original words. There is an error on the formal level, because ASST did not finish the sentence.

(4) a. i. And yes we take into account other accounts, the miscellaneous accounts, the representation expenses and the analysis of special accounts.

ii. A tady bereme v úvahu různé účty, osobní výdavky, výdaje a analýzy specifických…

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iii. F, G

iv. An inappropriate choice of the term "výdavky", the correct term is "výdaje" (G). There is an error on the formal level, because interpreter A did not finish the sentence.

b. i. Zde sledujeme různé související výdaje.

ii. S1

iii. Interpreter B used the general term "výdaje" instead of listing the individual items. This is a case of generalization, which is a well-chosen strategy, the message was not distorted.

c. i. A zde bereme v úvahu ostatní účty, různé účty, výdaje na reprezentaci a analýzu zvláštních účtů.

ii. F

iii. There is an error on the formal level, because ASST repeated the same word twice ("účty").

(5) a. i. And the second one gives us the short-term perspective.

ii. a druhý model nabízí krátkodobý pohled na finanční zdraví.

iii. P1

iv. Interpreter A chose to specify the term perspective and added the phrase "na finanční zdraví". Thit is a well-chosen strategy, considering the context; the message of the segment was not distorted.

b. i.

ii. Druhá část nám dává krátkodobou perspektivu.

iii. This passage corresponds to the original, no translation departure.

c. i.

ii. Druhá nám dává krátkodobou perspektivu.

iii. This passage corresponds to the original, no translation departure.

4.2 Audit speech 2

Audit speech 2 (11:27 minutes long) concerns the Open Data in the Netherlands.

For clarity, the number of words, characters and segments in the transcripts is given in the following Table 5:

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Table 5: The number of words, characters and segments in the transcripts A, B, and C (Audit speech 2, Dutch).

	source EN	transcript A (professional interpreter)	transcript B (trainee interpreter)	transcript C (ASST)
words	1,582	1,111	905	1,265
characters	9,294	7,196	5,701	8,381
segments	112	65	72	95

For the analysis of Audit speech 2, we compared 112 segments (source in English) with 65 (interpretation A in Czech), 72 (interpretation B in Czech), or 95 (transcript C, ASST) segments. The numbers alone show that there has been a reduction of the original textual volume during the interpretation (especially A and B), and thus various types of omissions (besides the usual text condensation) are to be expected in the interpretations. However, we expect that other (sub)categories of translation departures will also be represented. As for the transcript C, we can see that no such reduction of textual material has occurred, as the ASST system is very literal and does not use strategies such as human SI.

The following Table 6 shows the number of translation departures detected in the three analyzed transcripts of Audit speech 2. Items in italics indicate a larger number of given translation departures in A, B, or C transcript.

The Table 6 above presents the distribution of each translation departure in the performance of the professional (A), the trainee interpreter (B), and the ASST system (C). In the categories of omissions and substitutions and errors, in which we distinguish the severity of departures (scale 1 to 3), we can notice that in the least severe subcategory (O1, S1) the student has more departures (13 versus 7 in both cases). In comparison to ASST, the category O1 in ASST is negligible (1) and refers to one segment that is omitted in all three transcripts. In contrast, we see a large number in the S1 category for ASST (20), which is much higher than for human SI. Here, in most cases, it is a matter of translating the original structure too literally and not recognizing which word types are represented in the given syntactic structures. As in the previous speech, this results in errors of meaning – syntactically correct in the target language, but semantically meaningless. It is also related to the use of unusual word combinations instead of familiar collocations (*otevřené údaje* instead of *otevřená data*). This particular error is repeated many times (in parallel with the correct collocation); we had to count every occurrence, though.

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Table 6: The number of translation departures detected in the interpretations of Audit speech 2.

	A professional interpreter	B trainee interpreters	C ASST
O1	7	13	1
O2	8	18	0
O3	3	0	0
A1	1	0	0
A2	0	0	0
A3	0	0	0
S1	7	13	20
S2	13	4	3
S3	6	1	0
F	42	18	16
G	12	0	5

In the second level of severity, the student has more departures in the O2 category (18 versus 8), but fewer in the S2 category (4 versus 13) than the professional interpreter. What is very surprising, however, are the O3 and S3 categories, in which the student has fewer departures than the professional interpreter (0 versus 3 and 1 versus 6, respectively). ASST scores with 0 departures in O2, O3, and S3, and with 3 departures in S2 category; again, it has to do with a needlessly literal translation, for example: *innovative entrepreneurs can turn open data into new applications* (source) translated as *mohou inovativní podnikatelé z otevřených dat udělat nové aplikace*.

From the category of additions, we only recorded the least serious A1 grade, namely only one case in the professional interpreting.

The distribution of formal (F) and grammatical (G) errors is very noteworthy. In both categories, the professional interpreter has significantly more errors than the student: we recorded 18 formal errors in the student performance, whereas there are 42 errors in the professional performance (that is 24 more). The student made no grammatical errors, but the professional interpreter made a total of 12 errors in this category. As for ASST, 16 errors in F mean especially interferences which again can be seen as related to an overly literal translation; the G category includes erroneous plural endings of the neuter gender.

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Below, there is a sample of the annotation interface where we performed the analysis and assigned the (sub)categories of translation departures and errors to the text material (Table **??**). The description of the annotation interface is the same as the text describing Table **??**.

(6) Example of the annotation interface for the analysis with (sub)categories of translation departures and errors identified.

a. i. And here the United Kingdom and the United States are setting the pace.

ii. a tady ve Spojeném království a v Nizozemí je to důležité.

iii. S3

iv. *Interpreter A translates "the United States" as "Nizozemí" (= in Czech "the Netherlands", in fact). The overall meaning of this section is greatly affected.*

b. i.

ii. V1

iii. *Interpreter B omits this passage, but it is clear from the context what is meant, no relevant information is missing. We consider this to be an acceptable strategy.*

c. i. A tady Spojené království a Spojené státy udávají tempo případu.

ii. S1

iii. *ASST translates "udávají tempo případu", however, the usual collocation is just "udávají tempo" (= are setting the pace).*

(7) a. i. They see open data, em are clearly not a goal but a means to become transparent.

ii. A veřejné data nejsou cílem, ale je to opravdu prostředek k dosažení transparentnosti.

iii. F

iv. *Interpreter A uses an ungrammatical ending for the plural of the neuter gender: "veřejné", correct form in this case: "veřejná". The overall meaning of this section is not affected.*

b. i. Otevřená data jsou jedna z největších příležitostí, jak být transparentní.

ii. S1

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iii. *Interpreter B adds a bit of emphasis, stating that open data is "one of the biggest opportunities" (= in Czech "jedna z největších příležitostí") to be transparent. The overall meaning of this section is only slightly affected.*

c. i. A vidí, že otevřená data zjevně nejsou cílem, ale prostředkem, jak se stát transparentními.

ii.

iii. *This passage corresponds to the original, no translation departure.*

(8) a. i. An ini initiative to encourage government to be as open as possible.

ii. a snažíme se být co nejtransparentnějších.

iii. S3, F

iv. *Two (sub)categories applied in this section. The misunderstanding from the previous segment continues, intensified by the verb "snažíme se" (= "we try") (S3). Formal error in the form "nejtransparentnějších", correct form should be "nejtransparentnější" (F).*

b. i. A jsou tedy zde iniciativy, aby vlády byly co nejotevřenější.

ii.

iii. *This passage corresponds to the original, no translation departure.*

c. i. což je iniciativa, která má povzbudit vlády, aby byly co nejotevřenější.

ii.

iii. *This passage corresponds to the original, no translation departure.*

(9) a. i. And the Dutch government has open data ambitions and policies.

ii. A nizozemská vláda má zákony,

iii. S2

iv. *Interpreter A did not understand this segment and only captured and translated a part of the content that did not correspond to the original statement.*

b. i. A nizozemská vláda má velké ambice v tomto odvětví.

ii.

iii. *This passage corresponds to the original, no translation departure.*

c. i. A nizozemská vláda má otevřené datové ambice a politiky,

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ii. S1, F

iii. *ASST misinterpreted the structure "open data ambitions" and thought the first two words were adjectives. The correct translation is e.g. "má ambice v oblasti otevřených dat". The word "politiky" is used in plural (= politics).*

(10) a. i. But the actual implementation is still grass roots and it's in early stages.

ii. ale v této oblasti, ale v praxi ještě pokulhává.

iii. F

iv. *Interpreter A resorts to repeating the same thing and overgeneralizing.*

b. i.

ii.

iii. V2

Interpreter B omits this segment which slightly affects the meaning of this passage.

c. i. ale vlastní implementace je stále lidová a je v počáteční fázi.

ii. S2

iii. *ASST uses the word "lidová" (= plebeian, popular). "Zcela základní" or "elementární" (= elementary) would have been better suited. This misinterpretation slightly affects the meaning of this passage.*

4.3 Evaluation of the analysis

The following Table 7 summarizes both the professional's and trainees' performance in comparison with the ASST scores showing the number of translation departures in each category, for both analyzed speeches. All (sub)categories from our adopted analysis model with exceptions of A2 and A3 were present in the interpretations. Items in italics indicate the largest number in a given category in both speeches among the three transcripts analyzed.

The category O1 (omissions that are not essential for the meaning of the whole) is most represented in students' interpreting: there are 35 omissions in total in students' interpreting vs. 19 in professional interpreting. However, the ASST model made only 3 omissions defined as O1. As for the more serious omissions, students made 31 departures in the subcategory O2 (the professional interpreter scored 11 in the same subcategory). However, perhaps unexpected is the finding

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that in the subcategory O3 the professional interpreter scored 3, whereas students made 1 departure. On the other hand, it was to be expected that the ASST model made fewer omissions, which was confirmed (2 for O2, 1 for O3).

We can observe a very similar tendency as in the O1 subcategory also in the S1 subcategory of substitutions and errors: 28 occurrences in students' interpreting, whereas the professional interpreter made only 10 as S1 defined departures. But what could be seen as surprising are much higher numbers in the other two subcategories of substitutions and errors: S2 with 23 and S3 with 11 in professional interpreting, whereas in the students' interpretation there are only 6 and 1 occurrences, respectively, which are negligible amounts. The ASST model made 25 departures in S1 subcategory, 12 in S2, and 6 in S3 – it is therefore on the borderline between the results of the two human interpretations.

The analysis shows that the students made a large number of less serious departures (O1, O2, S1) in comparison with the professional interpreter, but they use them effectively, which allows them to avoid the more serious ones; the professional interpreter, on the other hand, made many of the more serious ones, which makes their interpreting of poorer quality. In conclusion, it should be noted that both evaluations of students' interpreting show that their interpreting is controlled and that they apply interpreting strategies consciously.

As for the A category (additions), only subcategory A1 occurred in the interpretations, twice in the professional performance and once in the students' interpretations. Addition can be considered a pragmatic category aimed at facilitating listeners' understanding, so it is logical that the ASST model did not commit it even once.

Surprisingly, the professional interpreter makes a large number of errors on the formal level (unfinished sentences, slips of tongue, incoherent and long sentences) and on the grammatical level, which spoils the overall impression of the interpreting (F 94, G 26). On the other hand, in students' interpreting we found far fewer occurrences in these subcategories (F 25, G 2). As for the performance of our ASST model, there are 50 occurrences in F and 29 occurrences in G (only 3 more than in the professional performance), many of which are actually repetitions of the same error.

Our analysis leads us to the conclusion that we were right in distinguishing not only the quantity, but also the severity scale. The more frequent occurrences in O1, A1, and S1 could be seen as functional translation departures that help both the interpreters and the clients to follow the basic semantic line of the original message.

Finally, we would like to comment on the number of interpreting segments. In Table 3 and Table 5 we have already mentioned that the students' interpreting

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has a higher number of segments than the professional interpreting (and lower than the original) in both speeches, despite the fact we found more omissions in the student interpreting (professional interpreter had 33 omissions, students 67). This is because the students, perhaps consciously, use the salami technique, i.e., they break down the long endless sentences into shorter and simpler utterances. Nevertheless, the highest number of segments was produced by the ASST model in both speeches. Obviously, the model translates the text literally and does not – with only minor exceptions – omit important information (the total number of omissions 6), which is an appropriate strategy for written translation (of written texts), but for live transcript translation, i.e., speech translation, this completeness can be considered counterproductive as it overwhelms the recipient (subtitles are read more slowly than speech is listened to).

Table 7: The number of translation departures in both speeches interpreted by a professional (A), by trainee interpreters (B), and by ASST model (C).

	A Professional interpreter			B Trainee interpreters			C ASST		
	Speech 1	Speech 2	Total	Speech 1	Speech 2	Total	Speech 1	Speech 2	Total
O1	12	7	19	22	13	35	2	1	3
O2	3	8	11	13	18	31	2	0	2
O3	0	3	3	1	0	1	1	0	1
A1	1	1	2	1	0	1	0	0	0
A2	0	0	0	0	0	0	0	0	0
A3	0	0	0	0	0	0	0	0	0
S1	3	7	10	15	13	28	5	20	25
S2	10	13	23	2	4	6	9	3	12
S3	5	6	11	0	1	1	6	0	6
F	52	42	94	7	18	25	34	16	50
G	14	12	26	2	0	2	24	5	29

We could see from all our analyses, interpreting is a complex task that requires not only linguistic expertise but also cultural sensitivity, pragmatic thinking, and quick decision-making ability. In general, professional interpreters and trainee interpreters tend to differ in their level of experience and ability to convey messages accurately and efficiently. Compared to human interpretation, whose output is the spoken word, the ASST model creates a translated transcription of the original.

To sum up our findings, our analysis revealed that the professional interpreter tended to copy the linguistic structure of the original message, resulting in an

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overuse of demonstrative pronouns and a lack of coherence. This led to a very redundant and ineffective language that was difficult for the audience to follow. Moreover, the interpreter was prone to interferences, which further compromized the accuracy of the interpretation. This led to confusion and misinterpretation of important information, which had a negative impact on the overall quality of the interpretation.

On the other hand, the trainee interpreters demonstrated a good understanding of the original message. They were able to filter the (relevant) information correctly and make functional generalizations and omissions to convey the intended meaning more efficiently. The trainee interpreters also completed their sentences, which helped maintain the coherence of the interpretation.

Nevertheless, to generalize the results of the analysis, we would obviously need a larger sample of interpreters. In our case, the professional interpreter appeared to lag behind the students.

To sum up, our findings suggest that while experience is important, the ability to analyse the original message and convey it accurately and efficiently is essential in interpreting. For individuals who do not understand the source language, relying on the services of a professional interpreter can be a necessary and even pleasant experience. However, the quality of the interpretation depends greatly on the interpreter's proficiency and level of experience. Both the professional and trainee interpreters may benefit from further training to improve their skills in these areas.

Regarding the ASST model, our analysis showed that the transfer of information is complete, almost literal: for long and complex sentences, it tends to copy the linguistic structure of the original message, which leads to redundant expressions that are incomprehensible and confusing in some passages for readers and would likely be incomprehensible for listeners if the output was actually spoken. Simple sentences are translated adequately. However, the ASST model is not able to use appropriate strategies, e.g. filtering of redundant information, corrections of slips of the tongue, etc. It converts lexical resources regardless of the context, mostly choosing the first dictionary meaning, which does not always work in a given sentence. Metaphors and established connections are translated literally, so the particular part of the translation does not make sense.

In our analysis, we have highlighted in particular those translation departures that can be perceived as inappropriate, but we would like to mention that the overall quality of the ASST outputs varies. Kocmi et al. (2021: 160) mention in this vein that automated systems are prone to "catastrophic errors".

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5 Discussion and future work

The production of human SI is a complex and multifaceted task that requires careful consideration of various criteria. In our study, we explored the differences between human SI and ASST and, specifically, we examined whether all relevant information was adequately interpreted and what strategies the interpreters used, we also assessed the linguistic competence of all three interpreters.

One of the challenges in measuring the similarity between human SI and ASST is the focus of automatic metrics on literal match, which is not always relevant for interpreting. Unlike translation, interpreting involves a more significant amount of interpreting strategies and pragmatic adaptation to the context and audience, which results in a greater variety of interpretations for a given source message.

We are aware of the fact that by including a diverse group of interpreters and assessors, we will be able to obtain a more comprehensive understanding of the quality of human SI in the future. In addition, we suggest that future studies include more speeches interpreted by both student and professional interpreters to increase the diversity of interpretations and assessors and to get more linguistic material that could be analyzed to identify specific linguistic solutions and interpreting strategies. The ASST system could then learn from such a corpus. This will allow us to obtain more comparable results and identify the factors that contribute to high-quality content in human SI.

The sound quality of the original speech is one of the input factors that affect interpreting performance, so the quality of the interpretation also depends on the sound quality of the original. Our analysis showed that the ASST system also needs the original speech to have good sound quality, as it does not work with context and confuses sound-alike words (the ASST system confused the word *class* with the word *glass* and used the term *sklo*, i.e., *glass*, instead of *type*. EN original: *And we try to compare the municipalities with the* class *of municipalities with the same size.* – ASST system: *A snažíme se srovnávat obce se* sklem *obcí se stejnou velikostí.*)

The ASST system is very literal and does not recognize metaphor or idioms, so there are quite serious content errors in its interpretation, because the interpretation is unintelligible to the listener and the sentences often do not make sense. (Instead of *it's your turn*, the system ASST translated *it's your floor(ing)*: EN original: *Mister Maybeurle,* the floor is yours. *Thank you.* – ASST system: *Pane Mayberle,* podlaha *je vaše. Děkuji.*)

Furthermore, the ASST system processes grammatical structures and collocations "as they are", so that unusual phrases arise in the target language, for example: EN original: *Ministry of Defence and Ministry of Security and Justice* had

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spent no time or money *on publication of public data.* – ASST system: *A ministerstvo obrany a ministerstvo bezpečnosti a spravedlnosti* nestrávily (lit. have not digested; sic!) čas ani peníze *na zveřejňování otevřených údajů.*

Another area for future research is a deeper and broader exploration of the differences in types of translation departures in human SI and ASST. This will enable us to identify the areas where ASST falls short and develop strategies to improve the accuracy and effectiveness of "machine SI". In other words, one of the open questions that could be answered in the future is if the integration of linguistic nuances and human-like strategies can amplify the usefulness of ASST.

Finally, we also need to consider the definition of a professional interpreter. While this term is often used to describe individuals with significant experience and training in interpreting, the criteria for defining a professional interpreter may vary depending on the context and requirements of the interpreting task. The scale of experience (number of years of experience) is somewhat misleading, it is not clear how often and how many hours the interpreter actually interprets and what is their exposure to the specific topic of the given task. The fact that the interpreter has many years of experience also means that he studied many years ago, when the learning of interpreting did not allow for frequent contact with native speakers or foreign internships as is the case today. Nowadays, students enrolled in interpreting programmes at the Prague Institute of Translation Studies (Charles University) benefit from a wealth of opportunities to speak and learn English, resulting in outstanding proficiency levels. Throughout their academic journey, they dedicate numerous hours to interpretation exercises (in most study plans today there are mandatory self-study seminars), amassing a depth of experience that far surpasses what was typical just two decades ago.

Considering all this, our study highlights the importance of evaluating human SI and ASST, too, based on a variety of criteria and using diverse assessors to obtain a more comprehensive understanding of its quality. Of course, different conditions apply to written translation (of written texts) than to live transcript translation, i.e., speech translation, because listening to the spoken word and reading written text are different cognitive processes. However, by addressing the challenges in measuring the similarity between human SI and ASST, we can develop strategies to improve the accuracy and effectiveness of ASST or even automated interpreting in the future.

6 Research limitations

This study has several limitations that should be acknowledged.

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It is clear from the model we have chosen that the assessment of some errors and segments must have been subjective, which is also acknowledged by Barik (1971) in his experiment. We emphasise that not all enumerated translation departures can always be classified as errors. They may be conscious departures that the interpreter decided to make as part of his/her interpreting strategy. Therefore, in the empirical part of our research we also highlighted the interpreting strategies the interpreters used to cope with the original speech. Furthermore, we are aware that there may have been inaccurate or overly subjective judgements made by the evaluators in the analysis.

The interpreters did not have the slides or the entire speech, only the topic and key words in advance. In the real world, it also happens that the interpreters do not get support/reference materials in advance, not even the words, they receive just the topic of the speech. We are aware of the fact that this may be considered contrary to good practice in professional settings: codes of ethics and recommended workflows require clients to provide professional interpreters with support/reference materials in advance. However, we wanted to simulate the usual course of interpreting in this respect, so the interpreters had to work under particularly harsh conditions.

Another point concerning the limits of our research is the fact that two interpretations by only one professional interpreter questions the representativeness of the research.

As mentioned in our study, quality in SI is measured in a very different manner (effect on listener, pragmatics, communication goal, acoustic features, impact, usefulness, adjustment to type of event) when compared to the present study.

Grammatical (G) and formal (F) errors were surprisingly predominant in the professional interpretation: this finding shows that the interpreter was not a "professional" one in the end. In the same vein, the condensation strategy (observed with trainee interpreters but lacking with the "professional" one) is yet another signal of this fact. If an interpreter claims to be professional, it doesn't necessarily mean anything. This issue needs to be addressed in the future. Furthermore, the translation system can't, by definition, produce "formal" errors as established here, so comparability of results in the present research design must be viewed with this in mind.

7 Conclusion

In this study, we compared the performance of a professional interpreter and two trainee interpreters with ASST outputs. As for the comparison of a professional

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interpreter and trainee interpreters, our findings suggest that, while experience is important in interpreting, analysis of the original speech is essential for delivering an effective interpretation. After comparing human SI and ASST, we confirm the expected difference in the nature of departures and errors. ASST struggles with difficult sound, pronunciation or syntactic conditions, leading to unnatural errors. At the same time, the processing capacity is technically higher than that of humans – given the limited task of translating words, not delving into the actual meaning of the message – and ASST is thus able to provide more detailed and literal translation. Looking at errors and translation departures only, ASST seems to perform better (fewer errors, and/or repeated errors where multiple instances "could be fixed at once"), but we are aware of the fact that the detailed and more literal output may not be better for the end user.

In conclusion, our study examined one of the aspects of quality in interpreting for SI as well as in real-time machine translation using Barik's and Kopczyński's taxonomy. This taxonomy of translation departures we used in our analysis is very strict, so that all the interpreting performances analyzed in our experiment appear to fall short of providing a correct rendering according to the assigned scores. However, it should be emphasized that the deficiencies marked in subcategories S1 and O1 are to be considered as conscious and adequate strategies from the interpreter's point of view, so the interpreter made the right decision to commit these departures. Our results revealed that human SI can be of poor quality, as indicated by numerous minus points. However, we also admit that various strategies can be employed to ensure that the interpretation meets the client's expectations, but the client's satisfaction was not the subject of our research.

Similarly, the difference in the time interpreters have for interpreting and the time we have as evaluators for evaluating was not the subject of our research. While we assessed the interpreting after the performance in sufficient ("infinite") time, interpreters typically have only a limited time to convey the message. Due to the time constraint, they are under more pressure, which of course can affect their decision-making process and can lead to departures and inaccuracies in the interpretation. These deficiencies can then be revealed only by careful analysis over a sufficient amount of time. Interpreting, unlike translation, is not primarily based on linguistic competence and the ability to stylize the text. The average listener (who is not a language expert) tolerates exaggerations, stylistic flaws or grammatical errors, moreover, he or she often does not even notice these deficiencies. If the transcripts were evaluated by multiple evaluators, primarily untrained laypersons without a background in linguistics, most of the errors in categories F and G would not be marked at all.

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To address these issues, further research is needed to expand the methods for assessing the quality of human interpretation in comparison with real time machine translation. By evaluating interpreting in a variety of contexts and with a larger sample of assessing subjects and along more criteria, we can gain a more comprehensive understanding of the factors that contribute to high-quality interpreting and develop strategies to improve the accuracy and effectiveness of human interpretation and real time machine translation.

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Abbreviations

ASR	automatic speech recognition
ASST	automatic simultaneous speech translation
RSI	remote simultaneous interpreting
SI	simultaneous interpreting
ÚFAL	Ústav formální a aplikované lingvistiky Matematicko-fyzikální fakulty Univerzity Karlovy (Institute of Formal and Applied Linguistics, Faculty of Mathematics and Physics, Charles University)
ÚTRL	Ústav translatologie Filozofické fakulty Univerzity Karlovy (Institute of Translation Studies, Faculty of Arts, Charles University)

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